

# Self-consistent Spatially-Resolved Cluster Dynamics of Irradiated Zirconium

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## Abstract

We present a self-consistent, spatially-resolved cluster-dynamics (SRCD) formulation for the irradiated microstructure of  $\alpha$ -zirconium, the construction by which its continuum fields are converted into a discrete dislocation population, and the two-time-scale method by which the whole is integrated. The formulation is self-consistent in a specific and restrictive sense: every capture efficiency that appears in a loop growth law, and every sink strength that the same loops impose on the mobile species, is generated from the *same* diffusion tensors that transport those species, rather than from an independent set of phenomenological bias factors. The immobile population is expanded from the lumped, stress-aligned pair of the earlier mean-field model to nine crystallographically resolved families: the three prismatic variants  $a_1$ ,  $a_2$ ,  $a_3$  of both interstitial and vacancy character, and a three-state basal vacancy chain — stacking-fault pyramid, faulted loop, perfect loop — whose two links are size-gated activated transfers that conserve vacancies exactly. The expansion is motivated by the experimental loop character — prismatic  $\langle a \rangle$  loops of both signs, and basal  $c$ -component vacancy loops that are predominantly *faulted* and that form by collapse of a pyramidal precursor rather than by accretion — and by the growth analysis, which shows that diffusional-anisotropy difference (DAD) opens a finite window of the mobile flux ratio  $x = \Phi_V/\Phi_I$  over which prismatic interstitial and basal vacancy loops grow together, but is degenerate in loop *character* and so cannot by itself support the observed coexistence of interstitial and vacancy loops on the same prismatic variant. Two resolutions of that degeneracy are carried: the promotion of  $x$  to a space-time field, which a spatially resolved model provides natively, and the character-resolved capture of March-Rico, Blondel and Wirth, which enters as a re-parameterization of the capture efficiencies and not as a structural change. Loop annealing is not represented by a fitted lifetime. Loops exchange vacancies across their own perimeter at a rate set by a size-dependent binding energy computed from the anisotropic loop self-energy, one expression serving both characters, so that the same mechanism shrinks a vacancy loop and grows an interstitial one and the classical climb threshold emerges rather than being imposed. Each family carries a second moment of its size distribution, closed on a log-normal built from the ratio of content to density and from the dispersion; this reduces every closure integral to a single exponent, makes the transfer gates and the dissolution current computable instead of assumed, and is what allows an annealing calculation to lose loops while its surviving mean size rises. The discrete handoff draws each loop's size from that same distribution and gives it an equal-area elliptical outline with the character- and size-dependent ellipticity the experiments report, so that stored defects are conserved exactly while line length, Burgers vector and dispersion are each accounted rather than assumed equal. The mobile species obey a steady anisotropic reaction-diffusion problem; the immobile families obey local ordinary differential equations at every finite-element node. The two are advanced by a Lie-Trotter operator split in which the mobile problem is solved to quasi-steady state on the mesh and the immobile problem is then integrated pointwise with the mobile species frozen. We give

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the split, its cost, its first-order error in the fast-solve cadence, and the open-system atom balance that replaces the closed 0D balance once the domain has absorbing surfaces. A thermal-annealing section evaluates the binding energy of every loop variant from the anisotropic self-energy, the fault energy and the applied stress tensor, integrates an isolated basal loop through to extinction, and sets out the two constructions by which a discrete population obtained at a chosen dose is annealed on the volume it came from. An appendix sets out the implementation path from the present code to this formulation, as an ordered sequence of steps each of which reproduces its predecessor exactly at a stated parameter value.

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# 1 Introduction

## 1.1 Scope

The first deliverable under the current SOW is Deliverable D1: *Multiscale Simulation Methodology*. The objective is to develop and extend a Spatially Resolved Cluster Dynamics (SRCD)–Discrete Dislocation Dynamics (DDD) framework for the nucleation and growth of irradiation-induced defect species in zirconium. The framework generates continuous spatial fields of defect-species concentration as functions of temperature and irradiation dose, with model parameters calibrated against representative defect species and behaviors observed in zirconium and its alloys. In milestone D1/M1, *Simulation of Initial Irradiation Damage*, those spatially resolved simulations predict microstructural evolution — nucleation and growth — and their outputs are continuous spatial fields of cluster concentration at prescribed temperatures and doses, to be validated against experiment across a range of irradiation conditions. This report documents the formulation and the solution method underlying that milestone.

Two things distinguish the model described here from its predecessor. The first is the *immobile population*: where the earlier mean-field model carried four lumped loop families, split phenomenologically into populations “aligned” and “non-aligned” with an applied stress, the present formulation carries the crystallographic families themselves — three prismatic variants of each character, and a three-state basal vacancy chain running from the stacking-fault pyramid through the faulted loop to the perfect loop. The second is *self-consistency*: the capture efficiencies that drive loop growth and the sink strengths those loops impose on the mobile species are generated from the same anisotropic diffusion tensors that transport the mobile species, so a single set of transport parameters governs both sides of the coupling. [Sections 1.2](#) and [1.3](#) give the experimental and the theoretical reasons for both changes.

## 1.2 Loop character in irradiated zirconium

At reactor-relevant temperatures zirconium is hexagonal close packed, with

$$a \simeq 0.323 \text{ nm}, \quad c \simeq 0.515 \text{ nm}, \quad c/a \simeq 1.5944, \quad (1)$$

that is, *below* the ideal ratio  $\sqrt{8/3} \simeq 1.6330$ . The deficit is what makes the prismatic rather than the basal planes the close-packed-competitive ones in  $\alpha$ -Zr, and it is consistent with prismatic loops being the population that forms first. The atomic volume follows from the same two constants,

$$\Omega = \frac{\sqrt{3}}{4} a^2 c = 2.3266 \times 10^{-29} \text{ m}^3 \quad \text{per atom}, \quad (2)$$

in agreement to 0.14% with the mass-density route  $M_{\text{Zr}}/(\rho N_A)$ . Neither  $c/a$  nor  $\Omega$  may be idealized anywhere in the model: consistency between the lattice basis, the loop Burgers vectors and  $\Omega$  is a requirement rather than a convenience, because every number density is  $C/\Omega$ , every line density is  $2\pi r C/\Omega$ , and every loop radius prefactor is  $\sqrt{\Omega/\pi b}$ .

Irradiation of  $\alpha$ -Zr by neutrons, protons, electrons and heavy ions produces two principal loop families [\[1, 2, 3, 4, 5\]](#).

**Prismatic  $\langle a \rangle$  loops.** These dominate at low and intermediate exposure. Their Burgers vector is

$$\mathbf{b}_a = \frac{1}{3} \langle 11\bar{2}0 \rangle, \quad |\mathbf{b}_a| = a \simeq 0.323 \text{ nm}, \quad (3)$$

they carry no  $c$  component, and they occur with *both* interstitial and vacancy character. The habit plane is a first- or second-order prismatic plane,  $\{10\bar{1}0\}$  or  $\{11\bar{2}0\}$ ; measured loop normals scatter about the ideal prism orientations and may be tilted by up to roughly  $30^\circ$ , so many resolved loops are not perfect edge loops. Below approximately 573 K the resolved  $\langle a \rangle$  population is generally interstitial-dominated, but mixed populations are well documented; near 623–673 K the vacancy fraction becomes comparable, and it increases with temperature and with prior deformation [\[1, 6, 3, 7\]](#). Total  $\langle a \rangle$  densities are commonly of order  $10^{22} \text{ m}^{-3}$  at diameters of 5–20 nm.

The  $\langle a \rangle$  shape is elliptical, with the major axis close to  $[0001]$  and the minor axis the basal direction contained in the habit plane. Writing  $e = 1 - d_{\min}/d_{\max}$ , the classic TEM measurements on neutron-irradiated high-purity Zr give  $e \simeq 0.1$  for large interstitial loops and  $e \simeq 0.4$  for large vacancy loops

[1]. Static energetics accounts for part of this and not for the difference between the two characters: orientation-dependent line energy  $\lambda(\xi)$  with  $\lambda_a > \lambda_c$  favors elongation along [0001], and atomistic calculations give  $\lambda_a/\lambda_c \simeq 1.1\text{--}1.3$  with ellipticity increasing with loop size — but they predict *similar* optimum ellipticity for both characters [8]. The difference is kinetic. Because the interstitial diffusion tensor in hcp Zr is far more anisotropic than the vacancy tensor, differently oriented segments of one prismatic loop do not receive identical point-defect fluxes, and Woo’s diffusional-anisotropy-difference (DAD) treatment predicts that this contribution acts in *opposite* directions on the two characters — amplifying vacancy-loop ellipticity and suppressing interstitial-loop ellipticity [9]. That combination reconciles  $e_{\text{vac}} \simeq 0.4$  with  $e_{\text{int}} \simeq 0.1$ , and it is the first appearance in this document of the mechanism that governs everything below: the same anisotropy difference that transports the defects also decides how they are partitioned among sinks.

**Basal  $c$ -component vacancy loops.** At higher exposure, vacancy loops with a  $c$  component appear on basal planes, and their emergence correlates with accelerated or “breakaway” irradiation growth [10, 3, 11, 12]. Three crystallographic states must be distinguished:

$$\mathbf{b}_{c/2} = \frac{1}{2}[0001], \quad |\mathbf{b}_{c/2}| = c/2 \simeq 0.257 \text{ nm}, \quad (4)$$

$$\mathbf{b}_{c/2+p} = \frac{1}{6}\langle 20\bar{2}3 \rangle, \quad |\mathbf{b}_{c/2+p}| = \sqrt{\frac{a^2}{3} + \frac{c^2}{4}} \simeq 0.318 \text{ nm}, \quad (5)$$

$$\mathbf{b}_c = [0001], \quad |\mathbf{b}_c| = c \simeq 0.515 \text{ nm}. \quad (6)$$

All three share the habit plane (0001) and the normal  $\mathbf{n} \parallel [0001]$ , and all three are vacancy in character. What separates them is the stacking fault. A basal vacancy platelet that removes one (0001) layer and collapses gives  $\frac{1}{2}[0001]$ , which is *not* a lattice translation of the hcp structure — half of [0001] carries an  $A$  layer onto a  $B$  position — so that loop bounds a high-energy basal fault. Shear by a basal partial removes most of the penalty,

$$\frac{1}{2}[0001] + \frac{1}{3}\langle 1\bar{1}00 \rangle = \frac{1}{6}\langle 20\bar{2}3 \rangle, \quad (7)$$

and removing a second layer — directly, or by the reaction of two faulted loops whose partials cancel — gives [0001], a lattice translation, so the fault vanishes and the loop is perfect.

Which of the three is observed follows from a competition between an energy that scales with the faulted *area* and one that scales with the loop *perimeter*. The perimeter term must not be taken from isotropic elasticity here.  $\alpha$ -Zr is elastically anisotropic —  $C_{33}/C_{11} = 1.15$  and  $2C_{44}/(C_{11} - C_{12}) = 0.91$  at 300 K [13] — and, more to the point, a basal  $\langle c \rangle$  loop and a prismatic  $\langle a \rangle$  loop present quite different line orientations to that anisotropy: every line element of a basal loop lies in the basal plane, whereas a prismatic loop runs from [0001] round to a basal direction and back. One isotropic prefactor  $\mu/(1 - \nu)$  cannot serve both. Writing the self-energy of a planar loop of radius  $R$  and Burgers vector  $\mathbf{b}_j$  with the anisotropic prelogarithmic energy factor averaged around its perimeter,

$$E_j(R) \simeq \underbrace{\gamma_j \pi R^2}_{\text{fault}} + \underbrace{\frac{\langle K \rangle_j |\mathbf{b}_j|^2 R}{2} \ln \frac{\alpha R}{|\mathbf{b}_j|}}_{\text{elastic self-energy}}, \quad \langle K \rangle_j = \frac{1}{2\pi} \int_0^{2\pi} K(\boldsymbol{\xi}_j(\theta), \mathbf{b}_j) d\theta, \quad (8)$$

with  $j \in \{a, c/2, c/2+p, c\}$ ,  $\gamma_a = \gamma_c = 0$  for the two unfaulted states,  $\boldsymbol{\xi}_j(\theta)$  the line tangent at angular position  $\theta$  on the loop, and  $\alpha$  absorbing the core cutoff. The average is taken because the character of the line, and its orientation relative to [0001], both change with  $\theta$ . The energy factor  $K$  is that of the Stroh sextic theory, in the Barnett–Lothe integral form [14, 15, 16]

$$K(\boldsymbol{\xi}, \mathbf{b}) = \frac{2\pi \mathbf{b} \cdot \mathbf{B}(\boldsymbol{\xi}) \cdot \mathbf{b}}{|\mathbf{b}|^2}, \quad \mathbf{B}(\boldsymbol{\xi}) = \frac{1}{4\pi^2} \int_0^{2\pi} [(\mathbf{m}\mathbf{m}) - (\mathbf{m}\mathbf{n})(\mathbf{n}\mathbf{n})^{-1}(\mathbf{n}\mathbf{m})] d\omega, \quad (9)$$

where  $(\mathbf{a}\mathbf{b})_{jk} = a_i C_{ijkl} b_l$  and  $\mathbf{m}(\omega), \mathbf{n}(\omega)$  is an orthonormal pair rotating about  $\boldsymbol{\xi}$  in the plane normal to it. Isotropy returns  $K = \mu$  for a screw segment and  $K = \mu/(1 - \nu)$  for an edge segment, so Eq. (8) reduces to the form usually quoted.

Hexagonal symmetry makes the factors needed here closed-form. With  $\bar{C}_{11} = (C_{11}C_{33})^{1/2}$ ,

$$K_c^\perp = (\bar{C}_{11} + C_{13}) \left[ \frac{C_{44}(\bar{C}_{11} - C_{13})}{C_{11}(\bar{C}_{11} + C_{13} + 2C_{44})} \right]^{1/2}, \quad \boldsymbol{\xi} \perp [0001], \boldsymbol{b} \parallel [0001], \quad (10)$$

$$K_a^\perp = (\bar{C}_{11} + C_{13}) \left[ \frac{C_{44}(\bar{C}_{11} - C_{13})}{C_{33}(\bar{C}_{11} + C_{13} + 2C_{44})} \right]^{1/2}, \quad \boldsymbol{\xi} \perp [0001], \boldsymbol{b} \perp \boldsymbol{\xi} \text{ basal}, \quad (11)$$

$$K_a^\parallel = \frac{C_{11}^2 - C_{12}^2}{2C_{11}}, \quad \boldsymbol{\xi} \parallel [0001], \boldsymbol{b} \perp [0001], \quad (12)$$

and the basal screw factor is  $\sqrt{C_{44}(C_{11} - C_{12})/2}$ . The first of these is the whole of the  $\langle c \rangle$  case: a basal loop with  $\boldsymbol{b} \parallel [0001]$  is pure edge at every point of its perimeter, and because the hexagonal elastic tensor is transversely isotropic its energy factor does not vary around the loop at all. Table 1 evaluates  $\langle K \rangle$  for all four families by quadrature of Eq. (9); the closed forms above are recovered to six figures, and the isotropic limits are the check on the normalization.

Table 1: Perimeter-averaged anisotropic energy factors of the  $\langle a \rangle$  and  $\langle c \rangle$  families, from Eq. (9) with the single-crystal constants of Fisher and Renken [13] at 300 K ( $C_{11} = 143.4$ ,  $C_{12} = 72.8$ ,  $C_{13} = 65.3$ ,  $C_{33} = 164.8$ ,  $C_{44} = 32.0$  GPa). “Range” is the variation of  $K$  around one loop, and is exactly zero for a basal loop of pure  $c$  character. The last column is the elastic weight that enters Eq. (8); it, and not  $|\boldsymbol{b}|^2$ , is what orders the states.

| Family                               | $\boldsymbol{b}$                 | $ \boldsymbol{b} $ (nm) | $\langle K \rangle$ (GPa) | Range (GPa) | $\langle K \rangle  \boldsymbol{b} ^2$ (GPa nm <sup>2</sup> ) |
|--------------------------------------|----------------------------------|-------------------------|---------------------------|-------------|---------------------------------------------------------------|
| $\langle a \rangle$ prismatic        | $1/3 \langle 11\bar{2}0 \rangle$ | 0.323                   | 53.1                      | 52.7–53.9   | 5.55                                                          |
| $\langle c \rangle$ faulted, $c/2$   | $1/2 [0001]$                     | 0.257                   | 57.8                      | —           | 3.83                                                          |
| $\langle c \rangle$ faulted, $c/2+p$ | $1/6 \langle 20\bar{2}3 \rangle$ | 0.318                   | 53.0                      | 49.5–56.5   | 5.36                                                          |
| $\langle c \rangle$ perfect          | $[0001]$                         | 0.515                   | 57.8                      | —           | 15.34                                                         |

Three consequences follow, of which the last is the one that acts on the fault competition.

1. **One averaged factor is adequate for the prismatic family and is not an approximation at all for the two pure- $c$  basal states.**  $K$  is constant around a basal loop with  $\boldsymbol{b} \parallel [0001]$  and varies by 2% around a prismatic  $\langle a \rangle$  loop. Only the mixed  $c/2 + p$  state, whose Burgers vector carries a basal component that runs from edge to screw as  $\theta$  advances, varies appreciably, 49.5–56.5 GPa.
2. **The isotropic prefactor is right for  $\langle a \rangle$  by coincidence and 7% low for  $\langle c \rangle$ .** With the polycrystalline averages  $\mu = 36$  GPa and  $\nu = 0.33$ ,  $\mu/(1 - \nu) = 53.7$  GPa, within 1% of  $\langle K \rangle_a$  but below  $K_c^\perp = 57.8$  GPa. Isotropic elasticity therefore understates basal loop energy *relative to* prismatic, and biases every energy comparison between the two families in the same direction.
3. **Anisotropy lowers the elastic cost of unfauling by 8%.** The  $c/2 + p$  state costs 1.40 times the  $c/2$  state in  $\langle K \rangle |\boldsymbol{b}|^2$  where  $|\boldsymbol{b}|^2$  alone gives 1.52, because its basal partial component samples the softer basal factors; the perfect loop remains exactly four times  $c/2$ , since the two share  $K$  and differ by a factor of two in  $|\boldsymbol{b}|$ .

The two terms of Eq. (8) order the states in opposite directions: the elastic term favors the smallest  $\langle K \rangle |\boldsymbol{b}|^2$ , which is  $c/2$ , while the fault term favors the perfect loop, whose elastic weight is four times that of  $c/2$ . Because the fault term grows as  $R^2$  and the elastic term only as  $R \ln R$ , unfauling becomes favorable above a critical radius, and it is the *intermediate* state  $c/2 + p$  — a small elastic weight together with an already reduced fault energy — that wins over the size range TEM resolves, the more so once the anisotropy of item 3 is accounted for. Table 2 collects the outcome. **The dominant observed basal loop is faulted, not perfect**, and being perfect is not by itself an advantage [17, 8].

Two consequences of Table 2 are load-bearing for the formulation. First, the quantity that converts loop area into stored defects is not  $|\boldsymbol{b}|$  but the edge component  $\boldsymbol{b} \cdot \boldsymbol{n}$ : the  $c/2 + p$  loop is not a pure edge loop, because its Burgers vector contains the basal partial of Eq. (7), so its vacancy content is that of

Table 2: Fault state of the three basal  $c$ -component loop types. All share the habit plane (0001) and the normal  $\mathbf{n} \parallel [0001]$ ; all are vacancy in character. The last column is the edge component that converts loop area into stored vacancies.

| State   | $\mathbf{b}$                            | Fault             | Experimental status                                              | $\mathbf{b} \cdot \mathbf{n}$ |
|---------|-----------------------------------------|-------------------|------------------------------------------------------------------|-------------------------------|
| $c/2$   | $\frac{1}{2}[0001]$                     | high-energy       | Reported; a precursor rather than a stable population.           | $c/2$                         |
| $c/2+p$ | $\frac{1}{6}\langle 20\bar{2}3 \rangle$ | <b>low-energy</b> | <b>The dominant experimentally observed <math>c</math> loop.</b> | $c/2$                         |
| $c$     | $[0001]$                                | <b>none</b>       | Observed, but relatively rarely identified.                      | $c$                           |

its  $c/2$  part alone. Second, the perfect state stores *twice* the vacancies per unit area that the faulted state does, which is why the two cannot be lumped into one family even though they share a habit plane. A model carrying a single basal family with  $\mathbf{b} \cdot \mathbf{n} = c/2$  — which is what the code implements today — represents the population that all three states share, namely the number of vacancies stored on basal planes, and does not distinguish among them.

**The basal stacking-fault pyramid, and why it precedes the loop.** The three states of Table 2 are all *planar*: each is what a basal vacancy platelet becomes once it has collapsed. Atomic-scale calculations of vacancy clustering in  $\alpha$ -Zr find that small vacancy clusters do not adopt that geometry at all. Below a crossover size the stable configuration is a compact three-dimensional cluster bounded by stacking faults: a pyramid, or bi-pyramid, *whose base rests on the basal plane* and carries the extrinsic basal fault of  $\frac{1}{2}[0001]$ , with prismatic facets closing it [18, 19]. Small basal vacancy clusters relax to these faulted pyramidal structures and become faulted  $\langle c \rangle$  loops only at larger sizes [19, 17], and Christiaen et al. draw the consequence that matters for the kinetics: a  $\langle c \rangle$  loop forms *not* by progressive accretion of vacancies onto a platelet but by *collapse of the pyramid once it exceeds a critical size* [18]. We call this configuration the basal *stacking-fault pyramid* (SFP). It is the hcp counterpart of the stacking-fault tetrahedron of fcc metals, and the energy competition that selects it has the same structure. Writing both energies for a cluster of  $m$  vacancies, the faulted area of either configuration is proportional to  $m$  and the energy of its bounding dislocation to  $\sqrt{m}$ , so

$$E_{\text{sfp}}(m) - E_{c_f}(m) \simeq \underbrace{(\Gamma_{\text{sfp}} - \Gamma_{c_f})m}_{\text{faulted area, } >0} - \underbrace{\Delta_{\text{el}}\sqrt{m} \ln \frac{\alpha\sqrt{m}}{|\mathbf{b}|}}_{\text{bounding dislocation, } >0}, \quad (13)$$

a difference that changes sign exactly once, at a crossover  $m^*$ . The pyramid wins below  $m^*$  because its edges are partials of small  $|\mathbf{b}|$  where the loop must carry  $\frac{1}{2}[0001]$ ; the loop wins above it because the pyramid folds more faulted area around the same stored vacancies ( $\Gamma_{\text{sfp}} > \Gamma_{c_f}$ ). Equation (13) is the size-selection statement that Eq. (8) is for the three planar states, and the two are independent: the first decides *whether* the cluster is planar, the second decides *which* planar state it is.

Experiment supports the same ordering. Liu, Beyerlein and Han resolved two-dimensional vacancy platelets in irradiated high-purity Zr and identified them as the precursors from which basal dislocation loops form, rather than as a separate terminal population [7]. The basal branch of the microstructure is therefore not a single population but a *chain*,

$$\text{SFP} \xrightarrow{E_{\text{col}}} c_f \xrightarrow{E_{\text{uf}}} c_p, \quad (14)$$

in which each step is an activated transition between states that all store vacancies on basal planes. **The chain is basal and has no prismatic counterpart.** The pyramid is built on a basal base and collapses to a basal loop, and on the prismatic branch there is nothing for it to be a precursor to: ab initio-parameterized formation energies place the prismatic vacancy  $\langle a \rangle$  loop *below* both the compact cavity and the faulted basal  $\langle c \rangle$  loop [17], so the planar prismatic loop is already the ground-state geometry rather than something a compact cluster must convert into. Prismatic vacancy loops are instead nucleated planar, by in-cascade vacancy clustering [20]. Section 2.11 draws the modeling consequence.

Two consequences make the distinction worth carrying as a separate species rather than as a small-size correction to  $c_f$ . First, the SFP is a route by which vacancies can be stored on basal planes *without* producing  $c$ -axis strain: its relaxation volume is close to isotropic, where a collapsed



platelet's is uniaxial along [0001], so vacancies held in pyramids contribute to the hydrostatic part of the eigenstrain and almost nothing to  $\varepsilon_c$ . Second, the conversion is barrier-limited, so the population spends a dose-dependent residence time in the pyramid state before any  $c$  loop exists at all. Together these offer a microstructural mechanism for the observation that  $c$  loops — and with them accelerated growth — appear only after an incubation exposure rather than from the first cascade [3, 10, 11]. That reading is a hypothesis this formulation makes testable, not an established attribution: [Section 2.8](#) carries the chain [Eq. \(14\)](#) explicitly so that the incubation it predicts can be compared against the measured onset dose.

Finally, basal loops are usually more nearly circular than prismatic ones, because the sixfold basal symmetry makes the in-plane line-energy variation small, and because [0001] is their *normal* rather than an in-plane axis. Large high-dose  $c$  loops need not remain circular — local stress, anisotropic segregation, coalescence, faceting and pinning all break the nominal symmetry, and strongly noncircular  $c$  loops are reported at high burnup — but no universal aspect ratio has emerged for them, and the prismatic ratios must not be transferred to them.

### 1.3 Coexistence, and why the formulation must be self-consistent

The experimental picture of [Section 1.2](#), whose full record is collected in [Appendix A](#), sets a problem that a mean-field model does not automatically solve. Define the mobile flux scales and their ratio,

$$\Phi_I = \sum_{n \in I_{\text{mob}}} n D_{I_n} C_{I_n}, \quad \Phi_V = \sum_{m \in V_{\text{mob}}} m D_{V_m} C_{V_m}, \quad x \equiv \frac{\Phi_V}{\Phi_I}, \quad (15)$$

which reduce to  $\Phi_I = D_i C_i$  and  $\Phi_V = D_v C_v$  in the monomer approximation. With  $Z_{s,m}$  the capture efficiency of sink family  $s$  for mobile species  $m$ , the net driving force on a loop of character sign  $\eta_s = \pm 1$  (+1 interstitial, -1 vacancy) is

$$F_s = \eta_s (Z_{s,I} \Phi_I - Z_{s,V} \Phi_V), \quad (16)$$

so that

$$s_i \text{L grows} \iff x < \frac{Z_{s,I}}{Z_{s,V}}, \quad s_v \text{L grows} \iff x > \frac{Z_{s,I}}{Z_{s,V}}. \quad (17)$$

Under identical local fluxes and identical capture coefficients, an interstitial and a vacancy loop of the same orientation therefore *cannot* both grow. Coexistence signals that at least one idealization has failed: homogeneity, identical capture coefficients, identical nucleation pathways, identical variants, or identical irradiation history.

**What DAD does, and where it stops.** Woo's DAD theory takes the tensors

$$\mathbf{D}_m = D_a^m (\mathbf{e}_1 \mathbf{e}_1 + \mathbf{e}_2 \mathbf{e}_2) + D_c^m \mathbf{e}_c \mathbf{e}_c, \quad p_m = \left( \frac{D_c^m}{D_a^m} \right)^{1/6}, \quad (18)$$

and assigns prismatic and basal sinks the orientation factors

$$Z_{a,m}^{\text{DAD}} = Z_m^0 \frac{p_m + p_m^{-2}}{2}, \quad Z_{c,m}^{\text{DAD}} = Z_m^0 p_m. \quad (19)$$

For basal-enhanced interstitial diffusion  $p_I < 1$ , and vacancies are usually near-isotropic,  $p_V \approx 1$ . The prismatic and basal thresholds are then  $x_a = Z_{a,I}^{\text{DAD}}/Z_{a,V}^{\text{DAD}}$  and  $x_c = Z_{c,I}^{\text{DAD}}/Z_{c,V}^{\text{DAD}}$ , and since  $(p + p^{-2})/2 > p$  for  $p < 1$  one has  $x_c < x_a$ : prismatic interstitial and basal vacancy loops grow together whenever

$$\boxed{x_c < x < x_a} \quad \text{with} \quad x_a - x_c = \frac{Z_I^0}{Z_V^0} \frac{1 - p_I^3}{2p_I^2}. \quad (20)$$

The window width is set *purely* by the anisotropy and vanishes as  $p_I \rightarrow 1$ . At 200 °C, with  $Z_I^0/Z_V^0 \approx 1.05$ –1.15 and  $p_I \approx 0.89$ –0.92 [9, 21, 22, 23], the window is open but narrow,  $x_c \approx 1.0 < x < x_a \approx 1.2$ . The elastic dislocation bias  $Z_I^0/Z_V^0 > 1$  shifts both thresholds together and opens nothing; what *positions*  $x$  inside the window is the production bias  $\eta = \varepsilon_V/\varepsilon_I \gtrsim 1$ , the excess of freely migrating vacancies left when more of the interstitial yield is locked into in-cascade clusters. A 0–20% excess suffices. DAD opens the interval; the production bias places the operating point inside it. Neither

substitutes for the other: with  $p_I = 1$  the thresholds coincide and no  $\eta$  can satisfy both conditions at once.

The limit of Eq. (20) is that it is an *orientation* statement. Equation (19) assigns one prismatic pair ( $Z_{a,I}^{\text{DAD}}, Z_{a,V}^{\text{DAD}}$ ) to interstitial and vacancy  $\langle a \rangle$  loops alike, so

$$\frac{Z_{aI,I}}{Z_{aI,V}} \xrightarrow{\text{DAD only}} \frac{Z_{aV,I}}{Z_{aV,V}}, \quad (21)$$

the prismatic window closes identically, and DAD is silent on the  $a_i\text{L}$ – $a_v\text{L}$  coexistence that Section 1.2 reports as an experimental fact. Two mechanisms remove that degeneracy, and the present formulation carries both.

**Resolution 1: the flux ratio is a field.** Promote  $x$  to  $x(\mathbf{r}, t) = \Phi_V(\mathbf{r}, t)/\Phi_I(\mathbf{r}, t)$ . Because interstitials migrate far faster than vacancies, the interstitial channel responds first and most sharply to any transient or gradient. In *time*, the nucleation transient — which in Zr can extend over of order 1 dpa — is strongly interstitial-rich, so  $x(t) \ll x_c$  and interstitial loops grow while vacancy loops merely nucleate and sit sub-critical; as the vacancy supersaturation builds,  $x(t)$  rises into  $(x_c, x_a)$  and the vacancy families begin to grow, with the interstitial loops established earlier persisting alongside them. In *space*, sinks — grain boundaries, free surfaces, precipitates, the dislocation network — make the mobile fields non-uniform, and because the interstitial diffusion length far exceeds the vacancy one, interstitials drain into the far field while vacancies pile up locally, so  $x(\mathbf{r})$  is depressed far from sinks and enhanced near them. A single specimen then contains regions on both sides of the prismatic threshold. This mechanism costs nothing to include here: **a spatially resolved formulation carries  $x(\mathbf{r}, t)$  natively**, and does not have to be told to. Its limitation is equally plain — it leaves  $Z_{aI,\cdot} = Z_{aV,\cdot}$  intact and buys coexistence by declining to evaluate the two growth conditions at the same point of space and time, so it predicts *segregated* coexistence and cannot give locally simultaneous growth in a homogeneous volume.

**Resolution 2: capture is character-resolved.** March-Rico, Blondel and Wirth [20] compute the defect-loop interaction energy atomistically and replace geometrical intersection by a drift-diffusion capture,

$$\mathbf{J}_\alpha = -D_\alpha \left[ \nabla C_\alpha + \frac{C_\alpha}{k_B T} \nabla E_{\text{int}}^{\alpha L} \right], \quad \alpha = i, v, \quad (22)$$

reduced in the rate equations to effective capture radii  $r_d$  entering additively in the reaction volume,

$$z_{ij}^S = 4\pi(r_i + r_j + r_d), \quad z_{ij}^T = \frac{4\pi^2(r_i + r_j + r_d)}{\ln[8r_i/(r_j + r_d)]}, \quad r_d^{\text{thermal}} = \delta_d \left( \frac{T_m}{T} \right)^{m_d}, \quad (23)$$

with  $T_m = 2128\text{K}$  the melting temperature of zirconium. *Spontaneous* capture radii are nearly the same for all loops and enlarge every reaction volume without separating characters; *thermal-drift* radii do separate them, because the dilatational field of an interstitial loop attracts interstitials over a longer range than it attracts vacancies, and conversely for a vacancy loop. The result is a **same-type capture bias**. Defining the character-splitting factor

$$\chi \equiv \frac{Z_{aI,I}/Z_{aI,V}}{Z_{aV,I}/Z_{aV,V}} = \frac{Z_{aI,I} Z_{aV,V}}{Z_{aI,V} Z_{aV,I}}, \quad (24)$$

the prismatic window  $Z_{aV,I}/Z_{aV,V} < x < Z_{aI,I}/Z_{aI,V}$  is non-empty if and only if  $\chi > 1$ , with logarithmic width  $\ln \chi$ . Equation (24) is the exact character analogue of the orientation width Eq. (20):  $\ln(x_a/x_c)$  measures how far DAD separates prismatic from basal, and  $\ln \chi$  how far thermal-drift capture separates interstitial from vacancy within the prismatic family. Because  $\chi > 1$  opens a genuine interval, a single uniform steady  $x$  satisfies both conditions at once, which is the stronger and the more falsifiable claim.

Two further ingredients are not optional. In-cascade vacancy clustering supplies the  $a_v\text{L}$  embryos, without which the capture bias has nothing to act on — homogeneous nucleation of a vacancy loop against the prevailing interstitial bias is prohibitively slow — and it is the *same* clustering asymmetry

that generates the production bias  $\eta$ , so the two must not be fitted as independent inputs. And cascade overlap reduces the effective generation rate at increasing dose,

$$\tilde{g}(\phi) = \varepsilon(\phi) g, \quad \varepsilon(\phi) = \frac{1 - \tanh(C_1\phi - C_2)}{2}(1 - C_3) + C_3, \quad (25)$$

with saturation in 0.01–0.1 dpa and a residual fraction  $C_3 \ll 1$ , which is what lets coarsening rather than unbounded nucleation dominate at high dose. Applied as a single scalar to both species, Eq. (25) rescales dose without moving  $x$ : it is a rate effect, not a threshold effect.

Table 3 summarizes the division of labor.

Table 3: What each ingredient does to the growth criterion. Only the last two separate loops of the same prismatic orientation; only character-resolved capture does so locally and instantaneously.

| Ingredient                        | Acts on                                      | What it does, and does not do                                                                                                                                                            |
|-----------------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Elastic bias $Z_I^0/Z_V^0$        | both thresholds together                     | Shifts $x_a$ and $x_c$ jointly; opens no window, and narrows the $a_iL-c_vL$ window by pushing $x_c$ toward unity.                                                                       |
| DAD, $p_I < 1$                    | orientation                                  | Opens the $a_iL-c_vL$ window of width Eq. (20). Cannot separate loops of one orientation, Eq. (21).                                                                                      |
| Production bias $\eta$            | position of $x$                              | Places $x$ inside whatever window exists. Creates no window; with $x_a = x_c$ it merely switches which family grows.                                                                     |
| Flux field $x(\mathbf{r}, t)$     | position of $x$ , resolved in space and time | Lets the two prismatic conditions be met at different doses or different places. Predicts <i>segregated</i> coexistence. Native to a spatially resolved model.                           |
| Thermal-drift capture, $\chi > 1$ | character                                    | Opens the prismatic window of logarithmic width $\ln \chi$ ; gives <i>local</i> simultaneous growth. Needs in-cascade clustering for the embryos, and still needs $x$ positioned inside. |

**Why this forces the formulation.** Three requirements follow, and together they are the content of Section 2.

1. **The immobile families must be crystallographic.** A scalar aligned/non-aligned split can express *how much* of a population is favored by a stress but not *which* prismatic system is, and it has no meaning at all under a multiaxial stress or a stress field. Resolving  $a_1$ ,  $a_2$ ,  $a_3$  makes the resolved stress on each habit plane a computed quantity. Carrying both characters on each variant is what makes Eq. (24) representable at all. Splitting the basal population into a faulted and a perfect state is forced by Table 2, since the two store different vacancy content per unit area.
2. **The capture efficiencies must come from the transport tensors.** A model that diffuses the mobile species with one anisotropy and biases the sinks with an independent phenomenological pair  $(\delta_i, \delta_v)$  is not merely unconstrained: it can hold simultaneously the two beliefs that diffusion is isotropic and that the sinks are biased by its anisotropy. Generating both from one  $p_m$  per species removes that possibility by construction, and it is the precise sense in which the present formulation is *self-consistent*. The phenomenological pair remains recoverable as the exact map

$$\delta_m = \frac{Z_{a,m}^{\text{DAD}} - Z_{c,m}^{\text{DAD}}}{Z_{a,m}^{\text{DAD}} + Z_{c,m}^{\text{DAD}}} = \frac{1 - p_m^3}{1 + 3p_m^3}, \quad (26)$$

which gives  $\delta_i \approx 0.07$ – $0.09$  and  $\delta_v \approx 0$  at the 200 °C anisotropies above. Note that  $Z_I^0/Z_V^0$  does not appear in  $\delta_m$  and must be carried separately, which is why Eq. (19) is the more faithful input.

3. **The mobile field must be spatially resolved.** Resolution 1 above is not an option available to a 0D model. It requires  $x(\mathbf{r}, t)$ , hence the mobile reaction–diffusion problem of Section 2.1 with real boundary conditions, and it is one of the two independent reasons for the spatial treatment — the other being that surfaces and grain boundaries are the dominant defect sink in any specimen small enough to be examined.

One caution, stated here so that it is not mistaken later for a prediction. The thermal-drift radii of Eq. (23) scale as  $(T_m/T)^{m_d}$ , so with  $m_d > 0$  the same-type bias, and hence  $\chi$ , is *strongest at low temperature*, while the observed vacancy  $\langle a \rangle$  fraction *increases* with temperature. These are not contradictory — the temperature dependence of the vacancy fraction is controlled largely by nucleation, by vacancy mobility, and by thermal emission from small vacancy clusters, none of which is contained in  $\chi$  — but reproducing the measured vacancy fraction against temperature is a genuine test of the combined model rather than something it obtains for free. A related hazard is double counting: the same-type capture bias *is* an elastic-interaction effect, and so is the conventional dislocation bias  $Z_I^0/Z_V^0 > 1$ . Only one of the two should carry the interstitial–vacancy asymmetry.

## 1.4 Outline

Section 2 gives the formulation: the mobile reaction–diffusion problem (Section 2.1), the reaction kinetics and cascade sources that both blocks share (Section 2.2), the expanded immobile family set with its growth, nucleation, coalescence and unfaulting laws (Section 2.3), and the size distribution that closes them (Section 2.9). Section 3 is the map from those fields to a discrete loop population — nodal volumes, loop counts, sizes drawn from the closed distribution, elliptical outlines, and what the domain refuses. Section 6 gives the solution method: the two-time-scale operator split, the quasi-steady mobile solve, the frozen-mobile pointwise immobile march, the splitting error and its control, the cost argument that makes of order  $10^6$  ordinary differential equations per substep tractable, and the open-system atom balance that replaces the closed 0D balance once the domain has absorbing surfaces. Five appendices follow. Appendix A collects the experimental record behind the coexistence of vacancy and interstitial  $\langle a \rangle$  loops; Appendix B gives the three theoretical results on capture bias that Section 1.3 takes as given; Appendix C is the record of what the code implements today against the formulation, including the two selectable loop models and the reconciliation of the 3D rate coefficients; and Appendix D is the step-by-step plan from that state to this formulation, with the regression and validation tests that make each step verifiable on its own.

# 2 The Self-consistent SRCD Formulation

The spatially resolved cluster-dynamics model represents the irradiation defect microstructure by continuum fields. It is called **reduced** because it resolves the concentration of a limited number of **mobile** defect clusters, each of fixed defect content, while clusters large enough to be **immobile** are described only by their number density and their total stored defect content, without an explicit size distribution. The mobile block is a steady anisotropic reaction–diffusion problem on the finite-element mesh (Section 2.1); the immobile block is a set of local ordinary differential equations, one system per node (Section 2.3). Section 2.2 collects the reaction kinetics, the cascade sources and the dislocation network that the two blocks share, so that each block can be read without reference to a mean-field appendix.

## 2.1 Mobile defect clusters

The mobile defect clusters are defined by the `mobileSpeciesVector`, a vector parameter set in the material file. We denote this vector  $\mathbf{m}_M$ ; its entries are the *signed* cluster sizes  $m^k$ , with the polarity  $p^k = m^k/|m^k| \in \{+1, -1\}$  selecting interstitial ( $p = +1$ ) or vacancy ( $p = -1$ ) character, and  $|m^k|$  the number of point defects in the cluster. For the example above  $\mathbf{m}_M = [-1, +1, +2, +3]^T$  and  $\mathbf{p}_M = [-1, +1, +1, +1]^T$ . As in Section 2.3, we use the boolean selectors  $\mathcal{M}_i = \frac{1}{2}(1 + \mathbf{p}_M)$  and  $\mathcal{M}_v = \frac{1}{2}(1 - \mathbf{p}_M)$ .

MoDELib2 solves the following steady-state reaction-diffusion partial differential equation (PDE) for the mobile content field  $\mathbf{c}_M = [c^v, c^i, c^{2i}, c^{3i}]^T$ , expressed in fractional (per-atom) units:

$$\mathbf{0} = -\nabla \cdot \mathbf{J} + \mathbf{P}_M + \frac{1}{2} (\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M + (\mathbf{E} - \mathbf{D}) \mathbf{c}_M + \mathbf{S}_{vL} \quad (27)$$

Throughout this work, we distinguish two kinds of non-scalar quantities by their typeface. Upright sans-serif bold denotes arrays in *species space* (indexed by the mobile defect species  $v, i, 2i, 3i$  or by loop family), with lowercase symbols for vectors, e.g. the concentration array  $\mathbf{c}_M$ , the stacked flux  $\mathbf{J}$ , and the vacancy-loop dissolution source  $\mathbf{S}_{vL}$  (Eq. 113), and uppercase symbols for matrices, e.g. the reaction matrix  $\mathbf{D}$ , the emission matrix  $\mathbf{E}$ , and the second-order reaction matrix  $\mathbf{R}_2$ . Bold-italic (e.g.

$\mathbf{J}^x$ ,  $\mathbf{D}^x$ ,  $\nabla$ ,  $\sigma$ ) is reserved for ordinary vectors and tensors in *Cartesian space*. Thus, the entries  $\mathbf{J}^x$  of the species-space flux array  $\mathbf{J}$  are themselves Cartesian vectors, and the divergence  $\nabla \cdot$  acts on each entry.

The stacked species flux collects the four Cartesian fluxes of the mobile species,

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}^v \\ \mathbf{J}^i \\ \mathbf{J}^{2i} \\ \mathbf{J}^{3i} \end{bmatrix}, \quad (28)$$

and the second-order reaction matrix, whose rows contract the concentration array with the per-species reaction matrices  $\mathbf{R}_2^x$  of Eq. (31), is

$$\mathbf{R}_2 = \begin{bmatrix} \mathbf{R}_2^v \\ \mathbf{R}_2^i \\ \mathbf{R}_2^{2i} \\ \mathbf{R}_2^{3i} \end{bmatrix}. \quad (29)$$

In Eq. (27) this stacked array enters through the contraction  $\frac{1}{2}(\mathbf{I} \otimes \mathbf{c}_M^T) \mathbf{R}_2 \mathbf{c}_M$ , where  $\mathbf{I}$  is the  $4 \times 4$  identity and  $\otimes$  the Kronecker product; its  $x$ -th entry is  $\frac{1}{2} \mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$ , so the term is a vector in species space.

Here  $\mathbf{J}^x = -\mathbf{D}^x \nabla c^x$  is the diffusive flux of species  $x$ , with a possibly-anisotropic diffusion tensor  $D_{ij}^x = D_{0ij}^x \exp(-E_{ij}^{xm}/k_B T)$  whose pre-exponential factors  $D_{0ij}^x$  and migration energies  $E_{ij}^{xm}$  are the material-file vectors `mobileSpeciesD0_SI` and `mobileSpeciesEnergyMigration_eV`. Consistent with Section 2.2, the scalar mobility of species  $x$  is the jump frequency  $\omega_x = z_c \nu_x \exp(-E_x^m/k_B T)$ , and the associated point-defect fluxes entering the loop coupling are  $\phi_v = \omega_v c^v$  and  $\phi_i = \omega_i(c^i + 2c^{2i} + 3c^{3i})$ .

The vector  $\mathbf{P}_M$  is the cascade production term,

$$\mathbf{P}_M = G \varepsilon \mathbf{f}, \quad (30)$$

where  $G$  is the nominal dpa rate (`doseRate_dpaPerSec`), and  $\varepsilon$  is the cascade surviving efficiency (`mobileSpeciesSurvivingEfficiency`), and  $\mathbf{f}$  the fraction vector partitioning cascade production among the mobile clusters. Expanded component-wise, and accounting for the per-cluster entry of the di- and tri-interstitial sources ( $G_{2i}/2$  and  $G_{3i}/3$ ),  $\mathbf{P}_M$  reproduces the cascade sources  $\{G_v, G_i, \frac{1}{2}G_{2i}, \frac{1}{3}G_{3i}\}$  of Section 2.2, with  $G_x = G \varepsilon_x$  the corresponding partition of the displacement rate.

The  $\mathbf{R}_2$ -term represents second-order reactions between mobile species, modeling the collision of two mobile species to form a third species. For the chosen vector of mobile species, we have

$$\begin{aligned} \mathbf{R}_2^v &= - \begin{bmatrix} 0 & \beta^{v,i} & \beta^{v,2i} & \beta^{v,3i} \\ \beta^{v,i} & 0 & 0 & 0 \\ \beta^{v,2i} & 0 & 0 & 0 \\ \beta^{v,3i} & 0 & 0 & 0 \end{bmatrix}, & \mathbf{R}_2^i &= - \begin{bmatrix} 0 & \beta^{v,i} & -\beta^{v,2i} & 0 \\ \beta^{v,i} & 2\beta^{i,i} & \beta^{i,2i} & \beta^{i,3i} \\ -\beta^{v,2i} & \beta^{i,2i} & 0 & 0 \\ 0 & \beta^{i,3i} & 0 & 0 \end{bmatrix}, \\ \mathbf{R}_2^{2i} &= -2 \begin{bmatrix} 0 & 0 & \beta^{v,2i} & -\beta^{v,3i} \\ 0 & -\beta^{i,i} & \beta^{i,2i} & 0 \\ \beta^{v,2i} & \beta^{i,2i} & 2\beta^{2i,2i} & \beta^{2i,3i} \\ -\beta^{v,3i} & 0 & \beta^{2i,3i} & 0 \end{bmatrix}, & \mathbf{R}_2^{3i} &= -3 \begin{bmatrix} 0 & 0 & 0 & \beta^{v,3i} \\ 0 & 0 & -\beta^{i,2i} & \beta^{i,3i} \\ 0 & -\beta^{i,2i} & 0 & \beta^{2i,3i} \\ \beta^{v,3i} & \beta^{i,3i} & \beta^{2i,3i} & 2\beta^{3i,3i} \end{bmatrix}, \end{aligned} \quad (31)$$

where  $\beta^{m,n}$  in Eq. (31) represents the reaction rates between species  $m$  and  $n$ . A positive reaction rate indicates production, while the negative sign means its consumption. The reaction rates are given by the equation

$$\beta^{m,n} = \frac{A^{m,n}}{\Omega |\mathcal{M}_m \mathcal{M}_n|} (\bar{D}^m + \bar{D}^n) (r^m + r^n), \quad (32)$$

The pairwise rate constants use the sum of the two reactant jump frequencies, consistently with the reaction rates  $R_{a,b} = (\omega_a + \omega_b) C_a C_b$  of Section 2.2,

$$\beta^{m,n} = (\omega_m + \omega_n), \quad (33)$$

with the recombination-volume factor  $\beta_r$  carried on the interstitial-vacancy channel ( $\beta^{v,i} \rightarrow \beta_r(\omega_i + \omega_v)$ ) and the combinatorial factors 2 and 4 retained on the like-species reactions ( $R_{i,i} = 2\omega_i C_i^2$ ,  $R_{2i,2i} = 4\omega_{2i} C_{2i}^2$ ). The capture radius  $r^m$  used to build  $\beta^{m,n}$  follows the point-defect / prismatic-loop geometry

$$r^m = \begin{cases} \left(\frac{3\Omega_0}{4\pi}\right)^{1/3}, & |m^m| = 1, \\ \left(\frac{|m^m|\Omega_0}{\pi b}\right)^{1/2}, & |m^m| > 1. \end{cases} \quad (34)$$

Expanding  $\frac{1}{2}\mathbf{c}_M^T \mathbf{R}_2^x \mathbf{c}_M$  component-wise reproduces the pair-reaction budget of each monomer/small-cluster equation, e.g. the interstitial line yields  $R_{2i,v} + 2\mathcal{E}_{2i} + 3\mathcal{E}_{3i} - (R_{i,v} + R_{i,i} + R_{i,2i} + R_{i,3i})$  once the emission and sink terms below are added.

The **E**-term represents the thermal dissociation (emission) of mobile species. It is the matrix

$$\mathbf{E} = - \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\alpha^{i,2i} & -\frac{1}{3}\alpha^{i,3i} \\ 0 & 0 & \alpha^{i,2i} & -\frac{2}{3}\alpha^{i,3i} \\ 0 & 0 & 0 & \alpha^{i,3i} \end{bmatrix}, \quad (35)$$

where  $\alpha^{x,nx}$  is the dissociation rate of an  $nx$ -cluster emitting a point defect of type  $x$ ,

$$\alpha^{m,n} = \beta^{m,n} \exp\left(-\frac{E_{m,n}^b}{k_B T}\right), \quad (36)$$

With  $E_{m,n}^b$ , the binding energy of the  $(m+n)$ -size cluster (`mobileSpeciesBindingEnergy_eV`). Component-wise,  $\mathbf{E}\mathbf{c}_M$  reproduces the thermal-emission terms  $\mathcal{E}_{2i} = \omega_{2i} e_{2i} C_{2i}$  and  $\mathcal{E}_{3i} = \omega_{3i} e_{3i} C_{3i}$  of [Section 2.2](#).

The **D**-term is the linear reaction matrix that governs the annihilation of mobile defects at immobile sinks. It absorbs the mobile diffusivities into the sink strengths, which themselves combine the network sinks with the mass-conserving loop absorption. The mobile-to-immobile reaction matrix is:

$$\mathbf{D} = \text{diag}\{\boldsymbol{\omega} \odot \mathbf{C}_s\} + \mathbf{D}_d \mathbf{K}_L, \quad \mathbf{D}_d = \text{diag}\{\bar{D}^v, \bar{D}^i, \bar{D}^{2i}, \bar{D}^{3i}\} \quad (37)$$

where  $\mathbf{C}_s = [C_s^v, C_s^i, C_s^{2i}, C_s^{3i}]^T$  are the network-only sink concentrations of [Eq. \(56\)](#) ( $C_s^v = \frac{a^2}{z_c} \rho_N$ ,  $C_s^i = \frac{a^2}{z_c} Z_N \rho_N$ ) and  $\boldsymbol{\omega} = [\omega_v, \omega_i, \omega_{2i}, \omega_{3i}]^T$ . The first term reproduces the network sink reactions  $R_{x,s} = \omega_x C_x C_s^x$ , while  $\mathbf{K}_L$  carries the loop-absorption debits `loop_abs_i`, `loop_abs_v` defined in [Eqs. \(71\)](#) and [\(76\)](#), so that mobile losses to loops match exactly the content gained by the immobile families and no defect is double-counted. The network term is already a reaction rate, whereas the loop sink strengths acquire the mobile diffusivities, so both contributions have units of frequency (Hz), and  $\bar{D}^m$  is the (orientationally averaged) diffusion coefficient of mobile species  $m$ .

The loop sink-strength block  $\mathbf{K}_L$  is diagonal in the mobile species and collects, for each mobile defect  $m$ , the sink strength of every immobile loop family,

$$\mathbf{K}_L = \text{diag}\{\mathbf{k}^2\}, \quad \mathbf{k}^2 = [k_v^2 \ k_i^2 \ k_{2i}^2 \ k_{3i}^2], \quad (38)$$

$$k_m^2 = \sum_{(s,k) \in \mathcal{F}_L} Z_{sk,m} \rho_{sL}^k + Z_{vc0,m} \alpha_{\text{sfp}} \pi R_{vL}^{c0} \frac{n_{vL}^{c0}}{\Omega} + Z_{N,m} \rho_N,$$

where  $\mathcal{F}_L$  is the eight-member loop subset of [Eq. \(64\)](#),  $\rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k$  the line density of family  $(s,k)$  ([Eq. \(65\)](#)), and  $Z_{sk,m}$  its capture efficiency for species  $m$ . The middle term is the basal stacking-fault pyramid, which has no line density and enters through the compact cross-section [Eq. \(39\)](#) instead; by [Eqs. \(73\)](#) and [\(74\)](#) it is exactly the cross-section that drives its growth. [Equation \(38\)](#) is the first place the resolved layout acts back on the mobile species: the six prismatic families are separate sinks with separate line densities, so a stress-driven imbalance among the variants changes  $k_m^2$  and therefore changes  $x = \Phi_V/\Phi_I$  itself. The basal stacking-fault pyramid has no line density, and enters instead through the compact cross-section

$$\langle k_{bp,m}^2 \rangle = \alpha_{bp} s_{bp} \pi r_{bp} n_{bp}, \quad (39)$$



with  $r_{bp} = R_{vL}^{c_0}$  of Eq. (66) and  $n_{bp} = n_{vL}^{c_0}/\Omega$ . This is a distinct species with a distinct geometry, *not* an interpolated limit of a loop family: every family in  $\mathcal{F}_L$  takes the line-density form at every size (Section 2.11).

**The efficiencies in Eq. (38) are the same numbers that drive loop growth, Eq. (68),** and this identity is the operational content of “self-consistent”: what a loop family absorbs is exactly what the mobile species lose, family by family and species by species, so no defect is created or destroyed at the interface between the two blocks. It also removes the possibility — present in any model that reads its sink bias from an independent parameter — of diffusing the species with one anisotropy while biasing their capture with another.

Both the growth and the sink efficiencies carry two independent effects. The first is the diffusional anisotropy difference, which depends on the habit plane alone and is described by Eq. (19),

$$Z_{a\mathcal{L},m}^{\text{DAD},0} = Z_m^0 \left( \frac{p^m + (p^m)^{-2}}{2} \right), \quad Z_{c\mathcal{L},m}^{\text{DAD},0} = Z_m^0 p^m, \quad (40)$$

with  $p^m = (D_c^m/D_a^m)^{1/6}$  the anisotropy of species  $m$  and  $Z_m^0$  its bias factor for an isotropically diffusing defect at zero stress. Unlike mean-field models, which presume a fixed DAD bias, the present formulation obtains  $p^m$  from the very tensor  $\mathbf{D}^m$  that appears in  $\mathbf{J}^m$ : the migration energies are split at fixed effective diffusivity,

$$E_m^{\langle 11 \rangle} = E_m^{\langle 22 \rangle} = E_m^{\text{eff}} + 2k_B T \ln p^m, \quad E_m^{\langle 33 \rangle} = E_m^{\text{eff}} - 4k_B T \ln p^m, \quad (41)$$

so that changing  $p^m$  alters directionality and not overall mobility, and the orientation-averaged diffusivity  $\bar{D}^m = (\det \mathbf{D}^m)^{1/3}$  that enters Eq. (37) is unchanged. The second effect is stress, entering through the stress-induced preferential absorption (SIPA) increment of Eq. (68).

At zero stress the partial sink strengths of the prismatic and basal families reduce to

$$\langle k_{a\mathcal{L},m}^2 \rangle = Z_m^0 (p^m + (p^m)^{-2}) \pi r^{a\mathcal{L}} n^{a\mathcal{L}} \quad (42)$$

$$\langle k_{c\mathcal{L},m}^2 \rangle = Z_m^0 p^m 2\pi r^{c\mathcal{L}} n^{c\mathcal{L}} \quad (43)$$

For isotropically diffusing defects ( $p = 1$ ) these recover the traditional dislocation sink strength, whereas for the di- and tri-interstitials, whose transport is basal-enhanced,  $p < 1$  gives  $\langle k_{c\mathcal{L},m}^2 \rangle < \langle k_{a\mathcal{L},m}^2 \rangle$ . Note that every mobile species carries its *own*  $p^m$  and hence its own pair of efficiencies. That resolution is what makes an anisotropy assigned to the clusters alone representable at all; a formulation that lumps  $c^i$ ,  $c^{2i}$  and  $c^{3i}$  into a single interstitial flux and applies one efficiency to all three cannot see it.

The mobile species PDE (27) is accompanied by the concentration boundary condition<sup>1</sup> (BC)

$$\mathbf{c}_M(\mathbf{x}) = \mathbf{c}_{\text{bnd,eq}}(\mathbf{x}) \quad (\text{mobile species BC}), \quad (45)$$

where  $c_{\text{bnd,eq}}^k(\mathbf{x})$  is the local concentration of the species in equilibrium with the boundary,

$$c_{\text{bnd,eq}}^k(\mathbf{x}) = c_0^k(\mathbf{x}) \exp \left( -\frac{m^k t_n(\mathbf{x}) \Omega}{k_B T} \right), \quad (46)$$

in which  $t_n$  is the normal traction on the boundary and  $c_0^k(\mathbf{x})$  is the thermal-equilibrium concentration

$$c_0^k(\mathbf{x}) = |m^k| \exp \left( -\frac{E_k^f + m^k q \Omega_k^r}{k_B T} \right), \quad (47)$$

where  $q(\mathbf{x}) = -\text{tr}(\boldsymbol{\sigma}(\mathbf{x}))/3$  is the local pressure,  $\Omega_k^r = \omega_k^r \Omega$  the relaxation volume of the  $k$ -th cluster (with a relative relaxation volume  $\omega_k^r$ , `mobileSpeciesRelaxVol`), and  $E_k^f$  its boundary formation energy (`mobileSpeciesEnergyFormation_eV`). The stress terms in Eqs. (46) and (47) are multiplied by the signed cluster size  $m^k$ , rendering opposite effects for interstitial and vacancy-type clusters.

<sup>1</sup>Eq. (45) is a pure essential (Dirichlet) BC representing a perfect sink. A more general Robin-type BC

$$-\mathbf{J}^k \cdot \mathbf{n} = h^k [c^k(\mathbf{x}) - c_{\text{bnd,eq}}^k(\mathbf{x})], \quad (44)$$

recovers the perfect sink condition as  $h^k \rightarrow \infty$  and the no-flux condition as  $h^k \rightarrow 0$ .

## 2.2 Reaction kinetics, cascade sources, and the dislocation network

This subsection collects the ingredients that the mobile and the immobile blocks share: the irradiation-modified jump frequencies that set the clustering kinetics, the pairwise and first-order reaction rates built from them, the network-only point-defect sinks, the cascade partition of the displacement rate, and the evolving network dislocation density. It is written once here so that neither block has to be read against a separate mean-field appendix.

**Jump frequencies.** The kinetics are built on Arrhenius jump frequencies. Small interstitial clusters are given their own migration energy, so the model carries *four* frequencies rather than two:

$$\omega_i = z_c \nu_i \exp\left(-\frac{E_i^m}{k_B T}\right), \quad \omega_{2i} = z_c \nu_i \exp\left(-\frac{E_{2i}^m}{k_B T}\right), \quad (48)$$

$$\omega_{3i} = \omega_{2i}, \quad \omega_v = z_c \nu_v \exp\left(-\frac{E_v^m}{k_B T}\right), \quad (49)$$

with  $z_c = 48$  the combinatorial number and  $\nu_i, \nu_v$  the Debye-scale attempt frequencies ( $\sim 5 \times 10^{12}$  Hz for Zr). The scalar diffusivities and the thermal-equilibrium vacancy concentration follow as

$$D_v = \frac{a^2}{z_c} \omega_v, \quad D_i = \frac{a^2}{z_c} \omega_i, \quad C_v^e = \exp\left(-\frac{E_v^f}{k_B T}\right). \quad (50)$$

Equation (50) is the bridge between the scalar mean-field kinetics and the tensorial transport of Section 2.1: the orientation-averaged diffusivity  $\bar{D}^m = (\det \mathbf{D}^m)^{1/3}$  equals the scalar  $D_m$  of Eq. (50) exactly, because Eq. (41) splits the migration energies at fixed effective diffusivity. No mobility is redefined by the anisotropy — only its directional weighting.

**Thermal emission.** The Boltzmann emission probabilities of the small clusters are

$$e_{2i} = \exp\left(-\frac{E_{2i}^B}{k_B T}\right), \quad e_{3i} = \exp\left(-\frac{E_{3i}^B}{k_B T}\right), \quad (51)$$

and the corresponding dissolution rates are  $\mathcal{E}_{2i} = \omega_{2i} e_{2i} C_{2i}$  and  $\mathcal{E}_{3i} = \omega_{3i} e_{3i} C_{3i}$ . Emission from a *loop* is a distinct channel, governed by the size-dependent vacancy binding energy of the mean-size loop, and is given in Eq. (110), whose binding energy Eq. (109) is computed from the loop energy rather than fitted. The current parameter set uses a constant 5 eV surrogate for the loop binding energy in the mean-field limit; the size-dependent form is carried by the spatially resolved code and is one of the known non-correspondences between the two.

**Pairwise and sink reaction rates.** Reaction rates use the *sum* of the two reactant frequencies,  $R_{a,b} = (\omega_a + \omega_b) C_a C_b$ , with the recombination-volume factor  $\beta_r$  carried on the interstitial-vacancy channel and the combinatorial factors retained on the like-species reactions:

$$R_{i,v} = \beta_r (\omega_i + \omega_v) C_i C_v, \quad R_{2i,v} = (\omega_{2i} + \omega_v) C_{2i} C_v, \quad (52)$$

$$R_{3i,v} = (\omega_{3i} + \omega_v) C_{3i} C_v, \quad R_{i,i} = 2 \omega_i C_i^2, \quad (53)$$

$$R_{i,2i} = (\omega_i + \omega_{2i}) C_i C_{2i}, \quad R_{i,3i} = (\omega_i + \omega_{3i}) C_i C_{3i}, \quad (54)$$

$$R_{2i,2i} = 4 \omega_{2i} C_{2i}^2, \quad R_{2i,3i} = 2 \omega_{2i} C_{2i} C_{3i}. \quad (55)$$

These are exactly the component-wise expansion of the second-order matrix  $\mathbf{R}_2$  of Eq. (31).

**Network-only sinks.** The point-defect sink concentrations carry the *network* alone; loop absorption is handled by the mass-conserving coupling of Section 2.3, and loops must therefore not also appear here or their absorption would be counted twice:

$$C_s^v = \frac{a^2}{z_c} \rho_N, \quad C_s^i = \frac{a^2}{z_c} Z_N \rho_N, \quad (56)$$

with  $Z_N$  the network interstitial bias and  $\rho_N$  the evolving network density. The corresponding reactions, using  $\omega_i$  for every interstitial species, are  $R_{v,s} = \omega_v C_v C_s^v$ ,  $R_{i,s} = \omega_i C_i C_s^i$ ,  $R_{2i,s} = \omega_i C_{2i} C_s^i$ ,  $R_{3i,s} = \omega_i C_{3i} C_s^i$ . The total dislocation density  $\rho_{tot} = \rho_N + \sum_{\mathcal{F}} \rho_{sL}^k$  is retained for diagnostics.



**Mobile flux scales.** The two aggregates that drive loop growth are

$$\phi_v = \Phi_V = \omega_v C_v, \quad \phi_i = \Phi_I = \omega_i (C_i + 2C_{2i} + 3C_{3i}), \quad (57)$$

which are Eq. (15) written for this mobile species set. Their ratio is the  $x$  whose thresholds Eq. (17) govern which families grow. In the self-consistent formulation the aggregate Eq. (57) is used only for reporting: each family's absorption is assembled species by species, with each species carrying its own efficiency and its own  $\bar{D}^m$  (Eq. (71)).

**Cascade source partition.** The displacement rate  $G$  (dpa s<sup>-1</sup>) is partitioned among free monomers, small clusters, and cascade-born loops so that the interstitial and the vacancy atom budgets each close to  $G$ :

$$G_v = G(1 - \varepsilon_{vL}), \quad G_i = G(1 - \varepsilon_{2i} - \varepsilon_{3i} - \varepsilon_{iL}), \quad (58)$$

$$G_{2i} = G\varepsilon_{2i}, \quad G_{3i} = G\varepsilon_{3i}, \quad (59)$$

$$G_{iL} = G\varepsilon_{iL}, \quad G_{avL} = G\varepsilon_{avL}, \quad G_{cvL} = G\varepsilon_{cvL}, \quad G_{\text{sfp}} = G\varepsilon_{\text{sfp}}, \quad (60)$$

where  $\varepsilon_{2i}, \varepsilon_{3i}$  are the di- and tri-interstitial cascade fractions,  $\varepsilon_{iL}$  the cascade prismatic interstitial loop fraction, and  $\varepsilon_{avL}, \varepsilon_{cvL}$  the in-cascade vacancy cluster yields feeding the prismatic vacancy and the basal faulted-loop families respectively, and  $\varepsilon_{\text{sfp}}$  the yield of basal stacking-fault pyramids, with  $\varepsilon_{vL} = \varepsilon_{avL} + \varepsilon_{cvL} + \varepsilon_{\text{sfp}}$ . In the pure-precursor limit of Section 2.8 the whole basal yield is assigned to the pyramid,  $\varepsilon_{cvL} = 0$ . The di- and tri-interstitial sources enter the monomer-cluster equations *per cluster*, that is as  $G_{2i}/2$  and  $G_{3i}/3$ . The atom balances satisfy

$$G_i + G_{2i} + G_{3i} + G_{iL} = G, \quad G_v + G_{avL} + G_{cvL} + G_{\text{sfp}} = G. \quad (61)$$

Equation (60) is where the two cascade-informed ingredients of Section 1.3 enter. The vacancy-loop yields  $\varepsilon_{avL}$  and  $\varepsilon_{cvL}$  are the in-cascade vacancy clustering that supplies the vacancy-loop embryos; the same clustering asymmetry sets the production bias  $\eta = \varepsilon_V/\varepsilon_I$  that positions  $x$  inside the growth window, so the two must be taken from one cascade database and not fitted independently. The cascade-overlap reduction Eq. (25) multiplies  $G$  itself, and being a single scalar applied to both species it rescales dose without moving  $x$ .

**Evolving network dislocation density.**  $\rho_N$  is a state variable, seeded at the grown-in, temperature-dependent material value (log-linear in  $T$  between anchors at 300 and 600 K),

$$\log_{10} \rho_N(0) = \log_{10} \rho_{300} + (\log_{10} \rho_{600} - \log_{10} \rho_{300}) \frac{T - 300}{600 - 300}. \quad (62)$$

It grows by the line length of the loops absorbed into it and recovers first-order toward the seed,

$$\frac{d\rho_N}{dt} = c_{\rho N} \frac{2\pi}{\Omega} \sum_{(s,k) \in \mathcal{F}} R_{sL}^k \Delta N_{LN,s}^k - k_{\rho N}^{\text{rec}} (\rho_N - \rho_N(0)), \quad (63)$$

with  $\Delta N_{LN}$  the loop-network coalescence number loss of Eq. (118). Through Eq. (56) the network then feeds back into the mobile sink strengths, closing the loop between the two blocks.

## 2.3 Immobile loop populations: the expanded family set

Interstitial clusters larger than  $m_{\text{max}}$  and vacancy clusters larger than  $m_{\text{min}}$  are immobile and are treated as dislocation loops. The earlier formulation carried four such families — one vacancy  $\langle c \rangle$  family and one interstitial  $\langle a \rangle$  family, each duplicated into populations “aligned” and “non-aligned” with an applied stress and partitioned between them by a scalar fraction  $f_a$ . Section 1.3 showed why that layout cannot carry the physics: an alignment fraction expresses how much of a population is favored but not which prismatic system is, it is meaningless under a multiaxial stress or a stress field, and having no vacancy family on the prismatic planes it cannot represent the  $a_iL$ – $a_vL$  coexistence at all. The present formulation replaces it.

## 2.4 The nine families and their geometry

Writing the character as  $s \in \{i, v\}$  and the habit as  $k \in \{a_1, a_2, a_3, c_f, c_p\}$ , the families carried are

$$\mathcal{F} = \underbrace{\{(i, a_1), (i, a_2), (i, a_3)\}}_{\text{prismatic interstitial}} \cup \underbrace{\{(v, a_1), (v, a_2), (v, a_3)\}}_{\text{prismatic vacancy}} \cup \underbrace{\{(v, c_0), (v, c_f), (v, c_p)\}}_{\text{basal vacancy chain}}, \quad (64)$$

nine in all, of which eight are dislocation loops and one is not:  $c_0$  is the basal stacking-fault pyramid of [Section 1.2](#), the precursor from which the basal loops are formed. Write  $\mathcal{F}_L = \mathcal{F} \setminus \{(v, c_0)\}$  for the eight loop families alone; every statement below that invokes a perimeter, a line density or a coalescence channel is a statement about  $\mathcal{F}_L$ , and the three places where the pyramid needs a form of its own are named as they arise. The subscript  $L$  on the state variables is retained for all nine as a label for *immobile*, so that a sum over  $\mathcal{F}$  may be written without qualification; it does not assert that  $c_0$  is a loop. Each family carries a character sign  $\eta_s$  (+1 interstitial,  $-1$  vacancy) and three spatially resolved fields — the *cluster density*  $n_{sL}^k(\mathbf{x})$ , clusters per host atom, the *defect content*  $q_{sL}^k(\mathbf{x})$ , point defects stored in that family per host atom, and the *second content*  $q_{sL}^k(\mathbf{x})$  of [Section 2.9](#), which carries the width of the size distribution and which the reader may set aside until then —  $q_{sL}^k = (c_{sL}^k)^2 / n_{sL}^k$  is the monodisperse family everything before [Section 2.9](#) describes. All three are dimensionless, so a number density per unit volume is  $n/\Omega$  and a stored defect density is  $c/\Omega$ . Three features distinguish [Eq. \(64\)](#) from the layout it replaces, and each is a statement about the crystallography rather than about the solver.

1. **The prismatic loops are resolved into their three variants  $a_1, a_2, a_3$ , of both characters.** The variants carry their own habit normals  $\hat{\mathbf{a}}_k$ , so the resolved stress on each is computed rather than parameterized, and the response to an arbitrary stress state is a redistribution among the three rather than a shift of one scalar. At zero deviatoric stress they are equivalent by symmetry, receive equal shares, and summing them recovers the lumped population exactly — which is the regression test on the change.
2. **There is no basal interstitial family.** Basal interstitial  $c$ -component loops are not a reported population in neutron-irradiated Zr. The formal  $c_L$  population retained in the growth inequalities of [Section 1.3](#) for symmetry is not carried as a state variable.
3. **The basal vacancy population is a three-state chain.** The chain [Eq. \(14\)](#) runs  $c_0 \rightarrow c_f \rightarrow c_p$ . The pyramid  $c_0$  is not planar and stores its vacancies with a nearly isotropic relaxation volume, where the faulted state  $c_f$  and the perfect state  $c_p$  are planar and, by [Table 2](#), store different vacancy content per unit area. Both transitions are thermally activated and both are one-way; they are given in [Section 2.8](#).

[Figure 1](#) places all nine families of [Eq. \(64\)](#) on the hexagonal cell, and the distinction between their two habits is the distinction between its two shaded orientations in panel (a). A prismatic family takes its Burgers vector from the horizontal  $\langle a \rangle$  arrow and its habit plane from one of the two vertical prismatic families; because either such plane is spanned by  $[0001]$  and one basal  $a$  direction, the vertical cell edge is the direction along which the loop outline elongates. A basal family lies instead in the horizontal face at the bottom and takes its Burgers vector from the vertical  $\langle c \rangle$  arrow, which is then the loop *normal* rather than an in-plane axis.

Two readings of the figure guard against errors the family set would otherwise invite. First,  $\langle c + a \rangle$  in panel (b) is close in magnitude to the basal Burgers vectors of [Eqs. \(4\) and \(5\)](#) but is a different vector of different character — it is the glide vector that accommodates  $c$ -axis strain in deformation — and the pyramidal planes containing it are not observed habit planes for irradiation-induced loops. [Equation \(64\)](#) therefore has no pyramidal family, and that is an experimental statement rather than a truncation. Second, the two habits are mutually *orthogonal*: the normals  $\hat{\mathbf{a}}_k$  of the three prismatic variants lie in  $(0001)$ , while the basal normal is  $[0001]$  itself. That orthogonality is why no single electron-beam direction images both populations at true size — a zone axis showing one family face-on views the other edge-on — so any comparison of [Eq. \(64\)](#) against TEM densities must combine at least two zone axes, and a single-axis micrograph constrains at most one habit.

The quantity converting loop area into stored defects is not  $|\mathbf{b}|$  but the edge component  $\mathbf{b} \cdot \mathbf{n}$ : a loop of radius  $R$  holds  $\pi R^2 (\mathbf{b} \cdot \mathbf{n}) / \Omega$  defects. The mean radius and the line density of family  $(s, k)$  therefore

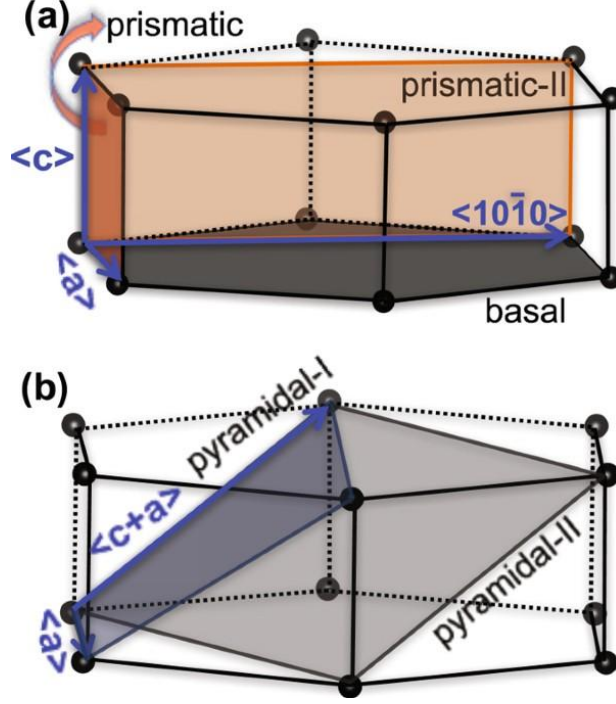


Figure 1: The principal planes and lattice translations of the hcp structure, on which the nine families of Eq. (64) are placed. (a) The families whose normals are perpendicular or parallel to  $[0001]$ : the *basal* plane  $\{0001\}$  (shaded, bottom), which is the habit of  $c_0$ ,  $c_f$  and  $c_p$ ; the first-order *prismatic* plane  $\{10\bar{1}0\}$ , a side face of the prism (the narrow shaded face at the left, indicated by the curved arrow), which is the habit of  $a_1, a_2, a_3$ ; and the second-order *prismatic-II* plane  $\{11\bar{2}0\}$ , a diagonal plane through the cell rather than a face of it (the wide shaded region), on which  $\langle a \rangle$  loops are also observed. The blue arrows are the two shortest lattice translations,  $\langle a \rangle = \frac{1}{3}\langle 11\bar{2}0 \rangle$  in the basal plane and  $\langle c \rangle = [0001]$  along the cell axis. (b) The inclined *pyramidal-I*  $\{10\bar{1}1\}$  and *pyramidal-II*  $\{11\bar{2}2\}$  planes, which contain  $\langle c+a \rangle = \frac{1}{3}\langle 11\bar{2}3 \rangle$  and which carry *no* family of Eq. (64). The cell is drawn at an arbitrary axial ratio, whereas  $\alpha$ -Zr has  $c/a \simeq 1.5944$  (Eq. (1)).

follow from its content and density as

$$R_{sL}^k = \lambda_k \sqrt{\frac{c_{sL}^k}{n_{sL}^k}}, \quad \lambda_k = \sqrt{\frac{\Omega}{\pi (\mathbf{b} \cdot \mathbf{n})_k}}, \quad \rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k, \quad (65)$$

with one geometric length per habit —  $\lambda_a$  shared by all six prismatic families,  $\lambda_{c_f}$  and  $\lambda_{c_p}$  for the two basal states — and the mean number of defects per loop is  $m_{sL}^k = c_{sL}^k / n_{sL}^k$ . Table 4 lists the nine families with the crystallography that fixes their geometry. Equation (65) describes a planar object and does not apply to the pyramid, whose stored vacancies fill a volume rather than bound an area. Its size follows instead from the volumetric relation already used for the compact limit in Eq. (129),

$$R_{vL}^{c_0} = \left( \frac{m_{vL}^{c_0} \Omega}{\sqrt{8}} \right)^{1/3}, \quad m_{vL}^{c_0} = \frac{c_{vL}^{c_0}}{n_{vL}^{c_0}}, \quad (66)$$

so the pyramid carries no  $\lambda_k$ , no Burgers vector and no line density. What it does carry is a capture cross-section, and Eq. (73) below shows that a cross-section is the only thing either the growth law or the sink strengths ever ask of a family.

Table 4: The nine families of Eq. (64); the first eight are dislocation loops and  $c_0$  is not. The three prismatic variants are related by the sixfold symmetry of the  $c$  axis and are crystallographically equivalent at zero stress.  $\lambda_k$  is evaluated with  $\Omega = 2.3266 \times 10^{-29} \text{ m}^3$ ,  $a = 0.323 \text{ nm}$ ,  $c = 0.515 \text{ nm}$ .

| Family          | Habit            | $\mathbf{b}$                            | Character | $\lambda_k$ (nm) | Comment                                                                                                                                       |
|-----------------|------------------|-----------------------------------------|-----------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| $a_1, a_2, a_3$ | $\{10\bar{1}0\}$ | $\frac{1}{3}\langle 11\bar{2}0 \rangle$ | $i$       | 0.151            | The dominant early population; $\mathbf{b} \cdot \mathbf{n} = a$ .                                                                            |
| $a_1, a_2, a_3$ | $\{10\bar{1}0\}$ | $\frac{1}{3}\langle 11\bar{2}0 \rangle$ | $v$       | 0.151            | Same habit and $ \mathbf{b} $ as the interstitial variants; separated from them only by capture character (Section 2.5).                      |
| $c_f$           | (0001)           | $\frac{1}{6}\langle 20\bar{2}3 \rangle$ | $v$       | 0.170            | Faulted, $\mathbf{b} \cdot \mathbf{n} = c/2$ , one removed basal layer. The dominant observed $c$ loop.                                       |
| $c_p$           | (0001)           | [0001]                                  | $v$       | 0.120            | Perfect, $\mathbf{b}\mathbf{n} = c$ , two removed layers, hence $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$ .                                    |
| $c_0$           | (0001) base      | partials                                | $v$       | —                | The stacking-fault pyramid: not planar, so Eq. (65) is replaced by Eq. (66) and it carries no $\lambda_k$ . Precursor to $c_f$ (Section 2.8). |

**The loops are elliptical; the model carries their equal-area radius.** Equation (65) treats every family as a circle. A real prismatic loop is not one: its outline is an ellipse with the major axis close to [0001] and the minor axis the basal direction contained in the habit plane (Fig. 2, left), and the ellipticity  $e = 1 - d_{\min}/d_{\max}$  reaches  $\simeq 0.1$  for large interstitial and  $\simeq 0.4$  for large vacancy  $\langle a \rangle$  loops, for the reasons given in Section 1.2. Three statements bound what the circular idealization costs, and none of them touches Eq. (71).

1. **The stored content is exact.**  $R_{sL}^k$  in Eq. (65) is defined through the loop *area*,  $\pi R^2 = \frac{\pi}{4} d_{\max} d_{\min}$ , so  $c_{sL}^k$  and  $m_{sL}^k = c_{sL}^k / n_{sL}^k$  are independent of the outline shape. Ellipticity redistributes a loop's perimeter; it does not change how many defects a loop of given area stores.
2. **Only the line density is affected, and by at most 5%.** The perimeter of an ellipse exceeds that of the circle of equal area, so  $\rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k$  understates the true line length by a factor

$$\frac{P}{2\pi R} = \frac{2}{\pi \sqrt{q}} E(1 - q^2), \quad q = \frac{d_{\min}}{d_{\max}} = 1 - e, \quad (67)$$

with  $E$  the complete elliptic integral of the second kind. This is 1.0021 at  $e = 0.1$  and 1.0490 at  $e = 0.4$ : the interstitial  $\langle a \rangle$  families are circular to two parts in a thousand, and even the most elliptical vacancy  $\langle a \rangle$  loops carry 4.9% more line than Eq. (65) credits them with. That is the single place shape enters the formulation, and it is an order of magnitude below the spread on the capture efficiencies it would multiply.

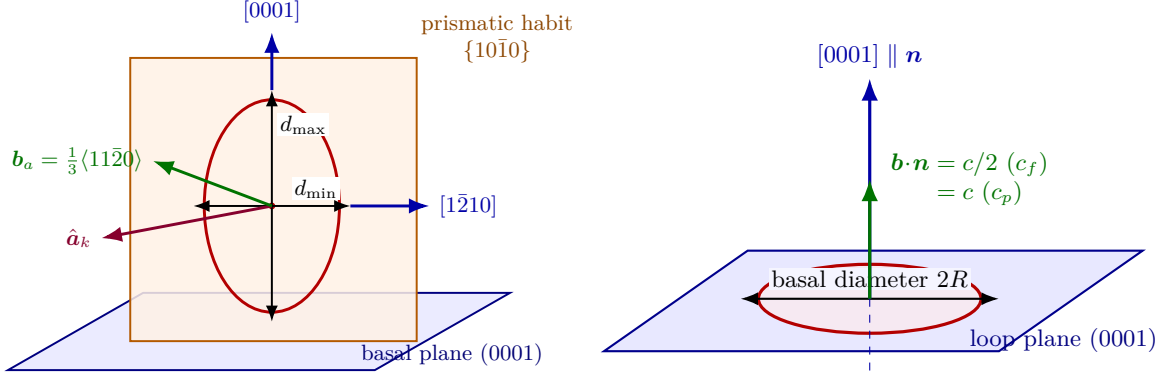


Figure 2: Geometry of the two habits of Eq. (64); both drawings are schematic in perspective rather than metrically crystallographic. *Left*: a prismatic  $\langle a \rangle$  family  $a_k$ . The loop plane contains  $[0001]$  and one basal  $a$  direction, the outline is elongated along  $[0001]$ , and the Burgers vector is basal while the measured habit normal  $\hat{a}_k$  — the vector on which Eqs. (68) and (77) resolve the stress — need not be exactly parallel to it. *Right*: a basal family  $c_f$  or  $c_p$ . The closed line and both shape axes lie in  $(0001)$  and  $[0001]$  is the loop *normal*; the apparent ellipse is the oblique projection of a circular basal loop and must not be read as intrinsic ellipticity. For  $c_f$  the Burgers vector additionally carries the basal partial of Eq. (7), so it is  $\mathbf{b} \cdot \mathbf{n}$ , and not  $|\mathbf{b}|$ , that appears in Eq. (65).

3. **The stress response and the basal families are untouched.** The habit normal  $\hat{a}_k$  on which Eqs. (68) and (77) resolve the stress is a property of the habit *plane*, not of the outline drawn in it, so SIPN and SIPA are independent of  $e$ . And for  $c_f$  and  $c_p$  the equal-area circle is not an idealization at all to within the sixfold basal symmetry:  $[0001]$  is their normal rather than an in-plane axis, so an isolated basal loop has no preferred in-plane direction to elongate along (Fig. 2, right).

What the family set does not carry is the ellipticity itself, and in particular its *divergence* between the two characters. That divergence is driven by the same anisotropy parameters  $p^m$  that generate the orientation factors  $A_{h(k)}(p^m)$  of Eq. (69), so a shape-resolved extension would be a second use of a quantity already in the model rather than a new mechanism: it would enter as one aspect ratio per family multiplying  $\rho_{sL}^k$ , and by item 2 it could move the sink strengths of Eq. (38) by no more than 5%. The expansion that matters for the coexistence problem is the one Eq. (64) makes — resolving the variants and the characters — not the shape of the individual loop.

## 2.5 Capture efficiencies: orientation, character, and stress

Every efficiency in the model is built as a product of three factors, one per degeneracy to be broken:

$$Z_{sk,m}(\boldsymbol{\sigma}) = \underbrace{Z_m^0}_{\text{elastic bias}} \underbrace{A_{h(k)}(p^m)}_{\text{DAD, orientation}} \underbrace{X_{s,m}}_{\text{character}} \underbrace{\left[ 1 + \frac{\Delta Z_{h(k),m}^{\text{SIPA}}(\sigma_{kk})}{Z_m^0 A_{h(k)}(p^m)} \right]}_{\text{stress, SIPA}}, \quad (68)$$

where  $h(k) \in \{a, c\}$  is the habit of family  $k$  and

$$A_a(p) = \frac{p + p^{-2}}{2}, \quad A_c(p) = p. \quad (69)$$

Each factor breaks exactly one degeneracy, and each is silent on the others.

- $A_{h(k)}(p^m)$  is the Woo DAD factor Eq. (19). It depends on the habit plane alone, through the angle between the loop normal and the  $c$  axis, and therefore assigns one pair to all six prismatic families and one pair to all three basal states, the pyramid included, whose base is  $(0001)$  so that  $h(c_0) = c$ . It separates prismatic from basal — opening the window Eq. (20) — and it separates nothing else.

- $X_{s,m}$  is the character factor. It is unity in the character-degenerate model, and  $X_{i,I}, X_{v,V} > X_{i,V}, X_{v,I}$  when the thermal-drift capture radii of Eq. (23) are used, so that

$$\chi = \frac{X_{i,I} X_{v,V}}{X_{i,V} X_{v,I}} > 1 \quad (70)$$

and the prismatic window acquires the finite logarithmic width  $\ln \chi$  of Eq. (24). It is the same for the three variants of a given character, since Eq. (24) resolves character and not orientation.  $\chi = 1$  **recovers the character-degenerate model exactly, which is the regression test on this factor.** Note the double-counting hazard of Section 1.3:  $X_{s,m}$  and  $Z_m^0$  describe overlapping elastic physics, so only one of them may carry the interstitial–vacancy asymmetry. The clean choice is to take  $X_{s,m}$  from the capture radii and set  $Z_m^0$  to unity; if  $Z_m^0$  is retained as a fitting parameter, it must be stated that it absorbs whatever the capture-radius database does not.

- The SIPA bracket carries the applied stress, resolved on each family’s own habit plane through  $\sigma_{kk} = \hat{\mathbf{a}}_k \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{a}}_k$  for the prismatic variants and  $\sigma_{33}$  on (0001) for the basal pair. It separates  $a_1$  from  $a_2$  from  $a_3$ , which nothing else in Eq. (68) can do. At  $\boldsymbol{\sigma} = \mathbf{0}$  the bracket is unity and every efficiency reduces to its DAD value.

Equation (68) is the whole of the self-consistency claim in one equation. Its  $p^m$  is the anisotropy of the very tensor  $\mathbf{D}^m$  that transports species  $m$  in Eq. (27); its  $\bar{D}^m$ , used in Eq. (71), is that tensor’s orientation average; and the same  $Z_{sk,m}$  appears in the sink strengths Eq. (38) that debit the mobile species. One set of transport parameters, one set of efficiencies, both sides of the coupling.

## 2.6 Growth of the stored content

Loop growth is the rate of change of stored content under the net biased flux of the correct sign for the family. All nine families obey one law,

$$\dot{c}_{sL}^k \Big|_{\text{growth}} = \underbrace{\Pi_{sL}^k}_{\text{geometry}} \eta_s \sum_{m \in \{v,i,2i,3i\}} |m| \bar{D}^m Z_{sk,m} c^m \varsigma_s(m) g_{sL}^k, \quad (71)$$

with  $\varsigma_s(m) = +1$  when the polarity of species  $m$  matches the character  $s$  and  $-1$  otherwise,  $|m|$  the number of point defects the mobile cluster carries, and

$$\Pi_{sL}^k = \frac{\lambda_k}{\ell} \varsigma_k^{\text{sink}} \sqrt{n_{sL}^k c_{sL}^k}, \quad \ell \equiv \frac{z_c \Omega}{2\pi a^2}, \quad (72)$$

the loop-perimeter prefactor implied by Eq. (65), in which  $\varsigma_k^{\text{sink}}$  is the per-family sink-scale calibration constant. Equation (72) is a special case of a more general statement, and writing that statement out is what carries Eq. (71) onto the pyramid unchanged. Because  $\rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k = 2\pi \lambda_k \sqrt{n_{sL}^k c_{sL}^k}$  by Eq. (65),

$$\Pi_{sL}^k = \varsigma_k^{\text{sink}} \frac{\rho_{sL}^k}{2\pi \ell} \quad \text{identically, for every } (s, k) \in \mathcal{F}_L, \quad (73)$$

that is, *the prefactor in the growth law is the family’s own sink density divided by  $2\pi\ell$* . A family therefore needs no geometry beyond the capture cross-section it presents to the mobile species, and the pyramid — which has no perimeter — supplies that cross-section in the compact form of Eq. (39):

$$\Pi_{vL}^{c_0} = \varsigma_{c_0}^{\text{sink}} \frac{\alpha_{\text{sfp}} \pi R_{vL}^{c_0} n_{vL}^{c_0}}{2\pi \ell} = \varsigma_{c_0}^{\text{sink}} \frac{\alpha_{\text{sfp}} R_{vL}^{c_0} n_{vL}^{c_0}}{2\ell}, \quad (74)$$

with  $R_{vL}^{c_0}$  from Eq. (66) and  $\alpha_{\text{sfp}}$  the pyramid capture prefactor of Eq. (39). With Eq. (74) substituted for Eq. (72), all nine families obey Eq. (71) in one form, and Eq. (73) guarantees in advance what Section 6.8 has to verify: what a family absorbs in Eq. (71) is exactly what it debits from the mobile pool in Eq. (38), the pyramid included. The gate  $g_{sL}^k \in [0, 1]$  is applied to the *shrinking* channel only and suppresses it as  $R_{sL}^k \rightarrow R_{\min}^k$ ,

$$g(R) = \begin{cases} 0, & u \leq 0, \\ u^2(3 - 2u), & 0 < u < 1, \\ 1, & u \geq 1, \end{cases} \quad u = \frac{R/R_{\min} - 1}{w_R}, \quad (75)$$

a  $C^1$ -continuous smoothstep chosen to avoid solver stiffness. Gating the whole opposite-polarity absorption term, rather than the net rate, keeps the point-defect balance closed: the un-absorbed defects stay in the free pool. Equation (75) is a device for keeping a *mean* size off a floor that a single size cannot cross, and Section 2.9 retires it: once the family carries a distribution, loops leave through the floor at the rate Eq. (127) determines, and there is nothing left for a smoothstep to prevent.

Two things about Eq. (71) are worth isolating.

**It is species-resolved, not flux-lumped.** The sum runs over the four mobile species with each carrying its own  $\bar{D}^m$  and its own  $Z_{sk,m}$ . Writing it instead as  $\Pi(Z_{s,I}\Phi_I - Z_{s,V}\Phi_V)$  with the aggregates Eq. (57) is the mean-field reduction, and it is lossy in a way that matters here: an anisotropy assigned to the clusters alone is invisible to it. In the present parameter set the di- and tri-interstitials carry 3.1% of the interstitial arrival, so the difference is small at these parameters and structural in general.

**Loop growth is a near-cancellation.** Equation (71) is the small residual of two much larger opposing fluxes, and this is the single most important numerical property of the immobile block. Measured on a prismatic interstitial family at 10 dpa, the interstitial gain and the vacancy loss agree to within a few percent, so the net content rate is a few percent of either. Small changes in a capture efficiency therefore produce large changes in net growth — which is exactly what makes the window analysis of Section 1.3 predictive, and equally what makes any parameter set fitted against one formulation stale for another.

Collecting the families, the debits on the mobile pool are the sums of the same species-resolved terms with the opposite sign,

$$\text{loop\_abs}_m = \sum_{(s,k) \in \mathcal{F}} \Pi_{sL}^k |m| \bar{D}^m Z_{sk,m} c^m g_{sL}^k, \quad \text{loop\_recomb} = \sum_{(s,k) \in \mathcal{F}} [\text{opposite-polarity terms}], \quad (76)$$

which enter Eq. (27) through  $\mathbf{Dc}_M$ . Every defect debited from the mobile pool by Eq. (76) is credited to a family by Eq. (71), term by term, so nothing is double-counted and nothing is lost.

## 2.7 Variant resolution and where the stress enters

The three prismatic variants are crystallographically equivalent about the  $c$  axis, so nothing in Eq. (69) separates them and an unstressed grain has no reason to prefer one. A stress does, and it acts in two places, both of them Boltzmann factors of the work the stress does on the defect volume the loop stores.

**Partition of the sources.** Production into variant  $k$  is weighted by the resolved normal stress on its habit plane and by the sign of the relaxation volume the loop carries, so that interstitial and vacancy loops of one variant respond in *opposite* senses:

$$w_k^s(\boldsymbol{\sigma}, T) = \frac{\exp(\eta_s \sigma_{kk} V_s / k_B T)}{\sum_{k'=1}^3 \exp(\eta_s \sigma_{k'k'} V_s / k_B T)}, \quad \sum_{k=1}^3 w_k^s = 1, \quad s \in \{i, v\}, \quad (77)$$

with  $V_s$  the formation (relaxation) volume of the clustering unit. A tensile normal stress on the habit plane favors interstitial loops of that variant and disfavors vacancy loops of it, which is stress-induced preferential nucleation (SIPN) in its simplest form. Equation (77) is the scalar-activation-volume case of the general partition through the formation-volume tensor  $\boldsymbol{\Omega}^k$ ,

$$\chi_s^k = \frac{\exp(\boldsymbol{\sigma} : \boldsymbol{\Omega}^k / k_B T) \mathcal{I}_s^k}{\sum_p \exp(\boldsymbol{\sigma} : \boldsymbol{\Omega}^p / k_B T) \mathcal{I}_s^p}, \quad (78)$$

in which  $\sigma_{kk} V_s$  is replaced by  $\boldsymbol{\sigma} : \boldsymbol{\Omega}^k$  and  $\mathcal{I}_s^k$  is the character selector. At zero deviatoric stress  $w_k^s = \frac{1}{3}$  for every variant and every character, and Eq. (77) replaces the scalar alignment fraction  $f_a$  of the lumped model entirely. Linearizing Eq. (77) for small stress gives  $w_k^s \simeq \frac{1}{3}(1 + \eta_s \zeta \hat{\sigma}_{kk})$ , which is the form the code carries.



**Stress in the growth law.** The same work term enters the vacancy concentration a loop is in equilibrium with, and hence the emission side of Eq. (71) and the annealing lifetime, through  $C_v^{\text{eq}}(k) \propto \exp(+\sigma_{kk}\Omega/k_B T)$ . This is the channel by which a variant under an unfavorable stress *shrinks* rather than merely being nucleated less often, and it is the one that produces irradiation creep at constant loop number. It acts in the same direction as Eq. (77), so the two contributions to the variant imbalance add.

Because all three basal states share a single habit plane, no such partition applies to them: a stress acts on  $c_0$ ,  $c_f$  and  $c_p$  only through the second channel, with  $\sigma_{33}$  resolved on (0001). **That asymmetry — three prismatic variants that redistribute, one basal orientation that cannot — is the origin of the texture dependence of irradiation growth**, and it is not representable at all in a model whose stress dependence is a single scalar alignment fraction.

## 2.8 The basal chain: pyramid, faulted loop, perfect loop

Section 1.2 established the crystallography of the chain Eq. (14); this subsection gives its kinetics. Four properties fix how the triple  $(c_0, c_f, c_p)$  is treated, and the fourth is the one that makes the chain more than a relabelling.

1. **Only  $c_0$  is nucleated.** Cascade collapse on basal planes produces pyramids, not loops. The faulted state is reached only through the collapse transition below and the perfect state only through unfaulting, so  $J_{vL}^{c_f} = J_{vL}^{c_p} = 0$  in the pure-precursor limit. A direct basal loop current  $J_{vL}^{c_f}$  is retained as an option in Eq. (100) because setting  $\varepsilon_{\text{sfp}} = 0$  and restoring it recovers the previous revision of the model exactly, which is the regression test on this addition.
2. **The planar states store different content per unit area.** A  $c_p$  loop removes two basal layers where a  $c_f$  loop removes one, so  $(\mathbf{b} \cdot \mathbf{n})_{c_p} = 2(\mathbf{b} \cdot \mathbf{n})_{c_f}$  and  $\lambda_{c_p} = \lambda_{c_f}/\sqrt{2}$  in Eq. (65). Two loops of equal radius differ by a factor of two in stored vacancies, which is why the two cannot be lumped even though they share a habit plane. Because  $c_f$  is not a pure edge loop — its Burgers vector contains the basal partial of Eq. (7) — it is the edge component  $\mathbf{b} \cdot \mathbf{n} = c/2$ , and not  $|\frac{1}{6}\langle 20\bar{2}3 \rangle|$ , that enters its content.
3. **All three share their capture efficiencies.** The pyramid's base and both loop habits are (0001), so Eq. (68) assigns  $h(c_0) = h(c_f) = h(c_p) = c$  and all three take  $A_c(p^m)$ , the same character factor  $X_{v,m}$  and the same resolved stress  $\sigma_{33}$ . They are three states of one population, not three competing sinks in the sense that the prismatic variants are. What differs among them is not  $Z$  but the cross-section that multiplies it — Eq. (74) for the pyramid, Eq. (72) for the loops.
4. **Only the planar states carry  $c$ -axis eigenstrain.** The pyramid's relaxation volume is nearly isotropic, Eq. (131) with  $\Omega_{\text{pyr}} = \mathcal{I}\text{IV}/3$ , so vacancies held in  $c_0$  contribute to the hydrostatic part of the eigenstrain and only  $\zeta_{c_0} \simeq \frac{1}{3}$  of their share to  $\varepsilon_c$  (Eq. (199)). Storing a vacancy in a pyramid is therefore *not* equivalent to storing it in a loop, and this is the property that turns the residence time in  $c_0$  into an incubation period for  $c$ -axis growth.

**The two transitions.** Both are one-way, thermally activated, and size-gated: a cluster converts once it is large enough for the destination state to be the lower-energy one (Eq. (13) for the first, Eq. (8) for the second) and once it has waited long enough to cross the barrier. Writing  $\theta \in \{\text{col}, \text{uf}\}$  for the two,

$$\nu_\theta(T) = \nu_\theta^0 \exp\left(-\frac{E_\theta}{k_B T}\right), \quad \Phi_\theta^{(j)} = \frac{1}{\langle m^j \rangle_k} \int_{R_\theta}^\infty m^j f_k(R) dR, \quad (79)$$

in which  $E_{\text{col}}$  is the pyramid-to-loop collapse barrier,  $E_{\text{uf}}$  the unfaulting barrier,  $R_{\text{col}}$  and  $R_{\text{uf}}$  the critical radii implied by Eqs. (8) and (13), and  $f_k(R)$  the size distribution of the source family — *not* an assumed one: Section 2.9 carries enough moments to construct it, and Eq. (126) evaluates  $\Phi_\theta^{(j)}$  in closed form. **The gate must be a distribution fraction and not a Heaviside step on the mean radius:** in a mean-field state  $R_{vL}^k$  is an average, and a hard threshold on an average transfers the entire family in one step. The three weights are needed because a transfer moves number, content and second content in different proportions — the converting subpopulation is larger than the family

mean. Setting  $\Phi_\theta^{(j)} \equiv 1$  makes a transition purely barrier-limited; setting  $\nu_\theta = 0$  removes it, and  $\nu_{\text{col}} = 0$  leaves the pyramids as a terminal population while  $\nu_{\text{uf}} = 0$  leaves the single basal loop family the model carried before.

The transfers themselves are written so that both number and content move together, with the *same* rate multiplying each:

$$\Gamma_\theta^n = \nu_\theta \Phi_\theta^{(0)} n_{vL}^{k(\theta)}, \quad \Gamma_\theta^c = \nu_\theta \Phi_\theta^{(1)} c_{vL}^{k(\theta)}, \quad \Gamma_\theta^q = \nu_\theta \Phi_\theta^{(2)} q_{vL}^{k(\theta)}, \quad (80)$$

with source  $k(\text{col}) = c_0$  and  $k(\text{uf}) = c_f$ . Whatever leaves one family arrives in the other, moment for moment, so no vacancy is created or destroyed by either step. Because  $\Phi_\theta^{(2)} > \Phi_\theta^{(1)} > \Phi_\theta^{(0)}$ , the converting subpopulation is larger than the family mean: a transfer therefore *lowers* the mean size of the source and delivers to the destination a population above the source mean, which is what a size-gated reaction does and what a single rate multiplying every moment could not express. For the unfaulting transfer that statement has a geometric reading — by Eq. (65) the transferred loops arrive with their radius reduced by  $1/\sqrt{2}$ , the same vacancies now occupying half the area on two layers — and for the collapse it has another: by Eq. (66) and Eq. (65) a pyramid of  $m$  vacancies arrives as a faulted loop of radius  $\lambda_{c_f} \sqrt{m}$ , which is larger than the pyramid it came from whenever  $m > (\Omega/\sqrt{8})^2 \lambda_{c_f}^{-6}$ , so collapse is a spreading of the same vacancies into a plane.

**Saturation of the pyramid population, in closed form.** The pyramid has no coalescence channel — it is a compact cluster and not a line sink, and the loop–loop and loop–network channels of Eq. (118) are defined on  $\mathcal{F}_L$  — so its density is controlled by exactly the two loss channels of Section 2.9: thermal dissolution at  $1/\tau_{\text{sfp}}$ , and conversion at  $\nu_{\text{col}} \Phi_{\text{col}}$ . Its number equation is therefore linear in  $n_{vL}^{c_0}$  and saturates at

$$n_{vL}^{c_0,*} = \frac{J_{vL}^{c_0}}{\tau_{\text{sfp}}^{-1} + \nu_{\text{col}} \Phi_{\text{col}}}, \quad f_{\text{col}} = \frac{\nu_{\text{col}} \Phi_{\text{col}}}{\tau_{\text{sfp}}^{-1} + \nu_{\text{col}} \Phi_{\text{col}}}, \quad (81)$$

where  $f_{\text{col}}$  is the branching ratio — the fraction of pyramids that become  $c_f$  loops rather than dissolving back into the vacancy pool. It is  $f_{\text{col}}$ , and not  $J_{vL}^{c_0}$ , that is the *effective* basal loop nucleation efficiency of the model, and the two Arrhenius factors combine in it into a single ratio,

$$\nu_{\text{col}} \tau_{\text{sfp}} = \nu_{\text{col}}^0 \tau_{\text{sfp}}^0 \exp\left(\frac{E_{\text{sfp}}^{\text{dis}} - E_{\text{col}}}{k_B T}\right), \quad (82)$$

so that  $f_{\text{col}}$  rises with temperature when  $E_{\text{col}} > E_{\text{sfp}}^{\text{dis}}$  and falls when the inequality reverses. That single sign is the model's prediction for the direction of the temperature dependence of the  $c$ -loop onset dose, and it is a falsifiable consequence of carrying the pyramid rather than nucleating  $c_f$  directly: with  $\varepsilon_{\text{sfp}} = 0$  the onset dose has no barrier in it at all.

## 2.9 The size distribution and its moments

A family in Eq. (64) is not a number of loops and a size. It is a *size distribution*  $f_k(m, t)$ , the density of loops of that family holding between  $m$  and  $m + dm$  defects, and everything the immobile block carries is a moment of it. This subsection introduces the distribution first, because the three fields the model advances are then one object seen three ways rather than three separate balances that happen to be written together, and because the closure that makes the third of them useful also makes the first two sharper.

**From the master equation.** A loop does not change size continuously. It changes by discrete events, each of which is the arrival of one mobile cluster or the emission of one point defect, and the exact statement is therefore a master equation on the integer size  $m$  rather than a differential equation. Let  $W_x^+(m)$  be the rate at which a loop of size  $m$  absorbs a mobile species  $x$  whose polarity *matches* the family's character, and  $W_x^-(m)$  the rate for the opposite polarity; each event moves the loop by

$$\mu_x = |m^x| \in \{1, 2, 3\}, \quad (83)$$

the number of point defects the arriving cluster carries, from `mobileSpeciesVector`. Thermal emission is folded into  $W^-$  with  $\mu = 1$ , since a loop emits one defect at a time. Then

$$\begin{aligned} \frac{\partial f_k(m, t)}{\partial t} = \sum_x \left[ W_x^+(m - \mu_x) f_k(m - \mu_x) + W_x^-(m + \mu_x) f_k(m + \mu_x) \right. \\ \left. - (W_x^+(m) + W_x^-(m)) f_k(m) \right] + J_{sL}^k \delta_{m, m_k^{\text{nuc}}}, \end{aligned} \quad (84)$$

a gain-loss balance in which the first two terms are loops arriving at  $m$  from below and from above and the third is loops leaving  $m$  in either direction. **Equation (84) is the exact statement;** Eq. (87) is what it becomes when the jumps are small.

The rates are not free. A loop captures on its perimeter, so  $W_x^\pm \propto R_{sL}^k = \lambda_k \sqrt{m}$ , and the constant of proportionality is the same capture efficiency and mobility that Eq. (71) already carries:

$$W_x^\pm(m) = \pi_k \sqrt{m} \bar{D}^x Z_{sk,x} c^x, \quad \pi_k = \frac{\lambda_k \varsigma_k^{\text{sink}}}{\ell}, \quad (85)$$

with the sign of the polarity match deciding whether a given  $x$  enters  $W^+$  or  $W^-$ . **No new parameter enters the size distribution.**

**The Kramers–Moyal expansion.** Treat  $m$  as continuous and expand the shifted terms of Eq. (84) about  $m$ . The zeroth-order terms cancel the loss term identically — which is the statement that the process conserves loops — and what survives is

$$\frac{\partial f_k}{\partial t} = \sum_{p \geq 1} \frac{(-1)^p}{p!} \frac{\partial^p}{\partial m^p} [a_p(m) f_k] + J_{sL}^k \delta(m - m_k^{\text{nuc}}), \quad a_p = \sum_x \mu_x^p [W_x^+ + (-1)^p W_x^-]. \quad (86)$$

The first two coefficients are the physically familiar ones and they are *different combinations of the same rates*:  $a_1$  is the difference of the two directions and  $a_2$  is their sum. **Truncating Eq. (86) at  $p = 2$  is Eq. (87)**, the Fokker–Planck equation for the size density,

$$\frac{\partial f_k}{\partial t} = - \frac{\partial}{\partial m} [F_k f_k] + \frac{\partial^2}{\partial m^2} [D_k f_k] + J_{sL}^k \delta(m - m_k^{\text{nuc}}), \quad (87)$$

in which the drift is the *net* flux of Eqs. (71) and (110) and the diffusion is half the *gross* traffic across the same perimeter,

$$F_k = a_1 = \pi_k \sqrt{m} \Sigma_k^{\text{net}}, \quad D_k = \frac{1}{2} a_2 = \frac{1}{2} \pi_k \sqrt{m} \Sigma_k^{\text{gro}}, \quad (88)$$

$$\Sigma_k^{\text{net}} = \sum_x \varsigma_s(x) \mu_x \bar{D}^x Z_{sk,x} c^x, \quad \Sigma_k^{\text{gro}} = \sum_x \mu_x^2 \bar{D}^x Z_{sk,x} c^x. \quad (89)$$

$\Sigma_k^{\text{net}}$  is exactly the signed sum that multiplies  $\Pi_{sL}^k$  in Eq. (71), so the drift of Eq. (87) is the growth law and carries no independent content.

**Note the weighting in  $\Sigma_k^{\text{gro}}$ .** It is not the net sum with the signs made positive: the second Kramers–Moyal coefficient weights each channel by  $\mu_x^2$ , one power more than the drift, because variance accumulates as the square of the step. For the monomer channels, which carry  $\mu = 1$ , the two weightings coincide and the distinction is invisible — and an earlier revision of this document asserted on that basis that “every absorption and every emission is a  $\pm 1$  step”. That is true of  $v$ ,  $i$  and of emission and false of  $2i$  and  $3i$ . Those clusters carry 3.1% of the interstitial arrival in  $\Sigma_k^{\text{net}}$ , so the extra power of  $\mu_x$  raises their share of  $\Sigma_k^{\text{gro}}$  to between 6% and 9%: **the spreading is under-estimated by 3–6% if the drift weighting is used for both.** Small, systematic, and in the direction that matters, since Eq. (92) is an argument that the spreading dominates.

**Why truncating at second order is justified.** Equation (86) is an infinite series and Eq. (87) keeps two terms of it. The expansion parameter is *not* the jump size itself but the jump size measured against the scale over which  $f_k$  varies: successive terms carry one more power of  $\mu_x \partial_m$ , so

$$\frac{\text{term } p+1}{\text{term } p} \sim \frac{\mu_x}{\sigma_m}, \quad \sigma_m^2 = \int (m - \bar{m}_{sL}^k)^2 f_k dm = (\bar{m}_{sL}^k)^2 (\Delta_k - 1), \quad (90)$$

with  $\sigma_m$  the width of the distribution. Three statements make this small here, and they are worth separating because only the first is about the absorption step alone.

1. **The step is small against the loop.** The mobile species are  $v$ ,  $i$ ,  $2i$ ,  $3i$ , so  $\mu_x \leq 3$  — and *this is a property of the species set, not an approximation*: there is no channel in this model that adds a large piece to a loop in one event. Against the sizes the families actually reach, measured on the reference march (interior medians, in defects per loop):

| dose [dpa] | $\bar{m}_{\langle c \rangle}$ | $\mu_{\max}/\bar{m}_{\langle c \rangle}$ | $\bar{m}_{\langle a \rangle}$ | $\mu_{\max}/\bar{m}_{\langle a \rangle}$ |
|------------|-------------------------------|------------------------------------------|-------------------------------|------------------------------------------|
| $10^{-4}$  | 584                           | $5.1 \times 10^{-3}$                     | 294                           | $1.0 \times 10^{-2}$                     |
| 0.1        | $4.0 \times 10^4$             | $7.4 \times 10^{-5}$                     | 183                           | $1.6 \times 10^{-2}$                     |
| 10         | $1.7 \times 10^5$             | $1.7 \times 10^{-5}$                     | 695                           | $4.3 \times 10^{-3}$                     |

The worst case in the whole table is  $1.6 \times 10^{-2}$ , and the  $\langle a \rangle$  value at  $10^{-4}$  dpa is 294 against a nucleus of  $n_{iL}^{\text{nuc}} = 300$  — the family has barely left its birth size, which is where this ratio is largest by construction.

2. **The rates vary slowly over a step.** Equation (86) also requires  $W_x^\pm(m \pm \mu_x) \simeq W_x^\pm(m)$  to the order retained. By Eq. (85) the rates go as  $\sqrt{m}$ , so the relative change over one event is  $\mu_x/2m$  — the same smallness again, and *half* it. A rate that varied exponentially in  $m$  would not permit this truncation however small the step.
3. **The distribution is broad in units of the step.** This is the condition Eq. (90) actually names, and it is the one that can fail while the other two hold. Writing  $N_{\text{ex}}$  for the number of exchange events a loop has made, a random walk of step  $\mu$  gives  $\sigma_m = \mu\sqrt{N_{\text{ex}}}$ , so

$$\frac{\mu}{\sigma_m} = \frac{1}{\sqrt{N_{\text{ex}}}}, \quad N_{\text{ex}} = \frac{2}{\mu^2} \frac{D_k}{F_k} \Delta m, \quad (91)$$

the second form following from  $\sigma_m^2 = 2D_k t$  and  $\Delta m = F_k t$ . **This is where Eq. (92) does double duty.** The same near-cancellation that makes the second moment necessary —  $D_k/F_k$  of 7.3 to 44 — also multiplies the number of exchanges a loop has made for a given net growth. With  $D_k/F_k = 44$  and the  $\langle c \rangle$  growth of the table above,  $N_{\text{ex}} \simeq 1.5 \times 10^7$  and  $\mu/\sigma_m \simeq 2.6 \times 10^{-4}$ ; for  $\langle a \rangle$ , which grows by only a few hundred defects,  $N_{\text{ex}} \simeq 3.5 \times 10^4$  and  $\mu/\sigma_m \simeq 5 \times 10^{-3}$ .

**Where the truncation is at its worst, and why it does not matter.** Equation (91) diverges at  $N_{\text{ex}} = 0$ : a loop that has just nucleated has made no exchanges, its distribution is a delta function, and no gradient expansion describes it. The model injects exactly that,  $J_{sL}^k \delta(m - m_k^{\text{nuc}})$ . The defect is real and it is bounded: it is confined to a region of width  $\sim \mu_x$  around  $m_k^{\text{nuc}}$ , which is 1% of the nucleus, and it is cured by the first few exchange events, which at these fluxes are immediate. **The second place the expansion is worst is the floor  $m_{\min}$ ,** where  $f_k$  is cut off and its gradient is largest, and the model does not rely on Eq. (87) there at all — Eq. (127) carries an explicit floor current instead, for exactly this reason.

**What the truncation is not.** It is not an assumption that the distribution is Gaussian, or narrow, or near equilibrium; Eq. (87) is solved in moments and the closure of Eq. (123) is a separate and later choice. And it is not a statement about the channels Eq. (87) omits: coalescence removes whole loops rather than moving them along  $m$ , so it is not a jump process in size at all and appears in Eq. (118) instead — which is also why  $\phi_{LL}$  and  $\phi_{LN}$  carry their own validity criterion in Section 3.9 rather than inheriting this one.

**Why the second moment is not a refinement here.** The two large opposing fluxes that *cancel* in  $F_k$  *add* in  $D_k$ , and Section 2.6 has already established that this cancellation is severe. Taking the measured prismatic fluxes at 10 dpa:

$$\frac{D_k}{F_k} = \frac{\Sigma_k^{\text{gro}}}{2 \Sigma_k^{\text{net}}} \simeq \begin{cases} 7.3, & \text{legacy loop model,} \\ 44, & \text{self-consistent loop model,} \end{cases} \quad (92)$$

so the distribution **spreads one to two orders of magnitude faster than it translates.** A model carrying only  $\bar{m}_{sL}^k$  is not making a small error in the shape of the distribution; it is discarding the dominant term of Eq. (87). The more self-consistent the capture efficiencies are made, the worse this gets, because self-consistency is what drove the net flux down.

**The three moments.** Carry the first three moments of  $f_k$ , all per host atom and all on the same footing:

$$n_{sL}^k = \int f_k dm, \quad c_{sL}^k = \int m f_k dm, \quad q_{sL}^k = \int m^2 f_k dm, \quad (93)$$

the *density*, the *content* and the *second content*. Multiplying Eq. (87) by  $m^j$  and integrating over  $[m_{\min}, \infty)$  gives one hierarchy,

$$\frac{d}{dt} \int m^j f_k dm = j \int m^{j-1} F_k f_k dm + j(j-1) \int m^{j-2} D_k f_k dm + J_{sL}^k (m_k^{\text{nuc}})^j - m_{\min}^j \Phi_{\min}^k, \quad (94)$$

in which the last two terms are the nucleation source and the flux of loops through the floor, each weighted by the  $m^j$  that the departing or arriving cluster carries. The three cases  $j = 0, 1, 2$  are taken in turn below, and to them are appended the channels that Eq. (87) does not describe — coalescence, which removes whole loops rather than moving them in size, and the activated transfers of Eq. (80), which move them between families.

**Zeroth moment: the density.** At  $j = 0$  both integrals in Eq. (94) vanish — drift and diffusion move loops along the size axis and cannot create or destroy one — so the density changes only by nucleation, by the floor current, by coalescence and by transfer: The immobile number densities evolve by nucleation, thermal annealing, coalescence, and — along the basal chain — the two activated transfers of Eq. (80):

$$\dot{n}_{iL}^{a_k} = J_{iL}^{a_k} - \Delta N_{iL}^{a_k}, \quad (95)$$

$$\dot{n}_{vL}^{a_k} = J_{vL}^{a_k} - \Delta N_{vL}^{a_k}, \quad (96)$$

$$\dot{n}_{vL}^{c_0} = J_{vL}^{c_0} - \frac{n_{vL}^{c_0}}{\tau_{\text{sfp}}} - \Gamma_{\text{col}}^n, \quad (97)$$

$$\dot{n}_{vL}^{c_f} = J_{vL}^{c_f} + \Gamma_{\text{col}}^n - \Delta N_{vL}^{c_f} - \Gamma_{\text{uf}}^n, \quad (98)$$

$$\dot{n}_{vL}^{c_p} = \Gamma_{\text{uf}}^n - \Delta N_{vL}^{c_p}, \quad (99)$$

for  $k = 1, 2, 3$ , with  $J_{vL}^{c_f} = 0$  in the pure-precursor limit of Section 2.8. Number changes only by nucleation, coalescence, transfer and — for the pyramid alone — dissolution: neither growth nor thermal emission creates or destroys a cluster, they only move vacancies into and out of the ones that exist. **No loop family carries a lifetime;** the  $\tau_{avL}$  and  $\tau_{cvL}$  of earlier revisions are gone, and what replaces them is the peripheral emission channel below, which acts on content alone. Note that  $\Gamma_{\text{col}}^n$  appears twice with opposite signs, and  $\Gamma_{\text{uf}}^n$  twice with opposite signs, so summing Eqs. (97) to (99) removes both: the total basal cluster number is blind to where along the chain the population sits, which is the number-side half of the conservation statement of Section 2.10.

$c_p$  is **TERMINAL**, and that is a structural property of the chain rather than a matter of calibration. Read Eq. (99) for what removes it: coalescence, and nothing else. It cannot unfault further, it carries no dissolution lifetime, and peripheral emission acts on its content and never on its number — so under a sustained vacancy supply its population is bounded only by the arrival of the transfer that feeds it. On the 200 nm demonstration march, whose  $\nu_{\text{uf}}$  is a placeholder,  $\tau_{\text{uf}}$  is  $3 \times 10^6$  s against a run length of  $10^8$  s — thirty  $e$ -foldings — and  $c_p$  duly ends up holding most of the stored basal content. That is the chain doing what it is written to do. **Whether a perfect basal loop should have a removal channel of its own is a modelling question this formulation has not answered**, and it should be settled before  $\nu_{\text{uf}}$  is fitted:  $\nu_{\text{uf}}$  and a  $c_p$  sink are not separately identifiable from a steady-state  $c_p$  density alone.

**Nucleation currents.** The interstitial variants nucleate from homogeneous clustering of the mobile species plus a cascade seed; the vacancy families nucleate from the cascade only, because homogeneous vacancy clustering against the prevailing interstitial bias is far too slow to compete. Each current is partitioned over the variants by the stress weights Eq. (77):

$$J_{iL}^{a_k} = w_k^i \left[ (R_{i,3i} + R_{2i,2i}) + \frac{G_{iL}}{n_{iL}^{\text{nuc}}} \right], \quad J_{vL}^{a_k} = w_k^v \frac{G_{avL}}{n_{avL}^{\text{nuc}}}, \quad J_{vL}^{c_0} = \frac{G_{\text{sfp}}}{n_{\text{sfp}}}, \quad J_{vL}^{c_f} = \frac{G_{cvL}}{n_{cvL}^{\text{nuc}}}, \quad (100)$$

with  $n^{\text{nuc}}$  the defects per nucleus. The basal current is written on the *pyramid* and not on the faulted loop: in the pure-precursor limit of Section 2.8  $\varepsilon_{cvL} = 0$ , so  $J_{vL}^{cf} = 0$  and every basal loop in the model has been a pyramid first. Retaining  $\varepsilon_{cvL} > 0$  alongside  $\varepsilon_{\text{sfp}}$  describes a cascade that produces both directly, and is the interpolation between the present model and the previous one. Nucleation conserves mass: the defects each nucleus carries are seeded as content in the corresponding  $c_{sL}^k$ ,

$$\dot{c}_{iL}^{ak}|_{\text{nuc}} = w_k^i (\mathcal{N}_c + G_{iL}), \quad \mathcal{N}_c = 4R_{i,3i} + 2R_{2i,2i} + 3R_{2i,3i}, \quad (101)$$

and likewise  $w_k^v G_{avL}$  for the prismatic vacancy variants and  $G_{\text{sfp}}$ ,  $G_{cvL}$  for the two nucleated basal states, written out in Eqs. (105) and (106). The cascade content seed is essential and not cosmetic: without finite content the  $\sqrt{n c}$  prefactor of Eq. (72) pins a family at the floor and it can never grow.

Equation (100) is where the March-Rico mechanism makes its second demand on the model, the first being  $\chi > 1$  in Eq. (68). The current  $J_{vL}^{ak}$  must be fed from the in-cascade vacancy cluster yield  $\varepsilon_{avL}$  of Eq. (60). Embryos without the same-type capture bias shrink back; the capture bias without embryos has nothing to act on. Neither ingredient alone is sufficient.

**First moment: the content.** At  $j = 1$  the drift integral survives and is the growth law Eq. (71) together with the emission channel derived below, while the diffusion integral still vanishes: spreading a distribution at fixed mean moves no defects. The corresponding content equations collect the growth law Eq. (71), the nucleation seed Eq. (101), the peripheral emission channel Eq. (110), the loop-network coalescence loss Eq. (118) and, on the basal chain alone, the two transfers Eq. (80). The six prismatic families carry no transfer and differ between the characters in exactly two places — which nucleation current feeds them, and the sign of the emission term:

$$\dot{c}_{iL}^{ak} = \dot{c}_{iL}^{ak}|_{\text{growth}} + w_k^i (\mathcal{N}_c + G_{iL}) + \Pi_{iL}^{ak} \bar{D}^v Z_{ia_k,v} c_{ia_k}^{v,\text{eq}} - \Delta C_{iL}^{ak}, \quad (102)$$

$$\dot{c}_{vL}^{ak} = \dot{c}_{vL}^{ak}|_{\text{growth}} + w_k^v G_{avL} - \Pi_{vL}^{ak} \bar{D}^v Z_{va_k,v} c_{va_k}^{v,\text{eq}} - \Delta C_{vL}^{ak}, \quad (103)$$

for  $k = 1, 2, 3$ . **The emission term enters Eq. (102) with a plus sign and Eq. (103) with a minus**, and that single sign —  $-\varsigma_s(v)$  of Eq. (110) — is the whole of the difference between an annealing vacancy loop and an annealing interstitial one on the same habit plane, with the same Burgers vector magnitude, at the same size. It is also the only place in Eqs. (102) and (103) where the two characters are treated differently at all:  $\Pi_{sL}^{ak}$ ,  $Z_{sa_k,v}$  and  $\Delta C_{sL}^{ak}$  have the same functional form for both, and  $\bar{D}^v$  is shared outright.

Two further readings of the pair are worth making explicit. First, the equilibrium concentrations  $c_{ia_k}^{v,\text{eq}}$  and  $c_{va_k}^{v,\text{eq}}$  are *not* equal even at equal size: by Eq. (111) they sit on opposite sides of  $c_{\infty}^{v,\text{eq}}$ , in the ratio

$$\frac{c_{va_k}^{v,\text{eq}}}{c_{ia_k}^{v,\text{eq}}} = \exp \left[ \frac{2}{k_B T} \left( \frac{\partial E_a}{\partial m} - \sigma_{kk} \Omega \right) \right], \quad (104)$$

so a single vacancy field  $c^v$  cannot be in equilibrium with both characters at once. For any  $c^v$  lying between the two — which is where a field sourced by one and drained by the other must sit — the vacancy family emits and the interstitial family absorbs, so vacancies flow through the lattice out of the vacancy loops and into the interstitial ones and *both* populations lose stored content. That is a cross-character annihilation channel, mediated by the mobile vacancies rather than by contact, and it is one the lumped, interstitial-only layout of the earlier model could not express at all. Second, summing Eqs. (102) and (103) over  $k$  and using  $\sum_k w_k^s = 1$  recovers the two lumped prismatic populations of that earlier model term by term, at zero deviatoric stress and with  $\chi = 1$ , which is the regression test on the variant resolution.

The basal chain adds the two transfers, and the pyramid the one lifetime the formulation retains:

$$\dot{c}_{vL}^{c_0} = \dot{c}_{vL}^{c_0}|_{\text{growth}} + n_{\text{sfp}}^{\text{nuc}} J_{vL}^{c_0} - \frac{n_{\text{sfp}}^{\text{nuc}} n_{vL}^{c_0}}{\tau_{\text{sfp}}} - \Pi_{vL}^{c_0} \bar{D}^v Z_{vc_0,v} c_{vc_0}^{v,\text{eq}} - \Gamma_{\text{col}}^c, \quad (105)$$

$$\dot{c}_{vL}^{cf} = \dot{c}_{vL}^{cf}|_{\text{growth}} + \Gamma_{\text{col}}^c + n_{cvL}^{\text{nuc}} J_{vL}^{cf} - \Pi_{vL}^{cf} \bar{D}^v Z_{vc_f,v} c_{vc_f}^{v,\text{eq}} - \Delta C_{vL}^{cf} - \Gamma_{\text{uf}}^c, \quad (106)$$

$$\dot{c}_{vL}^{cp} = \dot{c}_{vL}^{cp}|_{\text{growth}} + \Gamma_{\text{uf}}^c - \Pi_{vL}^{cp} \bar{D}^v Z_{vc_p,v} c_{vc_p}^{v,\text{eq}} - \Delta C_{vL}^{cp}, \quad (107)$$

with no  $\Delta C^{c_0}$ , the pyramid having no coalescence channel, and every emission term negative because all three basal states are vacancy in character. The two  $\Gamma^c$  pairs cancel on summation exactly as the  $\Gamma^n$  pairs do, so Eqs. (102), (103) and (105) to (107) together with the free pool close the balance Eq. (128).



**Thermal emission from the periphery.** Loops in this formulation do *not* anneal with a prescribed lifetime. They exchange vacancies with the lattice across their own perimeter, at a rate fixed by the size-dependent vacancy binding energy, and the shrinkage, the coarsening and the response to stress that a lifetime was standing in for all follow from that single channel. Only the pyramid keeps a lifetime  $\tau_{\text{sfp}}$  (Eq. (97)), because its loss channel is the collapse of a compact faulted cluster and not the climb of a dislocation line.

**The binding energy is computed, not fitted.** The energy of a planar cluster of  $m$  defects is Eq. (8) evaluated at  $R = \lambda_k \sqrt{m}$ , so the work to detach one defect follows by differentiation,

$$\frac{\partial E_k}{\partial m} = \underbrace{\frac{\gamma_k \Omega}{(\mathbf{b} \cdot \mathbf{n})_k}}_{\text{fault, size-independent}} + \underbrace{\frac{\langle K \rangle_k b_k^2 \lambda_k^2}{4R_{sL}^k} \left( 1 + \ln \frac{\alpha R_{sL}^k}{b_k} \right)}_{\text{capillarity, } \propto 1/R}, \quad (108)$$

in which  $\gamma_k$  is the family's fault energy, zero for the two unfaulted states, and  $\langle K \rangle_k$  its perimeter-averaged anisotropic energy factor from Table 1. Nothing here is new parameterization: Eq. (108) is built from the same  $\langle K \rangle_k$ ,  $\gamma_k$ ,  $\lambda_k$  and  $\Omega$  that already fix the family's geometry and its fault state. The vacancy binding energy of family  $(s, k)$  is then

$$E_b^{v,k}(\bar{m}^k) = E_v^f - \varsigma_s(v) \frac{\partial E_k}{\partial m} \Big|_{\bar{m}^k}, \quad \varsigma_s(v) = \begin{cases} +1, & s = v, \\ -1, & s = i, \end{cases} \quad (109)$$

with  $E_v^f$  the bulk vacancy formation energy, which is the  $R \rightarrow \infty$  limit of Eq. (109) for either character. **One expression, both characters**, and the sign of  $\varsigma_s(v)$  is the whole of the difference: detaching a vacancy from a vacancy loop *shrinks* it and releases line energy, so  $E_b < E_v^f$  and the smallest loops emit hardest; detaching a vacancy from an interstitial loop *grows* it and costs line energy, so  $E_b > E_v^f$  and it is the *largest* interstitial loops that emit most readily.

**Emission carries the same prefactor as absorption.** Because  $\Pi_{sL}^k$  of Eq. (73) is the family's perimeter, and emission is a peripheral process, detailed balance fixes the emission rate to the absorption rate coefficient times the vacancy concentration the loop core is in equilibrium with:

$$\dot{c}_{sL}^k \Big|_{\text{emit}} = -\varsigma_s(v) \Pi_{sL}^k \bar{D}^v Z_{sk,v} c_{sk}^{v,\text{eq}}(\bar{m}^k, \boldsymbol{\sigma}), \quad (110)$$

$$c_{sk}^{v,\text{eq}}(\bar{m}^k, \boldsymbol{\sigma}) = c_\infty^{v,\text{eq}} \exp \left[ \frac{1}{k_B T} \left( \varsigma_s(v) \frac{\partial E_k}{\partial m} + \sigma_{kk} \Omega \right) \right], \quad c_\infty^{v,\text{eq}} = \exp \left( -\frac{E_v^f}{k_B T} \right), \quad (111)$$

the stress entering through the work  $\sigma_{kk} \Omega$  done on the defect volume as the loop climbs by one defect — the same term, and the same resolved normal stress, as the emission channel of Section 2.7. **That term carries no**  $\varsigma_s(v)$ : emitting a vacancy inserts  $\Omega$  of material along  $\mathbf{n}_k$  whatever the family's character — it grows an interstitial loop and shrinks a vacancy one, and both insert — so a tensile  $\sigma_{kk}$  raises  $c_{sk}^{v,\text{eq}}$  for both, which is the same direction Eq. (77) partitions the sources in. Equation (174) replaces  $\sigma_{kk}$  by the resolution of the full stress tensor, which differs from it for the faulted basal state alone. Adding Eq. (110) to the vacancy term of Eq. (71) collapses the pair into the form the whole of this paragraph exists to produce,

$$\dot{c}_{sL}^k \Big|_v = \varsigma_s(v) \Pi_{sL}^k \bar{D}^v Z_{sk,v} \left[ c^v - c_{sk}^{v,\text{eq}}(\bar{m}^k, \boldsymbol{\sigma}) \right], \quad (112)$$

manifestly correct in sign for both characters and vanishing identically at  $c^v = c_{sk}^{v,\text{eq}}$  — note that the subscript carries *both* indices, because two families sharing a habit plane but differing in character are in equilibrium with different vacancy concentrations (Eq. (104)) — which is the check that Eqs. (109) and (110) are a detailed-balance pair rather than two independently fitted rates.

**Emission changes content, not number.** Equation (110) enters the content equation and leaves the density equation untouched, so under a pure anneal a family's mean size falls while its density stays exactly constant. That is a property of the reduced state rather than an approximation waiting to be repaired: the model carries one mean size  $\bar{m}^k$  per family and no size distribution, so there is no population of smallest members to reach zero and be lost, and a number-loss term proportional to the emission rate would smuggle in a distribution the state vector does not represent. Loop density changes by nucleation, by the two coalescence channels of Eq. (118), and — on the basal chain — by the transfers Eq. (80). Nothing else removes a loop. The gate Eq. (75) closes the shrinking channel as



$\bar{R}_{sL}^k \rightarrow R_{\min}^k$ , so a family that has emitted its content away parks at the floor with its density intact rather than running negative.

**That is also the one place the reduced state is visibly coarser than the physics**, and it is worth naming because an annealing experiment is exactly where it shows. A real distribution loses its smallest loops first, so a measured anneal reports the density *falling* and the surviving mean size *rising*, where Eq. (110) on two moments gives constant density and a falling mean size. Section 2.9 is the resolution, and it is the reason the second moment is carried at all: with a distribution in hand the density falls through a computed current Eq. (127) rather than through a fitted rate, and the surviving mean rises because it is the smallest loops that leave. Everything in the present paragraph is therefore the *two-moment* limit of what follows, recovered exactly at  $\Delta_k = 1$ .

Every emitted vacancy reaches the mobile pool, whatever the character of the family that released it, so Eq. (27) is sourced by

$$\mathbf{s}_{vL} = \left[ \sum_{(s,k) \in \mathcal{F}} \Pi_{sL}^k \bar{D}^v Z_{sk,v} c_{sk}^{v,\text{eq}} + \frac{n_{\text{sfp}}^{\text{nuc}} n_{vL}^{c_0}}{\tau_{\text{sfp}}}, 0, 0, 0 \right]^T, \quad (113)$$

the first sum running over all nine families and the second being the pyramid's dissolution. Note that this sum carries *no*  $\varsigma_s(v)$ : an interstitial loop emitting a vacancy puts that vacancy in the pool and simultaneously adds an interstitial to itself. That is thermal Frenkel-pair generation by climb, it is the mechanism by which an interstitial loop can grow with no irradiation at all, and it is booked as such in Eq. (135) — equally into both production integrals, so that  $I_{\text{st}} - V_{\text{st}}$  is untouched by it.

**What a pure anneal gives, and the threshold that reverses it.** Equation (112) reduces the sign of every family's response to one comparison,

$$\text{sign}(\dot{c}_{sL}^k|_v) = \varsigma_s(v) \text{sign}(c^v - c_k^{v,\text{eq}}), \quad (114)$$

and switching the source off makes it predictive at constant loop density. **Vacancy loops shrink at zero applied stress, and faster under tension:**  $\varsigma_v(v) = +1$  and  $\partial E_k / \partial m > 0$  put  $c_{vk}^{v,\text{eq}}$  above  $c_\infty^{v,\text{eq}}$  at every finite size, so  $c^v < c_{vk}^{v,\text{eq}}$  for any ambient field at or below bulk equilibrium, and the shrinkage accelerates as  $\bar{m}^k$  falls because  $\partial E_k / \partial m$  rises. That is the runaway a lifetime could only imitate, and it comes out of Eq. (108) with no annealing parameter at all. **Interstitial loops grow whenever**  $c^v < c_{ik}^{v,\text{eq}}$ , which is the same inequality read through  $\varsigma_i(v) = -1$ . At zero stress it fails by exactly the capillarity factor, so a genuinely unstressed anneal at  $c^v = c_\infty^{v,\text{eq}}$  has them creeping slowly *downward* rather than growing; what reverses them is the stress term of Eq. (111), which lifts  $c_{ik}^{v,\text{eq}}$  past the ambient field once the resolved normal stress on the habit plane exceeds

$$\sigma_k^* \equiv \frac{1}{\Omega} \frac{\partial E_k}{\partial m} \Big|_{\bar{m}^k} = \frac{\gamma_k}{(\mathbf{b} \cdot \mathbf{n})_k} + \frac{\langle K \rangle_k b_k^2 \lambda_k^2}{4 \Omega R_{iL}^k} \left( 1 + \ln \frac{\alpha R_{iL}^k}{b_k} \right). \quad (115)$$

Above  $\sigma_k^*$  the interstitial family climbs outward by emitting vacancies and grows with no interstitial supply whatever. Equation (115) is the classical climb threshold; it falls as  $1/R$ , so the largest loops pass it first and the family coarsens in size at fixed number, and it is the thermal-creep limit of the same channel that Eq. (77) drives under irradiation. It has a mirror on the vacancy character, which the corrected sign of Eq. (111) makes explicit: a *compressive* normal stress exceeding  $\sigma_k^*$  makes a vacancy loop grow, so Eq. (115) is one threshold read twice,  $\eta_s \sigma_{kk} > \sigma_k^*$ , and Section 4.1 collects both. A stress-free anneal is therefore one regression test on Eq. (110) — vacancy loops shrink, interstitial loops creep slowly downward, both densities constant — and an anneal under  $\sigma_{kk} > \sigma_k^*$  is the test that the emission channel is signed correctly for the interstitial character.

**Geometric coalescence.** Beyond flux-driven growth, loops coalesce through two channels, with rate scales set by the *absorbed-flux* climb speed — which, unlike the net growth velocity, stays positive at saturation, so coarsening persists. For a family of density  $n$ , content  $c$ , length scale  $\lambda$ , radius  $R = \lambda \sqrt{c/n}$  and number per volume  $N = n/\Omega$ , capping the radius at the relevant spacing ( $R_{LL} = \min(R, (\Omega/n)^{1/3})$ ,  $R_{LN} = \min(R, \rho_N^{-1/2})$ ) gives the Avrami overlap probabilities and rate scales

$$\phi_{LL} = 1 - \exp(-\kappa_{LL} \frac{4}{3} \pi R_{LL}^3 N), \quad \phi_{LN} = 1 - \exp(-\kappa_{LN} \pi R_{LN}^2 \rho_N), \quad (116)$$

$$\nu_{LL} = c_{LL} v_{\text{abs}} N^{1/3}, \quad \nu_{LN} = c_{LN} v_{\text{abs}} \sqrt{\rho_N}, \quad (117)$$

and the number and content losses

$$\Delta N_{LN} = \nu_{LN} \phi_{LN} n, \quad \Delta N = \nu_{LL} \phi_{LL} n + \Delta N_{LN}, \quad \Delta C = \nu_{LN} \phi_{LN} c. \quad (118)$$

The like-loop channel merges two loops into a larger one and therefore *conserves content*, removing density only; the loop–network channel removes both, and the removed content is booked as network sink absorption so the point-defect balance stays closed. The absorbed line length feeds  $\rho_N$  through Eq. (63). The capping is what makes Eq. (117) saturate rather than diverge as the number density collapses.

Opposite-character loop annihilation is available but is used cautiously, since a mean-field term overstates interactions that in reality require proximity, loop mobility, and a favorable elastic geometry. The caution applies with greatest force to  $(i, a_k) + (v, a_k)$  annihilation on the *same* variant: two prismatic loops of opposite character on one habit plane can annihilate outright, so a mean-field rate for that channel would suppress the very coexistence the model exists to reproduce. It should be gated on the variant index and on separation rather than applied as a bulk term — which is one concrete benefit of resolving the variants, since the lumped layout had no index on which to gate it.

**Second moment: the dispersion.** At  $j = 2$  both integrals survive, and the second is the term that has no counterpart in either equation above. Evaluating Eq. (94) at  $j = 2$  gives, with the closure integrals written in the compact form the next paragraph justifies,

$$\dot{q}_{sL}^k|_{\text{FP}} = 2\Pi_{sL}^k \left[ \underbrace{\Sigma_k^{\text{net}} \bar{m}_{sL}^k \Delta_k^{3/8}}_{\text{drift}} + \underbrace{\frac{1}{2} \Sigma_k^{\text{gro}} \Delta_k^{-1/8}}_{\text{spreading}} \right], \quad (119)$$

to which the same nucleation, transfer and floor terms are appended as in the lower two moments, weighted by  $m^2$  instead of  $m^1$  or  $m^0$ :

$$\dot{q}_{sL}^k = \dot{q}_{sL}^k|_{\text{FP}} + J_{sL}^k (m_k^{\text{nuc}})^2 - m_{\min}^2 \Phi_{\min}^k \pm \Gamma_{\theta}^q, \quad (120)$$

the transfer terms  $\Gamma_{\theta}^q$  appearing on the basal chain only and with the same signs as  $\Gamma_{\theta}^n$  and  $\Gamma_{\theta}^c$  in Eqs. (97) to (99). **The spreading term of Eq. (119) does not vanish when the family stops growing.** At  $\Sigma_k^{\text{net}} = 0$  — a family in flux balance, its mean size stationary — the drift term is zero and  $\dot{q}_{sL}^k = \Pi_{sL}^k \Sigma_k^{\text{gro}} \Delta_k^{-1/8} > 0$ : the distribution keeps broadening at fixed mean size and fixed number. There is no value of the two carried fields that expresses that.

**Equation (120) is incomplete, and the missing channel is the dominant one.** It lists the Fokker–Planck part, nucleation, the floor current and the transfers — but the density and content equations also carry *annealing* and *coalescence*, and a moment that ignores a channel the other two moments obey does not stay a moment. Coalescence is the one that matters, because it carries 98% of the loop-number loss. It must be written with care and it is easy to write backwards:

**A coalescing loop does not leave the family.** It *merges*, so its defects stay and only  $\dot{c}_{sL}^k|_{\text{coal}}$  is genuinely lost. Fewer loops holding the same content are *larger* loops, so coalescence **raises** the second moment. Debiting  $\dot{q}$  by  $\langle m^2 \rangle$  per departing loop — the natural-looking reading, and the one first implemented — has the sign inverted on the largest single term in the equation.

Neither channel carries a distribution of its own, so both are closed *shape-preserving*: from  $q = \Delta_k c^2/n$ , a channel moving  $(n, c)$  by  $(dn, dc)$  at fixed  $\Delta_k$  moves  $q$  by

$$dq = \Delta_k \left( 2\bar{m}_{sL}^k dc - (\bar{m}_{sL}^k)^2 dn \right), \quad (121)$$

which for coalescence at conserved content integrates to  $q \propto 1/n$  exactly. It is a conservative closure — merging really broadens a distribution, so the true dispersion grows faster than this — but it is what keeps the moment set inside  $\Delta_k \geq 1$ . Written with the inverted sign instead,  $q$  crosses zero at 1 dpa on the 200 nm demonstration march and is negative on 40% of the  $c_p$  nodes by 10 dpa.

**One channel cannot be made consistent at all, and it is worth saying why.** The annealing surrogate that Eq. (113) replaces removes loops at rate  $1/\tau$  while debiting content at the *fixed* size  $n_{vL}^{\text{nuc}}$ , so the pair it declares is a removed sub-population of mean  $M \neq \bar{m}_{sL}^k$ . Debiting  $q$  proportionally over-removes; debiting at  $M$  — the smallest value Cauchy–Schwarz permits for the removed part — lowers  $\Delta_k$  by  $a(M/\bar{m}_{sL}^k - 1)^2$  on a narrow family. No debit keeps the triple realizable, because the surrogate is removing loops of a size that is not there. That is one more argument for the peripheral-emission channel of Eq. (113): it deletes the lifetimes outright, and step 3 precedes step 5 for exactly this kind of reason.

**Log-normal closure, and the one exponent it produces.** Close Eq. (120) by taking  $f_k$  log-normal in  $m$ ,

$$f_k(m) = \frac{n_{sL}^k}{m s_k \sqrt{2\pi}} \exp \left[ -\frac{(\ln m - \mu_k)^2}{2s_k^2} \right]. \quad (122)$$

Its two parameters follow from the three carried moments, and specifically — this is the whole of the construction — from the *ratio of the content to the density* and from how far the second content departs from the square of that ratio:

$$\bar{m}_{sL}^k = \frac{c_{sL}^k}{n_{sL}^k}, \quad \Delta_k \equiv \frac{q_{sL}^k n_{sL}^k}{(c_{sL}^k)^2} = e^{s_k^2} \geq 1, \quad s_k^2 = \ln \Delta_k, \quad \mu_k = \ln \bar{m}_{sL}^k - \frac{1}{2}s_k^2. \quad (123)$$

$\Delta_k$  is the *dispersion*: the variance in size is  $\sigma_k^2 = (\bar{m}_{sL}^k)^2(\Delta_k - 1)$ , so  $\sqrt{\Delta_k - 1}$  is the relative width, and  $\Delta_k \rightarrow 1$  is the Dirac delta the previous formulation carried. Every closure integral the model needs is then a single expression,

$$\boxed{\langle m^j \rangle_k = (\bar{m}_{sL}^k)^j \Delta_k^{j(j-1)/2}} \quad \text{for every real } j, \quad (124)$$

which returns  $\bar{m}$  at  $j = 1$ ,  $q/n$  at  $j = 2$ , and  $(\bar{m})^j$  for every  $j$  when  $\Delta_k = 1$ . **Equation (124) at  $\Delta_k = 1$  is the regression test on this entire subsection:** it recovers the two-moment model exactly, for every term, without setting any rate to zero.

Two consequences of Eq. (124) are worth isolating.

1. **The half-integer moments the growth law needs are closed-form.** Loop growth scales with the perimeter, so it needs  $\langle m^{1/2} \rangle$  and the second moment needs  $\langle m^{3/2} \rangle$ ; a Gaussian gives neither in closed form and admits  $m < 0$ , where Eq. (124) gives  $\bar{m}^{1/2} \Delta^{-1/8}$  and  $\bar{m}^{3/2} \Delta^{3/8}$  on a strictly positive support. That, and not positivity alone, is why the closure is log-normal.
2. **The whole effect of the spread on growth and on the sink strengths is one factor  $\Delta_k^{-1/8}$ .** Since  $\Pi_{sL}^k \propto n \langle m^{1/2} \rangle$ ,

$$\Pi_{sL}^k \longrightarrow \Pi_{sL}^k \Delta_k^{-1/8}, \quad \rho_{sL}^k \longrightarrow \rho_{sL}^k \Delta_k^{-1/8}, \quad (125)$$

in Eqs. (38), (71) and (72) alike — a reduction, because  $\sqrt{\cdot}$  is concave, so a spread family presents *less* perimeter than a monodisperse one of the same density and content. It is a modest correction and it is not the reason to carry the second moment:  $\Delta_k = 1.25$  (a 50% relative width) gives 0.972 and  $\Delta_k = 2$  (a 100% width) gives 0.917. What the second moment is for is everything below.

**The gates and the dissolution current stop being assumptions.** Because  $R = \lambda_k \sqrt{m}$  and the log-normal is closed under the square-root map,  $R$  is log-normal too, with  $\mu_k^R = \ln \lambda_k + \mu_k/2$  and  $s_k^R = s_k/2$ . Every size-threshold integral in the formulation is therefore an error function of the carried moments. The transfer gates of Eq. (79) become

$$\Phi_\theta^{(j)} = \frac{1}{2} \operatorname{erfc} \left( \frac{\ln m_\theta - \mu_k - j s_k^2}{s_k \sqrt{2}} \right), \quad m_\theta = \left( \frac{R_\theta}{\lambda_k} \right)^2, \quad (126)$$

in which  $\Phi_\theta^{(0)}$  is the fraction of *loops* above the threshold,  $\Phi_\theta^{(1)}$  the fraction of *content* they carry and  $\Phi_\theta^{(2)}$  the fraction of the second moment. Since  $\Phi_\theta^{(2)} > \Phi_\theta^{(1)} > \Phi_\theta^{(0)}$ , a transfer removes a subpopulation

larger than the family mean, which is what a size-gated reaction physically does and what one rate multiplying all three moments could not express.

The dissolution current is the same statement at the other end of the distribution. It is the Fokker–Planck flux through the floor,

$$\Phi_{\min}^k = \left[ D_k \partial_m f_k + (\partial_m D_k - F_k) f_k \right]_{m=m_{\min}}, \quad f_k(m_{\min}) = \frac{n_{sL}^k \exp \left[ -\frac{(\ln m_{\min} - \mu_k)^2}{2s_k^2} \right]}{m_{\min} s_k \sqrt{2\pi}}, \quad (127)$$

every factor of which is analytic in  $(\mu_k, s_k)$  and hence in  $(n_{sL}^k, c_{sL}^k, q_{sL}^k)$ . Loops crossing the floor carry  $m_{\min}$  defects each, so they debit  $\Phi_{\min}^k$ ,  $m_{\min} \Phi_{\min}^k$  and  $m_{\min}^2 \Phi_{\min}^k$  from the three moments and release  $m_{\min} \Phi_{\min}^k$  vacancies into the pool through Eq. (113). That is exactly conservative by construction, and it replaces Eq. (75): the smoothstep gate was a device for preventing a mean size from crossing a floor it had no distribution to cross, and with Eq. (127) the family loses loops at a rate the distribution determines instead of being prevented from losing them at all. **Note what Eq. (127) gives at  $\Delta_k = 1$ :** a delta function has no tail, so  $\Phi_{\min}^k \equiv 0$  and the dissolution rate of the previous formulation was not small but identically zero.

**What this buys, which is an annealing simulation that behaves.** Section 2.9 recorded that a two-moment model under a pure anneal gives a falling mean size at exactly constant density, where a measurement reports the density falling and the surviving mean size rising. With Eqs. (120), (123) and (127) the model reproduces the measurement rather than contradicting it, and it does so without a fitted dissolution rate:  $\Sigma_k^{\text{net}} < 0$  drives  $\bar{m}_{sL}^k$  down, the spreading term of Eq. (119) drives  $\Delta_k$  up even as the family shrinks, the lower tail of Eq. (122) crosses  $m_{\min}$ , and  $\Phi_{\min}^k$  turns on. Because it is the *smallest* loops that are removed, the surviving mean rises against the drift — the two effects compete, and which wins is a prediction of Eq. (92) rather than an input. The measured  $D_k/F_k$  says that in this material the spreading is the stronger of the two, which is the quantitative form of the statement that irradiated loop populations coarsen.

**Cost, and what remains assumed.** The immobile state grows from two fields per family to three, 18 to 27 components per node, and the dense per-point factorization of Section 6.5 from  $18^3$  to  $27^3$  — a factor 3.4, which turns the measured 28.3s slow sweep into roughly 95s against a 925s fast solve. The spatial structure is untouched: Eq. (120) is local, like every other immobile equation, so the argument of Section 6.5 carries over unchanged and the third moment is affordable for the same reason the first two are. What remains assumed is the *shape*: a log-normal is a two-parameter family and cannot represent a bimodal size spectrum within one node. That limit is reachable — Ostwald ripening produces skewed and occasionally bimodal spectra, and  $F'_k(m) > 0$  here because  $F_k \propto \sqrt{m}$ , so the distribution broadens without bound rather than relaxing to a steady width. **The diagnostic is  $\Delta_k$  itself:** it is carried, it is dimensionless, and a family whose dispersion runs away is a family whose closure should be reported as out of range rather than trusted. The honest statement of what this subsection achieves is therefore comparative and not absolute — a two-parameter distribution in place of a delta function, with the parameter that measures the difference carried as a state variable.

## 2.10 The basal ledger: conservation across the chain

Adding a species to a chain is the point at which an atom balance is most easily broken, because a transfer that is written with two independent rates leaks at one end. Table 5 is the audit: every term that moves vacancies into or out of the three basal states, and the term that carries the opposite sign. Nothing in the chain is a source or a sink in its own right.

Summing Eqs. (105) to (107) with the prismatic vacancy families and the free vacancy pool, every internal transfer and every emission pair cancels and what survives is the open-system balance

$$\frac{d}{dt} \left( c^v + \sum_{(v,k) \in \mathcal{F}} c_{vL}^k \right) = \underbrace{G_v + G_{avL} + G_{cvL} + G_{\text{sfp}}}_{=G} - \dot{\Sigma}^{\text{rec}} - \dot{\Sigma}_v^{\text{sink}} - \mathcal{B}_v, \quad (128)$$

identical in form to the balance before the pyramid was added, with  $\mathcal{B}_v$  the flux through the absorbing surface of Section 6.8. **The chain changes where vacancies are held, never how many,** and

Table 5: Vacancy ledger for the basal chain Eq. (14). Each row is a term in Eqs. (105) to (107); the last column names where the compensating term sits. The two transfer rows cancel *within* the immobile block; every other row is matched outside it, and no row is unmatched.

| Channel                                         | Term                                                            | Compensating term                                                                                                                                                                                                                              |
|-------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cascade production of pyramids                  | $+n_{\text{sfp}}^{\text{nuc}} J_{vL}^{c_0} = G_{\text{sfp}}$    | Removed from the free-vacancy source $G_v$ in Eq. (60); counted once in $\dot{\Sigma}_v^{\text{prod}}$ .                                                                                                                                       |
| Absorption of mobile vacancies                  | $+ \text{terms of Eq. (71)}$                                    | The matching debit in $\text{loop\_abs}_v$ , Eq. (76), guaranteed term-by-term by Eq. (73).                                                                                                                                                    |
| Absorption of mobile interstitials              | $- \text{terms of Eq. (71)}$                                    | Booked as $\text{loop\_recomb}$ ; the annihilated pair leaves both pools.                                                                                                                                                                      |
| Pyramid dissolution                             | $-n_{\text{sfp}}^{\text{nuc}} n_{vL}^{c_0} / \tau_{\text{sfp}}$ | Returned to the mobile vacancy line by $\mathbf{S}_{vL}$ , Eq. (113).                                                                                                                                                                          |
| Peripheral emission, vacancy family             | $-\Pi_{vL}^k \overline{D}^v Z_{vk,v} c_{vk}^{v,\text{eq}}$      | Returned to the mobile vacancy line by $\mathbf{S}_{vL}$ , Eq. (113): a pure transfer, the family shrinking by what the pool gains.                                                                                                            |
| Peripheral emission, <i>interstitial</i> family | $+\Pi_{iL}^k \overline{D}^v Z_{ik,v} c_{ik}^{v,\text{eq}}$      | <b>Not</b> a transfer. The loop gains an interstitial and the pool gains a vacancy: thermal Frenkel-pair generation by climb, entered into <i>both</i> production integrals of Eq. (135) so that $I_{\text{st}} - V_{\text{st}}$ is conserved. |
| <b>Collapse</b> $c_0 \rightarrow c_f$           | $-\Gamma_{\text{col}}^c, +\Gamma_{\text{col}}^c$                | <b>Internal.</b> One rate multiplies both, Eq. (80): cancels exactly.                                                                                                                                                                          |
| <b>Unfaulting</b> $c_f \rightarrow c_p$         | $-\Gamma_{\text{uf}}^c, +\Gamma_{\text{uf}}^c$                  | <b>Internal.</b> One rate multiplies both, Eq. (80): cancels exactly.                                                                                                                                                                          |
| Loop-network coalescence                        | $-\Delta C^k$                                                   | Booked as network sink absorption in $\dot{\Sigma}_v^{\text{sink}}$ , Eq. (135); the line length feeds $\rho_N$ , Eq. (63).                                                                                                                    |
| Loop-loop coalescence                           | number only                                                     | Content-conserving by construction, Eq. (118).                                                                                                                                                                                                 |
| Dissolution through the floor                   | $-m_{\text{min}} \Phi_{\text{min}}^k$                           | Loops crossing $m_{\text{min}}$ carry $m_{\text{min}}$ defects each, released to the mobile vacancy line by $\mathbf{S}_{vL}$ , Eq. (113); the current itself is Eq. (127), not a fitted rate. Identically zero on two moments.                |

the sum Eq. (128) is the statement to assert on in a verification run: it must hold to solver tolerance at every substep and for any values of  $\nu_{\text{col}}$ ,  $\nu_{\text{uf}}$ ,  $\tau_{\text{sfp}}$ , the two gates and the dispersions  $\Delta_k$ , including the degenerate ones. Three limits are worth checking separately because each collapses the chain a different way:  $\nu_{\text{col}} \rightarrow \infty$  (the pyramid is transient and the model reduces to direct  $c_f$  nucleation),  $\tau_{\text{sfp}} \rightarrow \infty$  with  $\nu_{\text{col}} = 0$  (no basal loops ever form and all basal vacancies accumulate in  $c_0$ ), and  $\varepsilon_{\text{sfp}} = 0$  (the previous revision of the model, bit for bit).

## 2.11 Cluster morphology: the compact state is basal, and is a species

A vacancy cluster in  $\alpha$ -Zr has two possible geometries, and the model needs both. The compact one is the pyramid of Section 1.2 and the planar one is the loop:

$$R_{\text{pyr}}^k = \left( \frac{m^k \Omega}{\sqrt{8}} \right)^{1/3}, \quad R_{\text{loop}}^k = \sqrt{\frac{m^k \Omega}{\pi b^k}}, \quad R^k(m^k) = [1 - S^k] R_{\text{pyr}}^k + S^k R_{\text{loop}}^k, \quad (129)$$

the third form being the *interpolation* between them that earlier revisions of this model applied to every vacancy family, with the sigmoidal switch

$$S^k(m^k) = \frac{1}{1 + \exp \left[ -\frac{m^k - m_0^k}{w^k} \right]}, \quad m_0^k = \frac{m_{\min}^k + m_{\max}^k}{2} - m_s^k, \quad w^k = w_0 (m_{\max}^k - m_{\min}^k), \quad (130)$$

and a matching interpolation of the formation-volume tensor that Eq. (78) couples to the stress,

$$\Omega^k(m^k) = (1 - S^k) \Omega_{\text{pyr}}^k + S^k \Omega_{\text{loop}}^k, \quad \Omega_{\text{loop}}^k = \mathcal{I}^k \frac{\mathbf{b}^k \otimes \mathbf{b}^k}{\mathbf{b}^k \cdot \mathbf{b}^k} V^k, \quad \Omega_{\text{pyr}}^k = \mathcal{I}^k \frac{\mathbb{I}}{3} V^k, \quad (131)$$

with  $V^k = m^k \Omega$  the cluster volume and  $\mathcal{I}^k$  the family selector.

**The two geometries are kept and the interpolation between them is dropped.**  $R_{\text{pyr}}$  and  $\Omega_{\text{pyr}}$  are the geometry of  $c_0$  — Eq. (66) is  $R_{\text{pyr}}$ , and the isotropic  $\Omega_{\text{pyr}}$  is what gives  $\zeta_{c_0} \simeq \frac{1}{3}$  in Eq. (199) — while  $R_{\text{loop}}$  and  $\Omega_{\text{loop}}$  are the geometry of the eight families of  $\mathcal{F}_L$ . What is removed is the sigmoid:  $S^k \equiv 1$  for every  $(s, k) \in \mathcal{F}_L$ , so no loop family is ever assigned a partly compact shape, and no compact cluster is represented as a partly planar loop. The reason is that the compact state and the planar state are not two shapes of one population; they are two populations connected by a collapse, and only on the basal branch.

**Why the compact state is basal and nothing else.** Three results, taken together, leave no prismatic role for Eq. (130).

1. **The pyramid is built on the basal plane by construction.** Christiaen et al. find the compact faulted vacancy cluster in hcp Zr to be a pyramid or bi-pyramid *whose base rests on the basal plane*, carrying the extrinsic basal fault of  $\frac{1}{2}[0001]$  with prismatic facets closing it; and they find the  $\langle c \rangle$  vacancy loop to form not by accretion of vacancies onto a platelet but by *collapse of that pyramid beyond a critical size* [18]. Atomistic studies of vacancy clusters up to a few hundred vacancies report the same ordering: small basal vacancy clusters relax to faulted pyramidal structures and become faulted  $\langle c \rangle$  loops only at larger sizes [19, 17]. The object has a basal base and a basal collapse product. There is no prismatic counterpart of it to interpolate towards.
2. **On the prismatic branch the planar loop is already the ground state.** Formation energies for vacancy clusters of differing geometry, parameterized against ab initio data, place the prismatic vacancy  $\langle a \rangle$  loop *below* both the compact cavity and the faulted basal  $\langle c \rangle$  loop [17]. A compact prismatic precursor would therefore have to be a higher-energy state that a vacancy  $\langle a \rangle$  loop passes through on its way down, and where atomistics does see a pyramid-like cluster in a prismatic setting it *relaxes into* the prismatic loop rather than persisting as a metastable precursor to it [19]. A relaxation with no residence time is not a population, and a mean-field species is the wrong object to represent it.



3. **Prismatic vacancy loops are born planar, in the cascade.** The nucleation channel this formulation already uses for them,  $J_{vL}^{ak} = w_k^v G_{avL} / n_{avL}^{\text{nuc}}$  of Eq. (100), is in-cascade vacancy clustering, and it is the channel by which the atomistically informed cluster-dynamics treatment of March-Rico, Blondel and Wirth nucleates vacancy  $\langle a \rangle$  loops — assisted by in-cascade clustering and stabilized in growth by the thermal-drift capture radii, with no compact precursor anywhere in the chain [20]. A prismatic vacancy loop in this model is created as a loop, at  $n_{avL}^{\text{nuc}}$  vacancies, and is a loop for its whole life.

**What the sigmoid was doing, and what replaces it.** It was a surrogate for the basal precursor, and it was calibrated as one: the transition size of order 14 nm that made Eq. (130) reproduce the observed onset of the basal population [24] is the collapse size of Eq. (13) wearing a different name. Applying it to the prismatic vacancy families was an artifact of a layout in which those families did not exist separately from the basal one, not a claim about prismatic crystallography. The chain Eq. (14) now carries the same physics where it belongs and carries more of it: a barrier Eq. (79), a residence time, a branching ratio Eq. (81) and a conserved transfer Eq. (80), where the sigmoid had no barrier, no residence time and no separate state variable, and therefore could not produce an incubation period at all. Setting  $\varepsilon_{\text{sfp}} = 0$  and restoring Eq. (130) on the vacancy families returns the previous treatment exactly, which is the regression test that the two are alternatives and not additions.

## 2.12 Conservation: the atom-balance accumulators

**Atom-balance accumulators.** Six monotone integrals are carried alongside the physics purely as a conservation diagnostic; they do not feed back. With the stored interstitial and vacancy atoms

$$I_{\text{st}} = c^i + 2c^{2i} + 3c^{3i} + \sum_{(i,k) \in \mathcal{F}} c_{iL}^k, \quad V_{\text{st}} = c^v + \sum_{(v,k) \in \mathcal{F}} c_{vL}^k, \quad (132)$$

the production, recombination and sink integrals are

$$\dot{\Sigma}_i^{\text{prod}} = G_i + 2G_{2i} + 3G_{3i} + G_{iL} + \dot{\Sigma}^{\text{climb}}, \quad \dot{\Sigma}_v^{\text{prod}} = G_v + G_{avL} + G_{cvL} + G_{\text{sfp}} + \dot{\Sigma}^{\text{climb}}, \quad (133)$$

$$\dot{\Sigma}^{\text{rec}} = R_{i,v} + R_{2i,v} + R_{3i,v} + \text{loop\_recomb}, \quad (134)$$

$$\dot{\Sigma}_i^{\text{sink}} = R_{i,s} + 2R_{2i,s} + 3R_{3i,s} + \sum_{(i,k)} \Delta C^k, \quad \dot{\Sigma}_v^{\text{sink}} = R_{v,s} + \sum_{(v,k)} \Delta C^k. \quad (135)$$

in which

$$\dot{\Sigma}^{\text{climb}} = \sum_{(i,k) \in \mathcal{F}} \Pi_{iL}^k \bar{D}^v Z_{ik,v} c_{ik}^{v,\text{eq}} \quad (136)$$

is the thermal Frenkel-pair generation of Eq. (113): an interstitial loop emitting a vacancy adds one interstitial to itself and one vacancy to the pool, so it enters both integrals equally and leaves  $I_{\text{st}} - V_{\text{st}}$  — the quantity the balance actually constrains — untouched. Internal transfers — clustering, vacancy-family emission, and both steps of the basal chain Eq. (14) — cancel in the atom-weighted sums and are deliberately excluded; Table 5 is the term-by-term audit. **In a spatially resolved model Eq. (135) does not close**, and the residual is not solver error: it is the atoms that left through the absorbing surface. Section 6.8 treats this as the physical channel it is.

**What this subsection no longer contains.** Growth, creep and hardening were carried here in an earlier revision, which conflated a conservation diagnostic with the observables the model exists to predict. They are the subject of Section 5, where they can be stated for both the continuum and the discrete representation — a distinction that does not arise for Eq. (135), since the accumulators are integrals of the 0-D right-hand side and are indifferent to how the loops are represented.

## 3 From the Continuum Field to a Discrete Loop Population

The immobile block of Section 2.3 carries loops as fields. A discrete dislocation calculation carries them as line objects with a Burgers vector, a habit plane, a center and an outline. This section is the map between the two: given the nine fields  $(n_{sL}^k, c_{sL}^k, q_{sL}^k)$  at every node, it constructs a set of individual loops that reproduces them. The construction is used in two ways — offline, to export a configuration



for visualization, for TEM comparison or for a hardening calculation; and online, as the handoff at which the mean-field treatment of a family is abandoned and its loops are given to the discrete solver. Both need the same guarantees, and this section states them as conservation requirements rather than as implementation detail.

Two things distinguish what follows from the procedure the earlier note described. The first is that the size of each loop is now *drawn*, because [Section 2.9](#) carries a distribution and not a single size; the second is that the loops are *elliptical*, because a discrete loop has an outline where a continuum field has only an area. Neither was available before, and both change what the transfer conserves.

### 3.1 What the transfer must preserve

Order the requirements, because they cannot all be met exactly and the ranking is a modeling decision rather than a numerical one.

1. **Stored defects, exactly.**  $\sum_{\ell} m_{\ell}$  over the discrete loops of a family must equal the continuum content of the region they replace. This is the atom balance of [Section 2.12](#) and it is not negotiable: a transfer that loses defects appears downstream as a spurious sink.
2. **Loop number, to the integer draw.** A continuum density gives a real expected count and a discrete population an integer one, so the best available is  $|N_{\text{draw}} - \Lambda| \leq \frac{1}{2}$ .
3. **The dispersion  $\Delta_k$ , to the tail truncation of the size sampling.** This requirement did not exist before [Section 2.9](#), because a monodisperse family has nothing to reproduce. It is also the weakest of the four guarantees, and [Section 3.4](#) is explicit about why: the sampled distribution is truncated where the continuum's is not, and  $\Delta_k$  is the moment ratio that truncation reaches first.
4. **Sink strength, reported but not asserted on.** The line length the discrete population presents differs from  $\rho_{sL}^k$  for reasons that are physical rather than numerical — ellipticity is one of them — and the difference is quantified in [Section 3.7](#) instead of being forced to zero.

**Only the eight loop families convert.** The stacking-fault pyramid  $c_0$  of [Eq. \(64\)](#) has no Burgers vector and no line, so it has no discrete representation in a dislocation code at all. It stays in the continuum across every transfer, and a conversion that quietly folded its content into  $c_f$  would be reporting basal vacancies as dislocation content that the solver would then be free to move.

### 3.2 Nodal volumes: a restricted Voronoi tessellation

The fields are known at the  $M$  nodes  $\{\mathbf{x}_j\}$  of [Section 2.1](#), which are the second-order element nodes and are *not* uniformly spaced. Each node must therefore be given the material volume it represents. The natural assignment is the Voronoi cell

$$\mathcal{V}_j = \{\mathbf{x} : \|\mathbf{x} - \mathbf{x}_j\| \leq \|\mathbf{x} - \mathbf{x}_k\| \ \forall k\}, \quad (137)$$

restricted to the crystal,  $\mathcal{V}_j^{\Omega} = \mathcal{V}_j \cap \Omega$ , since the cells of boundary nodes are otherwise unbounded. The restricted cells partition the crystal exactly,  $\sum_j \mathcal{V}_j = V_{\Omega}$  with  $V_j = \text{vol}(\mathcal{V}_j^{\Omega})$ , which is the property every count below depends on.

**$\Omega$  is the convex hull of the node cloud, not its bounding box,** and on a non-cubic domain the distinction is not cosmetic. For a hexagonal prism the axis-aligned bounding box of a regular hexagon of circumradius  $a$  has area  $4\sqrt{3}a^2/2$  against the hexagon's  $3\sqrt{3}a^2/2$ , so

$$\frac{V_{\text{box}}}{V_{\text{crystal}}} = \frac{4}{3}, \quad (138)$$

and every loop count taken on the box is a third too high. The same hull supplies the faces used for the outline, the hidden-line test and the clip in the rendering, so one geometric object serves all of them and no part of the pipeline can disagree with another about where the crystal is.

Exact clipped three-dimensional Voronoi cells are expensive. Monte Carlo integration gives the volumes without constructing any faces: draw  $K$  points  $\mathbf{y}_1, \dots, \mathbf{y}_K$  uniformly in  $\Omega$  — by rejection

against the hull, so that samples landing in the six empty wedges of a prism are discarded rather than charged to the nearest boundary node — assign each to its nearest node, and count:

$$C_j = \sum_{\kappa=1}^K \mathbb{1}[\arg \min_i \|\mathbf{y}_\kappa - \mathbf{x}_i\| = j], \quad w_j = \frac{C_j}{K}, \quad V_j = w_j V_\Omega. \quad (139)$$

Every sample belongs to exactly one node, so  $\sum_j w_j = 1$  and  $\sum_j V_j = V_\Omega$  identically, whatever  $K$ . The counts are multinomial,  $\text{Var}(w_j) = w_j(1 - w_j)/K$ , so the per-node error falls as  $K^{-1/2}$  while the total stays exact — which is the right way round, since the total is what the conservation requirement of [Section 3.1](#) acts on. A KD-tree reduces the assignment from  $O(KM)$  to  $O(K \log M)$ .

### 3.3 From density to counts, and from counts to home nodes

The expected number of loops of family  $(s, k)$  associated with node  $j$  is the nodal quadrature of the density over its cell,

$$\lambda_{sj}^k = \int_{V_j^\Omega} \frac{n_{sL}^k(\mathbf{x})}{\Omega} dV \simeq \frac{n_{sL}^k(\mathbf{x}_j)}{\Omega} V_j, \quad \Lambda_s^k = \sum_j \lambda_{sj}^k, \quad (140)$$

the factor  $1/\Omega$  converting the per-host-atom density of [Eq. \(93\)](#) to a number per unit volume. **Both factors are needed and for different reasons:**  $n_{sL}^k$  says how many loops belong in unit material,  $V_j$  says how much material the node stands for. Sampling on  $n_{sL}^k$  alone would give every node equal geometric weight and would over-populate a refined region in exact proportion to how well it was resolved.

The total is taken as  $N_{\text{draw}} = \text{round}(\Lambda_s^k)$  rather than as a Poisson draw. This is deliberate: a Poisson total has  $\text{Var} = \Lambda$ , and a dose sweep in which the loop count fluctuates by  $\sqrt{\Lambda}$  between snapshots is unreadable for the trend it exists to show. Rounding bounds the count error at  $\frac{1}{2}$  and removes that fluctuation entirely. It also means the result is no longer an unconditioned Poisson process but a fixed-count spatial sampling whose intensity is still the Poisson measure — worth stating plainly, because the two are often conflated and only the second is being claimed.

Home nodes are then drawn from the normalized intensity,

$$p_{sj}^k = \frac{\lambda_{sj}^k}{\Lambda_s^k}, \quad (N_1, \dots, N_M) \sim \text{Multinomial}(N_{\text{draw}}; p_{s1}^k, \dots, p_{sM}^k), \quad (141)$$

so that  $\mathbb{E}[N_j] = \lambda_{sj}^k$  and the nodal populations are negatively correlated,  $\text{Cov}(N_j, N_i) = -N_{\text{draw}} p_j p_i$ , because the total is fixed. Spatial multinomial noise survives the rounding; where even that is unwanted in a sweep, systematic or residual allocation replaces the independent categorical draw at no cost to the expectation. Positions within a cell are drawn uniformly by rejection against the hull, so no loop center lands outside the crystal.

### 3.4 Sampling the size distribution

This is the step the earlier procedure left open, and it is where [Section 2.9](#) is used. A node carries three moments, so the log-normal [Eq. \(122\)](#) is fully determined there by [Eq. \(123\)](#):  $\bar{m}_{sL}^k = c_{sL}^k/n_{sL}^k$  and  $\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2$  give  $s_k = \sqrt{\ln \Delta_k}$  and  $\mu_k = \ln \bar{m}_{sL}^k - \frac{1}{2} s_k^2$ . **Every loop at that node is drawn from that distribution, not set to the mean.** With two moments there was no distribution to draw from and every discrete loop at a node was identical; the population that produced was monodisperse by construction, which is visible in any exported size histogram and wrong in a way no amount of spatial sampling could repair.

**Quantile sampling.** Independent draws  $m_\ell = \exp(\mu_k + s_k Z_\ell)$  with  $Z_\ell \sim \mathcal{N}(0, 1)$  are correct in expectation but noisy, and  $N_j$  is often small — a few loops per node is typical. Stratified sampling on the quantiles removes that noise deterministically,

$$m_\ell = \exp \left[ \mu_k + s_k \sqrt{2} \operatorname{erf}^{-1} \left( \frac{2\ell - 1}{N_j} - 1 \right) \right], \quad \ell = 1, \dots, N_j, \quad (142)$$

which is closed-form for a log-normal and is the size-space analogue of replacing the categorical draw of Eq. (141) by systematic allocation. It reproduces the intended distribution to the quadrature error of the midpoint rule,  $O(N_j^{-2})$ , in place of the  $O(N_j^{-1/2})$  of an independent draw. Sizes below  $m_{\min}$  are rejected and redrawn, so the sampled distribution is the log-normal truncated at the same floor the floor current Eq. (127) acts on — the two must use one truncation or the transfer will not conserve what Section 2.9 says the family holds.

**The conservation rescale, and why the log-normal survives it.** Neither draw reproduces  $c_{sL}^k$  exactly. Requirement 1 of Section 3.1 is met by rescaling the sample,

$$m_\ell \longrightarrow \alpha m_\ell^\beta, \quad \beta = \frac{s_k}{\hat{s}}, \quad \ln \alpha = \mu_k - \beta \hat{\mu}, \quad (143)$$

in which  $\hat{\mu}$  and  $\hat{s}$  are the sample's own log-mean and log-standard deviation. **Equation (143) is the log-normal family's own two-parameter group:**  $\ln m$  is affine in  $\ln m_\ell$ , so the rescaled sample is log-normal with exactly the target  $\mu_k$  and  $s_k$ , and both the mean size and the dispersion are restored without leaving the assumed family. A Gaussian admits no such map — a multiplicative rescale changes its variance and an additive one changes its mean, and neither preserves both — which is the third place, after the half-integer moments and the closure under  $R = \lambda\sqrt{m}$  of Section 2.9, where the choice of log-normal pays for itself. A single final multiplicative pass then makes  $\sum_\ell m_\ell$  equal the target content to machine precision, which is the ordering of Section 3.1 carried out in the obvious way.

**What the sample does *not* reproduce, and why it is not the content pass.** That pass cannot disturb the dispersion at all. Evaluated on a sample of  $N_j$  loops,  $\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2$  reads  $N_j \sum_\ell m_\ell^2 / (\sum_\ell m_\ell)^2$ , which is invariant under  $m_\ell \rightarrow \alpha m_\ell$ ; it is measured unchanged to ten digits across the pass.

$\Delta_k$  is nonetheless *not* reproduced exactly, and the mechanism is worth naming because it is neither quadrature error nor rounding. Equation (143) matches the *log* moments  $(\mu_k, s_k)$ , while  $\Delta_k$  is a ratio of *raw* moments, and the two coincide only in the limit. Equation (142) places no loop beyond

$$|z|_{\max} = \sqrt{2} \operatorname{erf}^{-1} \left( 1 - \frac{1}{N_j} \right), \quad |z|_{\max} = 1.28 \text{ at } N_j = 5, \quad (144)$$

so the sampled population is a log-normal *truncated in both tails*, where the continuum family it represents is not — and  $\int m^2 f(m) dm$  is precisely the moment those tails dominate. The shortfall is therefore governed by  $s_k$  as much as by  $N_j$ , and it decays far more slowly than the midpoint rule would suggest:

|                         |                      |                      |                      |                      |
|-------------------------|----------------------|----------------------|----------------------|----------------------|
| $\Delta_k$              | 1.02                 | 1.10                 | 1.35                 | 2.00                 |
| $s_k$                   | 0.141                | 0.309                | 0.548                | 0.833                |
| decay of the shortfall  | $N_j^{-0.70}$        | $N_j^{-0.67}$        | $N_j^{-0.60}$        | $N_j^{-0.49}$        |
| shortfall at $N_j = 10$ | $1.7 \times 10^{-4}$ | $3.8 \times 10^{-3}$ | $3.4 \times 10^{-2}$ | $1.5 \times 10^{-1}$ |

At  $N_j = 5$  and  $\Delta_k = 1.35$  the achieved dispersion is 1.287 against a target of 1.350, a 4.7% shortfall, and at  $\Delta_k = 2$  with  $N_j = 10$  it is 15%. Two consequences follow, and they point in opposite directions. The transferred population is always *under*-dispersed relative to the field it came from, so the line-length correction  $\Delta_k^{-1/8}$  of Eq. (125) does not cancel as exactly as Table 6 would otherwise claim — though at  $-4.7\%$  in  $\Delta_k$  that is  $0.6\%$  in line length, still an order below the ellipticity term. And the remedy is not more loops per node — the decay rates above make that expensive — but a quantile grid reaching further into the tails, or an analytic tail correction, either of which attacks the truncation rather than the count.

### 3.5 Ellipticity, orientation, and the polygon

A continuum loop has an area and no outline. A discrete loop has an outline, and Section 1.2 established that for prismatic  $\langle a \rangle$  loops that outline is an ellipse elongated along  $[0001]$ , with  $e = 1 - d_{\min}/d_{\max}$  reaching  $\simeq 0.1$  for interstitial and  $\simeq 0.4$  for vacancy loops at large size. The conversion is where that shape has to be constructed, because it is the first point in the model at which a loop has one.

**Equal-area axes.** Give the loop of content  $m_\ell$  the semi-axes

$$A_\ell = \frac{R_\ell}{\sqrt{\varrho_\ell}}, \quad B_\ell = R_\ell \sqrt{\varrho_\ell}, \quad \varrho_\ell = 1 - e_\ell, \quad R_\ell = \lambda_k \sqrt{m_\ell}, \quad (145)$$

so that  $A_\ell B_\ell = R_\ell^2$  and the ellipse has exactly the area of the circle it replaces. **Ellipticity therefore leaves the stored defects untouched**, which is what requirement 1 demands:  $\pi A_\ell B_\ell (\mathbf{b}\mathbf{n})_k / \Omega = m_\ell$  whatever  $e_\ell$  is. The shape is free precisely because the area is not.

**Orientation is fixed by the habit plane, not chosen.** For a prismatic family the habit plane is spanned by  $[0001]$  and the basal direction it contains, so

$$\hat{\mathbf{e}}_k^{\text{maj}} = [0001], \quad \hat{\mathbf{e}}_k^{\text{min}} = \hat{\mathbf{n}}_k \times [0001], \quad (146)$$

with  $\hat{\mathbf{n}}_k$  the variant's habit normal from Table 4. Both are exact unit vectors of the lattice, and  $\hat{\mathbf{e}}_k^{\text{maj}}$  lies in the plane for every prismatic variant, so no projection is needed. For the two basal families  $e \equiv 0$ : the sixfold symmetry of  $(0001)$  leaves no preferred in-plane direction,  $[0001]$  is their normal rather than an in-plane axis, and Section 1.2 is explicit that the prismatic aspect ratios must not be transferred to them. A basal loop is drawn circular unless a symmetry-breaking field is being modelled explicitly.

**Ellipticity depends on size, and the distribution now supplies the spread.** Atomistic calculations give nearly circular small loops and the anisotropic line-tension limit only for well separated segments (Section 1.2), so  $e$  saturates with size rather than being a constant:

$$e_\ell = e_s^\infty \frac{R_\ell}{R_\ell + R_e}, \quad e_i^\infty \simeq 0.1, \quad e_v^\infty \simeq 0.4, \quad (147)$$

with  $R_e$  a single length fitted to the measured  $e(d)$  trend, and the character-dependent limits of Section 1.2 entering through  $e_s^\infty$  — the one place in the whole conversion where the two prismatic characters are treated differently. **Because the sizes are drawn and not identical, Eq. (147) gives the discrete population a distribution of shapes:** the upper tail of Eq. (122) comes out elliptical and the small end nearly circular, in the proportion the dispersion  $\Delta_k$  dictates. A monodisperse conversion gives every loop of a family the same  $e$ , which is visibly wrong against a micrograph and was unavoidable before Section 2.9.

**The polygon must be equal-arc, not equal-angle.** A discrete loop is a closed polygon of  $N_{\text{side}}$  nodes. Placing them at equal increments of the parametric angle is correct for a circle and wrong for an ellipse, where it produces segments whose lengths differ by up to  $A/B = 1/\varrho$  — a factor 1.67 at  $e = 0.4$ . The remesh then splits and merges unevenly around the loop and the line tension is misrepresented exactly at the ends of the major axis, which is where the shape is. Place the vertices at equal increments of arc length instead,

$$s(\theta) = A_\ell E(\theta | 1 - \varrho_\ell^2), \quad \theta_\nu = s^{-1} \left( \frac{\nu}{N_{\text{side}}} P_\ell \right), \quad P_\ell = 4A_\ell E(1 - \varrho_\ell^2), \quad (148)$$

with  $E$  the incomplete and complete elliptic integrals of the second kind and  $s^{-1}$  evaluated by Newton iteration, which converges in two or three steps from the equal-angle guess.  $P_\ell$  is the same perimeter that Eq. (67) compares against the equal-area circle.

### 3.6 What the domain refuses, and what stays behind

A discrete loop is accepted only if every one of its nodes lies inside the grain. Large loops near a surface are therefore *refused*, silently, by the microstructure generator, and the conversion must predict that decision rather than discover it. With Eq. (145) the test is on the ellipse and not on a circle: a loop of major semi-axis  $A_\ell$  centered at  $\mathbf{x}_\ell$  fits when

$$\min_{\text{faces } F} \text{dist}(\mathbf{x}_\ell, F) \geq \max_\nu |\mathbf{r}_\nu \cdot \hat{\mathbf{m}}_F|, \quad (149)$$

$r_\nu$  running over the polygon vertices of Eq. (148); an elongated loop can fit where the circle of equal area would not, and can fail where it would have fitted, depending on how  $\hat{e}_k^{\text{maj}}$  lies against the nearest face.

Three consequences, of which the last is new.

1. **The refused share must be counted in defects, not in loops.** The refused loops are the large ones, so a refusal rate that is modest by count can be overwhelming by content — one refusal in six has been measured at 83% by count and 98% by defects. Only the defect share tells whether a transfer is usable.
2. **Refused defects stay in the continuum.** They are not discarded and not forced into smaller loops; the family’s fields are reduced by exactly what was accepted, and the remainder continues to be advanced by Section 2.3. That is what keeps the atom balance closed across a partial transfer.
3. **Refusal truncates the distribution, so it biases  $\Delta_k$ .** Because it removes the upper tail, the accepted population is a *doubly* truncated log-normal — at  $m_{\min}$  from Eq. (142) and at the fitting limit from Eq. (149) — and its dispersion is below the  $\Delta_k$  the continuum carried. The residue left behind is correspondingly *over*-dispersed. Both halves must be re-fitted after the split or the two will disagree about the same family, and neither will match the field it came from.

### 3.7 The transfer ledger

Table 6 states what the construction conserves and what it does not. The entries that are not unity are not errors to be tuned away; each is a definite statement about how the two representations differ.

Table 6: What a continuum-to-discrete transfer preserves. The first three rows are requirements met by construction; the last three are measured and reported.

| Quantity                     | Status                                   | Why                                                                                                                                                                                                                                  |
|------------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stored defects               | <b>exact</b>                             | Equation (143) plus the final multiplicative pass; the residue of Section 3.6 stays in the continuum.                                                                                                                                |
| Loop number                  | <b>exact to the draw</b>                 | $ N_{\text{draw}} - \Lambda  \leq \frac{1}{2}$ by rounding, Section 3.3.                                                                                                                                                             |
| Mean size                    | <b>exact</b>                             | Equations (142) and (143) land it within $\sim N_j^{-0.9}$ and are <i>exact</i> in $(\mu_k, s_k)$ ; the final content pass then makes $\sum_\ell m_\ell$ , and hence $\bar{m}_{sL}^k$ , exact outright.                              |
| Dispersion $\Delta_k$        | $\sim N_j^{-0.6}$ , and $s_k$ -dependent | Tail truncation, <i>not</i> quadrature error and <i>not</i> the content pass, which is exactly $\Delta_k$ -neutral: see Eq. (144). Measured $-4.7\%$ at $N_j = 5$ , $\Delta_k = 1.35$ , and $-15\%$ at $N_j = 10$ , $\Delta_k = 2$ . |
| Burgers vector               | $f_b = 1.000000$                         | Both sides now size a $\langle c \rangle$ loop with $\mathbf{b} \cdot \mathbf{n} = c/2$ (Table 2); this factor read $\sqrt{2}$ before that was reconciled.                                                                           |
| Line length from ellipticity | 1.002–1.049                              | The discrete loop carries its true elliptical perimeter Eq. (148), where $\rho_{sL}^k$ credited it with the equal-area circle (Eq. (67)); $e = 0.1$ and $e = 0.4$ bracket it.                                                        |
| Line length from dispersion  | $\Delta_k^{-1/8}$                        | Equation (125): the continuum already applies it, so it cancels — provided the discrete sizes are drawn from the same $\Delta_k$ , which is the reason Section 3.4 draws rather than assigns the mean.                               |
| Sink-strength convention     | $1/\zeta_k^{\text{sink}}$                | A calibration constant the two solvers do not share, outside the totals. It is a modeling decision to be settled before a transfer feeds a solve, not a crystallography error.                                                       |

The last row is the one to settle first. The three rows above it are now accounted: the Burgers discontinuity was removed by making both sides size a basal loop with the same edge component, the

ellipticity factor is bounded by 5% and computable per loop from Eq. (147), and the dispersion factor cancels identically once both sides carry the same distribution. **That cancellation is the practical argument for Section 2.9 in this context:** a monodisperse continuum handing loops to a discrete solver that draws them from a spread — or the reverse — disagrees about the sink strength by  $\Delta_k^{-1/8}$  with nothing in either representation to reveal it.

### 3.8 When the transfer fires

Everything above answers *how* a family is converted. It does not say *when*, and for the offline use — an export for a figure, a micrograph comparison or a hardening cell — it does not have to: the dose is whichever one is being looked at. For the online use it is the whole question, and it is not a free parameter. This subsection states the condition, which is automatic, and the second condition that can veto it.

**Three things must hold, and they are of different kinds.** The first is a choice; the second is a statement about when the continuum model stops being trustworthy; the third is a statement about whether the discrete representation is possible at all.

1. **Armed.** The transfer is off unless it is asked for, and the units permitted to switch are named. Nothing below runs otherwise.
2. **Necessary — the continuum has stopped being valid.** The mean-field treatment of coalescence has crossed its own error estimate, Eq. (151). This defines  $d_{\text{coarsen}}$ .
3. **Sufficient — the crystal can hold the loops.** The refusal test of Section 3.6 accepts a non-zero share of the family’s defects. If it does not, the transfer is *declined* rather than failed, and retried.

### 3.9 The detector is the model’s own coalescence estimate

The criterion is not imported from outside the model. Equation (117) already carries the two overlap probabilities, and the immobile block integrates them from the first dose step — **coarsening is modelled throughout, and the question is never when it starts.** The question is when the *mean-field* form of it stops being trustworthy, and the form answers that itself.

$1 - \exp(-\kappa\Lambda)$  is a Poisson estimate: it presumes the loops are placed independently at random. Its leading error is the two-body correlation, entering relatively at  $O(\Lambda)$  — and coalescence is precisely the process that correlates positions, because it removes close pairs. The estimate is therefore asymptotically exact as  $\Lambda \rightarrow 0$  and *progressively self-invalidating* as  $\Lambda \rightarrow 1$ . **That is what makes  $\phi$  the natural detector rather than some external criterion: it is the model’s own estimate of the very effect whose mean-field treatment is in question.**

It is also, identically, the crowding ratio of Eq. (171). With  $N = n_{sL}^k/\Omega$  and  $d_k = N^{-1/3}$ ,

$$\Lambda_{LL} = \frac{4}{3}\pi(R_{sL}^k)^3 N = \frac{\pi}{6} \left( \frac{2R_{sL}^k}{d_k} \right)^3 = \frac{\pi}{6} \Theta_k^3, \quad (150)$$

so the trigger of this subsection and the geometry selector of Table 8 are one measure read at two thresholds, not two criteria that might disagree.

**The gate.** Per interior node take  $\phi = \max(\phi_{LL}, \phi_{LN})$ , and fire when

$$\phi_{\text{gate}}(d) \equiv Q_{1-f^*}[\phi(\mathbf{x}_j, d)]_{j \in \mathcal{I}} \geq \phi^* \quad \text{on } H \text{ consecutive substeps,} \quad d_{\text{coarsen}} = \min d, \quad (151)$$

with  $Q_{1-f^*}$  the  $(1 - f^*)$  quantile and  $\mathcal{I}$  the innermost quartile by distance to the boundary. Four choices in Eq. (151) are not cosmetic:

- **The quantile, not the fraction.** “A share  $f^*$  of nodes exceed  $\phi^*$ ” and “ $\phi_{\text{gate}} \geq \phi^*$ ” are the same statement, but the first is a step and the second is smooth in dose. On the reference march the share goes from 0.000 at 0.1 dpa to 0.982 at 1 dpa, because the interior field is smooth enough that every node crosses at once; interpolating that step gives a meaningless crossing dose, while interpolating  $\phi_{\text{gate}}$  gives the right one.



- **Interior only**, by the rule of [Section 3.6](#). The boundary shell carries up to 99.8% of the  $\langle a \rangle$  density, for the reason given there, and a whole-domain reduction would fire the detector on that artifact at the first dose step.
- **The radius is the continuum’s own.**  $R_{sL}^k$  in [Eq. \(150\)](#) must be built from the **bn** the immobile block uses, so that  $\phi$  here is the  $\phi$  the solver computes. Before the Burgers reconciliation of [Table 6](#) the two sides differed by  $\sqrt{2}$  for  $\langle c \rangle$ , which would inflate  $\Lambda_{LL}$  by  $2^{3/2}$  and fire the detector roughly a decade of dose early; with  $f_b = 1.000000$  the distinction is now identically satisfied, and it is recorded here because the requirement outlives the coincidence.
- **Interpolation in  $\log d$** , since the fields move by decades across a decade of dose.

### 3.10 What it measures

On the two matched 500 nm hexagonal marches of [Appendix C](#), interior nodes,  $f^* = 0.10$ ,  $H = 2$ :

Table 7:  $d_{\text{coarsen}}$  in dpa, for the two formulations of [Table 21](#). Only  $\langle c \rangle$  ever crosses; the  $\langle a \rangle$  families plateau two orders below any usable threshold, so on this case they are nowhere near a transition. The operative default is  $\phi^* = 0.20$ .

| $\phi^*$                             | 0.10                                                                         | 0.15  | 0.20         | 0.25  | 0.30  |
|--------------------------------------|------------------------------------------------------------------------------|-------|--------------|-------|-------|
| legacy, $\langle c \rangle$          | 0.049                                                                        | 0.103 | <b>0.127</b> | 0.156 | 0.192 |
| self-consistent, $\langle c \rangle$ | 0.032                                                                        | 0.060 | <b>0.104</b> | 0.127 | 0.156 |
| ratio                                | 0.65                                                                         | 0.58  | 0.82         | 0.81  | 0.81  |
| $\langle a \rangle_{1-3}$ , either   | never; $\max_d \phi_{\text{gate}} = 0.007$ (legacy), 0.006 (self-consistent) |       |              |       |       |

Three readings, and the first is a warning about how the number must be quoted.

$d_{\text{coarsen}}$  **is meaningless without its  $\phi^*$** . Across the plausible threshold range the crossing dose moves by a factor of 4–5, and worse, a  $\phi^*$  set within about 0.05 of a family’s plateau makes the crossing unstable or unreachable altogether. The plateau in the last row of [Table 7](#) is what must be checked before any threshold is trusted.

**The switch dose is model-dependent, by 20–40%.** At matched  $\phi^*$  the self-consistent formulation crosses *earlier* — 0.104 against 0.127 dpa at the default, and 0.060 against 0.103 at  $\phi^* = 0.15$  — and the size of that gap is predicted by the formulations’ own difference rather than being a free outcome. The matched comparison of [Appendix C](#) gives the self-consistent model  $1.192\times$  the  $\langle c \rangle$  content at essentially unchanged number, so  $R \propto \sqrt{c/n}$  rises by 1.093 and  $\Lambda_{LL} \propto R^3 N$  by 1.30: a higher  $\phi$  at every dose, hence an earlier crossing. **A switch dose carried over from a legacy calibration is therefore late for a self-consistent run**, which matters because [Section 3.11](#) shows how little room being late leaves.

**Which family transitions is not model-dependent.** Both formulations put  $\langle c \rangle$  across and leave  $\langle a \rangle$  at  $\phi_{\text{gate}} \simeq 0.006$ – $0.007$ , two orders below threshold and flat in dose. That is the same conclusion [Table 8](#) reaches from  $\Theta_k$ , as [Eq. \(150\)](#) says it must, and it is why **c** is the only unit armed by default.

### 3.11 The feasibility gate closes as the coarsening gate opens

The second condition is the one to watch, and on the reference geometry the two nearly coincide. [Section 3.6](#) measured the accepted defect share for  $\langle c \rangle$  on the same march:

| dose [dpa]        | $10^{-2}$ | 0.1   | 1      | 2      | 5      | 10     |
|-------------------|-----------|-------|--------|--------|--------|--------|
| $f_{\text{kept}}$ | 0.990     | 0.757 | 0.0096 | 0.0005 | 0.0144 | 0.0232 |

against  $d_{\text{coarsen}} = 0.104$ – $0.127$  dpa. **The continuum becomes untrustworthy at almost exactly the dose at which the crystal stops being able to hold the loops**, and the usable window on a 500 nm domain is roughly 0.1–0.3 dpa wide. A transfer attempted late is not merely delayed; it is refused, and the ledger of [Section 3.7](#) then reports a population carrying a per-cent of the family’s defects.



This is a statement about the domain, not about the physics: the  $\langle c \rangle$  radius reaches 55 nm by 1 dpa, and no transfer of a 110 nm object into a 500 nm cell can be representative. It is the same refusal that [Section 3.6](#) quantifies, arriving here as a schedule constraint rather than as a loss of accuracy, and **it is the argument for sizing the domain from  $d_{\text{coarsen}}$  rather than choosing it first.**

When  $f_{\text{kept}} = 0$  the transfer is **declined**: the unit stays in the continuum, nothing is injected, and the attempt is repeated at the next boundary. Declining is the correct outcome and must not be recorded as a completed transfer — a transfer that removes a family from the continuum and gives the discrete side nothing is the one failure mode that would silently delete the population.

### 3.12 Where in the march it happens

At the **next fast-solve boundary** after [Eq. \(151\)](#) is satisfied, and *before* that solve. The ordering is forced: the fast solve which follows must already see both the injected network and the reduced continuum field, and performing the transfer immediately after one instead would leave a full interval in which the same loops act as sinks twice — once through  $k_m^2$  of [Eq. \(38\)](#) and once through the discrete field of [Eq. \(162\)](#). That double count is exactly what the removal step of [Section 3.6](#) exists to prevent, and it is undone by the wrong ordering alone.

The boundary is also where the state is already synchronized and where a solver call happens anyway, so the switch costs no extra solve. Three things are set at the same point: the discrete population is injected, the screened cutoff [Eq. \(170\)](#) is computed from the *current*  $k_m^2$  — it must be, since removing a family from the continuum lengthens  $L_s$  — and the network is handed to the fast solve.

**Status.** The detector runs on every march and is **reporting only** unless the transfer is armed, which is off by default; it can therefore be read on runs that predate it, which is how [Table 7](#) was obtained. The thresholds are  $\phi^* = 0.20$ ,  $f^* = 0.10$ ,  $H = 2$ .

$\phi^* = 0.20$  is not an arbitrary round number: it is where the neglected pair correlation of [Section 3.9](#) costs about 20% in the coalescence rate, and it lands at  $\Theta_k = 0.597$  in the like-loop channel. **It was until recently carried in two places at two different values** — the detector reporting on 0.20 while the march would have fired on 0.15, which by [Table 7](#) is a 23% gap in  $d_{\text{coarsen}}$  (0.127 against 0.103 dpa on the legacy march). Nothing revealed it, because the two are only ever compared when the transfer is armed and it is off by default. The five places that carry the threshold are now reconciled on the derived value, and the one that defines it names the other four.

### 3.13 Climbing the transferred loops: a two-parameter ellipse against a boundary-integral solve

The transfer ends with line objects: a Burgers vector, a habit plane, a center, and the polygon [Eq. \(148\)](#). Something must then move them, and the mean-field growth law [Eq. \(71\)](#) cannot, for a reason that is structural rather than numerical. [Equation \(71\)](#) is written for a *family*: its driving concentration is the nodal field  $c^m$ , whose shortest resolved length is the finite element. The concentration that sets a loop’s climb rate is the one at its *own line*, inside the depletion halo the loop itself creates, and that halo is invisible to the continuum block by construction — the loop is one of the sinks that field was computed with.

**Restoring the halo is the whole content of this subsection**, and it is where the offline and online uses of the conversion part company. An exported configuration only has to reproduce the fields it came from; a transferred population has to climb, and to climb it needs a concentration the continuum cannot supply.

Two constructions are carried. Both discretize the same pair of statements — a driving supersaturation at the line, and the concentration a climbing line emits — and they differ in exactly one place: whether the second statement is solved or replaced by a constant.

### 3.14 What both constructions must compute

A line element with tangent  $\hat{\boldsymbol{\xi}}$  on a loop of family  $(s, k)$  climbs in its habit plane along  $\hat{\boldsymbol{\eta}} = \mathbf{b} \times \hat{\boldsymbol{\xi}} / |\mathbf{b} \times \hat{\boldsymbol{\xi}}|$ , sweeping area and absorbing  $(\mathbf{b} \cdot \mathbf{n})_k / \Omega$  defects per unit area swept. Its normal velocity is therefore

$$v_n = \frac{\Omega}{(\mathbf{b} \cdot \mathbf{n})_k} \sum_{m \in \{v, i, 2i, 3i\}} \varsigma_s(m) |m| \Phi^m(\mathbf{r}), \quad (152)$$

with  $\varsigma_s(m)$  and  $|m|$  as in Eq. (71) and  $\Phi^m$  the arrival rate of species  $m$  per unit length of line, at the position  $\mathbf{r}$  of that element. Equation (152) is not in dispute;  $\Phi^m$  is. It requires the concentration *at the line*, which is

$$c^m(\mathbf{r}) = c_{\mathcal{C}}^m(\mathbf{r}) + c_{\mathcal{D}}^m(\mathbf{r}), \quad (153)$$

the continuum field of Eq. (27) plus the field the discrete loops themselves produce. The two constructions below are two answers to what  $c_{\mathcal{D}}^m$  is worth.

**Why the split in Eq. (153) is not optional, and how it was measured.**  $c_{\mathcal{D}}^m$  is not a small correction that may be folded into a fitted constant at all populations. Its magnitude relative to the smooth part depends on how the loop spacing  $d_k = (n_{sL}^k / \Omega)^{-1/3}$  compares with the screening length over which a line sink's gradient develops,

$$L_s^m = (k_m^2)^{-1/2}, \quad (154)$$

$k_m^2$  being the same total sink strength Eq. (38) the mobile block already assembles. Evaluated on a 500 nm hexagonal case, Eq. (154) gives  $L_s^m = 58.9\text{--}59.9\text{ nm}$  across the four species. Isolating  $c_{\mathcal{D}}^m$  on three populations in that same domain, by differencing the superposed and continuum fields at the nodes, gives the ratio of the mean  $|c_{\mathcal{D}}^m|$  within one screening length of a loop to its value in the far field:

|                     |          |      |             |
|---------------------|----------|------|-------------|
| loops in the domain | 1        | 5    | 62          |
| $d_k / L_s$         | isolated | 7.11 | 1.81        |
| near/far contrast   | 11.23    | 3.87 | <b>1.37</b> |

At  $d_k \gg L_s$  each loop carries a resolvable halo; by  $d_k \simeq 2L_s$  the halos have merged and what is left of  $c_{\mathcal{D}}^m$  is a smooth background — following the local loop density, since that is what produced it — carrying a 37% modulation at the scale of the individual loop. **It is the modulation, and only the modulation, that a discrete treatment of the field adds;** the background is what the spatially resolved  $k_m^2(\mathbf{x})$  of Eq. (38) already carries, and Section 3.17 argues that the modulation is below the resolution of the mobile problem. This is the measurement behind the selection rule, and it is also why the discrete sum below must be truncated rather than extended: carried past  $L_s$  it would re-count what  $k_m^2$  has already counted.

### 3.15 The reduced-order model: two degrees of freedom per loop

**The state.** Equation (145) gave the transferred loop a static outline. The reduced-order model (ROM) makes that outline dynamic and nothing else: the loop remains a centered ellipse on its own habit plane, with the axis directions Eq. (146) fixed by crystallography, and its state is the pair of semi-axes

$$(A_\ell, B_\ell), \quad A_\ell \parallel [0001], \quad B_\ell \parallel \hat{\mathbf{n}}_k \times [0001], \quad \varrho_\ell = \frac{B_\ell}{A_\ell}, \quad (155)$$

so that 2 numbers replace the  $2N_{\text{side}}$  of the polygon — 24 at  $N_{\text{side}} = 12$ . Equation (155) keeps  $\varrho_\ell$  as the same aspect ratio Eq. (145) carries, now free to pass through unity: the static conversion has  $\varrho_\ell = 1 - e_\ell \leq 1$ , elongation along  $[0001]$ , whereas the kinetics below produce  $\varrho_\ell > 1$  for a growing loop. The defect content is the invariant of the conversion,

$$m_\ell = \frac{\pi A_\ell B_\ell (\mathbf{b} \cdot \mathbf{n})_k}{\Omega}, \quad (156)$$

identical to the one Eq. (145) conserves, so the ROM and the conversion agree by construction and not by calibration.

**The two equations.** Evaluate Eq. (152) at the two axis endpoints and take  $\Phi^m$  from a capture length — the distance over which the gradient feeding the line develops. This gives one ordinary differential equation per axis:

$$\dot{A}_\ell = \frac{\Omega}{b^2 (\mathbf{b} \cdot \mathbf{n})_k} \sum_m \varsigma_s(m) |m| Z_{sk,m} \frac{D_m^{(A)}}{L_{\text{cap}}} \left[ c_\ell^m - c_{\text{eq}}^m \exp \left( -\frac{\varsigma_s(m) |m| \Omega \sigma_\ell^{(A)}}{k_B T} \right) \right], \quad (157)$$

$$\dot{B}_\ell = \frac{\Omega}{b^2 (\mathbf{b} \cdot \mathbf{n})_k} \sum_m \varsigma_s(m) |m| Z_{sk,m} \frac{D_m^{(B)}}{L_{\text{cap}}} \left[ c_\ell^m - c_{\text{eq}}^m \exp \left( -\frac{\varsigma_s(m) |m| \Omega \sigma_\ell^{(B)}}{k_B T} \right) \right]. \quad (158)$$

Every symbol, in order:

- $\Omega/[b^2 (\mathbf{b} \cdot \mathbf{n})_k]$  is the kinematic factor of Eq. (152), written in the units the code carries — concentrations are atom fractions and  $\Omega$ ,  $L_{\text{cap}}$  and  $(\mathbf{b} \cdot \mathbf{n})_k$  are in units of the reference Burgers magnitude  $b$  — so the group is dimensionless and  $D_m/L_{\text{cap}}$  carries the velocity.
- $\varsigma_s(m) = \pm 1$  and  $|m|$  are exactly those of Eq. (71): the polarity match against the family’s character, and the number of point defects the mobile cluster delivers. The same signed factor appears in the exponent, so a vacancy arriving at an interstitial loop both shrinks it and reverses its emission term. The implementation carries the two as the single signed vector `mobileSpeciesVector`, which is  $\varsigma_s(m)|m|$  for an interstitial family and its negative for a vacancy one; the ROM is applied only to the former.
- $Z_{sk,m}$  is the capture efficiency Eq. (68), unchanged. The ROM inherits the DAD, character and SIPA structure of the continuum model rather than introducing a discrete-specific bias.
- $c_\ell^m$  is the driving concentration at the loop, Eq. (153), evaluated as the continuum field at the loop center plus a *lagged monopole* sum over the loops within the cutoff Eq. (170). Lagged, so Eqs. (157) and (158) remain independent across loops and no coupled solve appears.
- $c_{\text{eq}}^m$  is the stress-free thermal equilibrium concentration of species  $m$ .
- $L_{\text{cap}}$  is the capture length. **It is the model’s one free parameter**, defaulted to the screening length Eq. (154). It enters both equations identically, so it cancels from the aspect ratio and sets the time scale alone: an error in  $L_{\text{cap}}$  gets the growth rate wrong and the ellipticity right.
- $\sigma_\ell^{(A)}, \sigma_\ell^{(B)}$  are the self-stress at the two endpoints, from the local curvature of the ellipse,

$$\sigma_\ell^{(A)} = -\frac{\mu (\mathbf{b} \cdot \mathbf{n})_k}{4\pi(1-\nu)} \kappa_A \ln \frac{1}{\kappa_A r_0}, \quad \kappa_A = \frac{B_\ell}{A_\ell^2}, \quad \kappa_B = \frac{A_\ell}{B_\ell^2}, \quad (159)$$

$\mu/(1-\nu)$  being the edge prelogarithmic factor already used in Section 1.2 and  $r_0$  the core radius, and  $\sigma_\ell^{(B)}$  the same expression evaluated at  $\kappa_B$ . ( $\nu$  is Poisson’s ratio here; as a subscript it indexes nodes, as in Eq. (148).)

- $D_m^{(A)}$  and  $D_m^{(B)}$  are the capture diffusivities, and they are the one place a modeling choice genuinely enters. This is developed next.

**The capture diffusivities, and the closure that is not settled.** The diffusivity feeding an endpoint is the one sampled in the plane the flux arrives through, that is, the plane perpendicular to the local tangent. One of the two is fixed by geometry and the other is not:

| endpoint                         | tangent $\hat{\xi}$ | capture plane                    | $D$                                         |
|----------------------------------|---------------------|----------------------------------|---------------------------------------------|
| end of $B_\ell$ (basal axis)     | $\parallel [0001]$  | basal                            | $D_m^{(B)} = D_a^m$                         |
| end of $A_\ell$ ( $[0001]$ axis) | basal               | spans $\hat{c}$ and $\mathbf{b}$ | $D_m^{(A)} = \sqrt{D_a^m D_c^m}$ or $D_c^m$ |

With  $p_m = (D_c^m/D_a^m)^{1/6}$  of Eq. (18), switching off the self-stress makes the ratio relax to a fixed point,

$$\varrho_\ell \longrightarrow \frac{D_m^{(B)}}{D_m^{(A)}} = \begin{cases} p_m^{-3}, & \text{geometric-mean closure,} \\ p_m^{-6}, & \text{normal-projection closure,} \end{cases} \quad (160)$$

so **both closures put the major axis in the basal plane whenever  $p_m < 1$** , which is the shape-resolved extension anticipated in Section 2.5 — a second use of  $p_m$  rather than a new mechanism. The geometric mean is the default, on the argument that a line sink in a transversely isotropic medium samples it, but *this is not settled*: at the fitted  $p_I = 0.91372$  the two give 1.311 and 1.718, and at  $p_m = 0.7$  they give 2.9 and 8.5. **The spread Eq. (160) must be quoted with any ellipticity the ROM produces**, and only a nodal solve of the same case can collapse it.

Three properties of Eqs. (157) and (158) that a steady-state reading would miss.

1. **Without Eq. (159) the model has no steady shape at all.** The ratios Eq. (160) are attractors of the ratio, not of the axes; the loop elongates without limit. The self-stress supplies the restoring term: as the loop elongates,  $\kappa$  rises at the major-axis ends, the back stress there raises  $c_{\text{eq}}^m$  and growth throttles. This is the mechanism by which a change of loop shape redistributes the Peach–Koehler force.
2. **The sign of growth decides which axis is major.**  $\varrho_\ell$  relaxes toward Eq. (160) only while the loop *grows*: a larger  $|\dot{B}_\ell|$  lengthens  $B_\ell$  when the net flux is positive and eats it when negative. Integrating the same  $p_m = 0.7$  loop gives  $\varrho : 1.00 \rightarrow 1.24 \rightarrow 1.43 \rightarrow 1.70$  growing and  $1.00 \rightarrow 0.86 \rightarrow 0.69 \rightarrow 0.27$  shrinking. Since Eq. (71) is a near cancellation, a family can sit on either side, and at the arrival ratio this model carries, the  $\langle a \rangle$  families are in the *shrinking* branch — where the ROM predicts elongation along [0001], the same sense as Eq. (147) but from a different mechanism. That coincidence of sense with divergence of mechanism is worth stating plainly: Eq. (147) is an elastic, line-tension statement fitted to atomistics, Eq. (160) a purely diffusional one, and they are not alternatives.
3. **The attractor is approached slowly, so ellipticity measures growth as much as it measures  $p_m$ .** From a circular nucleus of radius  $a_0$  whose two axes advance in the ratio  $q$ ,

$$\varrho(t) = \frac{a_0 + q\mathcal{R}t}{a_0 + \mathcal{R}t} \longrightarrow q \quad \text{only as } \mathcal{R}t \gg a_0, \quad (161)$$

so a loop grown to  $\varkappa$  times its nucleus shows only  $(1 + q(\varkappa - 1))/\varkappa$ : at  $q = 2.915$  that is 1.96 at  $\varkappa = 2$ , 2.53 at  $\varkappa = 5$  and 2.72 at  $\varkappa = 10$ . Integrated along a march whose  $\langle a \rangle$  loops reach  $1.12 \times$  their 300-defect nucleus, Eq. (161) yields 2.9% of a 31% asymptote. **A measured aspect ratio compared directly against Eq. (160) under-reads the anisotropy unless the loops are many times their nucleus size.**

**What the ROM cannot represent.** Coalescence by contact, junction formation, non-planar climb, and any departure from a centered ellipse. The first has a cheap runtime test and is checked rather than assumed; the others do not, and they are the reason the  $\langle c \rangle$  family does not use this construction.

### 3.16 The boundary-integral form: solving for the emitted field

The second construction removes  $L_{\text{cap}}$  by computing what it stands for. The unknowns are no longer two numbers per loop but a climb-velocity scalar  $v_\nu^m$  at every network node  $\nu$  and every mobile species —  $N_{\text{side}} \times 4 = 48$  per loop at  $N_{\text{side}} = 12$  — and the loop’s outline is whatever those velocities make it.

**The kernel.** For a straight segment  $\lambda$  from  $\mathbf{x}_0$  to  $\mathbf{x}_1$ , with chord  $\mathbf{L}_\lambda = \mathbf{x}_1 - \mathbf{x}_0$ , length  $L_\lambda$ , unit tangent  $\hat{\xi}_\lambda$  and Burgers vector  $\mathbf{b}_\lambda$ , the concentration of species  $m$  produced at a field point  $\mathbf{x}$  is linear in the climb velocities of its two nodes:

$$c_D^m(\mathbf{x}) = \sum_\lambda \sum_{s \in P} \left[ \mathcal{G}_{\lambda,0}^m(\mathbf{x} + \mathbf{s}) v_{\lambda_0}^m + \mathcal{G}_{\lambda,1}^m(\mathbf{x} + \mathbf{s}) v_{\lambda_1}^m \right], \quad (162)$$

$\mathcal{P}$  being the set of periodic images. The influence coefficients are the closed-form line integral of the *anisotropic* point-source Green's function along the chord, weighted by the segment's linear shape functions:

$$\mathcal{G}_{\lambda,\alpha}^m(\mathbf{x}) = \frac{(\mathbf{b}_\lambda \times \hat{\boldsymbol{\xi}}_\lambda) \cdot \hat{\boldsymbol{\eta}}_{\lambda_\alpha} L_\lambda}{4\pi Z_{sk,m}^D \sqrt{\mathcal{A}_m} \det \mathbf{D}_m} I_\alpha(\mathcal{A}_m, \mathcal{B}_m, \mathcal{C}_m), \quad \alpha \in \{0, 1\}, \quad (163)$$

with, writing  $\mathbf{R} = \mathbf{x} - \mathbf{x}_0$ ,

$$\mathcal{A}_m = \mathbf{L}_\lambda \cdot \mathbf{D}_m^{-1} \mathbf{L}_\lambda, \quad \mathcal{B}_m = -2 \mathbf{R} \cdot \mathbf{D}_m^{-1} \mathbf{L}_\lambda, \quad \mathcal{C}_m = \mathbf{R} \cdot \mathbf{D}_m^{-1} \mathbf{R} + \frac{r_0^2}{(\det \mathbf{D}_m)^{1/3}}, \quad (164)$$

and, with  $u = \mathcal{B}_m/\mathcal{A}_m$  and  $w = \mathcal{C}_m/\mathcal{A}_m$ ,

$$I_0 = \left(1 + \frac{u}{2}\right) \mathcal{Q} - \sqrt{1+u+w} + \sqrt{w}, \quad I_1 = -\frac{u}{2} \mathcal{Q} + \sqrt{1+u+w} - \sqrt{w}, \quad \mathcal{Q} = \ln \frac{2\sqrt{1+u+w} + 2 + u}{2\sqrt{w} + u}. \quad (165)$$

Term by term:  $(\mathbf{b}_\lambda \times \hat{\boldsymbol{\xi}}_\lambda) \cdot \hat{\boldsymbol{\eta}}$  is the volumetric strength of the climbing segment, the rate at which it removes material per unit climb velocity;  $\mathbf{D}_m^{-1}$  and  $\det \mathbf{D}_m$  carry the *full* diffusion tensor Eq. (18) at every field point, so no directional average and no closure of the kind Eq. (160) rests on is required anywhere;  $r_0$  regularizes the core, entering Eq. (164) scaled by the tensor's own determinant so that the regularization is isotropic in the transformed coordinates;  $I_0$  and  $I_1$  are the shape-function-weighted integrals, so that Eq. (162) distributes the source linearly along the segment as the network's own interpolation does; and  $Z_{sk,m}^D$  is the discrete-loop bias, selected by the sign of  $\mathbf{b} \cdot \mathbf{n}$  on the loop — the discrete counterpart of the character factor in Eq. (68).

**The linear system.** Requiring Eq. (152) to hold in a Galerkin sense on every segment closes the system. Let  $N_0(u) = 1 - u$  and  $N_1(u) = u$  be the shape functions of the field segment  $\lambda$  and  $\mathbf{r}_\lambda(u)$  its quadrature points. Then for each species  $m$ ,

$$\sum_{\nu'} \mathcal{K}_{\nu\nu'}^m v_{\nu'}^m = \mathcal{F}_\nu^m, \quad (166)$$

assembled from

$$\mathcal{K}_{\nu\nu'}^m = \sum_{\lambda \ni \nu} \sum_{\lambda' \ni \nu'} \int_0^1 N_\nu(u) (\mathbf{b}_\lambda \times \mathbf{L}_\lambda) \cdot \hat{\boldsymbol{\eta}}_\nu \mathcal{G}_{\lambda',\nu'}^m(\mathbf{r}_\lambda(u)) du, \quad (167)$$

$$\mathcal{F}_\nu^m = \sum_{\lambda \ni \nu} \int_0^1 N_\nu(u) (\mathbf{b}_\lambda \times \mathbf{L}_\lambda) \cdot \hat{\boldsymbol{\eta}}_\nu \left[ c_{\text{line}}^m(\mathbf{r}_\lambda(u)) - c_C^m(\mathbf{r}_\lambda(u)) \right] du. \quad (168)$$

Equation (167) is a double sum over *ordered pairs* of segments — field  $\lambda$  against source  $\lambda'$  — and it is what makes this a boundary-integral method rather than a differential one: no volume mesh appears, the unknowns live on the lines, and every line sees every other line. The right-hand side Eq. (168) is the driving supersaturation, the difference between the line's own local equilibrium and the continuum field at the same point:

$$c_{\text{line}}^m = c_{\text{eq}}^m(\text{tr } \boldsymbol{\sigma}) \exp \left( - \frac{m \Omega \mathbf{f}^{\text{PK}} \cdot (\mathbf{b} \times \hat{\boldsymbol{\xi}})}{k_B T (|\mathbf{b} \times \hat{\boldsymbol{\xi}}|^2 + 0.05 |\mathbf{b}|^2)} \right), \quad (169)$$

with  $\mathbf{f}^{\text{PK}}$  the Peach–Koehler force from the *actual* stress field — self-stress of every segment, applied load and image terms — and  $m$  the *signed* cluster size of Section 2.1, so that the exponent carries polarity and cluster size exactly as Eq. (157) does. **This is the term that replaces Eq. (159):** where the ROM evaluates a closed-form curvature back stress on an assumed ellipse, Eq. (169) reads the stress the network has already computed, on whatever shape the loop has actually taken. The  $0.05 |\mathbf{b}|^2$  regularizes the near-screw orientation, where  $\mathbf{b} \times \hat{\boldsymbol{\xi}}$  vanishes and climb is undefined.

**The screened cutoff.** Equation (167) is  $O(N_{\text{seg}}^2)$  and is 100% of the cost; the solve of Eq. (166) is negligible beside it. Truncation is legitimate here and would not be for a bare  $1/r$  kernel: a growing loop is a net sink, so its monopole does not vanish and a shell at radius  $r$  contributes  $\sim r$ , a sum that grows with the cutoff. The physical kernel is screened by the sink field as  $e^{-r/L_s}/r$  with  $L_s$  from Eq. (154), so

$$R_{\text{cut}} = n_L \max_m L_s^m, \quad (170)$$

and a pair is dropped only when no point of one segment can lie within  $R_{\text{cut}}$  of any point of the other, minimized over  $\mathcal{P}$ . The self term is never truncated. Measured against an untruncated assembly on a transferred  $\langle c \rangle$  population, the error in the climb strain rate is 19.3%, 9.6%, 3.9% and 0.25% at  $n_L = 1, 2, 3, 4$ :  $n_L = 4$ , **not** 3. There is a second reason the sum must be truncated and not merely may be:  $k_m^2$  in Eq. (38) already carries the mean-field response of the whole loop population, so Eq. (162) must supply only the near-field correction that mean field misses. Extending it past  $L_s$  double counts — the same failure the ledger of Section 3.7 guards against.

**Two consequences of Eq. (162) being linear in  $v_\nu^m$ .** First, the field cannot be evaluated offline: its amplitude is the output of Eq. (166), so there is no way to reconstruct  $c_D^m$  without reproducing the solve. Second, it vanishes identically on the first step of a solve, when the velocities are still zero. A single-step evaluation of Eq. (153) therefore returns  $c_C^m$  exactly, however many loops are present — a transferred population that is invisible in every diagnostic while being physically inert.

### 3.17 Selecting between them

The choice is made per family and on geometry rather than on preference, and it is really *two* choices rather than one. Two dimensionless numbers decide them, both computable from the continuum fields before any discrete object exists:

$$\Theta_k = \frac{2\bar{R}_{sL}^k}{d_k} \quad (\text{crowding}), \quad \Xi_k = \frac{d_k}{L_s} \quad (\text{halo separation}), \quad d_k = \left( \frac{n_{sL}^k}{\Omega} \right)^{-1/3}. \quad (171)$$

**They are independent, and reading them as one ranking is the mistake this paragraph exists to prevent.**  $\Theta_k$  asks whether the family needs discrete *geometry*: whether contact coalescence, junction formation and departure from a centered ellipse are live, which are the mechanisms Section 3.15 cannot represent. That is a failure of the immobile kinetics.  $\Xi_k$  asks whether the family needs to carry its own *concentration field*, by the measurement of Section 3.14. That is a question about the mobile coupling, and a family can need one without the other.

**What  $\Xi_k \lesssim 2$  does and does not say.** It does *not* say that the sink strength may be treated as uniform.  $k_m^2$  in Eq. (38) is already a spatially resolved field — it is built from  $n_{sL}^k(\mathbf{x})$  and  $\rho_{sL}^k(\mathbf{x})$  and evaluated at every quadrature point, exactly as the second-order reaction block is evaluated from the local mobile field — so “the mean field carries it” is a statement about the *fluctuation* and not about the field. What  $\Xi_k \lesssim 2$  says rests instead on a resolution argument:

a mobile defect diffuses a distance of order  $L_s$  before it is absorbed, so it responds to the average of  $k_m^2$  over  $L_s$  and not to the position of any individual loop. Structure in the sink strength is information down to  $L_s$  and is noise below it.

The relevant comparison is therefore not loop spacing against mesh, but  $L_s$  against mesh. On the reference case the mean CD node spacing is 10.5 nm against  $L_s \simeq 59$  nm: **the mesh already resolves the screening length 5.6 times over**, so the continuum  $k_m^2$  is nowhere near being the limiting approximation, and the 37% modulation of Section 3.14 is sub- $L_s$  structure that a diffusing defect averages away.

**On the reference case both families are in the merged-halo regime.** Interior medians at 10 dpa, from the same reduction the detector of Section 3.9 performs, with  $L_s$  from Eq. (154):



| family                                     | $R_{sL}^k$ [nm] | $d_k$ [nm] | $\Theta_k$ | $\Xi_k$ |
|--------------------------------------------|-----------------|------------|------------|---------|
| $\langle c \rangle$ basal                  | 57.6            | 109.5      | 1.05       | 1.86    |
| $\langle a \rangle$ prismatic, per variant | 3.6             | 88.6       | 0.081      | 1.50    |

and the drift is one-way: transferring a family out of the continuum lowers  $k_m^2$  and so *raises*  $L_s$ , which lowers  $\Xi_k$  further. A population that was marginal before a transfer is further inside the merged regime after one.

Table 8: The two questions of Eq. (171) are independent, so there are four cells and only two constructions in Sections 3.15 and 3.16 to fill them. Bracketed values are the measured position of each family on the reference 500 nm hexagonal case at 10 dpa.

|                                                                          | $\Xi_k \lesssim 2$ — halos merged                                                                                                                                                                                     | $\Xi_k \gtrsim 2$ — halos separated                                                                                                               |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Theta_k \ll 1$<br>shape freedom is enough                              | Continuum, Eq. (38). Nothing discrete is required in either respect. <b>Where</b> $\langle a \rangle$ sits ( $\Theta = 0.081$ , $\Xi = 1.50$ ), so the ROM is a <i>shape</i> refinement there and not a coupling one. | Ellipse ROM, Eqs. (157) and (158), with monopole neighbors. The field must be resolved; the geometry need not be.                                 |
| $\Theta_k \sim 1$<br>contact, junctions and non-elliptic shapes are live | <i>Nodal lines with a mean-field drive</i> — the missing cell, <b>not implemented</b> ; see below. <b>Where</b> $\langle c \rangle$ sits ( $\Theta = 1.05$ , $\Xi = 1.86$ ) — its loops overlap on average.           | Boundary integral, Eq. (166). Both the geometry and the field must be resolved, which is the only cell that genuinely requires the pair assembly. |

**The missing cell, and the spatial sink strength it needs.** The lower left of Table 8 is occupied and has no construction of its own. A family there needs full nodal freedom on its lines — contact and junctions are live — but does not need those lines to carry their own concentration field. Both requirements are met by taking the drive from the continuum alone, evaluating Eq. (152) node by node with  $c^m = c_C^m(\mathbf{x})$  interpolated to the line, and returning the population’s sink strength to the continuum as a nodal field built from where the loops actually are:

$$k_{m,D}^2(\mathbf{x}_j) = \sum_{\ell} Z_{sk,m} \varsigma_k^{\text{sink}} P_{\ell} W(\mathbf{x}_j - \mathbf{x}_{\ell}), \quad \sum_j V_j W(\mathbf{x}_j - \mathbf{x}_{\ell}) = 1, \quad (172)$$

with  $P_{\ell}$  the loop’s own elliptical perimeter Eq. (148),  $V_j$  the nodal volume Eq. (139), and  $W$  a partition-of-unity kernel of width  $\max(h, L_s)$  — *not* narrower, by the resolution argument above. Equation (172) adds to Eq. (38) exactly where Section 3.6 removed the transferred family from it, so the mobile solve sees the transferred loops through the same channel as every other sink. It costs  $O(N_{\text{seg}}^2)$  against the  $O(N_{\text{seg}}^2)$  of Eq. (167).

**An earlier revision of Eq. (172) carried a  $1/V_j$  in front of the sum as well**, which divides by the nodal volume twice: the normalization  $\sum_j V_j W = 1$  already makes  $W$  a density, so  $\sum_{\ell} P_{\ell} W$  is a length per volume and hence a sink strength, and the extra factor leaves length/volume<sup>2</sup>. Implementing it as printed made the population total come out low by exactly a factor of  $V$ . The two consistent readings — a dimensionless allocation  $\sum_j w_j = 1$  followed by  $k^2 = (1/V_j) \sum_{\ell} P_{\ell} w$ , or the density kernel above with no prefactor — give the same field under  $W = w/V_j$ , which is what identifies this as a transcription slip rather than a choice.

**The normalization must be imposed per loop, on the nodes actually within reach.** A kernel normalized once analytically leaks wherever the node cloud is irregular or truncated, and it is truncated at the surface — which is precisely where the refused loops sit. Imposed loop by loop, the total is conserved on any node set: measured at  $2 \times 10^{-16}$  relative, with  $\sum_j V_j W - 1$  below  $6 \times 10^{-16}$  for loops in the interior, on a face, at a corner, and outside the domain alike.

**And the gap it fills is now measured rather than argued.** Take a uniform population, refuse every loop within 150 b of a face — so  $f_{\text{kept}} = 0.346$  — and compare the field the surviving loops actually make against the scalar  $f_{\text{kept}}$  applied to the whole domain. The two carry the *same total* to four decimals, which is why the ledger of Section 3.7 closes on either; but the scalar route **overstates the sink in the shell by  $3.90\times$  and understates it in the interior by  $0.36\times$** . That is the spatial information the compensation discards, and it is a factor of eleven across the domain.



What remains of this step is the ingestion, not the field. `ImmobileSinks` builds  $k^2$  from the immobile field at the quadrature points and has no channel for an externally supplied nodal one. Adding it is a design decision — a trial function of its own, or an equivalent  $(n, c)$  written back into the family’s own slots, which would then also evolve as a continuum family and double-count the physics — and it is deliberately left open rather than settled by whichever is easier to write.

**Two things make this a real gap and not a refinement.** First, the two routes are not the same channel:  $c_D^m$  reaches the continuum problem only through the Dirichlet values on the outer surface, so a transferred population contributes to the mobile solve by superposition but contributes nothing to the finite-element operator, whereas [Eq. \(172\)](#) would. Second, and concretely, **the refusal of [Section 3.6](#) is a strongly spatial criterion whose compensation is currently spatially uniform.** Loops are refused *because they are near a face*; the defects they carried are returned to the continuum by scaling the whole family field by the single scalar  $1 - f_{\text{kept}}$ , at every node alike. In expectation that is the correct removal for a draw whose intensity was  $\lambda_{sj}^k$ , and it is why the ledger of [Section 3.7](#) closes. It stops being correct as soon as the accepted loops evolve, because their distribution then departs from  $n_{sL}^k(\mathbf{x})$ ; and it is already incorrect for the refused ones, which were concentrated in a boundary shell some 150 nm thick — a scale well above  $L_s$ , and therefore precisely where the resolution argument says spatial structure *is* information. Restoring it is what [Eq. \(172\)](#) is for.

**The cost this buys.** At the population of [Table 8](#), with the cutoff [Eq. \(170\)](#) and  $N_{\text{side}} = 12$ :

| scheme                                                  | work                                |
|---------------------------------------------------------|-------------------------------------|
| $\langle c \rangle$ boundary integral, 12 sides, cutoff | $4.7 \times 10^5$ segment pairs     |
| $\langle a \rangle$ ellipse ROM, monopole neighbors     | $1.0 \times 10^4$ evaluations       |
| <b>split total</b>                                      | <b><math>4.8 \times 10^5</math></b> |
| monolithic boundary integral, 12 sides, cutoff          | $2.0 \times 10^6$                   |
| monolithic boundary integral, 64 sides, all pairs       | $7.0 \times 10^8$                   |

The split with truncation is  $\sim 1450\times$  cheaper than the naive assembly on the rendering polygon and  $\sim 4\times$  cheaper than treating both families nodally. **The rendering polygon must not be handed to the climb solver:** 64 sides are free in a figure and cost a factor of 28 here, and the exported radius is area-matched at any  $N_{\text{side}}$ , so the two constants need not agree.

**The interface the split imposes.** The two families are neighbors of each other, so both representations must answer the same question — evaluate  $c_D^m$  at an arbitrary  $\mathbf{x}$  — the ellipse by its monopole, the nodal loop by [Eq. \(162\)](#). That common interface, and not either construction on its own, is the piece of structure the selection rule requires.

### 3.18 Implementation status

Stated in the register of [Appendix C](#), because the two constructions are not at the same stage.

- **The boundary-integral solve is implemented and in use.** [Equations \(163\)](#) and [\(165\)](#) are assembled per segment pair, [Eq. \(166\)](#) is factorized sparsely per species, and the cutoff [Eq. \(170\)](#) is an optional material key with  $n_L = 4$  as its default; absent or zero, no truncation is applied. A lumped-diagonal reduction of [Eq. \(167\)](#) is retained as a fallback. Because of the second consequence in [Section 3.16](#), a solve intended to exercise it must take at least two steps, and the superposed field [Eq. \(153\)](#) is published alongside  $c_C^m$  rather than folded into it, under a separate output key. **Every figure produced before that key existed is correct**, because  $c_D^m \equiv 0$  in those runs and the continuum block *was* the physical concentration; it is only once loops genuinely enter a solve that the block becomes the corrective part.
- **The ellipse ROM is implemented as a module and is not yet wired into the march.** [Equations \(157\)](#) and [\(158\)](#) are integrated by an explicit forward Euler step, which is the appropriate choice and not an economy: the system is two-dimensional per loop and the neighbor coupling is lagged, so there is no coupled solve and nothing for an implicit step to buy — in contrast to the immobile block, which is stiff and is integrated by BDF. Its material constants are read from the same material file the spatial solver uses, so the ROM and [Eq. \(68\)](#) cannot

disagree about  $p_m$  or  $Z_{sk,m}$ . What it has so far been used for is diagnosis: integrating Eqs. (157) and (158) along a completed march to produce the ellipticity trajectories quoted in Section 3.15. In that use it is a *post hoc* calculation — the concentrations it reads are not affected by the shapes it predicts — and those numbers should be read as what the ROM says about that march, not as a coupled result.

- **The transfer itself is armed by an option and is off by default**, for the reasons in Section 3.6: the accepted fraction must be checked in defects on every new geometry before a transfer is allowed to feed a solve, and the sink-strength convention in the last row of Table 6 is still an open modeling decision.

## 4 Thermal Annealing Simulations

Everything so far has been written for a crystal under irradiation. Switching the source off does not call for a different model, and that is the point of this section: the emission channel Eq. (110) was built as the detailed-balance partner of the absorption channel precisely so that a pure anneal is the same equations run at  $G = 0$ . What removing the source does is remove the term that has been hiding the thermal one. Under irradiation the mobile concentrations sit many orders of magnitude above their thermal values and the bracket of Eq. (112) is controlled by  $c^v$ ; with the source off,  $c^v$  relaxes to something of order  $c_\infty^{v,\text{eq}}$  and the bracket is controlled instead by the loop's own  $c_{sk}^{v,\text{eq}}$  — that is, by the binding energy. **An anneal is therefore the sharpest available test of Eqs. (108) and (109)**, because it is the only regime in which the binding energy is not a small correction to a much larger irradiation flux.

Four things are set out below, in order: the binding energy per vacancy of each loop variant, computed from the anisotropic elastic energy, the fault energy and the applied stress *tensor*; the emission rate each variant carries as a consequence; a worked isolated-loop calculation for the two basal states, with a helper that produces the numbers; and the two constructions by which an anneal is run on a discrete population obtained from an irradiation simulation at a chosen dose.

### 4.1 The binding energy per vacancy of the loop variants

**The energy of a loop under load.** Equation (8) gives the energy of a planar cluster in a stress-free crystal as the sum of a faulted-area term and an anisotropic elastic self-energy. An applied stress adds one term and one only. A loop of area  $A$ , Burgers vector  $\mathbf{b}_k$  and habit normal  $\mathbf{n}_k$  carries the plastic distortion  $\mathbf{b}_k \otimes \mathbf{n}_k A/V$ , so the work the applied stress does in creating it is  $\boldsymbol{\sigma} : (\mathbf{b}_k \otimes \mathbf{n}_k) A$  and

$$E_k(m; \boldsymbol{\sigma}) = \gamma_k \pi R^2 + \frac{\langle K \rangle_k |\mathbf{b}_k|^2 R}{2} \ln \frac{\alpha R}{|\mathbf{b}_k|} - \Sigma_k(\boldsymbol{\sigma}) \Omega m, \quad R = \lambda_k \sqrt{m}, \quad (173)$$

in which  $R$  and  $m$  are related by Eq. (65),  $\alpha$  absorbs the core cutoff, and

$$\Sigma_k(\boldsymbol{\sigma}) \equiv \frac{\mathbf{b}_k \cdot \boldsymbol{\sigma} \cdot \mathbf{n}_k}{(\mathbf{b} \cdot \mathbf{n})_k} \quad (174)$$

is the resolved stress the family actually feels, obtained by dividing the work per unit area by the defects stored per unit area. **Equation (174) is the whole of what the stress *tensor* contributes**, and it is not in general the normal stress  $\sigma_{kk}$  that Section 2.7 resolves.

- For every *pure edge* family — the six prismatic variants, and the perfect basal state  $c_p$  —  $\mathbf{b}_k \parallel \mathbf{n}_k$  and Eq. (174) collapses to  $\Sigma_k = \sigma_{kk}$  exactly, so nothing said in Section 2.7 changes.
- For the faulted basal state  $c_f$  it does not. Its Burgers vector  $\frac{1}{6}\langle 20\bar{2}3 \rangle$  of Eq. (5) carries the basal partial of Eq. (7) alongside its  $c/2$  edge component, and  $(\mathbf{b} \cdot \mathbf{n})_{c_f} = c/2$ , so

$$\Sigma_{c_f} = \sigma_{33} + \frac{2a}{\sqrt{3}c} \hat{\mathbf{p}} \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{c}}, \quad \frac{2a}{\sqrt{3}c} = 0.7242, \quad (175)$$

with  $\hat{\mathbf{p}}$  the partial's basal direction. **A basal shear acts on the faulted loop with 72% of the weight a  $c$ -axis normal stress carries, and acts on the perfect loop not at all.** That

is a character-independent, purely crystallographic consequence of Table 2 and it is testable: a basal shear should separate the annealing kinetics of the two basal states even where a uniaxial  $c$ -axis load does not distinguish them.

Equation (175) also exposes something Eq. (64) does not carry. There are three symmetry-equivalent choices of  $\hat{\mathbf{p}}$  for a given faulted loop, and a basal shear favors one of them; the model holds one  $c_f$  family and so cannot redistribute among them. That is the basal analogue of the prismatic variant resolution Eq. (77), it is absent for the same reason Section 2.4 gives — the three basal states share one habit plane — and it is the one place where sharing a habit plane is *not* sufficient to make a stress partition unnecessary.

**The work to detach one defect.** Differentiating Eq. (173) at fixed  $\sigma$  reproduces Eq. (108) with the stress term appended,

$$\left. \frac{\partial E_k}{\partial m} \right|_{\sigma} = \underbrace{\frac{\gamma_k \Omega}{(\mathbf{b} \cdot \mathbf{n})_k}}_{\text{fault}} + \underbrace{\frac{\langle K \rangle_k b_k^2 \lambda_k^2}{4R} \left( 1 + \ln \frac{\alpha R}{b_k} \right)}_{\text{anisotropic capillarity}} - \Sigma_k \Omega, \quad (176)$$

and the vacancy binding energy of family  $(s, k)$ , Eq. (109), becomes

$$E_b^{v,k}(\bar{m}^k, \sigma) = E_v^f - \varsigma_s(v) \left. \frac{\partial E_k}{\partial m} \right|_{\sigma=0} - \Sigma_k \Omega, \quad c_{sk}^{v,\text{eq}} = \exp \left( -\frac{E_b^{v,k}}{k_B T} \right). \quad (177)$$

Three properties of Eq. (177) are worth isolating before it is evaluated, and the second is a correction to the equation this document carried before.

1. **Nothing in Eq. (176) is a new parameter.**  $\gamma_k$  is the family's own fault energy from Table 2,  $\langle K \rangle_k$  its perimeter-averaged Stroh factor from Table 1,  $\lambda_k$  and  $(\mathbf{b} \cdot \mathbf{n})_k$  its geometry from Table 4, and  $\Omega$  and  $E_v^f$  are the lattice and point-defect constants of Eq. (2) and the material file. The anisotropic elasticity enters here and nowhere else, and it enters through the same average that ordered the fault states in Section 1.2: using  $\mu/(1-\nu)$  instead would understate the basal families' capillarity by 7% relative to the prismatic ones, which is a systematic error in the *contrast* this section measures.
2. **The stress term carries no character factor, and an earlier revision of Eq. (111) said otherwise.** Emitting a vacancy from a loop always adds  $\Omega$  of material back to the crystal along  $\mathbf{n}_k$ : an interstitial loop grows and inserts a further  $\Omega$ , a vacancy loop shrinks and returns the  $\Omega$  it had removed. Both insert. So the work the applied stress does per emitted vacancy is  $+\Sigma_k \Omega$  with the *same* sign for both characters, which is what Eq. (177) states and what Eq. (111) now carries. Written as  $\varsigma_s(v) (\partial E_k / \partial m - \sigma_{kk} \Omega)$  it is correct for the interstitial families and carries the wrong sign for the vacancy ones, so Eq. (115) — which is a statement about the interstitial character — is unaffected, and every statement about a  $\langle c \rangle$  loop under load is affected. The physical content of the correction is the ordinary one: a tensile stress across a habit plane favors inserting material there, hence favors interstitial loops of that variant and disfavors vacancy loops of it, which is exactly the direction Eq. (77) partitions the sources in.
3. **It has one threshold, not two.** A family grows, in the sense of its own character, when  $c_{sk}^{v,\text{eq}}$  crosses the ambient field, and collecting the signs gives a single condition covering both characters:

$$\eta_s \Sigma_k > \sigma_k^* \equiv \left. \frac{1}{\Omega} \frac{\partial E_k}{\partial m} \right|_{\sigma=0}, \quad (178)$$

with  $\sigma_k^*$  the classical climb threshold Eq. (115). Read for  $s = i$  this is Eq. (115) unchanged: an interstitial loop climbs outward under sufficient tension. Read for  $s = v$  it is its mirror, which the previous sign concealed: **a vacancy loop grows under a compressive normal stress exceeding  $\sigma_k^*$** , and is otherwise unconditionally shrinking. Both readings are the same statement that the loop's stored defect has a relaxation volume of one sign and the stress work has the other.

**Evaluated for the families.** Table 9 evaluates Eqs. (176) and (177) at  $\sigma = \mathbf{0}$  for the four distinct loop geometries of Eq. (64), each at the interior median radius the reference march reports for it in Table 8. Everything in the table is produced by `disloccluster_code.studies.loop_annealing`, whose `verify` target checks it against the closed forms of Section 1.2 — the three  $\lambda_k$  of Table 4, the exact ratio  $\lambda_{c_f}/\lambda_{c_p} = \sqrt{2}$ , the elastic-weight ordering of Table 1, the  $R \rightarrow \infty$  limits of Eq. (177), the threshold Eq. (178) and the shear weight of Eq. (175).

Table 9: Binding energy per vacancy, Eq. (177), at  $\sigma = \mathbf{0}$  and  $T = 773$  K, for the four distinct loop geometries. Each family is evaluated at the interior median radius of Table 8.  $E_v^f = 1.6$  eV and  $\Omega = 2.3266 \times 10^{-29} \text{ m}^3$  come from the material file;  $\gamma_{c_f} = 0.124 \text{ J m}^{-2}$  is the  $I_1$  basal fault. The last column is the supersaturation the loop imposes on its own surroundings, and is the quantity that drives an anneal.

| family               | $R$ [nm] | $\bar{m}$          | fault [eV] | capillarity [eV] | $E_b^{v,k}$ [eV] | $c_{sk}^{v,\text{eq}}/c_\infty^{v,\text{eq}}$ |
|----------------------|----------|--------------------|------------|------------------|------------------|-----------------------------------------------|
| $a_k$ , interstitial | 3.6      | 565                | 0          | 0.1922           | 1.7922           | 0.0559                                        |
| $a_k$ , vacancy      | 3.6      | 565                | 0          | 0.1922           | 1.4078           | 17.9                                          |
| $c_f$ faulted        | 57.6     | $1.15 \times 10^5$ | 0.0699     | 0.0262           | 1.5039           | 4.234                                         |
| $c_p$ perfect        | 57.6     | $2.31 \times 10^5$ | 0          | 0.0346           | 1.5654           | 1.681                                         |

Four readings, of which the third is the one the rest of this section rests on.

1. **One expression, both characters, and they land on opposite sides of  $E_v^f$ .** The two prismatic rows differ only in  $\varsigma_s(v)$  and straddle the bulk value symmetrically, 1.79 against 1.41 eV. The vacancy variant is in equilibrium with 17.9 times the bulk vacancy concentration and the interstitial variant with 0.056 of it, so at the same size and in the same field the first emits and the second absorbs — and both shrink, because absorbing a vacancy shrinks an interstitial loop. This is Eq. (114) made numerical.
2. **Size, not character, is what sets the magnitude.** The prismatic families sit at  $\bar{m} \simeq 565$  and the basal ones at  $10^5$ , and the capillary term falls as  $\ln R/R$ : it is 0.19 eV for the small loops and 0.03 eV for the large ones. A  $\langle c \rangle$  loop at the size the march produces is elastically almost a flat plate.
3. **The fault energy is what separates  $c_f$  from  $c_p$ , and it does not decay.** The fault term  $\gamma_k \Omega / (\mathbf{b} \cdot \mathbf{n})_k$  is independent of  $R$ , so at 57.6 nm it is 2.7 times the capillary term for  $c_f$  and is the entire driving force in the large-loop limit, where capillarity has gone to zero. The two basal states therefore differ by more than a factor of two in supersaturation (4.23 against 1.68) despite having the same habit plane, the same character and the same size, and Section 4.3 shows that this single number changes the *functional form* of the anneal and not merely its rate.
4.  **$\gamma_{c_f}$  is the one fitted-looking number here, and it is weak.** Ab initio basal fault energies for  $\alpha$ -Zr scatter over roughly 0.10–0.15  $\text{J m}^{-2}$  [17, 18], and the fault term is strictly linear in it. Carried through to the anneal-out time of Section 4.3, that whole spread moves a 25 nm faulted loop at 873 K from 7.4 to 4.8 h against 5.95 h at the central value — a  $\pm 25\%$  band on a quantity that varies by four orders of magnitude across the temperature range of interest.

The pyramid  $c_0$  is absent from Table 9 and from everything below. It has no perimeter, Eq. (108) is a statement about a bounding dislocation it does not possess, and its loss channel is the collapse Eq. (13) rather than peripheral emission — which is why Section 2.8 keeps a lifetime  $\tau_{\text{sfp}}$  for it alone. In an anneal it dissolves through Eq. (113)’s second term, and its vacancies join the same mobile pool the loops are emitting into.

## 4.2 Thermal emission from each family

Equation (110) gives the emission rate as a *content* rate for a whole family. For an anneal the useful form is per loop, and it follows from Eq. (73) without approximation: the prefactor in the growth law

is the family's own perimeter, so dividing by the number of loops leaves the perimeter of one,

$$\varepsilon_k = \underbrace{2\pi R_{sL}^k}_{\text{perimeter}} \frac{Z_{sk,v} \bar{D}^v}{\Omega} c_{sk}^{v,\text{eq}}(\bar{m}^k, \sigma) \quad [\text{vacancies emitted per loop per second}], \quad (179)$$

with the absorption rate the same coefficient acting on the ambient field  $c^v$ , so that the net rate is the bracket of Eq. (112) again. Two things about Eq. (179) decide what an annealing calculation can be compared against.

**The capture efficiency is not the irradiation one.**  $Z_{sk,v}$  in Eq. (68) is a *bias*: it measures how a loop's capture differs from a neutral sink's in a medium densely populated by other sinks, and its magnitude is fixed by the mean-field sink strength Eq. (38) that the population as a whole imposes. An isolated loop has no such surroundings, and its capture rate is the classical toroidal one — the absorption rate of a sink is  $4\pi\bar{D}C_{\text{cap}}$  with  $C_{\text{cap}} = \pi R/\ln(8R/r_0)$  the capacitance of a thin torus — so that per unit length of line,

$$Z_k^{\text{iso}} = \frac{2\pi}{\ln(8R_{sL}^k/r_0)}, \quad r_0 \simeq |\mathbf{b}_k|. \quad (180)$$

Equation (180) is not a competing model: it is what Eq. (68) reduces to when the screening length Eq. (154) exceeds the loop spacing so far that each loop draws on an unperturbed medium. It is close to unity over the whole size range here — 1.40 at 3.6 nm and 0.86 at 57.6 nm — which is why the mean-field  $Z$ 's of order unity are not absurd; but it *falls* with size where a bias does not, and it carries the only  $R$  dependence in Eq. (179) beyond the perimeter. Section 4.3 uses Eq. (180) because its loop is isolated by construction; Section 4.6 gives it up again as soon as more than one loop is present.

**The implemented code does not yet use Eq. (177).** The material file carries a constant 5 eV surrogate for the loop binding energy (`Eb_eV` index 0), inherited from the lumped model, where Eq. (177) gives 1.41 to 1.79 eV. At 773 K the surrogate therefore suppresses  $c_{sk}^{v,\text{eq}}$  by  $\exp(-3.5 \text{ eV}/k_B T) \simeq 1.6 \times 10^{-23}$ , i.e. **the emission channel is switched off entirely in every run made to date**. That is harmless under irradiation, where the bracket of Eq. (112) is dominated by  $c^v$  and the entire emission term is negligible — and it is the whole calculation in an anneal. Replacing the surrogate is step 3 of Appendix D.2; nothing in Section 4 can be run against the current binary until it lands.

Table 10 evaluates Eq. (179) at 773 K.

Table 10: Thermal emission and absorption per loop, Eq. (179), at  $T = 773 \text{ K}$  and  $\sigma = \mathbf{0}$ , with the ambient field at bulk equilibrium  $c^v = c_{\infty}^{v,\text{eq}} = 3.70 \times 10^{-11}$  and  $\bar{D}^v = 7.83 \times 10^{-15} \text{ m}^2 \text{ s}^{-1}$ . A negative net rate is a shrinking vacancy loop; the interstitial family's positive net rate is vacancy *absorption*, which shrinks it too. The last column is Eq. (181).

| family               | $R$ [nm] | $Z_k^{\text{iso}}$ | emitted [ $\text{s}^{-1}$ ] | absorbed [ $\text{s}^{-1}$ ] | $\dot{R}$ [ $\text{nm h}^{-1}$ ] |
|----------------------|----------|--------------------|-----------------------------|------------------------------|----------------------------------|
| $a_k$ , interstitial | 3.6      | 1.399              | $2.20 \times 10^{-5}$       | $3.94 \times 10^{-4}$        | −0.0043                          |
| $a_k$ , vacancy      | 3.6      | 1.399              | $7.06 \times 10^{-3}$       | $3.94 \times 10^{-4}$        | −0.0764                          |
| $c_f$ faulted        | 57.6     | 0.863              | $1.65 \times 10^{-2}$       | $3.89 \times 10^{-3}$        | −0.0113                          |
| $c_p$ perfect        | 57.6     | 0.925              | $7.01 \times 10^{-3}$       | $4.17 \times 10^{-3}$        | −0.0013                          |

Note what the last two columns do *not* do: they do not order the same way. The faulted ( $c$ ) loop emits 2.4 times as fast as the perfect one in absolute terms and shrinks 8.9 times as fast. At equal radius their perimeters are equal, so the whole of the difference is that the *net* rate — emission minus absorption, not emission — differs by 4.4, and that  $\dot{R}$  then divides by  $(\mathbf{b} \cdot \mathbf{n})_k$ , which is twice as large for the perfect state because it stores two removed layers per unit area rather than one. **Emission rate and shrinkage rate are different orderings of the same families**, and quoting one for the other is the easiest available error in reading an annealing calculation.

### 4.3 An isolated basal loop: the faulted and the perfect state

**The calculation.** Take one loop, alone, in an infinite crystal held at  $c^v = c_{\infty}^{v,\text{eq}}$ , with no irradiation and no other sink. This is the single-loop limit of Eq. (152): the arrival rate per unit length is

$Z_k^{\text{iso}} \bar{D}^v c^v / \Omega$  by Eq. (180), the loop is circular because a basal loop has no preferred in-plane direction to elongate along (Fig. 2, right), and both axes of Eqs. (157) and (158) collapse onto one:

$$\frac{dR}{dt} = \varsigma_s(v) \frac{Z_k^{\text{iso}} \bar{D}^v}{(\mathbf{b} \cdot \mathbf{n})_k} [c^v - c_{sk}^{v,\text{eq}}(m(R), \boldsymbol{\sigma})], \quad m(R) = \frac{\pi R^2 (\mathbf{b} \cdot \mathbf{n})_k}{\Omega}. \quad (181)$$

**Equation (181) is the growth law and nothing else.** No Green's function is evaluated, no pair is assembled, and the self-stress does not appear as a separate term: it is already inside  $c_{sk}^{v,\text{eq}}$  through Eq. (176), which is the same capillarity Eq. (159) represents but built from  $\langle K \rangle_k$  rather than from  $\mu/(1-\nu)$  and carrying the fault term that Eq. (159) has no place for. The equation is integrated as  $dt/dR$  rather than  $dR/dt$ , because  $R$  is the monotone variable and the rate diverges as the loop disappears; the trajectory ends at the model's own floor  $R_{\min} = 1$  nm from the material file, where a basal loop still holds 35 vacancies and the continuum expression Eq. (173) is at the edge of its validity.

**The parameters, and why these.** Equation (181) is evaluated for the two basal states at  $R_0 = 25$  nm (50 nm across) over 723–1073 K. Both numbers are chosen against the experimental record rather than for convenience. Basal  $c$ -component vacancy loops in neutron- and ion-irradiated Zr are reported at diameters from a few tens of nanometres upward once they appear at all, with densities of order  $10^{20} \text{ m}^{-3}$  [3, 10, 11], so 50 nm is at the small end of the observed distribution and is the size at which the model itself starts a transferred population. The temperature range is the one in which post-irradiation annealing of Zr alloys is actually carried out: damage recovery in  $\alpha$ -Zr sets in above roughly 700 K, the  $\langle a \rangle$  population is removed over hours in the 700–800 K range, and the  $c$ -component population is consistently reported as the more thermally robust of the two [3, 25, 2]. **The point of the comparison is the ordering and the shape, not the absolute lifetime**, which inherits  $E_v^f + E_v^m = 2.8$  eV from the fitted point-defect set and would move by an order of magnitude for a 0.15 eV change in either.

**What comes out.** Figure 3 and Table 11 give the result. Both states shrink monotonically to extinction at every temperature, as Eq. (114) requires with  $\varsigma_v(v) = +1$  and  $\partial E_k / \partial m > 0$ , and the faulted state is between 4.0 and 4.5 times the faster over the whole range.

Table 11: Anneal-out time of an isolated basal loop from  $R_0 = 25$  nm to  $R_{\min} = 1$  nm, at zero stress and  $c^v = c_{\infty}^{v,\text{eq}}$ , from Eq. (181). The last column is the ratio, which is nearly temperature-independent because both states are driven by the same  $\bar{D}^v c_{\infty}^{v,\text{eq}}$  and differ only in the size of the exponent.

| $T$ [K] | $c_f$ faulted | $c_p$ perfect | $t_{c_p}/t_{c_f}$ |
|---------|---------------|---------------|-------------------|
| 773     | 22.9 d        | 104 d         | 4.53              |
| 823     | 49.7 h        | 9.05 d        | 4.37              |
| 873     | 5.95 h        | 25.2 h        | 4.23              |
| 923     | 53.9 min      | 3.69 h        | 4.11              |
| 973     | 9.9 min       | 39.7 min      | 4.01              |
| 1023    | 2.15 min      | 8.44 min      | 3.92              |

Three results, and the first two are the ones worth carrying away.

1. **The two states obey different laws, because one of them has a driving force that does not vanish with size.** In the large-loop limit Eq. (176) tends to  $\gamma_k \Omega / (\mathbf{b} \cdot \mathbf{n})_k$  for the faulted state and to zero for the perfect one, so

$$\dot{R} \rightarrow \text{const} \Rightarrow t_{\text{anneal}} \propto R_0 \quad (c_f), \quad \dot{R} \propto \frac{\ln R}{R} \Rightarrow t_{\text{anneal}} \propto \frac{R_0^2}{\ln R_0} \quad (c_p). \quad (182)$$

Panel (c) measures the local exponent  $d \ln t / d \ln R_0$  at 873 K as  $2.43 \rightarrow 1.85 \rightarrow 1.50 \rightarrow 1.31$  for  $c_f$  at  $15 \rightarrow 45 \rightarrow 134 \rightarrow 400$  nm, and  $2.83 \rightarrow 2.35 \rightarrow 2.14 \rightarrow 2.05$  for  $c_p$  at the same sizes: both are converging on Eq. (182) and neither has arrived at 25 nm, which is why panel (a) shows two concave curves rather than a straight line and a curve. **A fault energy is not a rate constant, it is a change of kinetic law**, and no fitted annealing lifetime can represent both states.



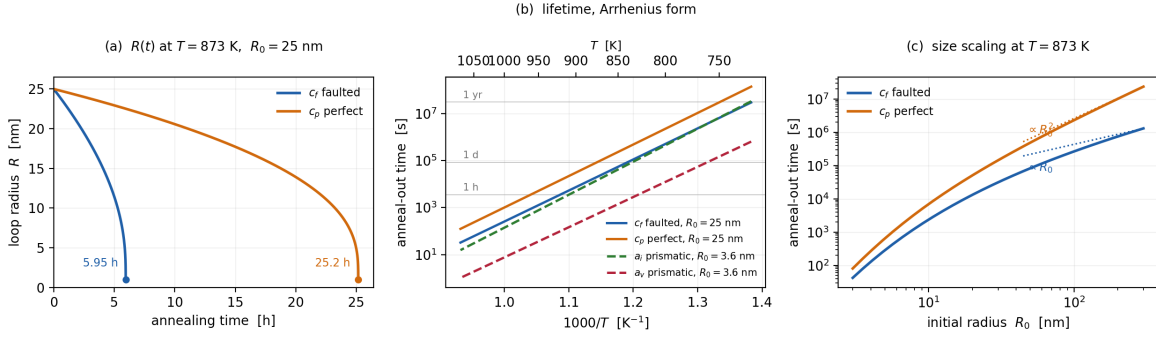


Figure 3: Thermal anneal of an isolated basal vacancy loop from Eq. (181), for the faulted state  $c_f$  and the perfect state  $c_p$  of Table 2. (a)  $R(t)$  at 873 K from  $R_0 = 25$  nm, on a linear time axis: the faulted loop reaches the floor in 5.95 h and the perfect one in 25.2 h, and the two curves have different *shapes* and not merely different rates. (b) The anneal-out time in Arrhenius form, with the two prismatic families added at their own reference size  $R_0 = 3.6$  nm; the slopes give  $Q_{\text{eff}} = 2.56\text{--}2.79$  eV against  $E_v^f + E_v^m = 2.80$  eV. (c) The size dependence at 873 K, with the two asymptotes of Eq. (182) drawn as guides. Produced by `disloccluster_code.studies.loop_annealing figure`.

2. **The lifetime is Arrhenius with an activation energy slightly below self-diffusion.** Panel (b) gives  $Q_{\text{eff}} = 2.63$  eV ( $c_f$ ), 2.67 ( $c_p$ ), 2.79 ( $a_i$ ) and 2.56 eV ( $a_v$ ) against  $E_v^f + E_v^m = 2.80$  eV. The deficit is not numerical: the rate carries  $\bar{D}^v c_{\infty}^{v,\text{eq}} \propto \exp[-(E_v^f + E_v^m)/k_B T]$  times the driving factor  $\exp(\mu_k/k_B T) - 1$ , which itself falls with temperature, and the size of the deficit is therefore a direct measure of  $\mu_k = \partial E_k / \partial m$ . **A measured annealing activation energy below the self-diffusion value is a measurement of the binding energy**, and the two prismatic rows bracket the basal ones from both sides because  $\zeta_s(v)$  puts their  $\mu_k$  on opposite sides of zero.
3. **Both prismatic characters anneal, and the vacancy one goes first.** At 873 K a 3.6 nm prismatic vacancy loop lasts 9.4 min and its interstitial counterpart 4.3 h, a factor of 27, entirely because  $\zeta_s(v)$  flips the sign of a 0.19 eV exponent. That is the quantitative form of the statement in Section 2.8 that one expression serves both characters, and it is a prediction: an anneal that removes the vacancy  $\langle a \rangle$  population well before the interstitial one, at the same size, is this model being right about a term it does not fit.

**What the applied stress does to it.** At 873 K and  $R_0 = 25$  nm, Eq. (181) gives 5.95 h at zero stress, 3.83 h under  $\sigma_{33} = +200$  MPa and 9.56 h under  $-200$  MPa. The sign is the corrected one of Eq. (177): a tensile stress across the basal plane opposes a vacancy platelet and accelerates its removal, and by Eq. (178) a compressive stress exceeding  $\sigma_{c_f}^* = \Omega^{-1} \partial E_k / \partial m = 842$  MPa would reverse it outright. The previous sign gives the opposite answer to both, and an annealing experiment under load is the cleanest way to tell them apart.

**Three limits of this calculation, which Section 4.6 exists to remove.** The loop is alone, so no neighbor re-absorbs what it emits; the ambient field is pinned at  $c_{\infty}^{v,\text{eq}}$ , where a real anneal has it determined by the balance of every source and sink in the volume; and the population is one loop, so nothing is said about how a *distribution* of sizes anneals — which is the observable, and which is where Section 2.9 predicts the density falls while the surviving mean size rises.

#### 4.4 The same anneal by the boundary-integral Green's function

Equation (181) compresses the whole of the transport into one closed-form number,  $Z_k^{\text{iso}}$ , and assumes the loop is a circle. Section 3.16 carries a construction that assumes neither: the boundary integral solves for the concentration at the line rather than parameterizing it, and the loop's outline is whatever the solved nodal velocities make it. The isolated loop is the one case in which that construction is cheap



enough to run all the way to extinction — one loop, no neighbors, so Eq. (170) never fires and the pair assembly is  $N_{\text{side}}^2$  with  $N_{\text{side}} \leq 128$  — and it is therefore the natural place to ask what Eq. (180) is worth and what the ellipse of Eq. (155) gives up. This subsection does both. **The conclusion is that Section 4.3 stands**, and that the two things which do move are worth knowing: the capture efficiency by a constant 9%, and the loop shape by a factor that turns out to expose an error in Eq. (159).

**The diffusion problem, written out.** One loop, alone, in an infinite crystal held at  $c_{\infty}^{v,\text{eq}}$ . The mobile vacancy field is quasi-steady on the climb time scale, so outside the core it satisfies the *anisotropic* steady diffusion equation with the line held at its own local equilibrium:

$$\nabla \cdot (\mathbf{D}_v \nabla c^v) = 0 \quad \text{outside the core tube of radius } r_0, \quad \begin{cases} c^v(\mathbf{x}) \rightarrow c_{\infty}^{v,\text{eq}}, & |\mathbf{x}| \rightarrow \infty, \\ c^v(\mathbf{x}) = c_{sk}^{v,\text{eq}}(\kappa(\mathbf{x}), \boldsymbol{\sigma}), & \mathbf{x} \text{ on the loop line,} \end{cases} \quad (183)$$

closed by the kinematic statement Eq. (152),  $v_n = \varsigma_s(v) \Omega \Phi^v / (\mathbf{b} \cdot \mathbf{n})_k$ . **Equation (183) is a Dirichlet problem on a closed line, and no capture efficiency appears in it anywhere** — which is exactly why it can be used to measure one. It is solved by the superposition Eq. (153), with  $c_c^v = c_{\infty}^{v,\text{eq}}$  uniform and  $c_{\mathcal{D}}^v$  the field the climbing line itself emits. An element climbing at  $v_n$  inserts material at the volumetric rate  $(\mathbf{b} \times \hat{\boldsymbol{\xi}}) \cdot \hat{\boldsymbol{\eta}}$  per unit length, so  $c_{\mathcal{D}}^v$  is the line integral of the anisotropic point-source Green's function

$$\nabla \cdot (\mathbf{D}_m \nabla \mathcal{G}_m) = -\delta(\mathbf{x}), \quad \mathcal{G}_m(\mathbf{x}) = \frac{1}{4\pi \sqrt{\det \mathbf{D}_m} \sqrt{\mathbf{x} \cdot \mathbf{D}_m^{-1} \mathbf{x}}}, \quad (184)$$

against that source. Carrying the network's own linear shape functions along each chord turns the integral into the closed form Eq. (163) with the elementary integrals Eq. (165), and requiring Eq. (183)'s Dirichlet condition on every segment in a Galerkin sense closes the system Eq. (166). The implementation follows `DislocationSegment::concentrationMatrices` term for term, core regularization Eq. (164) included, so what is solved here is the operator `GalerkinClimbSolver` assembles — evaluated on one loop instead of a population. One simplification is genuine and worth naming: a pure anneal has no irradiation, so only the vacancy species is carried, and the driving supersaturation is Eq. (111) rather than the Peach–Koehler form Eq. (169), because Section 4.3's capillarity is the thermodynamics being tested and substituting a second one would confound the comparison.

**What is held fixed, so that only the transport differs.** All three constructions compared below — Eq. (181), the boundary integral, and the ellipse ROM Eqs. (157) and (158) — are driven by the *same*  $c_{sk}^{v,\text{eq}}$ , built from the *same*  $\partial E_k / \partial m$  of Section 4.3, generalized from a circle to a local curvature:

$$\frac{\partial E_k}{\partial m}(\kappa) = \frac{\gamma_k \Omega}{(\mathbf{b} \cdot \mathbf{n})_k} + \frac{\langle K \rangle_k b_k^2 \lambda_k^2}{4} \kappa \ln \frac{e \alpha}{\kappa b_k}, \quad \kappa \rightarrow \frac{1}{R} \implies \text{Eq. (176) exactly}, \quad (185)$$

since  $1 + \ln x = \ln(ex)$ ; the reduction is checked to machine zero. So every difference reported below is a difference of **transport** and of nothing else. This is deliberate rather than convenient: Eq. (159) uses  $\mu/(1 - \nu)$  where Section 4.3 uses  $\langle K \rangle_k$  and has no place for the fault term at all, and letting the ROM carry its own thermodynamics would put a capillarity difference inside a transport comparison.

**The kernel test, which has to come first.** In an *isotropic* medium the boundary-integral solve of a circular loop must return Eq. (180) itself, because the self-field of a ring of uniform strength is  $Q \ln(8R/r_0)/2\pi R$  — the toroidal capacitance. Measured at  $N_{\text{side}} = 128$ , with  $\mathbf{D}$  replaced by  $\overline{\mathbf{D}}^v \mathbf{I}$ :

| family | $R$   | $Z$ solved | $2\pi / \ln(8R/ b_k )$ | error  |
|--------|-------|------------|------------------------|--------|
| $c_p$  | 5 nm  | 1.44469    | 1.44359                | 0.076% |
| $c_p$  | 25 nm | 1.05416    | 1.05389                | 0.026% |
| $a_i$  | 5 nm  | 1.30443    | 1.30384                | 0.045% |
| $a_i$  | 25 nm | 0.977648   | 0.977407               | 0.025% |

**This is what establishes that the boundary integral is the same physics as  $Z^{\text{iso}}$  and not a different model**, and it is the reason everything below can be attributed to the real diffusion

tensor and to shape freedom rather than to an implementation difference. It is also a non-trivial check on Eq. (163): the closed-form  $I_0, I_1$  of Eq. (165), the shape-function weighting and the core term of Eq. (164) all have to be right together for a 0.08% agreement with a result they were not built from.

$Z_k^{\text{iso}}$  **becomes an output, and is refitted.** With the kernel verified, the capture efficiency can be *read off* instead of assumed: solve the boundary integral on a circle of radius  $R$  with a unit driving supersaturation, take the (uniform) velocity, and invert Eq. (152) as Eq. (181) writes it,

$$Z_k^{\text{iso}} = \frac{v(\mathbf{b} \cdot \mathbf{n})_k}{\overline{D}^v \Delta c}, \quad \overline{D}^v = (\det \mathbf{D}_v)^{1/3}, \quad (186)$$

so that the  $Z$  returned absorbs both the geometry and the tensor's departure from the invariant  $\overline{D}^v$  that Eq. (181) carries. Fitted over  $R = 2\text{--}60\text{ nm}$  at 873 K, where  $D_a = 4.967 \times 10^{-14}$ ,  $D_c = 9.493 \times 10^{-14}\text{ m}^2/\text{s}$  and  $p_v = 1.11402$  by Eq. (18):

Table 12:  $Z_k^{\text{iso}}$  solved from Eq. (183) against the closed form Eq. (180), at 873 K over  $R = 2\text{--}60\text{ nm}$ . Two one-parameter refits are offered:  $\alpha$  shifts the logarithm's argument,  $f$  scales the whole efficiency. The last column is the discrepancy as Eq. (180) stands.

| family | $\alpha$ in $2\pi/\ln(\alpha R/r_0)$ | $f$ in $f \cdot \text{Eq. (180)}$ | err( $\alpha$ ) | err( $f$ )   | err as published |
|--------|--------------------------------------|-----------------------------------|-----------------|--------------|------------------|
| $c_f$  | 4.911                                | <b>1.09434</b>                    | 5.07%           | <b>0.66%</b> | 9.07%            |
| $c_p$  | 5.159                                | <b>1.09245</b>                    | 5.69%           | <b>0.72%</b> | 8.98%            |
| $a_v$  | 9.520                                | <b>0.97004</b>                    | 1.53%           | <b>0.22%</b> | 3.18%            |
| $a_i$  | 9.520                                | <b>0.97004</b>                    | 1.53%           | <b>0.22%</b> | 3.18%            |

Four things follow, and the second is the physical one.

1. **A constant factor fits; a shifted logarithm does not.** Scaling the efficiency leaves  $\leq 0.7\%$  over a factor of 30 in  $R$ , while shifting the logarithm's argument leaves 5%. The discrepancy is very nearly  $R$ -independent, which a multiplicative factor represents and a shift cannot. So the correction to Eq. (180) is a prefactor and not a core radius.
2. **The sign is opposite for the two habit planes, and that is the diffusion tensor speaking.** Basal loops capture 9% *more* than Eq. (180) says and prismatic loops 3% *less*. For vacancies  $D_c > D_a$ , so a basal loop — whose plane lies across the fast axis — is better fed than the orientation average, and a prismatic loop, whose plane contains it, is worse fed.  $\overline{D}^v = (\det \mathbf{D}_v)^{1/3}$  is an invariant and cannot know which. **This is the same  $p_m$  of Eq. (18) appearing in a third place**, after the diffusion tensor and the Woo efficiencies of Eq. (68), and its appearance here is not fitted.
3.  $a_v$  and  $a_i$  **return identical  $Z$** , to every digit, as they must: Eq. (186) is pure transport and the two families differ only in  $\zeta_s(v)$  and in their thermodynamics. That is a free consistency check on the whole construction.
4.  $f$  is temperature-dependent because  $D_c/D_a$  is. It is not a material constant, and quoting it without a temperature would be wrong.

**What that does to Fig. 3 and Table 11.** Substituting the solved  $Z$  into Eq. (181) and integrating as before gives Fig. 4(a) and the lifetimes

| $T$ [K] | $c_f$ faulted |             |       | $c_p$ perfect |             |       |
|---------|---------------|-------------|-------|---------------|-------------|-------|
|         | Eq. (180)     | Green's fn. | ratio | Eq. (180)     | Green's fn. | ratio |
| 773     | 22.9 d        | 20.6 d      | 0.902 | 0.284 yr      | 93.6 d      | 0.903 |
| 823     | 49.7 h        | 45.1 h      | 0.907 | 9.05 d        | 8.22 d      | 0.908 |
| 873     | 5.95 h        | 5.43 h      | 0.912 | 25.2 h        | 23.0 h      | 0.913 |
| 923     | 53.9 min      | 49.4 min    | 0.917 | 3.69 h        | 3.39 h      | 0.918 |
| 973     | 9.90 min      | 9.11 min    | 0.921 | 39.7 min      | 36.6 min    | 0.922 |
| 1023    | 2.15 min      | 1.99 min    | 0.924 | 8.44 min      | 7.81 min    | 0.925 |

Every conclusion of [Section 4.3](#) survives, because every one of them is a ratio or a shape and the correction is very nearly a constant. The absolute lifetimes fall by a uniform 8–10%; what the subsection actually asserts does not move:

| quantity                    | Eq. (180) | Green's function | change |
|-----------------------------|-----------|------------------|--------|
| $t_{c_p}/t_{c_f}$ at 773 K  | 4.5309    | 4.5377           | +0.15% |
| $t_{c_p}/t_{c_f}$ at 1023 K | 3.9240    | 3.9284           | +0.11% |
| $Q_{\text{eff}}, c_f$       | 2.6269 eV | 2.6200 eV        | −0.26% |
| $Q_{\text{eff}}, c_p$       | 2.6661 eV | 2.6593 eV        | −0.25% |
| $Q_{\text{eff}}, a_i$       | 2.7905 eV | 2.7922 eV        | +0.06% |
| $Q_{\text{eff}}, a_v$       | 2.5541 eV | 2.5558 eV        | +0.07% |

so [Table 11](#)'s last column stands at the precision it is quoted to,  $Q_{\text{eff}}$  still sits below  $E_v^f + E_v^m = 2.800$  eV by the same margin, and the  $a_i/a_v$  bracketing is untouched. The two *shapes* of  $R(t)$  and the asymptotes [Eq. \(182\)](#) are unchanged, because the fault term is what produces them. [Equation \(180\)](#) was never the load-bearing approximation in [Section 4.3](#); the fault term is.

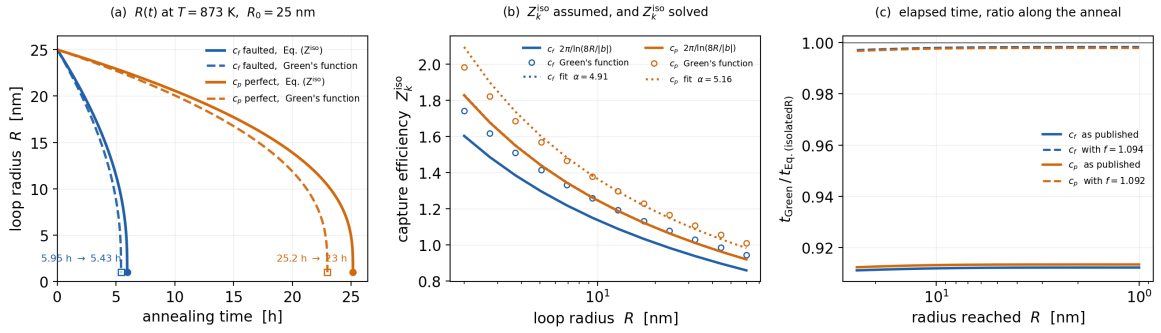


Figure 4: [Figure 3\(a\)](#) repeated with the boundary-integral solution superimposed. (a)  $R(t)$  at 873 K from  $R_0 = 25$  nm: solid is [Eq. \(181\)](#) with [Eq. \(180\)](#), dashed the same law with  $Z$  solved from [Eq. \(183\)](#). (b)  $Z_k^{\text{iso}}$  assumed against  $Z_k^{\text{iso}}$  solved, with the one-parameter refit of [Table 12](#). (c) The elapsed-time ratio along the anneal, before and after the constant refit  $f$  — it is flat, which is the statement of item 1 above, and  $f$  removes it to 0.2%. Produced by `disloccluster_code.studies.loop_annealing_greens figure-rt`.

**The loop shape during shrinkage.** The boundary integral carries no shape assumption, so it answers a question [Eq. \(181\)](#) cannot pose and [Eqs. \(157\)](#) and [\(158\)](#) can only answer inside an ellipse. Three families were marched nodally to extinction with  $N_{\text{side}} = 32$ , alongside the ROM on the same anneal and the same thermodynamics [Eq. \(185\)](#), and the aspect ratio  $\varrho_\ell = B_\ell/A_\ell$  of [Eq. \(155\)](#) compared ([Fig. 5](#), [Table 13](#)).

1. **A basal loop cannot become an ellipse, and both constructions say so.** Its habit plane is the basal plane, in which  $D_v$  restricted to the plane is isotropic, so the solved velocity is uniform around the line to  $4 \times 10^{-15}$  at every radius and both ROM capture diffusivities are identically equal. **This is the control the rest of the comparison rests on**, and it is also why [Section 4.3](#)'s circular assumption costs nothing for the family it is applied to.
2. **Equation (159) has the two axis curvatures swapped, and that is the whole of the  $a_v$  disagreement.** Parametrizing the habit-plane ellipse as  $(A_\ell \cos t)\hat{e}_1 + (B_\ell \sin t)\hat{e}_2$  gives  $\kappa(t) = A_\ell B_\ell / (A_\ell^2 \sin^2 t + B_\ell^2 \cos^2 t)^{3/2}$ , so

$$\kappa|_{A \text{ vertex}} = \frac{A_\ell}{B_\ell^2}, \quad \kappa|_{B \text{ vertex}} = \frac{B_\ell}{A_\ell^2}, \quad (187)$$

whereas [Eq. \(159\)](#) carries  $\kappa_A = B_\ell/A_\ell^2$  and  $\kappa_B = A_\ell/B_\ell^2$  — the pair interchanged, verified numerically against the polygon circumradius on a 1:3 ellipse (0.111111 and 3.00000 against

Table 13: Aspect ratio  $\varrho_\ell = B_\ell/A_\ell$  down each anneal at 873 K,  $A_\ell \parallel [0001]$  so that  $\varrho_\ell > 1$  is elongation *in* the basal plane. “Eq.” and “true” are the two curvature assignments discussed below; “geo” and “nrm” the two capture-diffusivity closures of Eq. (160). The basal row is the control: it must be 1, and is.

| family | $R/R_0$ | Green’s fn.   | ROM, $\kappa$ as Eq. (159) |        | ROM, true ellipse $\kappa$ |        |
|--------|---------|---------------|----------------------------|--------|----------------------------|--------|
|        |         |               | geo                        | nrm    | geo                        | nrm    |
| $a_v$  | 0.83    | 1.0102        | 1.1259                     | 1.2252 | 1.0393                     | 1.0793 |
| $a_v$  | 0.66    | 1.0081        | 1.6333                     | 1.8795 | 1.0521                     | 1.1068 |
| $a_v$  | 0.49    | 1.0065        | 2.9558                     | 3.4118 | 1.0495                     | 1.1015 |
| $a_v$  | 0.32    | <b>1.0002</b> | <b>4.7775</b>              | 4.7817 | <b>1.0415</b>              | 1.0848 |
| $a_i$  | 0.83    | 1.0472        | 1.0703                     | 1.1403 | 1.0654                     | 1.1310 |
| $a_i$  | 0.66    | 1.1072        | 1.1923                     | 1.3949 | 1.1664                     | 1.3452 |
| $a_i$  | 0.49    | 1.1929        | 1.4341                     | 1.9443 | 1.3551                     | 1.7710 |
| $a_i$  | 0.32    | <b>1.3264</b> | <b>2.0570</b>              | 3.5426 | <b>1.8132</b>              | 2.8847 |
| $c_p$  | all     | 1.0000        | 1.0000                     | 1.0000 | 1.0000                     | 1.0000 |

$A/B^2$  and  $B/A^2$ ). The swap is not cosmetic: it puts the large curvature on the *minor* axis and thereby *reverses* the sign of the capillary shape feedback. Its own prose is the geometric assignment — “as the loop elongates,  $\kappa$  rises at the major-axis ends, the back stress there raises  $c_{\text{eq}}^m$  and growth throttles” — and with that assignment restored the ROM behaves as described, settling at  $\varrho_\ell = 1.04$  against the boundary integral’s 1.00; as written it runs away to 4.78, a sliver. **This is a correction to Eq. (159), not a difference between the two constructions.**

- The two characters behave oppositely, and the exponential is why.** The driving supersaturation is  $c_\infty^{v,\text{eq}}(e^{\mu_k/k_B T} - 1)$  for a vacancy family and  $c_\infty^{i,\text{eq}}(1 - e^{-\mu_k/k_B T})$  for an interstitial one, so at  $\mu_k/k_B T \simeq 2.5$  the logarithmic sensitivity  $d \ln(\text{drive})/d\mu_k$  is  $1.09/k_B T$  against  $0.087/k_B T$ . The vacancy loop’s capillary shape control is  $12.5\times$  **stiffer relative to its own drive**, which is why it holds  $\varrho_\ell = 1.00$  while  $a_i$  reaches 1.33. It is the same exponent that makes  $a_v$  anneal  $27\times$  faster in Section 4.3, read on the shape instead of on the rate.
- Even corrected, the ROM over-reads the ellipticity** — by  $1.37\times$  on  $a_i$  at  $R/R_0 = 0.32$ , 1.81 against 1.33. That residual is not a parameter that could be tuned away, and the next paragraph measures where it comes from.

**Why the two shapes differ, and by how much.** Item 4 above is a statement that needs an account rather than an assertion, because the gap is large — 1.81 against 1.33 on  $a_i$  — and because the obvious explanation is the wrong one. The kinematics have to come first. Writing  $A_\ell = R/\sqrt{\varrho}$  and  $B_\ell = R\sqrt{\varrho}$  at fixed area, with both axes retreating,

$$\frac{d \ln \varrho_\ell}{dt} = \frac{|v_A|}{A_\ell} - \frac{|v_B|}{B_\ell} = \frac{|v_B|}{R} \varrho_\ell^{-1/2} \underbrace{\left[ \varrho_\ell \frac{v_A}{v_B} - 1 \right]}_{E(\varrho_\ell)}, \quad (188)$$

so it is  $E$ , not  $v_A/v_B$ , that drives elongation, and a steady shape needs  $E = 0$ . **In the ROM**  $v_A/v_B = D_m^{(A)}/D_m^{(B)} = p_m^3$  **is a constant**, fixed by crystallography and independent of the shape, so  $E_{\text{ROM}} = p_m^3 \varrho_\ell - 1$  grows without bound. In the boundary integral  $v_A/v_B$  is solved, and it does not stay constant.

- The dominant term is a shape-dependent capture efficiency, and the ROM has none.** Equation (183) imposes a *Dirichlet* condition on the line, so it is a capacitance problem, and the source density it induces concentrates where the curvature is high — the same point effect that puts charge on the sharp end of a conductor. As a loop elongates, the sharp end of its long axis therefore emits faster and retreats faster, and the elongation limits itself. The isotropic

Shape of an isolated loop as it anneals at  $T = 873$  K:  $D_c/D_a = 1.911$ ,  $p_v = 1.1140$ .  $A_I \parallel [0001]$ , so  $g_I > 1$  is elongation IN the basal plane

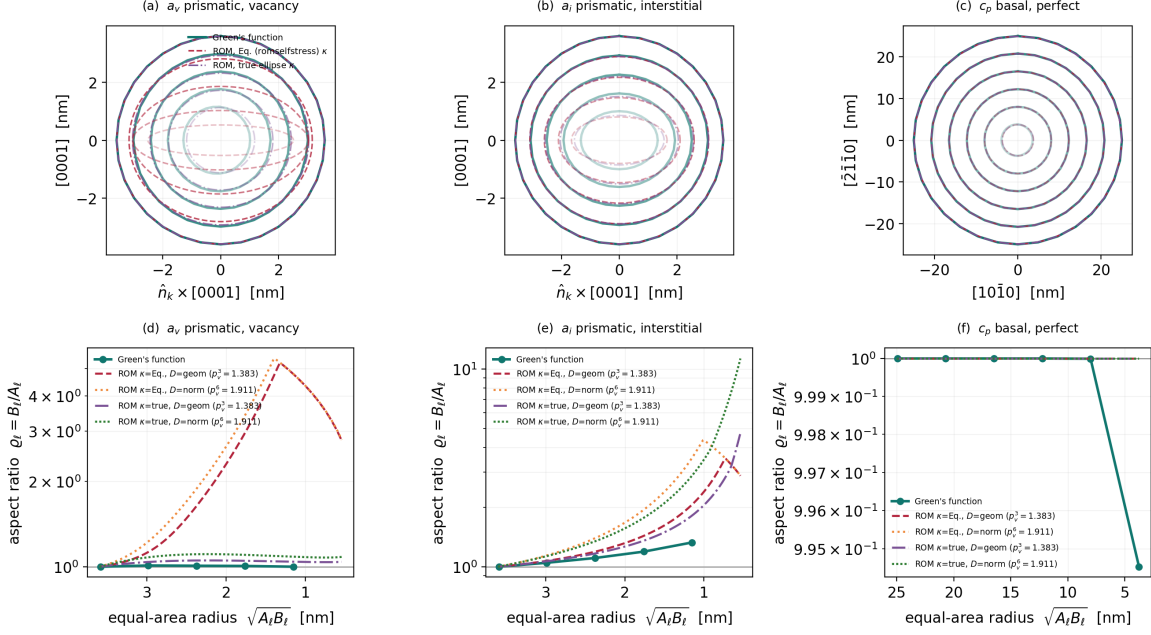


Figure 5: Loop shape during shrinkage at 873K, where  $D_c/D_a = 1.911$  and  $p_v = 1.1140$ . Top row, the outlines at six equal-area radii: solid is the nodal boundary-integral solve, dashed the ROM with Eq. (159)'s curvature assignment, dash-dotted the ROM with Eq. (187). Bottom row, the aspect ratio on a logarithmic axis, with both capture-diffusivity closures of Eq. (160). (a,d)  $a_v$ , where the uncorrected ROM collapses to a sliver and the solve stays circular; (b,e)  $a_i$ , the one family whose shape genuinely moves; (c,f)  $c_p$  basal, the control, circular in every column. Produced by `...loop_annealing_greens figure-shape`.

Table 14: Axis velocities on a *fixed* ellipse of equal-area radius 3.6 nm at 873 K under a *uniform* drive, so that capillarity plays no part and only transport and geometry act. The isotropic column replaces  $\mathbf{D}_v$  by  $\overline{\mathbf{D}}^v \mathbf{I}$ , where the ROM predicts  $v_A/v_B = 1$  at every aspect ratio: whatever the solve gives instead is the term the ROM does not have.  $E$  is the elongation drive of Eq. (188).

| $q_\ell$ | $v_A/v_B$ solved       |                   | elongation drive |                  |              |
|----------|------------------------|-------------------|------------------|------------------|--------------|
|          | isotropic $\mathbf{D}$ | true $\mathbf{D}$ | $E$ solved       | $E_{\text{ROM}}$ | ratio        |
| 1.00     | <b>1.0000</b>          | 1.2967            | 0.2967           | 0.3825           | 0.776        |
| 1.15     | 0.9743                 | 1.2552            | 0.4434           | 0.5899           | 0.752        |
| 1.35     | 0.9439                 | 1.2013            | 0.6217           | 0.8664           | 0.718        |
| 1.60     | 0.9086                 | 1.1343            | 0.8149           | 1.2121           | 0.672        |
| 2.00     | 0.8541                 | 1.0237            | 1.0475           | 1.7651           | 0.594        |
| 2.50     | 0.7860                 | 0.8685            | <b>1.1712</b>    | 2.4563           | 0.477        |
| 3.00     | 0.7147                 | 0.6689            | 1.0068           | 3.1476           | <b>0.320</b> |

column of Table 14 is that effect alone: it is 1.0000 at  $\varrho_\ell = 1$ , exactly, by symmetry — which is simultaneously a check on the solver and the reason this term does *not* contaminate the closure exponent measured on a circle — and it falls as  $\varrho_\ell^{-0.316}$  thereafter. **Nothing in Eqs. (157) and (158) can reproduce it**, because their two endpoints are coupled to each other only through the shared  $L_{\text{cap}}$  and never through a field.

2. **The consequence is that the two elongation drives diverge rather than differing by a constant.**  $E_{\text{ROM}}$  is linear in  $\varrho_\ell$  for ever; the solved  $E$  rises, peaks at 1.17 near  $\varrho_\ell = 2.5$  and turns over. Their ratio falls from 0.78 at  $\varrho_\ell = 1$  to 0.32 at  $\varrho_\ell = 3$ . **So the disagreement is not an offset but a compounding one, growing with how far the loop has already elongated** — which is exactly the pattern of Table 13, where  $a_v$  agrees to 3% at  $R/R_0 = 0.83$  and  $a_i$  disagrees by 37% at  $R/R_0 = 0.32$ .
3. **Capillarity is *not* the explanation, and this is worth establishing rather than assuming.** The natural suspicion is that the ROM resolves the capillary back stress at two vertices where the solve resolves it at every node. Re-running the nodal anneal with the driving supersaturation frozen at the equal-area curvature — which keeps its magnitude and removes its shape feedback entirely — gives

| $R/R_0$ | $\varrho_\ell$ nodal $\kappa$ | $\varrho_\ell$ frozen $\kappa$ | ratio        |
|---------|-------------------------------|--------------------------------|--------------|
| 0.82    | 1.0491                        | 1.0534                         | 1.004        |
| 0.65    | 1.1113                        | 1.1332                         | 1.020        |
| 0.47    | 1.2022                        | 1.2575                         | 1.046        |
| 0.30    | 1.3468                        | 1.4613                         | <b>1.085</b> |

so the capillary shape feedback is worth 8.5% in  $\varrho_\ell$  at the end of the anneal — **when the whole of the ROM’s excess is 37%**. Removing it *entirely* could account for at most a quarter of the gap, and the ROM does not remove it entirely; it carries it at two points. Capillary resolution is therefore a real but third-order contribution, and the geometric term of item 1 is the explanation.

4. **A residual 6% is present even at  $\varrho_\ell = 1$** , where the solved  $v_A/v_B = 1.2967$  against the ROM’s  $p_v^3 = 1.3825$ . That is the closure exponent of the next paragraph,  $q = 2.41$  rather than 3, and it is a statement about *transport* on a circle rather than about shape. It sets the initial rate at which the two constructions part company; items 1 and 2 are what make the parting grow.

**Read together, the four items say something about when the ROM may be used.** Its two ODEs are local — each endpoint sees its own diffusivity and its own curvature, and nothing else — whereas Eq. (166) makes every point of the line see every other. That difference is invisible on a circle, where the ROM’s error is the 6% of item 4, and it grows as the square of nothing in particular but monotonically with departure from a circle. **The ROM is therefore reliable for the  $\langle a \rangle$  population it was built for**, Table 8, whose loops sit at  $\Theta_k = 0.081$  and stay close to circular through a march, and it should not be trusted for a shape prediction once  $\varrho_\ell$  exceeds about 1.5 — which, by Eq. (161), is also where its own approach-to-attractor argument stops being the limiting consideration.

**Which capture-diffusivity closure, settled.** Section 3.15 leaves Eq. (160) open — “*this is not settled* ... only a nodal solve of the same case can collapse it” — because the diffusivity feeding the end of the [0001] axis is not fixed by geometry: the geometric mean gives an axis-velocity ratio  $D_m^{(A)}/D_m^{(B)} = p_m^3$  and the normal projection  $p_m^6$ . **The isolated loop is a nodal solve of exactly that case.** Put a *circular* prismatic loop in the boundary-integral solver with a *uniform* drive, so that the only thing which can make the two axis endpoints move at different speeds is the diffusion tensor, and read the ratio off as an effective exponent  $q = \ln(v_A/v_B)/\ln p_v$ :

| $T$ [K] | $p_v$   | $q$ at $R = 2$ nm | 3.6 nm | 10 nm | 25 nm |
|---------|---------|-------------------|--------|-------|-------|
| 773     | 1.12969 | 2.095             | 2.398  | 2.625 | 2.720 |
| 873     | 1.11402 | 2.116             | 2.408  | 2.630 | 2.724 |
| 1023    | 1.09652 | 2.137             | 2.419  | 2.635 | 2.727 |



against  $q = 3$  (geometric mean) and  $q = 6$  (normal projection).  $q$  is independent of temperature — hence of  $p_v$  — to three decimal places, and rises with  $R$  toward 3. The geometric mean is the large-loop limit and the normal projection is not a candidate at all; the shortfall at small  $R$  is the core regularization blunting the anisotropy, and it is a *size* effect rather than a material one, so it is a correction to the closure and not a competitor to it. **This removes the factor of  $p_m^3$  that Section 3.15 requires to be quoted with any ROM ellipticity.**

**One further correction the comparison forces.** Equation (157) writes the arrival rate as  $Z_{sk,m} D_m^{(A)} / L_{\text{cap}}$ , whereas Eq. (181) and the boundary integral both write it as  $Z_k^{\text{iso}} \bar{D}^v / \Omega$  — a concentration is an atom fraction, so  $c / \Omega$  is a number density and no length appears. Matching the two at the reference radius gives

$$L_{\text{cap}} = \frac{\Omega Z_{sk,v}}{b^2 Z_k^{\text{iso}}(R)} \simeq \frac{\Omega}{b^2} = 0.690 |b| = 0.223 \text{ nm}, \quad (189)$$

that is 0.16–0.23 nm across the four families, against the screening length Eq. (154) of  $\simeq 59$  nm that Section 3.15 defaults it to. **At that default the ROM’s isolated-loop climb rate is 260–370× too slow.** Section 3.15 is right that  $L_{\text{cap}}$  enters both axis equations identically and therefore sets the time scale alone and cancels from the aspect ratio — so Table 13 is untouched by this, and Eq. (189) is a calibration rather than a repair — but it means no ROM *lifetime* may be quoted at the screening-length default.

**Three limits of this comparison.** The loop is isolated, so nothing here tests the pair assembly Eq. (167) or the truncation Eq. (170), which are what make the boundary integral expensive on a population and are measured in Section 3.17 instead. Only the vacancy species is carried, which is exact for a pure anneal and not for a march. And the core convention is  $r_0 = |b_k|$  throughout, to match Eq. (180)’s own  $\ln(8R/|b|)$ , where the tutorials set `coreSize` =  $2b$ ; changing it moves  $Z$  by roughly  $\ln 2 / \ln(8R/b)$ , which is 11% at 25 nm and is a convention difference rather than a physical one. Below  $\simeq 3$  nm the polygon segments approach the core radius and the last snapshot of each shape march is discretization-limited; the  $c_p$  control measures the size of that as  $\varrho_\ell = 0.9945$  instead of 1 at the smallest radius drawn.

## 4.5 Annealing of loop populations

Section 4.3 closed by naming three limits of the isolated-loop calculation, and the first two are the same limit twice: the loop has no neighbors, so nothing re-absorbs what it emits, and the ambient field is *imposed* at  $c_\infty^{v,\text{eq}}$  rather than solved for. A real post-irradiation anneal has neither. When the source is switched off, thermal vacancies are emitted by some components of the microstructure and absorbed by others, and the concentration every loop actually sees is whatever that exchange settles at. This subsection solves for it, and the result changes not only the rates of Fig. 3 but the *separation* between the two families and, once a size distribution is admitted, the qualitative character of the anneal itself.

**The balance, and why it closes on vacancies alone.** A long time after the source is off, thermal emission of interstitials is negligible —  $E_i^f = 3.0$  against  $E_v^f = 1.6$  eV, so  $c_\infty^{i,\text{eq}} / c_\infty^{v,\text{eq}} = e^{-1.4/k_B T} = 1.0 \times 10^{-8}$  at 873 K — and with it mutual recombination. What remains is a pure vacancy balance [26]:

$$\left[ \text{total rate of thermal vacancy emission} \right] = \left[ \text{total rate of vacancy removal} \right]. \quad (190)$$

Component  $k$  of the microstructure exchanges vacancies with the matrix at the rate its own sink strength sets: it emits  $S_k \bar{D}^v c_k^{v,\text{eq}} / \Omega$  and absorbs  $S_k \bar{D}^v \bar{c}_v / \Omega$  per unit volume, so Eq. (190) reads  $\sum_k S_k c_k^{v,\text{eq}} = \bar{c}_v \sum_k S_k$  and

$$\boxed{\bar{c}_v = \frac{\sum_k S_k c_k^{v,\text{eq}}}{\sum_k S_k}}, \quad S_k = Z_{k,v} \rho_{sL}^k, \quad S_N = Z_{N,v} \rho_N. \quad (191)$$

Two things about Eq. (191) are worth stating before anything is computed from it. First,  $\bar{D}^v$  and  $\Omega$  cancel:  $\bar{c}_v$  is a **property of the microstructure and the temperature alone**, and carries no mobility, which is why it can be evaluated on a state without integrating anything. Second, Eq. (191) is not an imported construction — **its denominator has exactly the form of  $k_v^2$  in Eq. (38)**, the same sum of  $Z\rho_{sL}^k$  over  $\mathcal{F}_L$  plus the same  $Z_{N,v}\rho_N$ , and is that quantity identically when  $Z_{k,v}$  is taken to be the mean-field  $Z_{sk,v}$  of Eq. (68). The remote concentration is the sink-strength-weighted average of the levels the components would each impose on their own, and the model already assembles those weights for the mobile block. Nothing new is fitted here. Which  $Z$  is used, and the one thing that must not be done with the choice, is taken up at the end of this subsection.

**Each family’s own level, and the ladder they form.** Every component enters Eq. (191) through the concentration it is individually in equilibrium with, which is Eq. (111) unchanged:

$$c_k^{v,\text{eq}}(R) = c_\infty^{v,\text{eq}} \exp\left\{\frac{\varsigma_s(v)\mu_k(R) + \Sigma_k\Omega}{k_B T}\right\}, \quad \mu_k = \frac{\partial E_k}{\partial m} = \frac{\gamma_k\Omega}{(\mathbf{b}\cdot\mathbf{n})_k} + \frac{\langle K \rangle_k b_k^2 \lambda_k^2}{4R} \left[1 + \ln \frac{\alpha R}{b_k}\right]. \quad (192)$$

So the nine families of Eq. (64) arrange themselves on a ladder, and  $\varsigma_s(v)$  decides which rung each takes: the basal vacancy chain  $(v, c_f), (v, c_p)$  and the prismatic vacancy variants  $(v, a_{1,2,3})$  sit *above*  $c_\infty^{v,\text{eq}}$ , the prismatic interstitial variants  $(i, a_{1,2,3})$  sit *below* it, and the network dislocations sit *on* it, having neither fault nor curvature. This is (author?)’s Fig. (9.16) with voids removed —  $\alpha$ -Zr has none in this model — and the loop families of Eq. (64) in their place.

The growth law needs no modification at all: Eq. (181) with  $\bar{c}_v$  substituted for  $c_\infty^{v,\text{eq}}$ ,

$$\frac{dR_k}{dt} = \varsigma_s(v) \frac{Z_{k,v}\bar{D}^v}{(\mathbf{b}\cdot\mathbf{n})_k} [\bar{c}_v - c_k^{v,\text{eq}}(R_k)], \quad (193)$$

is the whole of the change. Because  $\bar{c}_v$  is a weighted average it always lies between the extreme rungs, so **in a mixed microstructure something is always emitting while something else is absorbing**. Which of the two a given family does follows from where its rung sits; what that does to its *size* then depends on its character, and the two questions must be kept apart:

| rung relative to $\bar{c}_v$ | role         | vacancy family | interstitial family |
|------------------------------|--------------|----------------|---------------------|
| above                        | net emitter  | shrinks        | grows               |
| below                        | net absorber | grows          | shrinks             |

**The three prismatic variants, and what a stress does to them.** At zero deviatoric stress the three variants are equivalent by symmetry (Section 2.4), share one rung and one fate. Under load they do not: the resolved stress  $\Sigma_k = \hat{\mathbf{a}}_k \cdot \boldsymbol{\sigma} \cdot \hat{\mathbf{a}}_k$  enters Eq. (192) directly, so each variant’s rung shifts by  $\exp(\Sigma_k\Omega/k_B T)$  while  $\bar{c}_v$  remains a single number. Under  $\sigma_{11} = 200$  MPa along  $[10\bar{1}0]$ ,  $a_1$  takes the full stress and  $a_2, a_3$  take  $\sigma \cos^2 60^\circ = \sigma/4$ , and their levels separate by a factor of 1.34 at 873 K. **On the reference microstructure that moves rates and not signs:** all three still shrink, at 2.4837 against  $2.5012 \times 10^{-13}$  m/s, a 0.7% split, because  $\bar{c}_v = 1.47 c_\infty^{v,\text{eq}}$  sits nearly two decades above every prismatic rung and lifting one of them to it would take 2065 MPa. Restricted to the prismatic family alone — which is what Eq. (191) gives for a microstructure with no  $\langle c \rangle$  population and no network —  $\bar{c}_v$  falls *inside* the variants’ own spread by construction, at  $0.0333 c_\infty^{v,\text{eq}}$  against rungs at 0.0401 and 0.0300, and the signs do separate:  $a_1$  **grows while  $a_2$  and  $a_3$  dissolve, under a purely thermal anneal with no irradiation at all.** That is variant selection by stress with no production term anywhere in it, and it is a prediction the resolved layout of Eq. (64) makes and a lumped  $\langle a \rangle$  population cannot.

**One size per family: the reference microstructure.** The microstructure annealed below is the one the 300 nm hardening cube of Section 5 carries, and it is specified by the continuum state it was drawn from rather than by the draw: `cube_population` samples the interior per-family state of the matched self-consistent march, so the two agree to the integer draw. At 0.1 dpa the cube holds 21  $\langle c \rangle$  and  $3 \times 98 \langle a \rangle$  loops in  $2.702 \times 10^{-20}$  m<sup>3</sup>, that is  $N_c = 7.77 \times 10^{20}$  and  $N_a = 1.088 \times 10^{22}$  m<sup>-3</sup>, against  $7.68 \times 10^{20}$  and  $1.092 \times 10^{22}$  in the state itself — 1.2% and 0.4%. Nothing is refused by the cube at this dose, so the anneal is run on the state, which carries the mean radii the draw needs anyway.

Figure 6 draws Eq. (191) for that state at 873 K. At 0.1 dpa the  $\langle c \rangle$  family sits at  $4.99 c_\infty^{v,\text{eq}}$  and carries 22% of the sink strength, the three  $\langle a \rangle$  variants together sit at 0.027 and carry 43%, and the network carries the remaining 34% at  $c_\infty^{v,\text{eq}}$  exactly. The result is  $\bar{c}_v = 1.46 c_\infty^{v,\text{eq}}$ : **the matrix is supersaturated in vacancies by half again for the whole anneal**, held there by the  $\langle c \rangle$  loops' own emission. Both families shrink — the  $\langle c \rangle$  loops because their rung is above the line, the  $\langle a \rangle$  loops because theirs is below it — and Table 15 is what that costs each of them.

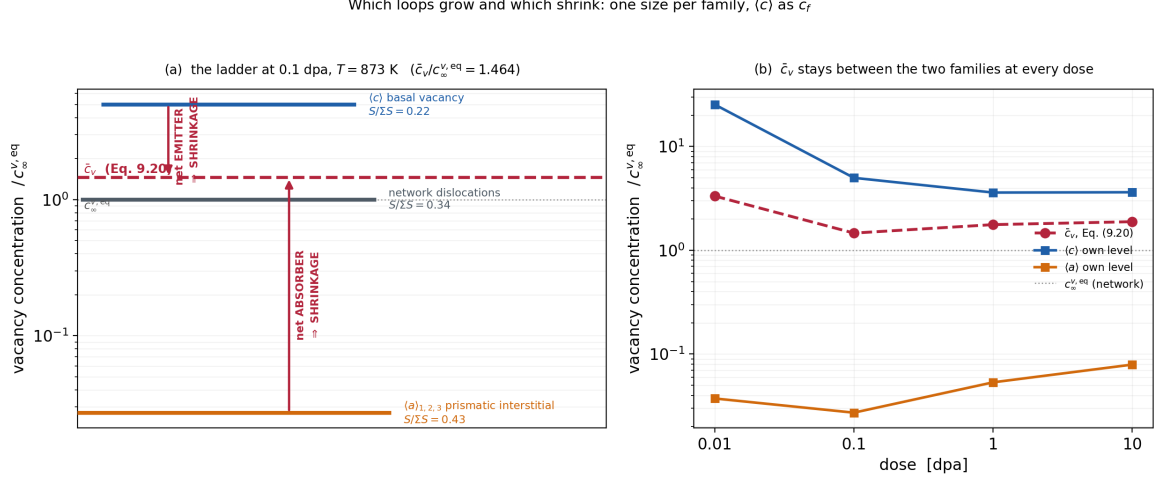


Figure 6: The concentration ladder of Eq. (191), one size per family, on the interior state of the reference march at 873 K. (a) Each rung is a component's own  $c_k^{v,\text{eq}}$  and its width is that component's weight  $S_k / \sum S_k$  in Eq. (191); the dashed line is the solved  $\bar{c}_v$ . The  $\langle c \rangle$  loops are net emitters and the  $\langle a \rangle$  loops net absorbers, and *both* shrink, because  $\varsigma_s(v)$  has opposite signs for them. (b) The same across dose:  $\bar{c}_v$  tracks between the two families at every one, never leaving the interval the rungs bracket. Produced by `dislocluster_code.studies.loop_annealing_population figure-ladder`.

Table 15: Section 4.3's anneal repeated on the reference microstructure at 873 K, with  $\bar{c}_v$  solved from Eq. (191) at every step instead of pinned at  $c_\infty^{v,\text{eq}}$ .  $\langle c \rangle$  is mapped onto the faulted state  $c_f$ , whose  $(\mathbf{b} \cdot \mathbf{n}) = c/2$  is the geometry the march carries. The last two columns are the same  $\bar{c}_v$  computed with the two capture efficiencies discussed at the end of this subsection; the calculation itself uses  $Z^{\text{iso}}$  throughout, and their spread is the honest uncertainty on the weighting.

| dpa  | state      |            | $\langle c \rangle$ lifetime |            | $\langle a \rangle$ lifetime |            | $\bar{c}_v / c_\infty^{v,\text{eq}}$ |                  |
|------|------------|------------|------------------------------|------------|------------------------------|------------|--------------------------------------|------------------|
|      | $R_c$ [nm] | $R_a$ [nm] | isolated                     | population | isolated                     | population | $Z_{sk,v}$                           | $Z^{\text{iso}}$ |
| 0.01 | 5.43       | 2.51       | 6.30 min                     | 6.89 min   | 2.29 h                       | 2.40 h     | 3.291                                | 3.330            |
| 0.1  | 25.84      | 2.19       | 6.37 h                       | 8.96 h     | 1.76 h                       | 65.8 min   | 1.857                                | 1.464            |
| 1    | 57.15      | 2.96       | 29.1 h                       | 53.6 h     | 3.10 h                       | 97.2 min   | 2.116                                | 1.766            |
| 10   | 55.82      | 3.64       | 27.9 h                       | 51.0 h     | 4.36 h                       | 2.14 h     | 2.188                                | 1.885            |

Three results, and the third is the one that matters for interpreting an experiment.

1. **The population coupling pushes the two families in opposite directions.** At 1 dpa the  $\langle c \rangle$  lifetime rises by  $1.84\times$  and the  $\langle a \rangle$  lifetime falls to  $0.52\times$ . The mechanism is the one (author?) states in a sentence: the vacancy loops emit vacancies, which holds the  $\langle c \rangle$  population up against its own dissolution and feeds the  $\langle a \rangle$  loops the vacancies that destroy them. Neither effect exists in Eq. (181), and neither is small.
2. **The separation between the two families is amplified, not merely shifted.** Isolated, the ratio  $t_{\langle c \rangle} / t_{\langle a \rangle}$  is 3.6 at 0.1 dpa; in the population it is 8.2. Whatever is inferred from a measured

pair of annealing times — and Section 4.3's  $Q_{\text{eff}}$  argument is exactly such an inference — the isolated calculation gets the separation wrong by more than a factor of two.

3. **The ambient is not a constant, because the microstructure that sets it is the one being dissolved.** Figure 7(b) tracks  $\bar{c}_v$  through the anneal: it steps up when the  $\langle a \rangle$  population vanishes at 66 min, removing the only rung below the line, and then runs away by more than an order of magnitude as the surviving  $\langle c \rangle$  loops shrink and their own capillary level climbs. The reverse ordering at 0.01 dpa is the same effect read backwards: there the  $\langle c \rangle$  loops are only 5.4 nm and are gone in seven minutes, after which the  $\langle a \rangle$  loops anneal in what is very nearly isolation, and their population lifetime comes back to  $1.05\times$  the isolated one. **An annealing model that fits one effective supersaturation to a whole anneal is fitting an average over that.**

The 300 nm reference microstructure annealed: Sec. 4.3's calculation with  $\bar{c}_v$  solved instead of pinned ((c) as  $c_l$ )

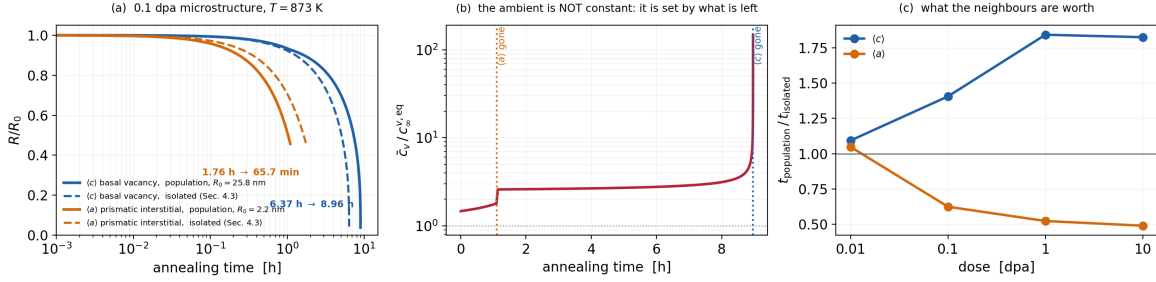


Figure 7: Section 4.3's calculation repeated on the reference microstructure. (a)  $R/R_0$  against annealing time on a logarithmic axis — the two families differ by a factor of 12 in size and 8 in lifetime, so no linear axis shows both — with the isolated trajectory of Eq. (181) dashed beside each. (b)  $\bar{c}_v$  through the anneal, with the two extinction times marked. (c) The lifetime ratio across dose: the neighbors are worth a factor of two, in opposite directions for the two families. Produced by `...loop_annealing_population_figure-reference`.

**Size distributions: the level becomes a curve, and a critical radius appears.** Everything above treats each family as monodisperse, which is what Section 2.4 carries before Section 2.9. Admit a distribution and one thing changes, but it changes the character of the answer rather than its value:  $c_k^{v,\text{eq}}$  in Eq. (192) is a function of  $R$ , whereas  $\bar{c}_v$  is a single number, so the two can *cross*. The crossing is a critical radius

$$c_k^{v,\text{eq}}(R_k^*) = \bar{c}_v \quad \Rightarrow \quad \left. \frac{dR_k}{dt} \right|_{R_k^*} = 0, \quad (194)$$

and it splits a family in two: for a vacancy family, loops below  $R_k^*$  dissolve and loops above it grow, which is Ostwald ripening *within one family* rather than the wholesale dissolution the monodisperse treatment gives.

Whether Eq. (194) has a root at all is decided by the fault term of Eq. (192), and this is where the two basal states of Table 2 part company. As  $R \rightarrow \infty$  the capillary part vanishes and  $c_k^{v,\text{eq}}$  tends to a floor,

$$c_k^{v,\text{eq}}(\infty) = c_\infty^{v,\text{eq}} \exp \left\{ \frac{\gamma_k \Omega}{(\mathbf{b} \cdot \mathbf{n})_k k_B T} \right\} = \begin{cases} 2.53 c_\infty^{v,\text{eq}}, & c_f \text{ (faulted)}, \\ 1.00 c_\infty^{v,\text{eq}}, & c_p \text{ (perfect)}, \end{cases} \quad (195)$$

at 873 K. If  $\bar{c}_v$  lies below a family's floor, no loop of that family is ever large enough to grow and the family dissolves at every size at once. On the reference microstructure that is exactly what happens to  $c_f$ :  $\bar{c}_v = 1.46$  against a floor of 2.53, so Eq. (194) has no root and the whole distribution is on the shrinking side (Fig. 8a,b). The unfaulted state  $c_p$  does have a root, but an order of magnitude beyond the distribution, so it gives the same answer by a different route. The  $\langle a \rangle$  families never have one either: their levels are capped at  $c_\infty^{v,\text{eq}}$  from below, and  $\bar{c}_v$  is above it.

The contrast is worth drawing explicitly, because it is easy to read Eq. (195) as saying that a faulted family cannot ripen. It does not. **A family on its own always ripens**, faulted or not:  $\bar{c}_v$  is then its own sink-weighted mean level, which necessarily lies inside its own spread, so  $R^*$  falls inside the distribution by construction. Taking the same 0.1 dpa  $\langle c \rangle$  population and removing the network and the  $\langle a \rangle$  loops puts  $R^*$  almost exactly at the mean radius and gives a textbook coarsening run:  $N$  falls while  $\bar{R}$  rises, with the total stored vacancy content conserved to the current through the floor (Fig. 8c). So what the fault energy changes is the *competition*, not the family: it lifts the whole rung, so that a network and an interstitial population holding  $\bar{c}_v$  near  $c_\infty^{v,eq}$  can put the entire faulted family above the line at once. **Ripening or wholesale dissolution is a property of the mixture, and Eq. (191) is what decides which.**

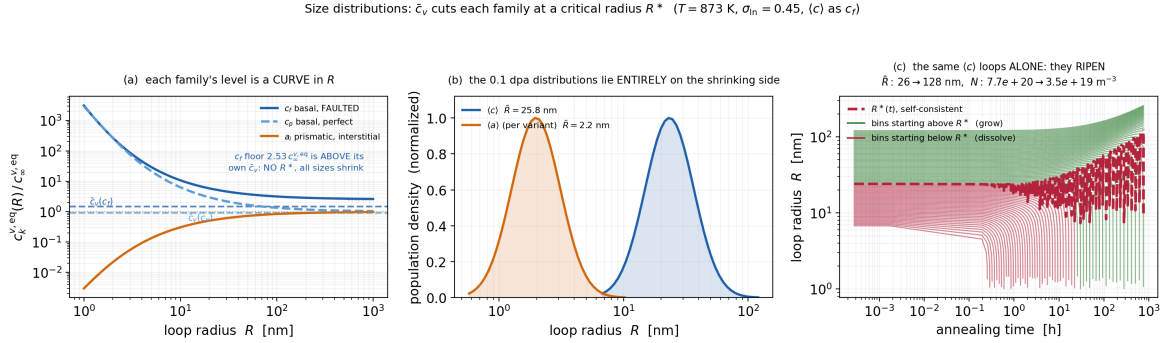


Figure 8: What a size distribution changes. (a) Each family's level is a curve in  $R$ , drawn against the  $\bar{c}_v$  its own mixture produces; the dotted horizontals are the floors Eq. (195). The faulted basal state's floor lies above its own  $\bar{c}_v$ , so Eq. (194) has no root. (b) The reference distributions at 0.1 dpa lie entirely on the shrinking side of  $\bar{c}_v$ . (c) The same  $\langle c \rangle$  population with the network and the  $\langle a \rangle$  loops removed, so that  $\bar{c}_v$  is its own: every size bin's trajectory, with the self-consistent  $R^*(t)$  drawn through them. Bins that start above  $R^*$  grow at the expense of those below, which is ripening and not annealing. The sawtooth on  $R^*(t)$  is the bin discretization —  $\bar{c}_v$  is a sum over delta functions, so every bin that dissolves steps it — and not a feature of the solution. The distribution width  $\sigma_{ln} = 0.45$  is the one assumed quantity in this figure. Produced by `...loop_annealing_population figure-ripening`.

**What is assumed, and what would move.** Four things, in decreasing order of how much they matter.

1. **One capture efficiency must serve both sides of Eq. (191), and which one is a choice worth 27%.** The formulation carries two that are not the same number: the mean-field  $Z_{sk,v}$  of Eq. (68), from which Eq. (38) assembles  $k_v^2$ , and the isolated-loop  $Z_k^{iso}$  of Eq. (180), which Eq. (193)'s own kinetics use. Either may be taken — they differ in their *ratio* between habits, which is the only thing  $\bar{c}_v$  sees, and on this microstructure that ratio differs by  $1.7\times$  and  $\bar{c}_v$  by up to 27% (last two columns of Table 15). **What may not be done is to take one in Eq. (191) and the other in Eq. (193).** The sum  $\sum_k S_k(\bar{c}_v - c_k^{v,eq})$  is the net vacancy exchange with the matrix *only* if the  $S_k$  that weight it are the  $S_k$  that drive the growth; mix them and vacancies are destroyed. That was caught by the conservation test above, at a factor of 85, in a ripening run that was otherwise entirely plausible — **a near-cancellation is unforgiving of a few per cent in its weights**, which is the same lesson Section 2.6 draws about Eq. (71).  $Z_k^{iso}$  is used throughout, because it makes Eq. (193) reduce to Eq. (181) exactly in the single-component limit, which is the comparison this subsection is about.
2. **The mapping of the march's  $\langle c \rangle$  family onto  $c_f$  rather than  $c_p$ .** The march carries  $(b \cdot n) = c/2$ , which is  $c_f$ 's geometry, but its loop-model kinetics carry no fault term. The choice sets the rung and therefore  $\bar{c}_v$ , and by Eq. (195) it sets whether a coarsening branch exists at all. Both are reported and neither is asserted to be the physical one; settling it is a question about the basal chain of Section 2.8, not about the anneal.

3.  $\bar{c}_v$  is **spatially uniform**. Equation (191) is a zero-dimensional balance, so it is a statement about a region large compared with the screening length Eq. (154) and small compared with the gradients of  $n_{sL}^k(\mathbf{x})$ . It is evaluated here on the interior mean, which is the same restriction Section 3.6 and the hardening study impose, and for the same reason.
4. **The distribution width is assumed**. The model carries one mean size per family before Section 2.9, so  $\sigma_{\text{ln}} = 0.45$  in Fig. 8 is an input. It re-partitions a fixed population —  $N$  and  $\bar{R}$  are held exactly — so it moves where  $R^*$  falls within the distribution and not whether one exists. Carrying  $q_{sL}^k$  through the anneal, as Section 2.9 does through the irradiation, would remove this assumption and is the natural next step.

## 4.6 Annealing a transferred discrete population

**The setup.** Run an irradiation march to a chosen dose — 1 dpa is used below — stop the source, convert the continuum immobile field to a discrete loop population by Section 3, and anneal that population *on the same volume*: the same mesh, the same CD nodes, the same nodal volumes Eq. (139), the same material file. Nothing is re-staged: the staged case is keyed on the geometry, the mesh and the material and not on the dose grid, so an annealing run reuses the irradiation run’s case directory outright.

Four things carry over unchanged and two change.

- **Unchanged:** the operator split Section 6.2 and its ordering; the mobile problem, which is still the steady anisotropic reaction–diffusion system Eq. (27); the immobile ODEs at every node; and the conservation ledger Eq. (135).
- **The source is switched off**,  $G = 0$ , so the production integrals of Eq. (135) stop accumulating and the mobile problem is driven *only* by the emission source  $\mathbf{S}_{vL}$  of Eq. (113). The sign of the whole problem inverts: under irradiation the loops are sinks drawing on a supersaturated pool, and under an anneal they are sources feeding a pool the boundary drains.
- **The dose axis becomes a time axis**. The march advances on  $\{\gamma_n\}$  with  $\Delta t = \Delta\gamma/G$ , which is undefined at  $G = 0$ ; the schedule is instead specified directly as a log-spaced grid in  $t$ , refined at the start where the fastest family is losing size quickest.

**The split is more comfortable here, not less.** Section 6.1 rests the quasi-steady approximation on the mobile field equilibrating fast against the immobile evolution. Removing the source slows the immobile evolution — there is no production to drive it, only the residual supersaturation of Table 9 — so the separation widens and the fast-solve cadence `fem_every` can be relaxed. One caution runs the other way and is specific to annealing: the emission source Eq. (113) depends on  $c_{sk}^{v,\text{eq}}(\bar{m}^k)$ , which is *exponential* in the loop size, so a family close to the floor changes its own source term by a factor of  $e$  over a size change of  $k_B T / (\partial^2 E_k / \partial m^2)$ . The splitting error of Section 6.6 is still first order in the fast-solve spacing, but the coefficient grows through the anneal rather than shrinking, and the cadence should be tightened at the end where it can be loosened at the start.

**The boundary condition is not a free choice.** On a domain with Dirichlet faces the emitted vacancies leave, the pool stays near  $c_\infty^{v,\text{eq}}$ , and the population anneals to extinction — which is the case Section 4.3 idealizes. On a *fully periodic* cell nothing leaves; the total vacancy content is conserved exactly, so the loops cannot all disappear, and the calculation converges instead on an Ostwald ripening steady state in which large loops grow at the expense of small ones at constant total content. Both are physical and they are different experiments: a thin TEM foil or a fine-grained specimen is the first, a well-annealed single-crystal interior is closer to the second. `BOUNDARY['periodic_face_ids']` **therefore selects which annealing experiment is being simulated**, and an empty list — the default — is the extinction case.

**Option 1: annealing through the growth equation.** Every loop is advanced by its own copy of Eq. (181), or of Eqs. (157) and (158) where the outline is allowed to be elliptical, with two substitutions that acknowledge the loop is no longer alone. The ambient field  $c^v$  becomes the continuum field  $c_C^v(\mathbf{x}_\ell)$  at the loop’s center plus the lagged monopole sum over the neighbors inside the cutoff Eq. (170),



exactly as in Section 3.15; and  $Z_k^{\text{iso}}$  returns to  $Z_{sk,v}$  of Eq. (68) wherever the population is dense enough for a bias to mean anything. The cost is  $O(N_{\text{loop}})$  per step plus the neighbor sum, the loops decouple, and forward Euler is adequate for the same reason Section 3.18 gives.

**Its one free parameter loses its default.**  $L_{\text{cap}}$  in Eqs. (157) and (158) is set to the screening length Eq. (154), on the argument that a loop draws on the medium out to the distance at which some other sink intercepts the defect first. That argument survives an anneal; the *number* does not survive the loop shrinking. Equation (154) is a property of the surroundings and is essentially fixed here — see below — while  $R_{sL}^k$  falls to the floor, so  $L_s/R$  runs from about 1 at the transfer state to 73 at  $R_{\text{min}}$  on the reference case. Once  $L_s \gg R$  the gradient feeding the line develops over the loop's own size and not over  $L_s$ : the outer cutoff of the capture problem is set by the sink itself, which is what Eq. (180) states in its logarithm, and  $L_{\text{cap}}$  must cross over from  $L_s$  to  $O(R_{sL}^k)$ . **An annealing run that keeps  $L_{\text{cap}} = L_s$  throughout under-reads the late-time shrinkage rate by that same ratio**, and the error is largest exactly where the loops are disappearing — which is the part of the trajectory an annealing experiment measures.

**Option 2: annealing through the Green's-function solve.** The boundary-integral construction of Section 3.16 is used unchanged. Neither the kernel Eq. (163) nor the assembly Eqs. (167) and (168) contains anything that assumes a growing loop: the driving supersaturation Eq. (169) simply changes sign, the climb velocities  $v_v^m$  come out negative, and  $c_D^m$  of Eq. (162) is a positive hump around each loop rather than a depletion halo. What this buys over Option 1 is the one mechanism a monopole cannot carry: **a shrinking loop's vacancies are re-absorbed preferentially by its nearest neighbors**, which slows the anneal locally and by an amount that depends on where the neighbors are. A mean field carries the average of that through  $k_m^2$  and cannot carry its variance, and it is precisely at the small separations where the effect is largest that the average is least representative.

**The cutoff survives an anneal, and the reason is the dislocation network.** Equation (170) sets  $R_{\text{cut}} = n_L L_s$  with  $n_L = 4$ , and the obvious worry is that  $L_s$  runs away as the loops vanish, taking the pair count with it. It does not, because the loops are not the only sink:  $k_m^2$  in Eq. (38) carries the network density  $\rho_N$ , which an anneal of the loop population does not remove. On the reference case at 10 dpa  $L_s = 59.0 \text{ nm}$ , i.e.  $k_m^2 = 2.87 \times 10^{14} \text{ m}^{-2}$ , of which  $\rho_N = 1.88 \times 10^{14} \text{ m}^{-2}$  is 65%; removing the loops entirely therefore takes  $L_s$  only to  $\rho_N^{-1/2} = 72.9 \text{ nm}$ , a rise of 24%, and  $R_{\text{cut}}$  from 236 to 292 nm. **The pair count per loop grows by at most  $(1.24)^3 \simeq 1.9$  while the loop count falls to zero**, so the cost of Option 2 falls monotonically through the anneal. The same bound is what keeps the truncation *legitimate*: the justification for cutting the sum is that  $k_m^2$  already carries the mean-field response, and  $k_m^2$  retains two thirds of its magnitude no matter how completely the loops anneal. Both statements fail together in a crystal whose network has also recovered, and that is the case in which Eq. (170) must be re-derived rather than re-used.

**The selection rule runs backwards during an anneal, and that is the operational result.** Equation (171) defines the two independent selectors, and both fall as the loops shrink — but by wildly different amounts, and it is the disparity rather than the direction that decides the run. With the loop number fixed  $d_k$  is fixed, the loops' own contribution to  $k_m^2$  is proportional to  $\rho_{sL}^k = 2\pi R_{sL}^k n_{sL}^k$  and so to  $R_{sL}^k$ , and the network's contribution is not, so

$$\Theta_k = \frac{2R_{sL}^k}{d_k} \propto R_{sL}^k \longrightarrow 0, \quad \Xi_k = d_k \sqrt{k_N^2 + \varkappa R_{sL}^k} \longrightarrow d_k \sqrt{k_N^2}, \quad (196)$$

with  $k_N^2$  everything in Eq. (38) that is not the annealing family.  $\Theta_k$  goes to zero and  $\Xi_k$  barely moves. On the reference case  $\Xi_k$  for the  $\langle c \rangle$  family falls only from 1.856 to 1.501, by the factor  $\sqrt{\rho_N/k_m^2} = 0.809$ , and stays inside the merged-halo column of Table 8 for the whole anneal. So the family crosses *one* cell boundary and not two: it leaves the crowded row and never leaves the merged-halo column, which is exactly the trajectory Table 16 is built around. What it does not do is cross it early.

The reference  $\langle c \rangle$  family enters at  $R = 57.6 \text{ nm}$  and  $d_k = 109.5 \text{ nm}$ , i.e.  $\Theta_k = 1.05$ , in the lower-left cell of Table 8 where its loops overlap on average.  $\Theta_k$  falls below 0.5 only at  $R = 27.4 \text{ nm}$ , and integrating Eq. (181) for the faulted state gives that at 22.3 h of a 29.5 h anneal-out at 873 K, and at 35.5 min of 47.4 min at 973 K. **Three quarters of the anneal is spent in the crowded regime**, because  $\dot{R}$  is smallest when the loop is largest, and the population spends most of its life near the

size it started at. So the representation must be switched *during* the run, on  $\Theta_k$  evaluated from the discrete population itself, and the switch pays for only the last quarter.

Table 16: The two annealing constructions. Both are exact statements of Eq. (152); they differ in what supplies the concentration at the line, which is the same distinction Section 3.14 draws under irradiation. The last row is the one that decides a run.

|                   | Option 1 — growth equation                                                                                     | Option 2 — Green’s functions                                                                            |
|-------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| unknowns per loop | 1 (circular) or 2 (elliptical), Eqs. (157), (158) and (181)                                                    | $N_{\text{side}} \times 4$ climb velocities, Eq. (166)                                                  |
| field at the line | continuum $c_C^v$ plus a lagged monopole sum                                                                   | solved: $c_D^v$ from Eq. (162), linear in the velocities                                                |
| capillarity       | inside $c_{sk}^{v,\text{eq}}$ , Eq. (176), with the fault term                                                 | from the network’s own stress, Eq. (169); the fault term must be added to it                            |
| cost              | $O(N_{\text{loop}})$ , falling through the anneal                                                              | $O(N_{\text{seg}}^2)$ inside $R_{\text{cut}} = n_L L_s$ , and $R_{\text{cut}}$ grows through the anneal |
| misses            | neighbor re-absorption beyond the monopole; contact coalescence; departure from an ellipse                     | nothing structural; it is the cost that limits it                                                       |
| when to use       | $\Theta_k \lesssim 0.5$ , i.e. the last quarter of the anneal for the reference $\langle c \rangle$ population | $\Theta_k \gtrsim 0.5$ , i.e. the first three quarters                                                  |

**Losing loops, and why this is the calibration experiment for Section 2.9.** In the continuum representation a family loses loops through the floor current Eq. (127), which is an integral of the assumed log-normal Eq. (123) below  $m_{\text{min}}$  — a closure, evaluated from two moments. In the discrete representation a loop reaching  $R_{\text{min}}$  is deleted and its remaining defects are returned to the mobile pool, one loop at a time, by counting. **The two are the same physical process computed two entirely different ways, and an anneal is the only regime in which the process is the dominant one.** Running both on the same transferred state therefore measures the closure directly:  $n_{sL}^k(t)$  from the continuum against the surviving loop count from the discrete population, on the same volume, with no fitted quantity between them. Two features make the comparison sharp rather than merely suggestive.

- **The signature is qualitative.** Section 2.8 noted that the two-moment model gives constant density and falling mean size under a pure anneal, where experiment reports density *falling* and the surviving mean size *rising*. A discrete population reproduces the experimental signature by construction — the smallest loops are the ones that reach the floor — so the test is whether Eq. (127) reproduces it too, and by how much.
- **The dispersion is transferred, not assumed.** Section 3.4 draws each loop’s size from the same log-normal the closure integrates, so at  $t = 0$  the two representations agree on the distribution exactly. Any subsequent divergence is the closure failing to track a distribution that Eq. (181) is reshaping — and it will reshape it, because  $\dot{R}$  is size-dependent and, by Eq. (182), size-dependent differently for the two basal states.

**Conservation during an anneal is a stronger check than during irradiation.** The ledger Eq. (135) loses its largest term when the source is switched off, so what remains is a balance among quantities of comparable size:

$$\underbrace{\Delta\left(\sum_{(s,k)} \eta_s c_{sL}^k\right)}_{\text{stored, falling}} + \underbrace{\Delta\left(\sum_m |m| c^m\right)}_{\text{mobile}} + \underbrace{\int_0^t \oint_{\partial\Omega} \mathbf{J} \cdot \mathbf{n} \, dA \, dt'}_{\text{grain boundary}} = 0. \quad (197)$$

Under irradiation the boundary term is 71% of cumulative production and the stored term is 0.1% of it (Section 6.8), so closure tests the large channels and says almost nothing about the small one. In

an anneal the stored term *is* the source, and Eq. (197) becomes a direct test of whether every vacancy that left a loop was accounted for — including the ones returned by loops deleted at the floor, which is the step most likely to lose them. On a periodic cell the boundary term is identically zero and Eq. (197) reduces to a two-term conservation statement that must hold to solver tolerance, which is the cheapest available regression test on the whole annealing path.

**What is implemented.** Equations (177), (179) and (181) and every number in Tables 9 to 11 are produced by `dislocluster_code.studies.loop_annealing`, which reads its constants from the material file and is verified against the closed forms of Section 1.2. It is a standalone evaluation of this section’s equations and does not run a march. The march-level annealing path requires, in order: the computed binding energy in place of the 5 eV surrogate (Section 4.2), a  $G = 0$  schedule specified in time rather than dose, and the  $\Theta_k$ -triggered switch between the two constructions of Table 16. The first is already step 3 of Appendix D.2; the other two are steps 14 and 16 of Appendix D.6. **A fourth requirement is not visible from the formulation and was found in the assembled system:** Eq. (27)’s emission source  $\mathbf{S}_{vL}$ , Eq. (113), has no counterpart in the mobile solve at all, which under irradiation is a small correction and under an anneal is the entire drive. It is step 13, it comes before the other two, and Eq. (191) is its acceptance test. The test cases for annealing a *distribution* — in both representations, which is the comparison this subsection argues is the calibration experiment for Section 2.9 — are Table 24.

## 5 Macroscopic Observables and Mechanics Coupling

Three quantities are asked of this model by anyone who has to use it: irradiation *growth*, irradiation *creep*, and radiation *hardening*. This section states how each is computed, and it must state it twice, because past  $d_{\text{coarsen}}$  the loop population exists in two representations at once and no observable may be taken from one of them alone.

**The three observables are two objects.** Growth and creep are the same tensor — the plastic distortion carried by the loop population — read at two different times: growth is its accumulated value under no applied load, creep the rate of its deviator under one. Hardening is not a distortion at all but a threshold stress, and it is nonlinear in the population, which is why it composes differently from the other two.

**The population is split, and the split is not a detail.** From Section 3.8, a family that crosses  $\phi^*$  has the fraction  $f_{\text{kept}}$  of its defects converted to discrete loops while  $1 - f_{\text{kept}}$  stays in the continuum, and Section 3.6 measures  $f_{\text{kept}}$  falling from 0.99 to below 0.01 across a decade of dose on a 500 nm domain. An observable evaluated on the continuum fields alone is therefore too small by  $f_{\text{kept}}$ , and one evaluated on the pre-transfer fields plus the discrete population is too large by the same factor. Section 5.2 states the composition rules; they are not the same rule for all three observables.

### 5.1 The continuum representation

**One tensor underlies both growth and creep.** A loop of family  $(s, k)$  is a platelet: an area  $A_\ell$  of habit plane  $\hat{\mathbf{n}}_k$  across which the crystal has gained or lost one layer of atoms of thickness  $(\mathbf{b} \cdot \mathbf{n})_k$ . Its contribution to the plastic distortion of a volume  $V$  is  $\eta_s A_\ell (\mathbf{b} \cdot \mathbf{n})_k \hat{\mathbf{n}}_k \otimes \hat{\mathbf{n}}_k / V$ , and the platelet volume per unit crystal volume summed over a family is by construction its stored content,  $\sum_\ell A_\ell (\mathbf{b} \cdot \mathbf{n})_k / V = c_{sL}^k$ . Hence

$$\beta^P = \sum_{(s,k) \in \mathcal{F}} \eta_s \Upsilon_{h(k)} c_{sL}^k \hat{\mathbf{n}}_k \otimes \hat{\mathbf{n}}_k, \quad (198)$$

with  $\eta_s = \pm 1$  the character sign of Section 2.4. **The content fields are the plastic distortion, up to one factor per habit.**  $\Upsilon_{h(k)}$  is that factor: it is unity if a stored defect displaces exactly  $\Omega$ , and departs from unity only by the ratio of the defect’s relaxation volume to  $\Omega$ . It is written  $\Upsilon$  and not  $A$  deliberately —  $A_{h(k)}(p_m)$  in Eq. (69) is the Woo orientation factor, an unrelated quantity, and the two were carried under one symbol in an earlier revision of this document.

Equation (198) is the only constitutive statement on the continuum side. Growth and creep are two readings of it.

**Growth.** At zero applied stress the variants are equally populated and Eq. (198) is diagonal in the crystal frame. Its prismatic and basal components are

$$\varepsilon_a = \Upsilon_a \sum_{(i,a_k)} c_{iL}^{a_k} - \Upsilon_a \sum_{(v,a_k)} c_{vL}^{a_k}, \quad \varepsilon_c = -\Upsilon_c (c_{vL}^{c_f} + c_{vL}^{c_p} + \zeta_{c_0} c_{vL}^{c_0}), \quad (199)$$

the prismatic vacancy families entering with the opposite sign to their interstitial counterparts, which is the direct macroscopic consequence of carrying both characters — and the reason the coexistence question of Section 1.3 is an observable question and not only a microstructural one.  $\zeta_{c_0} \simeq \frac{1}{3}$  is the fraction of the pyramid’s nearly isotropic relaxation volume that resolves on [0001], against unity for the two planar states. **That single coefficient is where the incubation appears in an observable:** while the basal vacancies sit in  $c_0$  they contribute a third of their share to  $\varepsilon_c$ , and the accelerated  $c$ -axis growth begins only as Eq. (80) moves them onto planes.

**Creep.** Under load the variants are no longer equally populated, and what an experiment measures is the deviator of Eq. (198) evolving in time,

$$\dot{\varepsilon}^{\text{creep}} = \sum_{(s,k) \in \mathcal{F}} \eta_s \Upsilon_{h(k)} \dot{c}_{sL}^k (\hat{\mathbf{n}}_k \otimes \hat{\mathbf{n}}_k - \frac{1}{3} \mathbb{I}), \quad (200)$$

a tensor rather than the scalar an alignment fraction could produce. **The variant resolution of Eq. (64) is what makes Eq. (200) possible at all:** with the three prismatic variants lumped there is no imbalance to take a deviator of, and SIPA appears only as a scalar bias. The stress enters through the variant weights and the SIPA bracket of Eq. (68), so nothing in Eq. (200) is fitted to creep data.

The trace of Eq. (198) is not discarded: it is the swelling, and it is small here because the two characters store nearly equal numbers of atoms (Section 6.8), not because either term is small.

**Where it enters the solver.** The rate form is supplied per point as  $\mathbf{L}^P = \sum_X \dot{c}^X (\hat{\mathbf{n}}_X \otimes \hat{\mathbf{n}}_X)$ , differenced across the immobile step from the endpoint states rather than differentiated — the states are what the slow step returns, and a one-sided difference over the substep is consistent with the operator split of Section 6 to the same order as the split itself.

**Hardening.** Hardening is not a distortion and does not follow from Eq. (198). Each family is an obstacle population of number density  $N_k = n_{sL}^k / \Omega$  and obstacle *diameter*  $d_k = 2R_{sL}^k$ , and the dispersed-barrier law gives

$$\Delta\tau_k = \alpha_k \mu b \sqrt{N_k d_k}, \quad \Delta\tau_{SR} = \left[ \sum_k \Delta\tau_k^2 \right]^{1/2}, \quad \Delta\tau_{LR} = \alpha_{LR} \mu b \sqrt{\rho N}, \quad (201)$$

the short-range contributions adding in quadrature and the long-range network term separately, and the yield increment following by the Taylor factor  $M$ ,

$$\Delta\sigma_y = M (\Delta\tau_{SR} + \Delta\tau_{LR}). \quad (202)$$

**Equation (202) multiplies by  $M$ ; an earlier revision of this document divided by it.** The Taylor factor converts a resolved shear stress into the macroscopic stress that produces it,  $\sigma = M\tau$  with  $M \simeq 3$ , so dividing understates the yield increment by an order of magnitude. The implementation has always multiplied.

**Neither  $\alpha_k$  nor  $M$  is computed by this model,** and neither has a default that would be distinguishable in the output from a fitted one. Both must be stated whenever a hardening number is quoted. Section 5.2 is where  $\alpha_k$  comes from.

## 5.2 The discrete representation

Once a family is discrete, none of Eqs. (198) and (201) is needed for it: the mechanics machinery that the dislocation code already carries computes the same observables without a constitutive step, and the two subsections differ in exactly that.

**Growth and creep are measured, not modelled.** The dislocation code accumulates the plastic distortion of the network as its segments move, and writes the full tensor and its rate at every step,

$$\beta_{\mathcal{D}}^P = \frac{1}{V} \sum_{\lambda} \frac{1}{2} (\mathbf{b}_{\lambda} \otimes \mathbf{A}_{\lambda} + \mathbf{A}_{\lambda} \otimes \mathbf{b}_{\lambda}), \quad \mathbf{A}_{\lambda} = \text{area swept by segment } \lambda, \quad (203)$$

alongside  $\dot{\beta}_{\mathcal{D}}^P$ , together with  $\text{tr}$  and  $\|\cdot\|$  of each. Three things follow, and they are the reason to want the discrete representation for an observable at all.

- **No strain factor appears.** Equation (203) is a sum over the segments that actually moved, with their actual Burgers vectors; there is no  $\Upsilon_{h(k)}$  to assign and no habit-plane idealization. Comparing it against Eq. (198) on the same population is therefore a *measurement* of  $\Upsilon_{h(k)}$ , which is the one clean route to a quantity the continuum side has to assume.
- **Climb and glide separate by the trace.** A climbing prismatic loop has  $\mathbf{b} \parallel \hat{\mathbf{n}}$  and contributes to  $\text{tr} \beta^P$ ; glide is volume-conserving and does not. So the trace isolates the irradiation-driven part from the deformation-driven part with no tagging of segments.
- **It is a rate, natively.**  $\dot{\beta}^P$  is what the creep measurement wants and is written directly, where the continuum side must difference Eq. (198) across a substep.

The cost is that Eq. (203) is only as good as the climb velocities feeding it, which is the accuracy question Section 3.16 settles through the cutoff convergence — the climb strain rate is precisely the quantity that sweep is flat in.

**Hardening is measured by loading the cell.** The discrete route does not evaluate Eq. (201); it rebuilds the interior population as loops in a periodic cell, loads it, and reads the critical resolved shear stress off the response. Three routes are staged and they are not alternatives of equal cost:

| route | what it does                                                       | cost and role                                                                                        |
|-------|--------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| A     | stress ramp on a frozen obstacle field; $\Delta\tau$ read directly | the reference measurement, but affordable only at selected doses                                     |
| C     | Eq. (201) with $\alpha_k$ fitted on route A                        | free at every dose once $\alpha_k$ is known — this is how the continuum side acquires its $\alpha_k$ |
| D     | the same cell seeded by density rather than per-loop export        | the smoke test, not a measurement                                                                    |

Route B — the full stress–strain curve at imposed strain rate — is days per dose and is deliberately not staged. **The relationship between routes A and C is the whole point of the pairing:** A is the only thing here that measures  $\alpha_k$ , and C is what makes the measurement usable at doses A cannot reach.

Two properties of the measurement must be quoted with any number it produces. **The criterion is an offset**, the stress at which the plastic shear first reaches a stated value; an irradiated cell creeps below its breakaway stress, so the number depends on the offset and the offset is a choice, not a property of the material. And **the loops are frozen**: they are inserted as sessile loops with no climb solver attached, so they neither absorb nor are cut, and nothing channels. That is the largest modeling assumption in the measurement and it biases hardening high as strain accumulates.

**Composing the two representations.** Past  $d_{\text{coarsen}}$  both representations hold part of the same family, and the rule for combining them is different for the distortion observables and for hardening. It follows from whether the observable is linear in the population.

**Why root-sum-square is not an approximation here.** Split one family into a continuum part of density  $N_C$  and a discrete part  $N_D$  at the same obstacle diameter  $d$  and strength  $\alpha$ . Then

$$\sqrt{\Delta\tau_C^2 + \Delta\tau_D^2} = \alpha\mu b\sqrt{(N_C + N_D)d} = \Delta\tau[N_C + N_D], \quad (204)$$

that is, quadrature over the two pieces reproduces *exactly* the single population they were cut from. Linear addition would instead give  $\alpha\mu b(\sqrt{N_C d} + \sqrt{N_D d})$ , which overstates the result by up to  $\sqrt{2}$  at

Table 17: How an observable is composed when a family is split between the two representations, with  $f_{\text{kept}}$  the discrete share of [Section 3.6](#). The continuum fields used must always be the *post*-transfer ones, which [Section 3.8](#) has already scaled by  $1 - f_{\text{kept}}$ .

| observable             | composition                                                        | why                                                                                                                                                     |
|------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| growth $\beta^P$       | $\beta_C^P + \beta_D^P$ , <b>linear</b>                            | <a href="#">Equation (198)</a> is linear in $c_{sL}^k$ and <a href="#">Eq. (203)</a> is a sum over segments; the two partition the same stored content. |
| creep $\dot{\epsilon}$ | $\dot{\epsilon}_C + \dot{\epsilon}_D$ , <b>linear</b>              | Same argument on the rate. Both must be expressed in the same frame and unit system first; the discrete side is written in the code's own $c_s/b$ .     |
| hardening $\Delta\tau$ | $[\Delta\tau_C^2 + \Delta\tau_D^2]^{1/2}$ , <b>root-sum-square</b> | $\Delta\tau \propto \sqrt{N}$ , so linear addition over-counts. See <a href="#">Eq. (204)</a> .                                                         |
| sink strength $k_m^2$  | continuum part only, <b>plus</b> <a href="#">Eq. (172)</a>         | The transferred loops were removed from <a href="#">Eq. (38)</a> at the transfer; adding them back is what <a href="#">Section 3.17</a> is about.       |

an even split. [Equation \(204\)](#) is the same rule [Eq. \(201\)](#) already applies across families, and it is what makes a split family harmless *provided both pieces are scored by the same law*.

**That proviso is where the care is needed.** A route-A measurement of the discrete part is not a  $\Delta\tau_k$  computed from [Eq. \(201\)](#): it is a response of a cell containing those loops, and it already includes whatever  $\alpha$  they actually have. Combining a measured  $\Delta\tau_D$  with a law-derived  $\Delta\tau_C$  by [Eq. \(204\)](#) therefore assumes the continuum remainder has the same obstacle strength as the exported loops — which is the assumption route C exists to test and not one to make silently. The clean procedure is the one the routes were built for: fit  $\alpha_k$  on route A at a dose where the transfer is feasible, then score *both* pieces with [Eq. \(201\)](#).

**Three failure modes, all of them silent.** A transfer leaves no marker in the fields it reduced, so each of these produces a plausible number:

1. **Using the pre-transfer continuum field.** The observable is then too large by exactly  $f_{\text{kept}}$  in the distortion channels — a factor of two at  $f_{\text{kept}} = 0.5$ , and at 0.99 it is a doubling.
2. **Reporting a domain mean rather than an interior mean.** The boundary shell carries up to 99.8% of the  $\langle a \rangle$  density ([Section 3.6](#)), so a domain mean of any loop observable measures the shell. Every number in [Section 5.2](#) is an interior mean, by the same innermost-quartile rule the transfer itself uses.
3. **Counting the network twice.** The loop-network channel of [Eq. \(118\)](#) feeds absorbed loop line length into  $\rho_N$ , and  $\rho_N$  then appears in  $\Delta\tau_{LR}$  of [Eq. \(201\)](#). Once a family is discrete, that absorption happens in the dislocation code instead, and a  $\rho_N$  still being advanced by [Eq. \(118\)](#) for the transferred share double counts it.

**The refused share is misplaced, not lost.** One further caution follows from [Section 3.17](#): refusal is a spatial criterion — loops are refused for being near a face — while the defects they carried are returned to the continuum by a single scalar applied at every node alike. The atom balance closes, so [Eq. \(198\)](#) evaluated over the whole domain is right; a *locally* resolved observable in the boundary shell is not.

### 5.3 Mechanics observables options

[Sections 5.1](#) and [5.2](#) describe two representations. They do not by themselves say which to use, and that is a separate decision with three answers, because a representation is not the same thing as a *route to a number*. [Table 18](#) states the three and what each costs; the rest of this subsection is why the third exists.



Table 18: The three routes to a mechanics observable. “Assumed” means a coefficient the route cannot itself determine and which must therefore be quoted with any number it produces.

| Route         | Measured                                               | Assumed                                                   | Dose range                   |
|---------------|--------------------------------------------------------|-----------------------------------------------------------|------------------------------|
| (1) continuum | nothing — everything follows from $c_{sL}^k, n_{sL}^k$ | $\Upsilon_{h(k)}, \alpha_k, \alpha_{LR}, M$               | all                          |
| (2) discrete  | $\beta^P, \dot{\beta}^P, \Delta\tau$                   | $M$ only                                                  | $d \lesssim d_{\text{feas}}$ |
| (3) hybrid    | $\Upsilon_{h(k)}, \alpha_k$ at the calibration doses   | $M$ ; that $\alpha_k, \Upsilon_{h(k)}$ carry over in dose | all                          |

**(1) Continuum.** Evaluate Eqs. (198) to (202) on the fields the march already carries. No transfer is performed, no dislocation calculation is run, and the cost is a post-processing pass over the stored snapshots.

Its strength is that it is defined at every dose, including doses at which Section 3.11 shows no transfer is possible at all, and it is the only route with that property. Its weakness is the “assumed” column:  $\Upsilon_{h(k)}$  converts stored content to strain, and  $\alpha_k$  converts an obstacle population to a stress, and **neither is computed anywhere in the continuum model**. A continuum hardening curve is therefore a statement about  $N_k(d)$  and  $d_k(d)$  with the constitutive step supplied from outside.

There is a second and subtler limitation. Past  $d_{\text{coarsen}}$  the mean-field treatment of coalescence is by construction no longer reliable (Section 3.9), and  $N_k$  and  $R_{sL}^k$  are exactly the quantities that treatment sets. **So route 1 is not merely unvalidated above  $d_{\text{coarsen}}$ ; its inputs are the ones the detector has flagged.** That is a sharper objection than the missing coefficients, because no amount of external calibration repairs it.

**(2) Discrete.** Convert the population to loops at selected doses (Section 3) and compute the observables entirely within the dislocation code:  $\beta^P$  and  $\dot{\beta}^P$  from Eq. (203), and  $\Delta\tau$  by loading the cell. Nothing is assumed but the Taylor factor, which is a polycrystal-averaging statement and not a property of this model at all.

**The binding constraint is feasibility, not cost.** Section 3.6 measures the accepted defect share  $f_{\text{kept}}$  falling from 0.99 at  $10^{-2}$  dpa to 0.0096 at 1 dpa on a 500 nm domain, because the  $\langle c \rangle$  radius outgrows the cell. Write  $d_{\text{feas}}$  for the dose at which  $f_{\text{kept}}$  leaves unity. Then:

$$d_{\text{feas}} \gtrsim d_{\text{coarsen}} \quad \text{is required for route 2 to be available where it is needed,} \quad (205)$$

and on the reference geometry the two are  $\simeq 0.1$  dpa and 0.104–0.127 dpa respectively — **they coincide, and the window in which route 2 is both necessary and possible is a factor of two or three in dose.** Above it a “fully discrete” calculation is not expensive, it is unavailable: the population that would carry the observable is the part the crystal refuses.

Two further properties must be quoted with any route-2 number. The exported loops are inserted sessile with no climb solver attached, so they neither absorb nor are cut and nothing channels, which biases hardening high as strain accumulates; and the yield point is an offset criterion, so the number depends on an offset that is a choice.

**(3) Hybrid.** Use route 2 where it is feasible to *measure the coefficients route 1 assumes*, then use route 1 with those coefficients everywhere, adding the discrete contribution wherever a transfer has actually been performed. Concretely:

- 1. Calibrate small.** At one or more doses below  $d_{\text{feas}}$ , where  $f_{\text{kept}} \simeq 1$  and the cell can hold the population, export and run. Two coefficients come out, and they are the two route 1 cannot produce:
  - $\alpha_k$ , by fitting Eq. (201) to the measured  $\Delta\tau$  — this is exactly the route A/route C pairing of Section 5.2;
  - $\Upsilon_{h(k)}$ , by comparing  $\beta_D^P$  of Eq. (203) against Eq. (198) evaluated on the same population. **The discrete side carries no strain factor**, so the ratio of the two *is* the factor.

A small cell suffices, because both coefficients are properties of the obstacle and the defect, not of the domain.

2. **Compose past the coarsening dose.** Where a transfer has fired, the family is split and Table 17 applies: distortions add linearly,  $\Delta\tau$  adds in quadrature by Eq. (204), and the continuum fields used are the post-transfer ones. Where no transfer has fired — because  $f_{\text{kept}} = 0$  and it was declined — the continuum term is the whole of it, but now with a measured  $\alpha_k$ .

**This is what makes the quadrature rule legitimate.** Section 5.2 noted that combining a *measured*  $\Delta\tau_{\mathcal{D}}$  with a *law-derived*  $\Delta\tau_{\mathcal{C}}$  by Eq. (204) assumes the two pieces share an obstacle strength. Route 3 removes the assumption by scoring *both* pieces with Eq. (201) at an  $\alpha_k$  that was measured rather than posited — the measurement is used to fix the coefficient, not to supply one of the two terms. A route-2 number and a route-1 number are never added.

**What route 3 still assumes, and it is the one to test.** That  $\alpha_k$  and  $\Upsilon_{h(k)}$  measured at  $d \lesssim d_{\text{feas}}$  carry over to higher dose. For  $\Upsilon_{h(k)}$  this is mild: it is a relaxation volume per stored defect and has no obvious dose dependence. For  $\alpha_k$  it is not mild — obstacle strength depends on obstacle size, the loops grow by an order of magnitude across the dose range, and  $\alpha$  is known to vary with  $d$  for loops. **The honest form of route 3 therefore fits  $\alpha_k(d_k)$  rather than a constant**, which needs calibration at more than one dose and is the argument for spending the route-2 budget on a spread of small cases rather than on one large one.

**Which to use.** Route 1 for a survey across dose, for a parameter sweep, or for any comparison in which the constitutive coefficients cancel — and it is the only choice below  $d_{\text{coarsen}}$ , where the continuum is the model. Route 2 to measure, never to survey: it is the reference the other two are calibrated and checked against, and Eq. (205) bounds where it can be run at all. Route 3 for a number that is to be compared with experiment, because it is the only one of the three that is defined at every dose *and* carries no unmeasured obstacle coefficient.

**Status.** Route 1 is implemented for hardening and for the distortion rate, and route 2 is implemented for hardening (all three of its sub-routes) and supplied natively for  $\beta^P$  by the dislocation code. Route 3's  $\alpha_k$  fit exists. **The  $\Upsilon_{h(k)}$  measurement of step 1 above has been run and returns unity by construction** — the algebra is set out at step 11 of the implementation plan — so it is not a measurement of  $\Upsilon$  at all, and the distortion observables still carry an assumed strain factor in every route but 2. What the comparison *is* good for is checking the handoff, which it does sharply. Obtaining a physical  $\Upsilon_{h(k)}$  requires putting `immobileSpeciesRelRelaxVol` into one representation and not the other, and that is a modelling decision this document has not made.

## 6 Solution Method: the Two-Time-Scale Operator Split

### 6.1 Scale separation

The system of Section 2 spans time scales that differ by many orders of magnitude. The mobile species recombine, cluster and reach the absorbing boundaries on the time scale  $\tau_M \sim (k^2 D)^{-1}$ , which for interstitials at 573 K and the sink densities of interest is of order microseconds. The immobile families evolve on the dose scale: at  $G = 10^{-7}$  dpa s<sup>-1</sup>, one dpa is 10<sup>7</sup> s. Thirteen decades separate the two.

That separation is what makes the problem tractable, and it is exploited in the standard way. On the slow manifold the mobile species are *slaved*: they sit, at every instant, on the quasi-steady state determined by the immobile population currently present. Formally, writing the mobile block as  $\dot{\mathbf{c}}_M = \varepsilon^{-1} \mathcal{R}_M(\mathbf{c}_M; \mathbf{Q})$  and the immobile block as  $\dot{\mathbf{Q}} = \mathbf{g}(\mathbf{Q}; \mathbf{c}_M)$  with  $\varepsilon = \tau_M / \tau_Q \ll 1$ , the singular limit  $\varepsilon \rightarrow 0$  replaces the first by the algebraic constraint  $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}) = \mathbf{0}$ . This is the quasi-steady-state assumption, and the spatial code implements it *exactly*: its mobile solver `ClusterDynamicsFEM::solveMobileClusters` has no time derivative at all. Equation (27) is not a discretization of a transient problem taken to large times; it is the steady problem itself.

### 6.2 Structure of the split

The two blocks are advanced by a Lie–Trotter operator split. Over one dose interval  $[\gamma_n, \gamma_{n+1}]$ , with  $\Delta\gamma = \gamma_{n+1} - \gamma_n$  and  $\Delta t = \Delta\gamma / G$ , the scheme alternates:

- S1. Mechanics.** Solve the elastic boundary-value problem with the current inelastic deformation to obtain  $\sigma_n(\mathbf{x})$ . Constant in the cases reported here, retained for generality.
- S2. Assemble local coefficients** from the current immobile state  $\mathbf{Q}_n(\mathbf{x})$  and  $\sigma_n$ : the diffusion tensors  $\mathbf{D}^m$ , the efficiencies  $Z_{sk,m}$  of Eq. (68), the sink operator  $\mathbf{D}(\mathbf{Q}_n)$  of Eq. (37), the quadratic  $\mathbf{R}_2$ , the variant weights  $w_k^s(\sigma_n)$  of Eq. (77), and the boundary data.
- S3. Fast solve (FEM, Newton, spatially coupled).** With  $\mathbf{Q}_n$  *fixed*, solve  $\mathcal{R}_M(\mathbf{c}_M^*; \mathbf{Q}_n) = \mathbf{0}$  for the quasi-steady mobile field  $\mathbf{c}_{M,n}^*(\mathbf{x})$ .
- S4. Slow march (pointwise, decoupled).** At every node  $q$ , integrate  $d\mathbf{Q}/dt = \mathbf{g}(\mathbf{Q}; \mathbf{c}_{M,n}^*(\mathbf{x}_q))$  over  $[t_n, t_n + \Delta t]$  with the mobile species *frozen*, giving  $\mathbf{Q}_{n+1}(\mathbf{x}_q)$ .
- S5. Post-process per point.** Radii, sink strengths, the inelastic strain rate of Eq. (200), and the update of the inelastic deformation.

Steps S2–S3 couple space, through diffusion. **Step S4 does not**, and that is the structural fact the rest of this section rests on.

### 6.3 The fast step

The fast step is the spatial code’s own steady mobile solve, driven on demand. The immobile state currently held is written into the CD block of the configuration file; the solver is invoked for a single step with its immobile solver disabled, so the immobile field is passed through untouched; and the relaxed mobile field is read back. Because the mobile problem carries no time derivative and its cascade production is built from the constant generation rate, nothing about the solve depends on  $\Delta t$  or on the accumulated dose.

Three details of this arrangement are load-bearing.

**The fast step must be inside the loop.** If the mobile field is instead replayed from a previously recorded run, the scheme is not an operator split at all:  $\mathbf{c}_M$  never responds to the immobile state the march is building, so a starting state away from quasi-steady state can never relax. Any spatially uniform seed is such a state, because it carries no boundary layer while the true  $\mathbf{c}_M^*(\mathbf{x})$  is pinned to thermal equilibrium on every Dirichlet face. The replay mode is retained in the code only so that the effect of closing the loop can be measured.

**Both routes must share the fast step’s solver settings.** The spatial code’s default makes the mobile solve a *projected* Newton iteration that does not converge on this problem: it caps out at 53.6% error in  $C_i$ . That projection is disabled for the coupled route and for the standalone comparison alike. Left at the default on one side only, a route-to-route comparison measures that solver defect rather than the physics.

**Warm starting.** The Newton loop is started from whatever mobile field is already in the CD block, so passing the previous  $\mathbf{c}_M^*$  costs nothing and saves iterations. The first solve of a march starts instead from the diffusion-free local reaction balance.

### 6.4 The slow step, and why the mobile field must be frozen

With  $\mathbf{c}_M^*(\mathbf{x})$  known and held fixed, the immobile ordinary differential equations at different nodes are *independent*. This is not an approximation introduced for speed: the only thing coupling neighboring points is diffusion of the mobile species, and freezing them removes it by construction. A point’s immobile equations then depend on its own loop populations and on the four mobile values pinned into its state, and on nothing else.

The step is therefore implemented by reusing the existing stiff integrator with a `freeze_mobile` flag that zeroes the four mobile derivatives *after* every reaction term has been evaluated. The immobile equations, the conservation accumulators and  $\rho_N$  all continue to read the mobile values as frozen parameters; nothing is dropped. Each node becomes one case of an OpenMP batch mode, integrated concurrently in a single subprocess with its own solver workspace, and identical states are deduplicated

before the batch is dispatched (a measured factor of 1.25 at high dose on a 500 nm case, and considerably higher early in a march, when much of the domain still shares one state).

**The integrator.** CVODE with backward differentiation formulas and a dense linear solver, at  $\text{rtol} = 10^{-6}$ , is used throughout. The choice is measured rather than assumed: against a tight reference, BDF with a dense or a full-band solver gives a maximum relative error of  $3.0 \times 10^{-4}$ , a Krylov solver  $1.7 \times 10^{-2}$ , and the implicit Runge–Kutta tables designed as IMEX pairs fail outright in pure implicit mode, by four to seven orders of magnitude. The system is 15-dimensional after freezing, so a dense factorization is  $\sim 1100$  flops and there is nothing to gain from sparsity.

## 6.5 Why of order $10^6$ ODEs per substep is tractable

The counts below are quoted for the two-moment immobile block. Carrying the second moment of [Section 2.9](#) raises the per-node immobile state from 18 components to 27 and the dense factorization by  $27^3/18^3 = 3.4$ ; because [Eq. \(120\)](#) is local in exactly the way every other immobile equation is, the argument of this subsection is unchanged by it and only the block dimension moves.

Read the totals carelessly and the scheme looks impossible. A 500 nm case advances 90 617 nodes  $\times 15$  integrated equations, that is  $1.09 \times 10^6$  scalar ordinary differential equations, per substep, and a tightly coupled stiff system of that size would be hopeless. **It is not one system.** It is 72 494 independent 15-dimensional systems after deduplication, and the right-hand side takes a single point’s state vector with no neighbor state in it anywhere.

The cost consequence, for one implicit factorization:

|                              | flops                                  |
|------------------------------|----------------------------------------|
| split: $72\,494 \times 15^3$ | $2.4 \times 10^8$                      |
| coupled: $(1\,359\,255)^3$   | $2.5 \times 10^{18}$                   |
| <b>ratio</b>                 | <b><math>1.0 \times 10^{10}</math></b> |

Sparsity softens the coupled figure without rescuing it: a three-dimensional finite-element Jacobian still factorizes at  $\sim O(N^2)$  with fill growing as  $O(N^{4/3})$ . Stiffness localizes the same way. Each point’s system genuinely is stiff — of order 100 internal CVODE steps per point per substep, which is why BDF — but stiffness cost scales with the *dimension of the coupled block*, not with how many independent blocks there are.

**The coupling has not vanished; it has moved to where it is affordable.** The tightly coupled problem still exists — it is the fast solve, 362 468 unknowns, 925 s measured, against 28.3 s for an entire slow sweep over all 90 617 points. What the split buys is *frequency*: because `solveMobileClusters` has no time derivative, the coupled problem is solved a handful of times over a march instead of at every timestep. The counterfactual follows from the same measurements, at 925 s/5 Newton iterations = 185 s per linear solve:

$$\begin{aligned}
 100 \text{ internal steps} \times 24 \text{ substeps} / 5 \text{ steps per refactorization} &\approx 480 \text{ refactorizations,} \\
 480 \times 185 \text{ s} &\approx 25 \text{ h,}
 \end{aligned}
 \tag{206}$$

for the four-species mobile block *alone*, against the 11.3 min the slow side actually takes for all 24 substeps. Three things stack to produce that: freezing the mobile field removes the spatial coupling; the coupled part has no time derivative, so it is solved a few times rather than thousands; and batch parallelism with deduplication spreads what remains over the available cores. The slow side is linear in node count, as the structure predicts — 9.3 s per substep at 27 720 nodes and 28.3 s at 90 617, that is  $3.35 \times 10^{-4}$  against  $3.12 \times 10^{-4}$  s per node.

The bill arrives as splitting error.

## 6.6 Splitting error and its control

Freezing  $\mathbf{c}_M^*$  over a substep ignores the drift of  $\mathbf{Q}$ , and hence of the sinks, within that substep. This is a first-order Lie–Trotter error,  $O(\Delta\gamma)$  per step and one-sided: with strengthening sinks the loops are slightly over-grown.

**The accuracy dial is the fast-solve cadence, not the dose-snapshot spacing.** The march advances the immobile block in  $n_{\text{sub}}$  substeps per dose interval and re-solves the mobile field every  $n_{\text{fem}}$  substeps, so the splitting error is first order in the interval between fast solves and not in the interval between output snapshots. Two practical notes follow. The cadence counter must run over the *whole march*, not within an interval: counted within an interval,  $n_{\text{fem}}$  becomes silently inoperative whenever an interval holds fewer substeps than the cadence, and with one substep per interval the test fires every time. And the substep count is fixed *per dose interval* rather than by a wall-clock cap on  $\Delta t$ : a 5 dpa interval at  $10^{-7}$  dpa s $^{-1}$  is  $5 \times 10^7$  s, which a fixed  $5 \times 10^4$  s cap would have split into a thousand batch subprocesses.

Three refinements are available and are not used by default. A step cap accepts a step only if the relative change in sink strength is below a tolerance,  $\max_q \Delta \langle k^2 \rangle / \langle k^2 \rangle \leq \eta_{\text{split}} \sim 0.05\text{--}0.1$ , and halves  $\Delta\gamma$  otherwise. Strang splitting — half an immobile step, a mobile re-solve, half an immobile step — gives  $O(\Delta\gamma^2)$  for one extra fast solve. An in-step Picard iteration between S3 and S4 removes the splitting error entirely and is the natural fallback if a regime ever needs it.

## 6.7 The algorithm

---

**Algorithm 1** Operator-split QSSA march: quasi-steady mobile FEM  $\oplus$  pointwise immobile CVODE.

---

- 1: **Initialize**  $\mathbf{Q}_0(\mathbf{x})$ , either pristine or from a seed written uniformly onto the nodes;  $\sigma_0$ ; the dose grid  $\{\gamma_n\}$ , refined through the nucleation transient.
  - 2: **for**  $n = 0, 1, \dots, N - 1$  **do**
  - 3:    $\Delta\gamma = \gamma_{n+1} - \gamma_n$ ;  $\Delta t = \Delta\gamma/G$ ; partition  $[t_n, t_n + \Delta t]$  into  $n_{\text{sub}}$  substeps.
  - 4:   [S1] Solve mechanics  $\Rightarrow \sigma_n(\mathbf{x})$ .
  - 5:   **for** each substep  $k$  **do**
  - 6:     **if**  $k_{\text{global}} \bmod n_{\text{fem}} = 0$  **then**
  - 7:       [S2–S3] assemble local coefficients from  $\mathbf{Q}$ ; Newton-solve  $\mathcal{R}_M = \mathbf{0}$  warm-started from the previous  $\mathbf{c}_M^* \Rightarrow \mathbf{c}_M^*(\mathbf{x})$ .
  - 8:     **end if**
  - 9:     [S4] for all nodes  $q$  in parallel, integrate the immobile block with `freeze_mobile`,  $\mathbf{y}_q[0:4] = \mathbf{c}_M^*(\mathbf{x}_q)$ , over the substep  $\Rightarrow \mathbf{Q}(\mathbf{x}_q)$ .
  - 10:    [checkpoint]  $\mathbf{Q}$  has just been rebound wholesale and the fast solve is either fully applied or was not scheduled; this is the only consistent point in the body.
  - 11:   **end for**
  - 12:   [S5] per node: radii, sink strengths, inelastic strain rate; write the snapshot.
  - 13: **end for**
- 

Algorithm 1 is what the driver executes. Three operational points are worth recording, because each was established by running it rather than by design.

**A point can fail the slow step, and where it fails is diagnostic.** On a 500 nm anisotropic march, one node of 90617 failed the immobile integration at the  $0.01 \rightarrow 0.1$  dpa substep, and failed *deterministically* — in the full batch, again when re-run alone, twice, across two separate invocations. It sits 51  $b$  from the basal-pole face, inside the boundary layer, with entirely unremarkable concentrations (30th to 50th percentile in every mobile species). It is not an extreme-value node. The location is the tell: anisotropic interstitial diffusion narrows the boundary layer along  $\langle c \rangle$  and steepens its gradient, which is what puts that point beyond CVODE’s reach at  $\text{rtol} = 10^{-6}$ . Re-running failed points alone is worth doing — it costs a handful of cases against tens of thousands and distinguishes a transient from a real failure — but it does not help here, and the retry log says so. A point that fails alone needs a solver or a model answer, not another attempt. Note what the failure tolerance does before reaching for it: it substitutes an *identity* step, freezing the point for that substep. It does not recover the point; it declares the march successful without it.

**Checkpointing has exactly one consistent point.** It is after the immobile batch has returned and the state has been rebound wholesale, with the substep counter incremented to count substeps *completed*, so that a resume is exactly “skip the first  $k_{\text{global}}$  substeps” and the fast-solve cadence

reproduces. A resumed march re-enters at the state it left, but both the fast solve and the batch are OpenMP-parallel, so the trajectory after a resume is not bit-identical to an uninterrupted one unless the thread count is pinned.

**The staged case is keyed on the domain, not on the run.** The geometry, mesh, boundary conditions, material and temperature are hashed; the dose grid is not. Changing only the doses therefore reuses the staged case, its node set and its bootstrap, and a material change — including one as small as a lattice constant — re-stages everything automatically.

## 6.8 Conservation across the split, and the open-system balance

In quasi-steady state the mobile pool does not accumulate, so production balances recombination, loop absorption, *and outflux through the boundary*:

$$\int_V G^{\text{tot}} dV = \int_V \left( \mathcal{R}_{\text{rec}} + \sum_{(s,k) \in \mathcal{F}} \dot{c}_{sL}^k \right) dV + \underbrace{\oint_{\partial V} \sum_m \mathbf{J}^m \cdot \mathbf{n} dS}_{\text{boundary loss}}. \quad (207)$$

Because the immobile absorption uses the same  $Z_{sk,m} \bar{D}^m c^m$  that the mobile solve balanced, atoms are conserved to splitting order *provided* the volume check includes the boundary integral.

Equation (207) is the point at which the mean-field bookkeeping stops applying. The accumulators Eq. (135) integrate a closed atom balance,  $d(\text{stored})/dt = \text{production} - \text{recombination} - \text{sink absorption}$ , with *no transport term*. On a spatially resolved march that cannot close, and the residual is not solver error — it is the atoms that left through the absorbing surface. Two things force it: the domains here are Dirichlet over their whole surface unless faces are made periodic, and the slow step freezes the mobile species while production keeps accumulating into them, the mobile pool’s own balance being closed by the *fast* solve, which is where the flux to the boundary lives.

The magnitude settles the interpretation. A 200 nm march reports a residual of  $-71\%$  of cumulative production. Nothing in a CVODE run at  $\text{rtol} = 10^{-6}$  is 71% wrong. The residual is therefore published as a physical channel, `grain_boundary` =  $-\text{residual}$ , positive when atoms have left; in defect counts at 10 dpa on that case, the boundary absorbs 71.4% of the interstitials and 71.9% of the vacancies produced, against 22.9% recombined, 5–6% absorbed at network sinks and 0.1% stored in the microstructure. Equal Frenkel production and equal recombination are exact; the storage asymmetry ( $4.7 \times 10^6$  interstitials against  $1.3 \times 10^6$  vacancies) is the bias-driven excess that drives growth.

**An independent measurement of the same channel.** The closure above is a difference, and inherits every other channel’s error. It can be checked independently, without touching the accumulators, because the fast solve returns a *steady* field: for each mobile species  $0 = \nabla \cdot (D \nabla c) + R$ , so by the divergence theorem

$$\int_V R dV = \oint_{\partial V} (-D \nabla c) \cdot \mathbf{n} dS = \text{net outward flux}. \quad (208)$$

Here  $R$  is exactly the mean-field right-hand side for the mobile components — that model has no diffusion, so its mobile equations *are* the reaction terms, and the boundary appears nowhere in it — which is why the volume integral of its mobile right-hand side measures the flux to the boundary. On the 200 nm march the two agree to 3.7% at 10 dpa (measured/closure = 0.963, and 1.008 and 0.970 at 0.1 and 1 dpa), an independent confirmation rather than an identity; the 11% discrepancy at 0.01 dpa is the trapezoid rule on a coarse logarithmic grid across which the rate falls by a third.

**That agreement is not general, and must be re-established on any new geometry.** On a 500 nm hexagonal case with nine dose points the same diagnostic gives ratios of 0.153, 0.173 and 0.279 at 0.1, 1 and 10 dpa — a factor of 3.6, not 3.7%. Three things are established about that disagreement and one is not. It is not a regression: re-run today, the 200 nm ratios are 0.900–1.116 against the 0.963–1.110 recorded above, both columns having moved together with the  $\Omega$  correction, so the ratio survived. It is not the quadrature: the 500 nm grid is the finer of the two, so grid error would go the other way. It is not a difference of parameter sets: with the node states fixed, the two give bit-identical mobile rows. What it *is* remains open, and the leading candidate is the fast-solve cadence — that march performs one mobile solve per dose interval and freezes the field across three substeps, so the



snapshot field the measured side integrates is not the field the accumulators saw. That predicts the right sign and has not been shown to give a factor of 3.6. A surface integral computed inside the spatial code, where the finite-element gradients already exist, would be the gold standard, and this disagreement is the argument for building it.

## 6.9 Verification

**Recovery of the mean-field model.** The primary regression test removes the spatial content: uniform fields, periodic boundaries, zero stress so that  $w_k^s = \frac{1}{3}$  exactly, and the boundary sinks lumped away. The fast boundary-value problem then has no transport gradient and Eq. (27) collapses to the mean-field mobile right-hand side, while summing the three prism variants recovers the lumped population. What this test establishes is that the spatial discretization, the split and the unit conversions are correct. What it cannot establish is anything about the boundary layer, which is precisely the structure the test removes.

**Route-to-route comparison.** The coupled route can be compared against the spatial code’s own internal split, in which its nodal semi-implicit immobile update replaces the CVODE march. With the fast step closed, both routes solve the same fast problem with the same code, each against its own immobile state, so two differences remain and neither is the integrator alone: the two immobile right-hand sides are different *models*, not two integrations of one model; and the splitting frequencies differ, the internal route refreshing the mobile field once per output step.

**Where a domain mean is not quotable.** On any geometry whose surface is a Dirichlet sink, **a whole-domain mean of a loop quantity measures the boundary shell and not the material.** Cascade nucleation is spatially uniform but the only loop-removal channel is coalescence, driven by the absorbed mobile flux, which vanishes where the mobile concentrations are pinned. The boundary is a sink for mobile defects but *not* for loops, so the  $\langle a \rangle$  density climbs steeply toward it. On a  $1\ \mu\text{m}$  cube with second-order elements, 27% of the nodes lie on the six faces and at 21 dpa they carry 99.8% of the total  $\langle a \rangle$  density, with  $c_a/N_a = 300.8$  against a nucleus size of 300 — every loop there is a fresh nucleus that never absorbed an interstitial. Quote the *interior* mean, the innermost quartile by distance to the nearest face. The difference is not cosmetic: at 21 dpa the coupled march’s  $\langle a \rangle$  density is  $474\times$  the mean-field value as a domain mean and  $0.75\times$  it in the interior. That the coupled march reproduces this artifact is a point *in favor* of the coupling — the same boundary behavior emerges whether the nodal scheme or the CVODE march advances the immobile population.

## A Experimental evidence for the coexistence of vacancy and interstitial $\langle a \rangle$ loops in irradiated zirconium

### A.1 Overview

One of the unusual features of irradiation damage in  $\alpha$ -zirconium is the simultaneous presence of vacancy- and interstitial-type  $\langle a \rangle$  dislocation loops. The irradiation-induced  $\langle a \rangle$  loops have Burgers vectors

$$\mathbf{b}_a = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad (\text{A.1})$$

and lie predominantly on, or close to, prismatic habit planes, particularly the first-order prism family

$$\{10\bar{1}0\}. \quad (\text{A.2})$$

In contrast to the common situation in cubic metals, where the elastic dislocation bias generally favors the survival and growth of interstitial loops while vacancy loops tend to shrink, experiments in Zr have repeatedly demonstrated populations containing both

$$a_i\text{L} \quad \text{and} \quad a_v\text{L} \quad (\text{A.3})$$

loops.

The relative populations are not universal. They depend strongly on irradiation temperature, neutron or charged-particle irradiation, fluence, alloy composition, specimen geometry, and local microstructure. Nevertheless, the coexistence itself is firmly established experimentally. This appendix collects the record behind that statement; it is the experimental premise on which [Section 1.3](#) rests, and the observation that [Appendix B](#) exists to explain.

## A.2 Early neutron-irradiation evidence

One of the earliest quantitative characterizations was performed by Kelly and Blake on neutron-irradiated zirconium [27]. The loops had

$$\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle \quad (\text{A.4})$$

and were not ideal pure-edge prismatic loops; their habit-plane normals were tilted away from the ideal orientation.

After irradiation followed by annealing at approximately 490 °C, they found that approximately

$$\boxed{f_V \simeq \frac{2}{3}, \quad f_I \simeq \frac{1}{3}.} \quad (\text{A.5})$$

Thus vacancy  $\langle a \rangle$  loops were not a small anomalous population; they constituted the majority of the observed  $\langle a \rangle$  loops under these conditions.

## A.3 Jostsons–Kelly–Blake neutron experiments

A particularly important experimental study was subsequently performed by Jostsons, Kelly, and Blake [1]. Their TEM analysis of neutron-irradiated zirconium showed that

$$\boxed{\text{mixed vacancy/interstitial } \langle a \rangle \text{ loop populations occurred under all irradiation conditions investigated.}} \quad (\text{A.6})$$

The loops possessed

$$\mathbf{b} = \frac{1}{3}\langle 11\bar{2}0 \rangle, \quad (\text{A.7})$$

and were generally non-edge in character.

An additional experimentally important distinction was observed in the loop morphology. Vacancy  $\langle a \rangle$  loops, particularly the larger loops, were substantially more elliptical, whereas interstitial loops tended to remain more nearly circular.

For the elliptical vacancy loops,

$$\text{major axis} \approx [0001], \quad (\text{A.8})$$

while the minor axis lies in the basal plane. This is the observation quantified in [Section 1.2](#) as  $e_{\text{vac}} \simeq 0.4$  against  $e_{\text{int}} \simeq 0.1$ , and drawn in [Fig. 2](#). Schematically,

$$\boxed{a_i\text{L} : \text{more nearly circular}, \quad a_v\text{L} : \text{more elliptical.}} \quad (\text{A.9})$$

This morphological distinction provides independent evidence that the two populations are physically different sink structures rather than TEM contrast variants of a single loop population.

## A.4 International neutron-irradiation “round-robin” experiment

An especially convincing confirmation came from the international TEM round-robin study of Northwood et al. [2]. Multiple laboratories independently examined neutron-irradiated zirconium and Zircaloy-2.

For specimens irradiated at approximately

$$T \simeq 400^\circ\text{C} \simeq 673\text{ K}, \quad (\text{A.10})$$

inside–outside contrast analysis established that

$$\boxed{a_i\text{L and } a_v\text{L loops were both present, with an excess of vacancy loops.}} \quad (\text{A.11})$$

This result is particularly important because it arose from *bulk neutron irradiation followed by post-irradiation TEM*, rather than from irradiation of a thin foil inside an electron microscope.

The round-robin study therefore provides strong evidence that the coexistence is not simply an artifact associated with free surfaces or electron irradiation of TEM foils.

## A.5 Electron-irradiation experiments

Electron irradiation in a high-voltage electron microscope has produced some of the clearest direct observations of loop nucleation and growth, although thin-foil effects must be considered carefully.

Griffiths, Loretto, and Smallman irradiated high-purity Zr with 1 MeV electrons over approximately

$$273 \text{ K} \leq T \leq 773 \text{ K} \quad (\text{A.12})$$

and demonstrated that both vacancy and interstitial

$$\frac{1}{3}\langle 11\bar{2}0 \rangle \quad (\text{A.13})$$

loops could nucleate and grow during irradiation [28].

Thus the experimental phenomenon is not merely that vacancy loops survive after neutron irradiation; vacancy  $\langle a \rangle$  loops can be observed *growing under continuing irradiation*. That is a stronger statement than survival, and it is the one Eq. (71) must be able to reproduce.

## A.6 Effect of foil orientation and stress

The electron-irradiation experiments also revealed that loop character can depend on specimen orientation and stress.

Griffiths et al. showed at approximately 715 K that both vacancy and interstitial

$$\frac{1}{3}\langle 11\bar{2}0 \rangle \quad (\text{A.14})$$

loops nucleated and grew, but that their crystallographic populations were correlated with the TEM foil normal [29].

Vacancy loops tended to select Burgers vectors more nearly parallel to the foil normal, whereas interstitial loops tended to select Burgers vectors more nearly perpendicular to it. The authors associated this behavior with tensile stress generated in the foil by surface oxidation. This is stress-induced preferential nucleation acting on the variants, and it is the effect Eq. (77) carries: the two characters respond in *opposite* senses to the same resolved normal stress.

Consequently, HVEM observations provide compelling evidence that both loop types can grow, but the quantitative vacancy/interstitial ratio in thin foils should not automatically be interpreted as the corresponding bulk ratio.

## A.7 Experiments showing predominantly interstitial loops

Not every irradiation experiment produces a nearly equal vacancy and interstitial population.

Carpenter and Watters studied 1 MeV electron irradiation of Zr and Zircaloy-2 over approximately

$$315 \text{ K} \leq T \leq 725 \text{ K} \quad (\text{A.15})$$

and reported that the low-dose damage consisted predominantly of perfect interstitial  $\langle a \rangle$  loops [30].

Similarly, Nakamichi et al. irradiated Zircaloy-2 and Zircaloy-4 with 1 MeV electrons and found interstitial

$$\frac{1}{3}\langle 11\bar{2}0 \rangle \quad (\text{A.16})$$

loops; the large loops observed at approximately 770 K were identified as interstitial using  $\mathbf{g} \cdot \mathbf{b}$  and inside–outside contrast analysis [31].

These observations demonstrate that coexistence does not imply

$$N_{a_i\text{L}} = N_{a_v\text{L}} \quad (\text{A.17})$$

under every irradiation condition.

## A.8 Influence of alloying and irradiation particle

Hellio, de Novion, and Boulanger examined pure Zr and several Zr-based alloys under both 500 keV  $\text{Zr}^+$  irradiation and 1 MeV electron irradiation [32]. The loops had  $a$ -type Burgers vectors and were preferentially associated with

$$\{10\bar{1}0\} \quad (\text{A.18})$$

prismatic planes. Alloying additions substantially affected the loop density and growth kinetics.

More recently, Gaumé et al. performed a direct comparison using recrystallized Zircaloy-4 [33]. After 1 MeV electron irradiation at

$$T = 673 \text{ K} \quad \text{and} \quad 723 \text{ K}, \quad (\text{A.19})$$

they observed

$$\boxed{a_i\text{L} + a_v\text{L}} \quad (\text{A.20})$$

populations at both temperatures.

Furthermore, the fraction of interstitial loops increased as the temperature was raised from 673 to 723 K. In contrast, after 600 keV  $\text{Zr}^+$  irradiation at 723 K, they identified only interstitial  $\langle a \rangle$  loops.

Thus,

$$\boxed{\text{loop character depends not only on } T, \text{ but also on the irradiation particle and defect-production mechanism.}} \quad (\text{A.21})$$

This distinction is important because electron irradiation produces predominantly isolated Frenkel pairs, whereas neutron and heavy-ion irradiation additionally produce displacement cascades and in-cascade defect clusters — the two source terms that [Section 2.2](#) separates.

## A.9 Modern quantitative ion-irradiation measurements

Liu, Beyerlein, and Han performed a detailed TEM analysis of high-purity Zr irradiated with He ions [7]. Loop character was determined using diffraction contrast and the inside–outside method.

At

$$T = 350^\circ\text{C} = 623 \text{ K}, \quad (\text{A.22})$$

approximately 274  $\langle a \rangle$  loops were analyzed. About

$$\boxed{90\% \ a_i\text{L}, \quad 10\% \ a_v\text{L}} \quad (\text{A.23})$$

were observed.

At

$$T = 400^\circ\text{C} = 673 \text{ K}, \quad (\text{A.24})$$

the vacancy fraction increased to approximately

$$\boxed{70\% \ a_i\text{L}, \quad 30\% \ a_v\text{L}.} \quad (\text{A.25})$$

Thus modern quantitative TEM confirms that the vacancy-loop fraction can increase substantially with temperature while the two loop characters continue to coexist.

The study also confirmed the morphology reported in the older neutron experiments: larger vacancy  $\langle a \rangle$  loops were distinctly elliptical with their long axes approximately parallel to  $[0001]$ .

## A.10 Temperature dependence

The experimental literature is sometimes summarized by the approximate trend

$$\begin{aligned} T &\lesssim 573 \text{ K} : & a_i\text{L} \text{ usually predominates,} \\ 573 \text{ K} &\lesssim T \lesssim 673 \text{ K} : & a_i\text{L} \text{ and } a_v\text{L} \text{ may become comparable,} \\ T &\gtrsim 673 \text{ K} : & \text{large vacancy fractions can occur.} \end{aligned} \quad (\text{A.26})$$

This trend has been summarized from the neutron-irradiation literature by Dai et al. [34].

Equation (A.26) should *not* be interpreted as a universal phase diagram. Electron- and ion-irradiation experiments show substantial departures from it, because foil surfaces, irradiation-induced stress, alloy chemistry, cascade production, and irradiation flux all influence the defect balance.

## A.11 Dependence on neutron fluence and local microstructure

Measurements on reactor-irradiated zirconium alloys further indicate that the vacancy/interstitial ratio is not spatially uniform.

Eriksson et al. summarize observations in neutron-irradiated Zr and Zr alloys showing that

$$\boxed{f_{a_v\text{L}}/f_{a_i\text{L}} = f(T, \Phi, \text{grain, chemistry}),} \quad (\text{A.27})$$

where  $\Phi$  denotes neutron fluence [35].

Large grain-to-grain variations have been reported. The  $\langle a \rangle$  loops also tend to form arrays or layers approximately parallel to the basal plane.

This spatial organization implies that the populations should not necessarily be treated as two uniformly mixed mean-field sink populations — which is the argument for resolving them in space, as Section 6 does, rather than for adding further lumped families.

## A.12 Summary of representative experimental observations

Table 19 collects the observations above.

## A.13 Experimental conclusions

The experimental literature therefore supports several robust conclusions.

First,

$$\boxed{a_i\text{L} \text{ and } a_v\text{L} \text{ can coexist as stable, growing irradiation defects in Zr.}} \quad (\text{A.28})$$

This conclusion is particularly strong for neutron irradiation, where the mixed population has been independently confirmed by several TEM studies.

Second, there is no universal fixed fraction

$$f_I : f_V. \quad (\text{A.29})$$

Depending upon irradiation conditions, the observed population ranges from strongly interstitial dominated to vacancy dominated.

Third, the two loop populations have distinguishable morphology:

$$a_v\text{L} : \text{larger loops often strongly elliptical,} \quad (\text{A.30})$$

while

$$a_i\text{L} : \text{generally more nearly circular.} \quad (\text{A.31})$$

Fourth, irradiation conditions influence the population strongly:

$$f_{I,V} = f(T, \text{fluence, flux, irradiation particle, chemistry, stress, local sink structure}). \quad (\text{A.32})$$

Finally, thin-foil electron-irradiation experiments demonstrate that both loop characters can nucleate and grow, but their quantitative population fractions can be influenced by free surfaces and oxidation-induced stresses. The neutron-irradiation experiments are therefore the more important evidence that the coexistence is an intrinsic bulk irradiation phenomenon.

Table 19: Representative experimental observations of vacancy and interstitial  $\langle a \rangle$  loops in irradiated Zr and Zr alloys.

| Reference              | Material /<br>irradiation                                    | Temperature              | Observation                                                                                                                                                                                             |
|------------------------|--------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kelly–Blake [27]       | Neutron-irradiated Zr<br>followed by annealing               | Anneal near 490°C        | Approximately 2/3 vacancy and 1/3 interstitial $\langle a \rangle$ loops.                                                                                                                               |
| Jostsons et al. [1]    | Neutron-irradiated Zr                                        | Several conditions       | Mixed $a_i\text{L}/a_v\text{L}$ populations under all irradiation conditions examined; large $a_v\text{L}$ loops more elliptical and $a_i\text{L}$ loops more circular.                                 |
| Northwood et al. [2]   | Neutron-irradiated Zr<br>and Zircaloy-2                      | $\sim 400^\circ\text{C}$ | Both vacancy and interstitial $\langle a \rangle$ loops, with an excess of vacancy loops.                                                                                                               |
| Carpenter–Watters [30] | 1 MeV electron, Zr<br>and Zircaloy-2                         | 315–725 K                | Low-dose population predominantly interstitial.                                                                                                                                                         |
| Griffiths et al. [28]  | 1 MeV electron,<br>high-purity Zr                            | 273–773 K                | Both vacancy and interstitial $\langle a \rangle$ loops nucleated and grew.                                                                                                                             |
| Griffiths et al. [29]  | 1 MeV electron, Zr                                           | $\sim 715$ K             | Both types observed; loop character and Burgers-vector population strongly influenced by foil orientation/stress.                                                                                       |
| Nakamichi et al. [31]  | 1 MeV electron,<br>Zircalloys                                | 320–970 K                | Interstitial loops dominate; large loops at 770 K identified as interstitial.                                                                                                                           |
| Gaumé et al. [33]      | Electron and $\text{Zr}^+$<br>irradiation, RXA<br>Zircaloy-4 | 673 and 723 K            | Electron irradiation produced both $a_i\text{L}$ and $a_v\text{L}$ ; interstitial fraction increased with temperature. $\text{Zr}^+$ irradiation at 723 K produced only identified $a_i\text{L}$ loops. |
| Liu et al. [7]         | He-ion irradiated<br>high-purity Zr                          | 350 and 400°C            | Approximately 90/10 $a_i\text{L}/a_v\text{L}$ at 350°C and 70/30 at 400°C.                                                                                                                              |



## A.14 Implication for microstructure-evolution models

Any model of irradiation damage in Zr that permits only interstitial  $\langle a \rangle$  loops,

$$a_i\text{L} \quad \text{only}, \quad (\text{A.33})$$

cannot reproduce the full experimental microstructure.

At minimum, the state space should contain two independent populations,

$$\boxed{C_{a_i\text{L}}(R, \mathbf{n}, \mathbf{x}, t), \quad C_{a_v\text{L}}(R, \mathbf{n}, \mathbf{x}, t),} \quad (\text{A.34})$$

with independent nucleation and capture kinetics. That requirement is what [Eq. \(64\)](#) meets: the three prismatic variants are carried of *both* characters, which the lumped, stress-aligned pair of the earlier model could not represent at all.

The experiments further imply that their capture coefficients cannot in general be assumed identical:

$$K_{\alpha, a_i\text{L}} \neq K_{\alpha, a_v\text{L}}. \quad (\text{A.35})$$

This is the character factor  $X_{s,m}$  of [Eq. \(68\)](#); [Appendix B](#) sets out the mechanism that makes it differ from unity.

The coexistence data therefore provide a direct experimental constraint on the finite-loop transport calculation: a successful model must produce a region of parameter space in which

$$\boxed{\dot{R}_{a_i\text{L}} > 0, \quad \dot{R}_{a_v\text{L}} > 0} \quad (\text{A.36})$$

simultaneously. [Section 1.3](#) shows that diffusional anisotropy alone does not open such a region, because it is degenerate in loop character, and identifies the two resolutions the formulation carries.

## B Point-defect capture bias of dislocation loops in zirconium

[Section 1.3](#) takes three results as given: that a loop capture efficiency can be *calculated* from the loop elastic field rather than fitted; that something breaks the symmetry between a vacancy and an interstitial loop of the same size and Burgers-vector magnitude; and that in  $\alpha$ -Zr the symmetry is broken in the particular direction that lets both characters grow at once. This appendix sets out where those results come from. Three studies supply them, and they are best read as one line of argument rather than three: the effective-medium calculation of Woo [\[36\]](#), which showed that the bias is calculable; the saddle-point analysis of Woo [\[37\]](#) one year later, which identified what makes it differ between characters; and the molecular-statics capture radii of March-Rico and Wirth [\[38\]](#), which realize the prediction quantitatively in zirconium. None is re-derived here. The purpose is to make explicit which assumptions the character-splitting factor of [Eq. \(23\)](#) inherits, and which of them have been tested.

**Two bias conventions.** The first study uses the ratio form

$$B_L = \frac{Z_{i,L}}{Z_{v,L}} - 1, \quad (\text{B.1})$$

and the second the bounded form natural to a character-resolved discussion,

$$B_{jL} = 1 - \frac{Z_{v,jL}}{Z_{i,jL}}, \quad j = i, v, \quad (\text{B.2})$$

where  $Z_{i,jL}$  and  $Z_{v,jL}$  are the SIA and vacancy capture efficiencies of a loop of *character*  $j$ . The two agree in sign — both are positive for preferential SIA capture and vanish together — but not in magnitude, and [Eq. \(B.2\)](#) is bounded above by unity where [Eq. \(B.1\)](#) is not. Only the sign is used below.

## B.1 Woo (1981): the bias is calculable

A dislocation loop is not merely a geometrical absorber. Its elastic field interacts differently with vacancies and self-interstitial atoms, producing a drift flux in addition to ordinary diffusion, so the effective sink strength for SIAs need not equal that for vacancies. Woo developed a continuum treatment of this for a *finite* loop, for which the interaction field and the capture geometry are intrinsically three-dimensional, and applied it to zirconium [36].

For a mobile species  $\alpha \in \{i, v\}$  the flux is

$$\mathbf{J}_\alpha = -D_\alpha \left[ \nabla C_\alpha + \frac{C_\alpha}{k_B T} \nabla U_\alpha(\mathbf{r}) \right], \quad (\text{B.3})$$

which is Eq. (22) with the interaction energy supplied by continuum elasticity. The second term is the elastic drift: capture is enhanced where  $U_\alpha$  decreases toward the loop and suppressed where it increases. The interaction itself is

$$U_\alpha(\mathbf{r}) = -P_{ij}^\alpha \epsilon_{ij}^L(\mathbf{r}), \quad (\text{B.4})$$

with  $P_{ij}^\alpha$  the elastic dipole tensor of the point defect and  $\epsilon_{ij}^L$  the loop strain field; in an isotropic dilatational approximation this reduces to

$$U_\alpha(\mathbf{r}) \simeq -\Delta\Omega_\alpha p_L(\mathbf{r}), \quad (\text{B.5})$$

$\Delta\Omega_\alpha$  being the defect relaxation volume and  $p_L$  the hydrostatic part of the loop stress.

The exact diffusion problem around a finite loop is awkward because the absorbing region is toroidal and the potential strongly position-dependent. Woo adopted the effective-medium framework of Brailsford, Bullough and Hayns, replacing the detailed boundary condition on the torus by a rate-limiting condition on an enclosing sphere of radius  $r_s$ , while retaining the elastic drift field outside it explicitly. The steady state  $\nabla \cdot \mathbf{J}_\alpha = 0$  then reads

$$\nabla \cdot \left[ D_\alpha \left( \nabla C_\alpha + \frac{C_\alpha}{k_B T} \nabla U_\alpha \right) \right] = 0, \quad (\text{B.6})$$

and its solution gives the total flux into the sink, hence  $Z_{i,L}$  and  $Z_{v,L}$  and the bias Eq. (B.1). **The conceptual result is that the bias need not be an empirical parameter:** loop geometry and Burgers vector give the elastic field, which gives  $U_i$  and  $U_v$ , which give the drift, the fluxes, the two efficiencies and their ratio. That chain is what Section 2.5 relies on when it generates every efficiency from the same diffusion tensors that transport the species.

## B.2 Woo (1982): what breaks the symmetry between characters

The 1981 calculation leaves one question open. If the point defect is represented as an isotropic center of dilatation and the loop within isotropic linear elasticity, then vacancy and interstitial loops of equal size carry essentially the *same* bias,

$$B_{vL} \simeq B_{iL} > 0. \quad (\text{B.7})$$

This is the elastic counterpart of the diffusional degeneracy Eq. (21), and it creates the same difficulty: a positive SIA bias at both loop types makes interstitial loops grow and vacancy loops shrink, yet the experiments of Section 1.2 and Appendix A show stable and growing vacancy loops in irradiated Zr and Ti. Earlier explanations, the effective-medium treatment among them, invoked a *size-dependent* bias differential — small loops less biased than large ones, so vacancies flow to the smaller population — which requires vacancy loops to be systematically smaller than neighboring interstitial loops, for which there is little evidence. What is needed instead is an *intrinsic* difference at equal size.

Woo's resolution is that the migrating defect should not be represented as an isotropic dilatation center at all [37]. The elastic dipole tensor at the migration *saddle point*,  $P_{ij}^s(\hat{\mathbf{h}})$  with  $\hat{\mathbf{h}}$  the jump direction, can be strongly anisotropic, and the interaction

$$E_s(\mathbf{r}, \hat{\mathbf{h}}) = -P_{ij}^s(\hat{\mathbf{h}}) \epsilon_{ij}^L(\mathbf{r}) \quad (\text{B.8})$$

then retains the coupling between the *shear* components of the defect dipole and those of the loop strain field, which Eq. (B.5) discards. An isotropic saddle-point tensor returns the harmonic form and no intrinsic character difference; an anisotropic one permits  $B_{vL} \neq B_{iL}$ .

The symmetry breaking follows from a sign. The strength  $Q$  of an infinitesimal edge loop reverses with character,  $Q = +Q_0$  for vacancy and  $-Q_0$  for interstitial, and the angular-averaged anisotropic contribution scales as  $\langle \beta E_s \rangle \propto PQ(1 - p_1)$  with  $p_1$  a normalized eigenvalue of the saddle-point dipole tensor. Hence

$$\boxed{A^{j,a_i L} = -A^{j,a_v L}, \quad j = i, v,} \quad (\text{B.9})$$

where  $A^{j,L}$  is the anisotropic shape contribution controlling the reaction radius for defect  $j$  at loop  $L$ . **Equation (B.9) is the symmetry-breaking statement of this appendix:** an anisotropic defect configuration that contributes attractively at one loop character contributes repulsively at the other. Saddle-point configurations fall into three classes —  $M$ , elongated along the jump direction;  $F$ , flattened along it; and  $N$ , approximately isotropic — and an  $M$ -type defect fits the strain field of a vacancy loop more favourably than an  $F$ -type, while at an interstitial loop the preference reverses.

Carried into a drift-diffusion calculation of the kind above, the dilute-sink reaction radius is

$$R_a = |k_0 \beta PQ|^{1/3} S^{-1}(A, B), \quad S(A, B) = \int_1^\infty t^{-2} \exp(At^{-3} - Bt^{-6}) dt, \quad (\text{B.10})$$

in which  $A$  carries the character-sensitive saddle-point shape contribution and  $B$  collects the terms even under a reversal of character. Because  $A^{j,a_i L} = -A^{j,a_v L}$ , loops of *identical size* acquire different reaction radii and therefore different biases. The analysis identifies a class of materials satisfying

$$\boxed{B_{iL} > 0, \quad B_{vL} < 0,} \quad (\text{B.11})$$

— interstitial loops capturing SIAs preferentially, vacancy loops capturing vacancies preferentially, so that both populations grow without any systematic difference in size — and proposes zirconium and titanium as members of it. The prediction was not carried through quantitatively for Zr, because the saddle-point dipole tensors were unavailable; the worked examples were Cu and Fe, for which both characters come out positively biased.

### B.3 March-Rico and Wirth (2023): the atomistic realization

The same physical question was revisited at the atomistic scale [38], with the experimentally observed coexistence of Appendix A as the motivation. Conventional cluster dynamics assigns the same geometrical capture radius to both  $\langle a \rangle$  characters because they share a Burgers-vector magnitude — exactly the degeneracy Eq. (21) records — and the defect-loop interaction was therefore computed directly rather than assumed.

Molecular statics with the BMD19.1 potential was used to build interstitial  $\langle a \rangle$  loops, vacancy  $\langle a \rangle$  loops, vacancy  $\langle c \rangle$  loops and the faulted basal-pyramid configurations of Section 1.2. Introducing a vacancy or an SIA at many positions and relaxing gives the binding-energy map

$$E_b^{\alpha,L}(\mathbf{r}) = E_f^{\alpha,\text{bulk}} - E_f^{\alpha,L}(\mathbf{r}), \quad (\text{B.12})$$

and the accompanying atomic stress fields show vacancy and interstitial loops to produce stress states of approximately *opposite* character — short-range radial compression with longer-range tension normal to the loop for one, the corresponding trends with opposite signs for the other. That opposition is the atomistic counterpart of the sign reversal of  $Q$  in Eq. (B.9).

Two definitions of capture follow from the same maps, and they do not agree. Under *spontaneous* capture — the region within which a defect is absorbed without long-range thermally assisted drift — SIAs are preferentially captured by every loop type,  $Z_i > Z_v$ , which reproduces the conventional dislocation-bias picture and the degeneracy Eq. (B.7). It cannot explain coexistence, for the same reason Section 1.3 gives: enlarging every reaction volume alike does not separate characters. Under *thermal-drift* capture the capture distance is fixed by comparing the binding interaction with thermal energy,

$$E_b^{\alpha,L}(r_c) \sim \frac{3}{2} k_B T, \quad r_c = r_c(\alpha, L, R_L, T), \quad (\text{B.13})$$

and the landscape then produces an inherent preference for defects of the *same* character as the loop,

$$\boxed{a_i L \rightarrow I, \quad a_v L \rightarrow V,} \quad (\text{B.14})$$

so that  $r_c^{i,a_iL} > r_c^{v,a_iL}$  and  $r_c^{v,a_vL} > r_c^{i,a_vL}$ , and therefore  $B_{a_iL} > 0$  with  $B_{a_vL} < 0$  — **exactly the sign pattern Eq. (B.11) predicted**. Two  $\langle a \rangle$  loops of essentially the same crystallography carry very different capture efficiencies because their elastic fields have opposite signs, and it is this separation of the four radii  $r_c^{i,a_iL}$ ,  $r_c^{v,a_iL}$ ,  $r_c^{i,a_vL}$ ,  $r_c^{v,a_vL}$  that Eq. (23) carries into the present formulation.

For a loop large enough that the reaction volume is approximately toroidal, the capture radius converts to a sink strength as

$$Z^T = \frac{4\pi^2 R_L \rho_L}{\ln \left[ 1 + \frac{8R_L}{r_1 + r_d} \right]}, \quad (\text{B.15})$$

with  $r_1$  and  $r_d$  the toroidal capture dimensions; since  $r_d^i \neq r_d^v$  one obtains  $Z_{i,L} \neq Z_{v,L}$  and a bias

$$B_L = \frac{Z_{i,L}}{Z_{v,L}} - 1 \quad (\text{B.16})$$

parameterized from atomistic energetics rather than from a continuum model.

## B.4 How the three fit together, and where they do not

Table 20 sets the three side by side. All address one mechanism — the loop stress field modifies the point-defect interaction, which biases transport and capture, so that  $Z_i \neq Z_v$  — and none treats the loop as a geometrical absorbing surface. They differ in the level at which the interaction is obtained: continuum elasticity giving  $U_{\alpha,L}(\mathbf{r})$  and a drift–diffusion solution, against an interatomic potential giving  $E_b^{\alpha,L}(\mathbf{r})$  and a capture radius.

Table 20: Three routes to the point-defect capture bias of a dislocation loop in zirconium. The last row is the reason the three belong in one appendix.

| Feature              | Woo (1981) [36]                           | Woo (1982) [37]                             | March-Rico and Wirth (2023) [38]                     |
|----------------------|-------------------------------------------|---------------------------------------------|------------------------------------------------------|
| Description          | Continuum elastic interaction             | Continuum elastic interaction               | Atomistic defect–loop interaction                    |
| Loop                 | Continuum finite loop                     | Infinitesimal edge loop, strength $\pm Q_0$ | Explicit relaxed atomic structure                    |
| Point defect         | Continuum elastic dilatation center       | Elastic dipole tensor at the saddle point   | Explicit vacancy or SIA                              |
| Operative anisotropy | None (isotropic defect)                   | Saddle-point anisotropy <i>shape</i>        | Defect–loop anisotropy <i>binding</i>                |
| Interaction          | $U_{\alpha,L}(\mathbf{r})$ , Eq. (B.4)    | $-P_{ij}^s \epsilon_{ij}^L$ , Eq. (B.8)     | $E_b^{\alpha,L}(\mathbf{r})$ , Eq. (B.12)            |
| Capture              | Drift–diffusion on a rate-limiting sphere | Drift–diffusion reaction radius Eq. (B.10)  | Thermal-drift radius Eq. (B.13)                      |
| Temperature          | Through $U/k_B T$                         | Through $\beta P Q$                         | Through $\frac{3}{2} k_B T$ in the capture criterion |
| Near-core structure  | Not resolved                              | Not resolved                                | Resolved by the potential                            |
| Result for Zr        | $Z_i \neq Z_v$ , bias <i>calculable</i>   | $B_{iL} > 0$ , $B_{vL} < 0$ <i>proposed</i> | $B_{a_iL} > 0$ , $B_{a_vL} < 0$ <i>obtained</i>      |

The distinction matters most for coexistence. Both  $\langle a \rangle$  characters share a Burgers-vector magnitude and a prismatic habit, so a geometrical capture model assigns them the same radius; the atomistic calculations show  $r_c^{\alpha,a_iL} \neq r_c^{\alpha,a_vL}$ , and it is the opposite stress states that do it. That is the same-type preference which permits both populations to grow, and the mechanism Section 1.3 invokes to remove the degeneracy Eq. (21).

**The agreement in sign is evidence, not proof**, and the reason is that the two mechanisms act on different quantities. The elastic mechanism acts at the migration *saddle point*: for a jump,

$$E_m(\mathbf{r}, \hat{\mathbf{h}}) = E_m^0 + E_s(\mathbf{r}, \hat{\mathbf{h}}) - E_g(\mathbf{r}), \quad \Gamma_{\hat{\mathbf{h}}} \propto \exp \left[ -\frac{E_m(\mathbf{r}, \hat{\mathbf{h}})}{k_B T} \right], \quad (\text{B.17})$$

so the critical object is  $P_{ij}^s$ . The atomistic calculations instead supply relaxed *ground-state* binding maps, and the thermal-drift criterion rests on the depth and extent of that landscape. One modifies a

migration barrier, the other a binding energy and a capture volume; they are related through the same elastic defect–strain coupling but are not mathematically identical. The 2023 result should therefore be read as a modern atomistic confirmation and quantitative extension of the 1982 concept rather than as an independent discovery — the symmetry-breaking ingredient, an anisotropic point-defect response combined with opposite loop strain fields, was already identified — but the attribution of the measured same-type preference to saddle-point anisotropy specifically remains untested.

## B.5 A unified formulation, and the calculation that would settle it

The three are complementary rather than competing, and suggest a multiscale decomposition. Far from the core the interaction should be describable by anisotropic elasticity,

$$E_{\text{el}}^{\alpha,L}(\mathbf{r}) = -P_{ij}^{\alpha} \epsilon_{ij}^L(\mathbf{r}), \quad (\text{B.18})$$

while near it, nonlinear elasticity, atomic reconstruction and the detailed SIA configuration matter; the difference  $E_{\text{core}}^{\alpha,L} = E_{\text{MS}}^{\alpha,L} - E_{\text{el}}^{\alpha,L}$  is what molecular statics supplies, so that

$$E_{\text{int}}^{\alpha,L}(\mathbf{r}) = E_{\text{anisotropic elasticity}}^{\alpha,L}(\mathbf{r}) + E_{\text{core}}^{\alpha,L}(\mathbf{r}). \quad (\text{B.19})$$

This keeps the correct long-range behavior and the efficiency of continuum elasticity together with the accurate short-range capture physics of the atomistic calculation. The capture coefficient of a microstructure-evolution model should then not be a universal empirical bias factor but

$$K_{\alpha L} = K_{\alpha L}(R_L, \mathbf{b}_L, \mathbf{n}_L, s_L, T, \mathbf{D}_{\alpha}, P_{ij}^{\alpha}), \quad (\text{B.20})$$

with  $s_L = \pm 1$  the loop character, entering a spatially resolved growth law as

$$\dot{N}_L = s_L [K_{iL} C_i(\mathbf{x}_L) - K_{vL} C_v(\mathbf{x}_L)]. \quad (\text{B.21})$$

Equation (B.20) is the general form of which the efficiencies of Section 2.5 are a particular parameterization: the effective-medium treatment supplies the long-range elastic contribution through  $\mathbf{D}_{\alpha}$  and  $P_{ij}^{\alpha}$ , the saddle-point analysis the character dependence through  $s_L$ , and the atomistic radii the near-loop magnitudes.

Carried to its conclusion, the transport problem becomes  $\nabla \cdot \mathbf{J}_{\alpha} = 0$  with

$$\mathbf{J}_{\alpha} = -\mathbf{D}_{\alpha}(\mathbf{r}) \left[ \nabla C_{\alpha} + \frac{C_{\alpha}}{k_B T} \nabla E_g^{\alpha,L}(\mathbf{r}) \right], \quad (\text{B.22})$$

which differs from Eq. (B.3) in one essential respect: *the diffusivity is itself a field*, because the barrier of Eq. (B.17) depends on position and jump direction. The equilibrium binding emphasized by the atomistic work enters through  $E_g$ , and the saddle-point anisotropy through  $\mathbf{D}_{\alpha}(\mathbf{r})$ .

A direct atomistic test in  $\alpha$ -Zr would therefore need  $E_g(\mathbf{r})$ ,  $E_s(\mathbf{r}, \hat{\mathbf{h}})$  and  $E_m = E_s - E_g$  for vacancies and SIAs around both  $\langle a \rangle$  characters. Nudged-elastic-band or equivalent saddle-point calculations would then determine whether the same-type thermal-drift bias arises primarily from the saddle-point anisotropy of Eq. (B.9) or from additional core-scale effects. **That calculation has not been done**, and until it is, the sign agreement in the last row of Table 20 is a consistency check and not an attribution.

## B.6 Summary

Point-defect capture by a dislocation loop depends not only on the magnitude of the loop elastic field but on the *character and anisotropy of the defect-loop interaction*. Three results follow one from another. The bias is calculable rather than empirical: the elastic potential produces a drift component of the defect flux and the resulting drift–diffusion problem determines  $Z_i$  and  $Z_v$ . The classical isotropic-defect approximation makes that bias the same for both loop characters,  $B_{iL} \simeq B_{vL} > 0$ , and saddle-point shape anisotropy is what breaks the degeneracy, permitting  $B_{iL} \neq B_{vL}$  and specifically the zirconium-relevant regime  $B_{iL} > 0$ ,  $B_{vL} < 0$ . Molecular statics subsequently obtained precisely that same-type capture tendency for finite  $\langle a \rangle$  loops in  $\alpha$ -Zr, in a form — defect-specific capture radii

— that can be carried into a cluster-dynamics model, which is how it enters Eq. (23) here and the mean-field study of March-Rico et al. [20].

The attractive formulation for the present model is the multiscale one: anisotropic elasticity for the long-range defect-loop interaction, molecular statics for the near-core correction, retaining the physical basis of the elastic drift theory while incorporating the atomistic capture asymmetry. That is the form Eq. (B.20) anticipates, and the one a fully self-consistent capture kernel would take. It is also the result on which the character-splitting factor of Eq. (23), and with it the coexistence argument of Section 1.3, ultimately rests.

## C Current code structure

The formulation of Section 2.3 is the target. This appendix states plainly what the present code carries against it, because the difference is a matter of record rather than of interpretation. Nothing here is part of the formulation; it is the record of the implementation as it stands, and every place where the two diverge is named.

### C.1 The implemented family set, and the two loop models

ZrMicro-3d carries **four** immobile families, not eight: one basal vacancy family and three prismatic interstitial variants,

$$\mathcal{F}_{\text{impl}} = \{(v, c), (i, a_1), (i, a_2), (i, a_3)\}, \quad (\text{C.1})$$

each with a number density  $N_k$  (per  $b^3$ ) and a stored content  $c_k$ . This is the reduction of Eq. (64) obtained by setting  $\chi = 1$  and  $J_{vL}^{a_k} = 0$  — which removes the three prismatic vacancy families — and  $\nu_{\text{uf}} = 0$  with  $c_p$  merged into  $c_f$ , which collapses the basal pair to the single family carrying  $\mathbf{b} \cdot \mathbf{n} = c/2$ . The prismatic vacancy variants, the faulted/perfect split and the unfaulting reaction are *formulated but not yet implemented*; Appendix C.5 states what completing them requires.

Within the implemented four, the slow step’s *model* is selectable and is orthogonal to the solver and to the route:

Table 21: The two immobile formulations carried by the code. The integrator is chosen independently.

|                         | loop_model = 0 — legacy                                                                                                                                         | loop_model = 1 — self-consistent                                                                           |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| families                | $iL, aiL, vL, avL$ (aligned / non-aligned split)                                                                                                                | $c, a_1, a_2, a_3$ — the CD block of the spatial code                                                      |
| capture                 | phenomenological $Z_I^a = 1 + \delta_i$ , $Z_V^a = 1 - \delta_v$ and the basal counterparts; <i>one</i> interstitial efficiency for $C_i, C_{2i}, C_{3i}$ alike | Woo, Eq. (68), built from the same $p_m$ that sets the diffusion tensor; <i>one efficiency per species</i> |
| bridge to the 3-D block | a lumping followed by a variant split, both lossy                                                                                                               | an identity, up to the $1/\Omega$ unit conversion; the round trip is exact                                 |
| status                  | <b>the fitted model</b> ; bit-identical to the pre-change binary                                                                                                | consistent with the fast solve, <b>not yet calibrated</b>                                                  |

Table 21 is the operational form of the self-consistency argument of Section 1.3. In `loop_model = 0` the diffusion tensor reaches loop growth *only* through the mobile concentrations, never through the capture efficiencies, which are workbook constants: the co-growth criterion Eq. (20) then predicts the spatial solver’s behavior but predicts a coupled march only indirectly. In `loop_model = 1` the criterion applies to the coupled march directly, because the same  $p_m$  generates both. Two consequences must be stated together with any result:

- **The legacy path is bit-identical to the fitted binary**, and that is why its right-hand-side assembly is kept verbatim inside its branch. Regrouping it into an algebraically identical form was measured to move the tenth significant digit, and the 28-parameter set was fitted against those exact operations.
- **The parameter set is fitted against `loop_model = 0` and is therefore stale for `loop_model = 1`**. It is equally stale with respect to two corrections made since: the atomic



volume  $\Omega$ , which had been carrying  $V_{\text{cell}}/4$  — four atoms in a primitive cell holding two — and the  $\langle c \rangle$  Burgers vector, which is  $\mathbf{b} \cdot \mathbf{n} = c/2$  and had been entered in three mutually inconsistent ways across the three codes. Both corrections move the loop densities by factors of order two, and they compound. A refit is required before any comparison with experimental densities.

## C.2 Immobile equations as implemented

For the four families of Eq. (C.1), the geometry and the evolution equations the code integrates are

$$m_k = \frac{c_k}{N_k \Omega}, \quad r_k = \sqrt{\frac{m_k \Omega}{\pi b |b_k|}}, \quad S_k = 2\pi r_k N_k, \quad (\text{C.2})$$

with the sigmoidal bi-pyramid/loop interpolation of Eqs. (129) and (130) applied to the vacancy family — which Section 2.11 supersedes, the compact state having become the species  $c_0$  and the interpolation having been dropped — and

$$\frac{dN_k}{dt} = \underbrace{\frac{G \varepsilon_k}{n_k^{\text{nuc}} \Omega}}_{\text{cascade nucleation}} - \underbrace{(\nu_{LL} \phi_{LL} + \nu_{LN} \phi_{LN}) N_k}_{\text{coalescence}} - \underbrace{\frac{N_k}{\tau_{vL}} \delta_{k,v}}_{\text{dissolution}}, \quad (\text{C.3})$$

$$\frac{dc_k}{dt} = \underbrace{G \varepsilon_k}_{\text{cascade seed}} + \underbrace{\dot{g}_k}_{\text{absorption}} - \underbrace{\nu_{LN} \phi_{LN} c_k}_{\text{loop-network}} - \underbrace{\frac{n_k^{\text{nuc}} N_k \Omega}{\tau_{vL}} \delta_{k,v}}_{\text{annealing release}} - \underbrace{\alpha_v(m_k) \delta_{k,v}}_{\text{emission}}, \quad (\text{C.4})$$

where the absorption flux is the species-resolved sum of Eq. (71),

$$\dot{g}_k = \sum_m \bar{D}_m Z_{km} S_k c_m |m| \text{sgn}_k(m), \quad (\text{C.5})$$

gated on the shrinking channel by Eq. (75), and the Avrami factors are Eq. (117). The annealing release term in Eq. (C.4) is a lifetime channel proportional to  $N_k$ , which Section 2.9 supersedes: loop lifetimes have been replaced by the peripheral emission Eq. (110), whose binding energy Eq. (109) is computed from the loop energy and which acts on content alone. It is the single term in Eq. (C.4) that most departs from the present formulation, and the one that controlled the mean  $\langle c \rangle$  size in the lifetime model. The DAD efficiencies are generated from one anisotropy parameter per mobile species,

$$Z_m^{(c)} = Z_m^0 p_m, \quad Z_m^{(a)} = \frac{Z_m^0}{2} (p_m + p_m^{-2}), \quad (\text{C.6})$$

which is Eq. (19) again, and is generated from the same source as the migration-energy tensor by Eq. (41), so a change to  $p_m$  cannot alter one without the other.

## C.3 Mapping the rate coefficients between the two codes

MoDELib2-NNL builds its binary reaction coefficients as

$$K_{ab} = p_{ab} \frac{4\pi(r_a + r_b)(D_a + D_b)}{\Omega}, \quad (\text{C.7})$$

whereas the mean-field kinetics of Eq. (55) use  $K_{ab} = f_{ab}(\omega_a + \omega_b)$ . Equating the two and using Eq. (50) gives the prefactors that must be supplied to the spatial code:

$$p_{ab} = f_{ab} \frac{z_c}{a^2} \frac{\Omega}{4\pi(r_a + r_b)} \frac{D_a^{0D} + D_b^{0D}}{D_a^{3D} + D_b^{3D}} \quad (\text{C.8})$$

The final ratio is unity except for the di- and tri-interstitials, which transport at the monomer diffusivity for the reason given in Appendix C.4. The mapping is directly verifiable, because MoDELib2-NNL prints its assembled interaction matrices in  $\text{s}^{-1}$  at start-up; Table 22 compares them.

Table 22: Assembled second-order interaction coefficients at 573 K, spatial code against the mean-field expressions of Eq. (55).

| entry                     | assembled [s <sup>-1</sup> ] | expression [s <sup>-1</sup> ]                            | difference |
|---------------------------|------------------------------|----------------------------------------------------------|------------|
| ( <i>v</i> , <i>i</i> )   | $2.526876 \times 10^7$       | $\frac{1}{2}(\omega_i + \omega_v) = 2.52560 \times 10^7$ | 0.05 %     |
| ( <i>i</i> , <i>i</i> )   | $-2.021161 \times 10^8$      | $-4\omega_i = -2.020216 \times 10^8$                     | 0.05 %     |
| ( <i>v</i> , 2 <i>i</i> ) | $6.940722 \times 10^3$       | $\omega_{2i} + \omega_v = 6.93473 \times 10^3$           | 0.09 %     |

## C.4 Changes made to the spatial code

The `ZrMicro-3d` branch modifies `MoDElib2-NNL` in three categories. The distributed `Zr4.txt` material file is left untouched; the reconciled definition is a separate file, `Zr3d_ghoniem.txt`.

**(a) Physics that was absent.** The immobile population was not simplified in the baseline — it was absent. `iSize` was 0, `solveImmobileClusters()` was an empty function body, there was no immobile integrator of any kind, and the source carried a literal `// Missing immobile sinks` placeholder.

- `iSize` raised from 0 to 8, enabling four immobile families, each with a density and a content.
- `solveImmobileClusters()` implements Eqs. (117), (C.3) and (C.4).
- A new evaluator supplies the loop-absorption sink acting on the mobile species. The grain-boundary contribution is deliberately excluded, since it is imposed spatially by the Dirichlet condition; the network dislocation sink reuses the existing `otherSinks` mechanism.
- `loopDADbias()` implements Eq. (C.6).
- The finite-element node coordinates are written once per run, without which the fields cannot be located in space: the cluster-dynamics trial functions live on second-order elements, so the field rows do *not* correspond to the mesh vertices.
- An optional key makes the solve also write the *superposed* mobile field  $c_{\text{FEM}} + c_{\text{DD}}$ . The spatial code solves the mobile species by superposition, and the CD block of the output is the *corrective* part  $c_{\text{FEM}}$  alone; a figure drawn from it is smooth by construction however many discrete loops are present, because the loop-localized depletion lives entirely in  $c_{\text{DD}}$ .

### (b) Corrections.

- **Binding energies read before initialization.** The routine assembling the first-order reaction matrix read the binding-energy array to build the dissociation rates, but that array was declared *after* the matrix in the class. Since C++ initializes members in declaration order this was an uninitialized read: the Boltzmann factor evaluated to unity instead of  $3.8 \times 10^{-9}$ , inflating the dissociation rate to the bare attempt frequency and suppressing  $C_{2i}$  and  $C_{3i}$  by a factor  $2.4 \times 10^6$ . It was invisible in the distributed calibration because every binding energy was set to 5 eV, for which the factor is negligible whether applied or not.
- **Atomic volume.** The polycrystal class returned `det(lattice basis)`, the volume of the lattice *cell*, which for a hexagonal lattice contains two atoms. Every content–radius–density conversion inherited the error; an optional material key now supplies  $\Omega$  explicitly.
- **Consistency of the diffusion operator.** The flux is formed as  $-D/\Omega$  and multiplied back by an atomic volume in the weak form. These were the same quantity until the cluster-dynamics  $\Omega$  became independently settable, after which the diffusion operator carried a spurious factor of 3.98 relative to every reaction term. Both now use the same  $\Omega$ .
- **Stiff loss terms.** Vacancy-loop dissolution and Avrami coalescence were integrated explicitly, and both violate the stability limit at the dose step used ( $\Delta t/\tau_{vL} \approx 35$  and  $\nu_{LN}\phi_{LN}\Delta t \approx 3.6$ ), so each sub-step overshoot and clamped to the positivity floor. Both are first order in their own variable and are now integrated implicitly.

- **Crystallography.** The immobile Burgers vectors are multiplied by the lattice basis, that is, they are lattice components; Cartesian components had been supplied, giving  $|b| = 1.0, 0.753, 1.197$  for the three prismatic families instead of all unity. Separately, the hexagonal lattice basis took  $c/a$  from the ideal  $\sqrt{8/3}$ ; it now reads the physical  $c$  from the material file, and the  $\langle c \rangle$  loop is built with  $\mathbf{b} \cdot \mathbf{n} = c/2$  in every one of the four places that value appears. **A hard-coded ideal  $c/a$  elsewhere made that change a silent no-op**, and the failure mode is worth recording: the basal-loop generator chose its branch by comparing the plane spacing against  $\sqrt{8/3}$ , so with the physical ratio present the spacing matched neither branch and every requested basal loop vanished with no message and no non-zero exit code. The ratio is now taken from the lattice basis itself, and the fall-through throws.
- **Units.** The minimum stable radius was compared in meters against radii in units of  $b$ , leaving the shrinking gate permanently open; and the loop number density was accumulated per atom while the geometric helpers expect it per  $b^3$ , inflating the inferred mean loop size by a factor 2.82.

(c) **Parameter reconciliation.** Table 23 lists the material-file entries that differ from the distributed values. Every original value is preserved in a comment in the file.

Table 23: Material parameters changed to reconcile the two codes.

| quantity             | distributed                        | reconciled                           | basis                                          |
|----------------------|------------------------------------|--------------------------------------|------------------------------------------------|
| surviving efficiency | 0.01                               | 1.0                                  | defects lost only to in-cascade clustering     |
| cascade fractions    | 1, 0.6, 0.3212, 0.0788             | 0.995, 0.9779, 0.005, 0.00333        | Eq. (60)                                       |
| $E^m$ (SIA)          | 0.09 eV                            | 0.759101 eV                          | fitted value                                   |
| $E^m$ (vacancy)      | 0.9 eV                             | 1.2 eV                               | fitted value                                   |
| $D_0$                | $4.96/8.19 \times 10^{-10}$        | $5.21645 \times 10^{-7}$             | $D_0 = a^2 \nu$                                |
| $E_v^f$              | 1.5 eV                             | 1.6 eV                               | gives $C_v^{\text{eq}} = 8.45 \times 10^{-15}$ |
| $\Omega$             | $4.78 \times 10^{-29} \text{ m}^3$ | $2.3266 \times 10^{-29} \text{ m}^3$ | Eq. (2)                                        |
| reaction prefactors  | 2.856, 1, 0                        | Eq. (C.8)                            | Table 22                                       |
| $E_b$ (2i, 3i)       | 5 eV                               | 0.957, 2.0 eV                        | fitted values                                  |
| network sink         | 0                                  | $\rho_N, Z_N \rho_N$                 | Eq. (56)                                       |

**Cluster transport.** The di- and tri-interstitials transport at the *monomer* diffusivity, not at the diffusivity implied by  $E_{2i}^m = 1.36 \text{ eV}$ . This follows the mean-field model, which is inconsistent on the point but unambiguous about where each frequency is used: the network-sink and loop-absorption terms for clusters both use  $\omega_i$ , while  $\omega_{2i}$  appears only in the cluster-cluster reactions, and those rates are recovered exactly through the last factor of Eq. (C.8). The alternative — using  $E_{2i}^m$  for transport — makes  $D_{2i} \sim 5 \times 10^{-6} D_i$  and, since Eq. (27) is a pure steady-state system with no mass term, leaves the cluster block of the Newton matrix singular.

**Loop sink convention.** The mean-field model applies the  $\langle c \rangle$  radius prefactor  $\ell_c$  to all four families and carries an additional fitted factor  $Q$  on the  $\langle c \rangle$  channel, whereas the spatial code builds  $r_k$  from each family’s own  $|b_k|$ . A per-family multiplier  $\zeta_k^{\text{sink}}$  in Eq. (72) restores the mean-field convention exactly: 0.2915 for  $\langle c \rangle$  and 0.7923 for  $\langle a \rangle$ . It is applied identically to the sink seen by the mobile species and to the growth flux it causes, mirroring the mean-field code where both derive from a single prefactor. One near-identity is worth recording:  $\lambda_a/\lambda_c = \sqrt{b_c/b_a} = 1.2627$  and the fitted  $\langle a \rangle$  sink scale 0.7923 are reciprocals to 0.05%, so the spatial code’s sink calibration silently undoes the mean-field model’s use of  $\ell_c$  for  $\langle a \rangle$  sinks.

## C.5 Status of the expanded family set

Completing Eq. (64) in the code requires four changes, none of which alters the bilinear structure of Eq. (71):

1. **Three prismatic vacancy families**, raising `iSize` from 8 to 16 and the immobile state at each node from eight to sixteen components; the pyramid of item 5 adds two more, for eighteen.

2. **Character-resolved capture efficiencies:** the factor  $X_{s,m}$  of Eq. (68), taken from the thermal-drift capture radii through Eq. (23), with  $\chi = 1$  reproducing the present model bit-for-bit as the regression test.
3. **A vacancy-loop nucleation current on the prismatic families,**  $J_{vL}^{ak} = w_k^v G_{avL} / n_{avL}^{\text{nuc}}$ , fed from the in-cascade vacancy cluster yield and not from homogeneous clustering.
4. **The faulted/perfect basal split and the unfaulting transfer** Eq. (80), with  $\nu_{\text{uf}} = 0$  recovering the present single basal family.
5. **The basal stacking-fault pyramid**  $c_0$  as a separate species with its own cascade yield  $\varepsilon_{\text{sfp}}$ , dissolution lifetime  $\tau_{\text{sfp}}$  and collapse transfer Eq. (80), adding two further state components per node and the compact sink form Eq. (74). Setting  $\varepsilon_{\text{sfp}} = 0$  recovers the present model, and Eq. (128) is the assertion that makes the addition verifiable independently of any parameter value.

Each has a regression test that returns the present model exactly, which is the property that makes the expansion verifiable. The cost is calibration: the existing parameter set was fitted with the prismatic families purely interstitial and with one lumped basal family, so both additions redistribute stored content between families and the fit must be redone before the model is compared with measured loop densities.

## D Code implementation plan

Appendix C is the record of what the code carries today; the body of this report is what it should carry. This appendix is the path between them, written as an ordered sequence of steps with a test at the end of each. It is a plan for the implementation, not a summary of the physics, and it is separated from Appendix C for the same reason that appendix is separated from Section 2: the formulation should not have to be re-read to find out what state the software is in.

**The ordering principle.** Every step below changes exactly one thing and carries a **regression test that returns the previous step bit-for-bit** at a stated parameter value. That is what makes the sequence safe to execute: a step that fails its regression is wrong, independently of whether its new physics looks plausible, and a step that passes can be left in place while the next is attempted. The formulation was written with these degeneracies in it deliberately —  $\chi = 1$ ,  $\varepsilon_{\text{sfp}} = 0$ ,  $\nu_{\theta} = 0$ ,  $\Delta_k = 1$  — and they are the reason the expansion is verifiable at all rather than being a rewrite that must be judged as a whole.

**Two things executing Phase I corrected in this plan.** Both concern the regressions rather than the physics, and both were measured rather than argued.

**A parameter set to zero does not always recover the predecessor's bits.** Step 1's stated regression is  $\varepsilon_{avL} = 0$ . That recovers the previous *physics* exactly and not the previous *arithmetic*, for two independent reasons. A family carrying no nucleation still sits at the concentration floor, and a family at the floor is **not inert**: the coalescence climb speed is formed as  $\dot{r} = \ell \Phi^{\text{gain}} / 2\sqrt{n}c$  while  $\Phi^{\text{gain}}$  itself carries  $\sqrt{n}c$  through the sink prefactor, so the floor *cancels* and an empty family reports a finite climb velocity, feeding  $c_{\rho}(2\pi/\Omega) r \dot{n}_{LN}$  into  $\rho_N$  — where the  $1/\Omega$  turns an apparent  $10^{-20}$  into a real source. Measured before it was guarded: the basal loop content came out  $387\times$  different. And even guarded, adding components to the implicit block changes CVODE's step sequence *when those components are identically zero*, because the WRMS error norm divides by the block size and padding 9 to 17 shrinks it by  $\sqrt{9/17}$ . The residual tracks the tolerance —  $1.05 \times 10^{-6}$ ,  $1.16 \times 10^{-8}$ ,  $1.56 \times 10^{-10}$  at  $\text{rtol} = 10^{-6}, 10^{-8}, 10^{-10}$  — which is what identifies it as discretization rather than physics. **The exact regression for step 1 is therefore the family COUNT, not the cascade yield.** Step 2's  $\chi = 1$  *is* bit-for-bit, because it adds a factor and not a state.

**Habit and polarity stop coinciding at step 1, in three places at once.**  $A_{h(k)}$  of Eq. (68) selects on the habit plane, and before step 1 the one vacancy family was also the one basal family — so every implementation selected it with a *polarity* test instead, correctly by coincidence. Adding prismatic vacancy variants would have handed them the *basal* capture efficiency, silently, both rows

being populated and neither obviously wrong: in the 0-D growth term, in `ClusterDynamicsFEM`'s immobile solve, and in the  $k_m^2$  that `ImmobileSinks` hands the fast solve. The same separation is required for the length scale, the coalescence coefficients and the minimum-radius gate.

**Three assertions that must hold at every step.** These are not step-specific and should be wired in once, at the start, as run-time checks that can be enabled in a debug build:

1. **Atom balance.** Equation (128) and its interstitial counterpart, evaluated per substep, must close to solver tolerance for arbitrary values of every rate coefficient. Table 5 is the term-by-term map of what pairs with what.
2. **Non-negativity.**  $n_{sL}^k$ ,  $c_{sL}^k$  and  $q_{sL}^k$  are non-negative, and  $\Delta_k = q_{sL}^k n_{sL}^k / (c_{sL}^k)^2 \geq 1$  by the Cauchy–Schwarz inequality. A  $\Delta_k < 1$  is a solver failure, not a physical state, and should abort rather than be clamped.
3. **Self-consistency of the efficiencies.** The  $Z_{sk,m}$  used in Eq. (71) and the  $Z_{sk,m}$  used in Eq. (38) must be the same object in memory, not two evaluations of one formula. This is the claim of Section 2.5 and the cheapest possible way to keep it true is to make it structurally impossible to violate.

## D.1 Phase I — the family set

**Step 0: lock the baseline.** Before anything changes, record a reference trajectory of the present model — a fixed case, fixed seed, fixed dose grid — and hash it. **Goal:** every regression test below is a comparison against this artifact, so it must exist first and must be reproducible on a second machine. **Test:** two runs on different hosts agree to the last stored digit.

**Step 1: three prismatic vacancy families.** Raise `iSize` from 8 to 16, add  $(v, a_1), (v, a_2), (v, a_3)$  with their own densities and contents, and give them the nucleation current  $J_{vL}^{a_k} = w_k^v G_{avL} / n_{avL}^{\text{nuc}}$  of Eq. (100). **Regression:**  $\varepsilon_{avL} = 0$  reproduces step 0 bit-for-bit. **Validation goals:** (i) at zero deviatoric stress the three variants carry exactly 1/3 of their character's population each, to round-off — this is the symmetry the variant resolution must not break; (ii) summing the three recovers the lumped prismatic population of step 0 term by term at  $\varepsilon_{avL} = 0$ ; (iii) under a uniaxial stress the variant weights Eq. (77) sum to unity and move in *opposite* directions for the two characters.

**Step 2: character-resolved capture.** Insert the factor  $X_{s,m}$  of Eq. (68), taken from the thermal-drift capture radii. **Regression:**  $\chi = 1$  reproduces step 1 bit-for-bit. **Validation goal:** the coexistence window opens. Sweep the mobile flux ratio  $x = \Phi_V / \Phi_I$  and confirm a finite interval over which  $\dot{c}_{iL}^{a_k} > 0$  and  $\dot{c}_{vL}^{a_k} > 0$  *simultaneously*, of logarithmic width  $\ln \chi$  as Section 1.3 predicts. A window that does not appear, or whose width does not track  $\ln \chi$ , means the factor has been applied to the wrong index.

## D.2 Phase II — thermal channels

**Step 3: peripheral emission replaces the lifetimes.** Delete  $\tau_{avL}$  and  $\tau_{cvL}$  and every term proportional to them. Implement  $\partial E_k / \partial m$  from Eq. (108) using the  $\langle K \rangle_k$  of Table 1, the binding energy Eq. (109), the equilibrium concentration Eq. (111) and the net form Eq. (112). **Regression:** *one was manufactured, and the claim that none exists was wrong.* This step removes a channel rather than generalizing one, so no value of any *existing* parameter recovers its predecessor — but keeping the deleted channel behind a switch does. `emission_model` selects the lifetimes (0) or the computed binding energy (1), and 0 reproduces step 2 bit-for-bit, verified on 192 states. The step is still placed early, because the switch is scaffolding to be removed at the refit and not a permanent alternative; but it is no longer irreversible while it is being tested. **Validation goals:** (i) *detailed balance* — setting  $c^v = c_{sk}^{v,\text{eq}}(\bar{m}^k, \sigma)$  makes the net vacancy exchange of Eq. (112) vanish to machine precision, for both characters, at arbitrary size and stress. This is the sharpest available test that emission and absorption are one mechanism and not two; (ii) *the bulk limit* —  $E_b^{v,k} \rightarrow E_v^f$  and  $c_{sk}^{v,\text{eq}} \rightarrow c_\infty^{v,\text{eq}}$  as  $R \rightarrow \infty$ , from both characters, which checks the sign of  $\varsigma_s(v)$  in Eq. (109); (iii) *the stress-free anneal* — with the source off, vacancy loops shrink, interstitial loops creep slowly downward, and both densities are exactly constant; (iv) *the stressed anneal* — above the threshold  $\sigma_k^*$  of Eq. (115) the interstitial

families grow with no interstitial supply, and the measured threshold agrees with the closed form to the accuracy of  $\langle K \rangle_k$ .

**What running it measured.** All four goals hold. Detailed balance is the sharpest and it is exact to the resolution of the probe: reading the solver’s own  $c_k^{v,\text{eq}}$  back out of it — by bisecting for the  $c^v$  at which a family’s content rate changes sign, so that nothing is instrumented — and comparing against Eq. (111) gives  $\Delta E_b$  of +0.000, +0.019, −0.000, +0.000 and −0.005 meV across both characters and  $R = 5\text{--}25$  nm. The bulk limit is approached from *below* by both vacancy families and from *above* by the interstitial one, which is what checks the sign of  $\varsigma_s(v)$ ; the stress-free anneal leaves every density *exactly* constant, as it must once the lifetimes are the only channel that removed loops; and the climb threshold reproduces  $\Sigma^* = (\partial E / \partial m) / \Omega = 615.8$  MPa, the family shrinking below it and growing above with no interstitial supply at all.

**And a consequence of this step that is not a test failure but is easy to mistake for one.** Goal (iii) is stated as a virtue — every density exactly constant — and it is, as a test. In a *driven* run it is a gap: **after step 3 no vacancy loop family has any direct number sink at all.** The lifetimes were it; peripheral emission replaces them on *content* and never on number, because emission moves vacancies out of a loop and does not remove the loop. What is left is coalescence.

Measured on one interior node of the 200 nm case, integrated  $0 \rightarrow 10$  dpa with the mobile species frozen, changing one switch at a time:

|                              | $N_{(a)}$             | $N_{(c)}$             | ratio |
|------------------------------|-----------------------|-----------------------|-------|
| step 2 (emission_model= 0)   | $1.04 \times 10^{-7}$ | $1.18 \times 10^{-8}$ | 8.83  |
| step 3 (emission on)         | $5.51 \times 10^{-7}$ | $5.45 \times 10^{-7}$ | 1.01  |
| steps 4–6 (chain on as well) | $2.13 \times 10^{-7}$ | $1.14 \times 10^{-6}$ | 0.19  |

so step 3 alone raises  $N_{(c)}$  by  $46\times$  and inverts a ratio that should be  $\sim 10$ . **The basal chain is a minor contributor**, and  $\varepsilon_{\text{sfp}}$  is almost irrelevant — two decades of it move  $N_{(c)}$  by  $1.5\times$ .

**Coalescence is working, and it is doing the whole job — but only after this step.** Switching it off raises  $N_{(a)}$  by  $371\times$  and  $N_{(c)}$  by  $116\times$ ; multiplying every coefficient by 10 lowers them by  $7.5\times$  and  $2.8\times$ . Resolved per family, and with the emission switch as the second variable, it says something sharper — the factor by which switching coalescence off raises each density:

|                   | $c_f$         | $a_1$ (i)   | $a_1^v$ (v)  |
|-------------------|---------------|-------------|--------------|
| emission_model= 0 | $1.02\times$  | $806\times$ | $1.01\times$ |
| emission_model= 1 | $152.9\times$ | $807\times$ | $91.9\times$ |

**Before this step, coalescence did essentially nothing to a vacancy loop family.** Not because the channel was missing — it has always been there, on every loop family — but because  $\tau_{cvL}$  removed the loops first and the population never built up enough to overlap. The interstitial family, which never carried a lifetime, is untouched by the switch. **So the reason the basal coefficients are small is that the fit never had to make them large**, and the  $\sim 10^3$  the table below asks for is not a correction to a fitted number so much as the first time that number has had to carry the channel at all. The problem is not that the channel is absent but that its  $\langle c \rangle$  coefficients are the wrong size for the job it has just inherited:  $c_{LL}$  and  $c_{LN}$  read 0.162 and 2.96 for the basal families against 121 and 1131 for the prismatic ones, roughly  $700\times$  smaller, because they were fitted in a model where  $\tau_{cvL}$  removed the basal loops and they did not have to.

**This gives the refit a testable prediction rather than a free parameter.** Scaling the two basal coefficients alone —  $N_{(a)}$  is insensitive to them to five digits, so the families decouple:

| scale on $c_{LL}, c_{LN}$ (basal) | 1    | 10   | 100  | 1000 |
|-----------------------------------|------|------|------|------|
| $N_{(a)} / N_{(c)}$               | 0.19 | 0.53 | 1.37 | 5.92 |

Recovering the step-2 ratio needs a factor of order  $10^3$ , which lands the basal coefficients at  $\sim 160$  and  $\sim 2960$  — within about 30% of the prismatic values. That is a coincidence worth taking seriously: it says the basal coefficients were suppressed by exactly the amount  $\tau_{cvL}$  was supplying, and that the geometric channel does not in fact distinguish the two habits by much.



The refit has now been run against the experimental targets, and the sink is physical. At 573 K and 24.5 dpa ( $N = 8.0 \times 10^{20} \text{ m}^{-3}$ ,  $d = 95 \text{ nm}$ ), scaling both basal coefficients together:

| scale              | 1     | 30    | 100   | 1000        | $10^4$ |
|--------------------|-------|-------|-------|-------------|--------|
| $N/N_{\text{exp}}$ | 49.1  | 12.0  | 6.9   | <b>2.0</b>  | 0.35   |
| $d/d_{\text{exp}}$ | 0.57  | 0.24  | 0.18  | <b>0.10</b> | 0.08   |
| $J_C$              | 1.617 | 0.925 | 0.737 | 0.495       | 0.597  |

The density prediction is confirmed and it is not a fix. A factor  $10^3$  does bring  $N$  from  $49\times$  experiment to  $2\times$  — but  $d$  collapses from  $0.57\times$  to  $0.10\times$ . Coalescence removes loops *faster than they grow*, so the survivors are young; it cannot buy fewer loops without also buying smaller ones, and the measurements want fewer *and larger*. An unconstrained two-lever search stops at  $7\times$  and  $4\times$  ( $J$   $0.779 \rightarrow 0.588$ ) because the total objective is nearly flat over three decades: the density gain is paid for in diameter.

Nor can the dissolution current buy it back. At the  $10^3$  coalescence, sweeping  $m_{\text{min}}$  over 1–100 moves the basal family not at all —  $N/N_{\text{exp}}$   $2.02 \rightarrow 2.00$ ,  $d$   $9.8 \rightarrow 9.7 \text{ nm}$ ,  $J_C$  flat at 0.494. It is a strong lever on the *prismatic* families — at that same coalescence,  $J_A$  falls  $0.370 \rightarrow 0.272$  as  $m_{\text{min}}$  goes  $1 \rightarrow 30$  — and none on the basal one, because the flux through the floor is set by the log-normal’s lower tail rather than by loops actually reaching zero size.

A search over the new levers confirms it, with  $m_{\text{min}}$  and  $\chi$  added to the candidate set:

| lever set                                                | $J$           | $J_A$  | $J_C$  |
|----------------------------------------------------------|---------------|--------|--------|
| baseline, unrefitted                                     | 0.7790        | 0.4814 | 1.6166 |
| $\{m_{\text{min}}\}$                                     | 0.8794        | 0.4271 | 3.5063 |
| $\{m_{\text{min}}, \chi\}$                               | 0.8719        | 0.4230 | 3.4672 |
| $\{m_{\text{min}}, \chi, c_{LL}, c_{LN}\}$               | <b>0.3683</b> | 0.2816 | 0.5331 |
| $+\{Z_{v,c}, \varepsilon_{vL}\}$                         | 0.5625        | 0.3456 | 1.4000 |
| hand-picked: coal. $\times 10^3$ , $m_{\text{min}} = 30$ | <b>0.3264</b> | 0.2720 | 0.4944 |

$d_c$  stays at  $0.07\text{--}0.11\times$  experiment in *every* fit while  $N_c$  is brought to  $1.5\text{--}7\times$ : the density can always be bought and the diameter never follows.  $\chi$  comes out inert (0.9887, 0.9987 — it sits at 1), and  $Z_{v,c}$  and  $\varepsilon_{vL}$  go to their bounds, so neither is a lever here. **Two of these rows also say the optimizer is not to be trusted on this landscape:** a superset scored *worse* than its subset, which cannot happen at a genuine optimum, and a hand-picked corner beat the fitted one. The numbers are exploratory; the statement about  $d_c$  is not, because it holds across all of them.

So the missing channel is physical: something that removes vacancy loop NUMBER without shortening the survivors’ lives. That is what  $\tau_{cvL}$  did, and it is what neither channel replacing it does — coalescence shortens lives by construction, and peripheral emission shrinks a loop continuously and never removes one. Equation (99)’s terminal  $c_p$  is one place to look; the more likely one is that a loop shrinking under emission needs a *disappearance* channel, which the floor current — tied to a distribution tail rather than to a vanishing radius — does not supply.

One caution the refit also produced.  $J_A$  moved with the basal coefficients,  $0.481 \rightarrow 0.358$ , so the two families do *not* decouple here. The claim that they do came from a frozen-mobile probe at a single node; in the 0-D the vacancy pool is free and shared, so removing  $\langle c \rangle$ s returns vacancies that change the  $\langle a \rangle$ s. Any statement of the form “this lever touches only that family” must be re-checked with the mobile species free.

Two things about the probe, since the first two versions of it were wrong in ways that looked like physics. A one-second window moves the content by  $\sim 10^{-12}$  of itself, below what the solver’s printed output resolves, so the rate reads as *exactly* zero and the bisection returns whichever bracket end it started from — to the digit, which reads as a confident answer rather than a failed measurement. And  $C_i$ ,  $C_{2i}$ ,  $C_{3i}$  cannot be set below the concentration floor, so they sit at it and keep absorbing; the crossing then lands at  $c^{v,\text{eq}} + \Phi_I/(\text{vacancy coefficient})$  rather than at  $c^{v,\text{eq}}$ , biased high by 19% at  $R = 5 \text{ nm}$  and 3.5% at  $25 \text{ nm}$  — *shrinking with  $R$  exactly as an additive contaminant should*, while the two vacancy families were already right to  $6 \times 10^{-4}$ . That asymmetry between the characters is what identified it.

Goal (iii) now has numbers to hit rather than signs. Section 4.3 integrates that anneal in closed form and Section 4.4 repeats it as a nodal solve of Eq. (183), so a single-loop anneal run

against the implemented emission must reproduce Table 11 and not merely move in the right direction. Two corrections come with it and belong to this step rather than to a later one. **First**, the capture efficiency the mean-field growth law uses is wrong in the isolated limit by 9.1% for the basal families and 3.2% for the prismatic ones (Table 12, last column), and the correction is a *prefactor*  $f = 1.094$  on  $c_f$ , 0.970 on the  $\langle a \rangle$  families — not a core radius, and it has *opposite sign for the two habits*, so it cannot be absorbed into a single fitted constant. **Second**, the emission term now in the binary is size-independent: `ClusterDynamicsFEM.cpp:748` evaluates  $S_k \bar{D}^v e^{-E_b/k_B T}$  at a constant  $E_b$ , which is the “constant surrogate” this step exists to remove, and it is a pure loss — it has no  $-c^v$  partner, so it does *not* vanish at  $c^v = c_{sk}^{v,eq}$ . Goal (i) is therefore a test the present code fails by construction, which is what makes it worth running first.

**Step 4: the basal chain.** Add the pyramid  $c_0$  as a species with its own cascade yield  $\varepsilon_{sfp}$ , dissolution lifetime  $\tau_{sfp}$ , compact sink form Eq. (74) and volumetric geometry Eq. (66); add the two activated transfers Eq. (80). Remove the morphology sigmoid from every loop family,  $S^k \equiv 1$  (Section 2.11). **Regression:**  $\varepsilon_{sfp} = 0$  with  $\varepsilon_{cvL}$  restored and the sigmoid re-enabled reproduces step 3 bit-for-bit. **Validation goals:** (i) *conservation across the chain* — Eq. (128) closes for arbitrary  $\nu_{col}$ ,  $\nu_{uf}$ ,  $\tau_{sfp}$ , including the three degenerate limits named in Section 2.10; (ii) *the branching ratio* — the measured steady pyramid density matches the closed form Eq. (81) and the fraction converting matches  $f_{col}$ ; (iii) *incubation* — the dose at which  $c_f$  first becomes measurable shifts with  $E_{col}$  in the direction Eq. (82) predicts, and the  $c$ -axis strain Eq. (199) lags the basal vacancy storage by the residence time. This third goal is the point of the whole step and is the one to report.

### D.3 Phase III — the size distribution

**Step 5: the second moment.** Add the third field  $q_{sL}^k$  per family, Eq. (120), and the log-normal closure Eqs. (123) and (124). Replace the smoothstep Eq. (75) by the floor current Eq. (127), and the scalar transfer gates by the moment-weighted fractions Eq. (126). Apply  $\Delta_k^{-1/8}$  wherever Eq. (125) says it belongs. **Regression:**  $\Delta_k = 1$  reproduces step 4 for every term, without setting any rate to zero. **Validation goals:** (i) *the closure identity* — Eq. (124) checked against direct numerical quadrature of Eq. (122) at  $j = \frac{1}{2}, 1, \frac{3}{2}, 2, 3$  over a range of  $\Delta_k$ , to a relative tolerance of  $10^{-12}$ ; (ii) *pure spreading* — with  $\Sigma_k^{net}$  forced to zero,  $\bar{m}_{sL}^k$  and  $n_{sL}^k$  are constant while  $\Delta_k$  grows at the rate Eq. (119) gives; (iii) *the annealing signature* — a pure anneal now produces a *falling* density together with a *rising* surviving mean size, which is the behavior the two-moment model provably could not produce and is the reason for the step; (iv) *the floor current vanishes correctly* —  $\Phi_{min}^k \rightarrow 0$  exponentially as  $\Delta_k \rightarrow 1$  at fixed  $\bar{m}_{sL}^k > m_{min}$ .

**Goal (iii) must be tested on a mixture that can produce the signature, and against one that cannot.** Section 4.5 shows the signature is a property of the *mixture* rather than of the model: it appears only where Eq. (194) has a root, and Eq. (195) decides whether it does. On the reference microstructure it does not —  $\bar{c}_v = 1.46$  against the faulted basal floor 2.53 — so the whole  $\langle c \rangle$  distribution sits on the shrinking side and a *correct* implementation returns a falling density with a *falling* mean size. Testing goal (iii) on that mixture alone would reject a working step 5; testing it on a single family alone would accept a step 5 that ripens unconditionally. The pair of cases is specified as A1/A2 of Appendix D.6, and the reason the pair is needed rather than either case is that the two differ only in what *else* is in the cell.

**Six things executing step 5 corrected, and one measurement.** The first is the largest and is recorded where it belongs, beside Eq. (120): **that equation omits the coalescence and annealing channels the other two moments carry**, and coalescence is 98% of the loop-number loss. Written the natural-looking way — debiting  $\dot{q}$  by  $\langle m^2 \rangle$  per departing loop — the sign is inverted, because a coalescing loop *merges* rather than leaving, so its defects stay and the second moment must *rise*. With the inverted sign,  $q$  crosses zero at 1 dpa on the demonstration march and is negative on 40% of the  $c_p$  nodes by 10 dpa. Both channels are closed shape-preserving, Eq. (121).

The second is a sign in this document too. Equation (127) is written  $D \partial_m f + (\partial_m D - F)f$ , which is *minus* the Fokker–Planck flux in  $+m$  — that is, it is *already* the outflow through the floor and must be subtracted from the moments as it stands. Reading it as the flux and negating it removes loops

only when they are *arriving* from below, where there is nothing to arrive from, so  $\Phi_{\min}^k$  measured identically zero at every  $\Delta_k$  and goal (iv) passed for the wrong reason.

The third is that goal (iv)’s *exponentially* is literal rather than rhetorical, and a test has to be written for it. At  $\Delta_k = 1.001$  the floor sits 94 standard deviations into the lower tail,  $\exp(-94^2/2)$  underflows, and the current is exactly 0.0 in double precision — so a criterion demanding strict monotonicity in  $\Delta_k$  fails on  $0 < 0$ . The measured sequence over  $\Delta_k = 1.001, 1.05, 1.5, 4$  is 0, 0,  $5.0 \times 10^{-27}$ ,  $4.8 \times 10^{-24}$ , and *non-decreasing, vanishing at the left end* is the statement the formulation actually makes.

The fourth is a spelling. Every family count in the 3-D code was written as `iSize/2`, correct only while there were two fields per family; the third moment makes it 13 by integer division and every per-family loop then runs off the end of its array. `iFam` and `iMom` are now named separately and `iSize` is derived from them. The block is laid out moment-major — nine  $n$ , nine  $c$ , nine  $q$  — so the first eighteen components stay where they were and an `evl` written before this step widens into it with  $q = 0$ , which reads back as  $\Delta_k = 1$ : the monodisperse limit those runs were computed in.

The fifth is that  $\Delta_k$  is *not*  $qn/c^2$  on the 3-D side. The 0-D carries all three moments per atom, so the ratio is unit-free there; the CD block carries the number per  $b^3$  and the content as an atom fraction, which is why MoDELlib spells the mean size  $CI/N/\omega$ . Exactly one factor of  $\omega$  is therefore needed, and without it the dispersion — and with it the  $\Delta_k^{-1/8}$  correction, the one place the second moment enters the fast solve at all — is a different number on the two sides of the coupling.

The sixth is that three of the validation probes were under-driven, each in a way that read as a failure of the step. Vacancies left at the concentration floor give no traffic at all, so every current measures zero; a prismatic family used for the spreading test has an equilibrium concentration four decades below the basal one, so  $\Delta_k$  moves in the ninth digit; and driving the annealing probe *below* equilibrium shrinks every loop, so the mean falls for a reason that has nothing to do with the distribution ( $400.0 \rightarrow 398.46$ , density *and* mean falling, which is not the signature). Putting the matrix at the family’s own  $c_k^{v,eq}$  — which is what step 3 bought — isolates it, and the signature then reads  $n : 1.0000 \times 10^{-10} \rightarrow 9.9997 \times 10^{-11}$  with  $\bar{m} : 400.0000 \rightarrow 400.0088$ .

The measurement is the gates. Equation (126) is checked twice over: the closed form against direct quadrature of the log-normal, to  $10^{-9}$  or better at every  $j$ ; and the C++ *operationally*, as the fraction of each moment a short transfer actually removes, inverted through  $1 - e^{-\nu\Phi dt}$  so the linearization bias does not survive. At  $\bar{m} = 400$ ,  $\Delta = 1.6$ ,  $m_\theta = 800$ :

$$\Phi^{(0)} = 0.087894, \quad \Phi^{(1)} = 0.251980, \quad \Phi^{(2)} = 0.506900,$$

solver against closed form to  $3 \times 10^{-5}$ , and the transferred population has mean size 1146.7 against the family’s 400 — a factor of 2.87. The source mean therefore *falls* although nothing shrank, which is the whole content of  $\Phi^{(2)} > \Phi^{(1)} > \Phi^{(0)}$  and is what one rate multiplying all three moments could not produce.

**The step in a coupled march.** The demonstration case is a 200 nm cube at 573 K and  $10^{-7}$  dpa/s, 47 828 CD nodes, carrying all nine families and all three moments — 38 components per quadrature point on the 0-D side and 27 in the CD block, the first march in which the size distribution exists rather than being assumed. **Its basal-chain rates are placeholders:** nothing in the material file or the workbook supplies  $\varepsilon_{\text{sfp}}$ ,  $\tau_{\text{sfp}}$ ,  $\nu_{\text{col}}$  or  $\nu_{\text{uf}}$ , and they were chosen on the single criterion that the transfers resolve on the run’s own timescale. No loop density should be read off it. What it does show is that every mechanism the last five steps added is live together, and two of them are visible in the state itself:

| family       | $n$ [1/atom]           | $c$ [1/atom]           | $\bar{m}$ | $\Delta_k$ |
|--------------|------------------------|------------------------|-----------|------------|
| $c_f$        | $2.150 \times 10^{-6}$ | $5.909 \times 10^{-3}$ | 2748      | 9.38       |
| $a_{1..3}$   | $4.791 \times 10^{-8}$ | $6.403 \times 10^{-5}$ | 1336      | 2.95       |
| $a_{1..3}^v$ | $3.739 \times 10^{-8}$ | $4.839 \times 10^{-6}$ | 129.4     | 2.64       |
| $c_p$        | $2.756 \times 10^{-6}$ | $1.003 \times 10^{-1}$ | 36397     | 81.9       |
| $c_0$        | $3.911 \times 10^{-7}$ | $6.807 \times 10^{-5}$ | 174.1     | 2.07       |

interior mean at 10 dpa, the innermost quartile by distance to a face. Every  $\Delta_k$  is strictly greater than 1 — no family has collapsed back to the monodisperse limit — and the bound holds far more strongly than at the means: over all 47 828 nodes, all nine doses and all nine families, some  $3.4 \times 10^6$

triples, the *minimum*  $\Delta_k$  is 1.000000000. That is the realizability statement of Eq. (123) surviving a real march, and it is what the corrected coalescence closure buys: written the other way,  $q$  is negative on 40% of the  $c_p$  nodes at this dose.

The prismatic variants are the clean case. All three carry the same field, as they must at zero applied stress, and the interstitial and vacancy variants sit at  $\bar{m} = 1336$  and 129 with  $\Delta$  of 2.95 and 2.64 — relative widths near 1.4 and 1.3, which is a genuinely broad distribution and not a perturbation about a mean.

**The basal families say something else, and it is about the chain rather than about the numbers.**  $c_p$  ends the march holding almost all of the stored basal content, at  $\bar{m} = 36397$  and  $\Delta = 82$ . That is Eq. (99) doing what it says:  $c_p$  is terminal, coalescence is its only number loss, and with  $\tau_{\text{uf}} = 3 \times 10^6$  s against a  $10^8$  s run — thirty  $e$ -foldings, and a placeholder — everything upstream of it arrives and stays. **No basal loop density should be read off this run.** What the run does establish is that the mechanism is live end to end, and that the open question identified after Eq. (99) is not academic.

## D.4 Phase IV — the discrete handoff

Phase IV is the one whose status has changed most since this plan was first written, and the steps below are marked accordingly: much of it is built and measured, and what remains is smaller and more specific than the phase heading suggests.

**Step 6: conversion with a drawn size and an elliptical outline.** *Done, except the export.* The sizes are drawn from Eq. (142) and rescaled by Eq. (143); the ellipse of Eqs. (145) to (148) is built, and Eq. (149) tests the refusal on its *vertices*, so the shape reaches the geometry that decides what the domain keeps.

**It does not reach dislocation dynamics.** `aLoopIndividualSpecification` carries one `loopRadii` and one `loopSides` per loop and no aspect ratio, so `write_microstructure` can only emit a circle of the equal-area radius. Exporting the shape needs either a new field on that specification — an aspect and a major-axis direction, which `aLoopGenerator` would then have to honour — or writing explicit vertices instead of a radius. Until one of those exists the discrete population is elliptical everywhere except in the solver, which is the one place the shape would change an answer.

*What follows is the specification as it was written.* The construction of Section 3 is in place — hull-restricted nodal volumes Eq. (139), counts and home nodes Eqs. (140) and (141), the conservation rescale Eq. (143), the refusal test Eq. (149) and the transfer ledger — but it runs monodisperse and circular, because quantile sampling Eq. (142) needs the third moment of step 5 and the ellipse Eqs. (145), (147) and (148) has not been wired into the export. **Regression:**  $e \equiv 0$  and  $\Delta_k = 1$  reproduce the present export, which is what it does today. **Validation goals:** (i) *the ledger* — Table 6 row by row: defects exact to  $10^{-16}$ , number to  $\frac{1}{2}$ , mean size exact after the content pass, and the dispersion to the *tail-truncation* bound of Eq. (144) —  $\sim N_j^{-0.6}$  and  $s_k$ -dependent, **not** the  $O(N_j^{-2})$  an earlier revision of this plan asserted; (ii) *the round trip* — re-binning the exported loops back into  $(n, c, q)$  recovers the fields to the sampling error, the single strongest test because it exercises every stage at once; (iii) *ellipticity accounting* — drawn line length against  $\rho_{sL}^k$  in 1.002–1.049, tracking Eq. (67) loop by loop; (iv) *refusal* — Eq. (149) agrees with the generator on every loop and the refused share is reported in *defects*.

**Step 7: the trigger.** *Implemented.* Section 3.8: the Avrami gate Eq. (151) on interior nodes, the hold, the log-dose interpolation, firing at the next fast-solve boundary and before it, and the feasibility veto when  $f_{\text{kept}} = 0$ . **Regression:** with the transfer disarmed the detector is reporting-only and every previous result stands unchanged, which is how Table 7 was measured on runs that predate it. **Validation goals:** (i) the identity Eq. (150) holds numerically, so the trigger and the crowding selector are one measure — verified to four digits; (ii)  $d_{\text{coarsen}}$  is quoted with its  $\phi^*$ , since Table 7 moves it by a factor 4–5 across the plausible range; (iii) the five copies of  $\phi^*$  agree — they did not until recently, and the detector reported on one value while the march would have fired on another. **Remaining:** nothing in the mechanism. The open question is a modeling one, Eq. (205): on the reference geometry  $d_{\text{feas}} \simeq d_{\text{coarsen}}$ , so the domain must be sized from the trigger rather than chosen first.

**Step 8: the climb constructions.** *Split.* The boundary-integral solve of Section 3.16 is implemented and converged — the kernel Eqs. (163) and (165), the Galerkin system Eq. (166), and the screened cutoff Eq. (170) at  $n_L = 4$ , which is the smallest value meeting its own flatness requirement. The ellipse ROM of Section 3.15 exists as a module and **is not wired into the march**; it has so far been used only *post hoc*. **Validation goals:** (i) the cutoff sweep is flat in the climb strain rate, which is the only quantity truncation error shows up in — done, 0.25% at  $n_L = 4$ ; (ii) *the closure* — run the  $\langle a \rangle$  family nodally and through the ROM on one case and measure which of Eq. (160) is right — **done, and it was the isolated-loop anneal that did it**: see below; (iii) the ROM’s invariant Eq. (156) agrees with the export’s by construction, not by calibration.

**Goal (ii) is settled and it settled two further things with it.** The anneal of Section 4.4 is a nodal solve of exactly the case Eq. (160) is ambiguous about — a circular prismatic loop under a uniform drive, where only the diffusion tensor can make the two axis endpoints differ — so it reads the exponent off directly:  $q = 2.10\text{--}2.73$ , rising with  $R$  toward the geometric mean’s 3 and *independent of temperature to three decimals*, which retires the normal projection’s  $q = 6$  and with it the factor-of-1.31 ambiguity. It also forces the calibration Eq. (189):  $L_{\text{cap}} = \Omega Z_{sk,v}/b^2 Z_k^{\text{iso}} \simeq 0.16\text{--}0.23$  nm, against the screening length  $\simeq 59$  nm the module defaults to, **so wiring the ROM into the march at its present default would run the climb 260–370× too slow**. That is now a blocking defect on this step rather than a documented default. **And it bounds where the ROM may be used at all:** Table 14 shows the ROM carries no shape-dependent capture efficiency — the solve’s  $v_A/v_B$  falls as  $\varrho_\ell^{-0.316}$  where the ROM’s is constant — so the two elongation drives Eq. (188) diverge rather than offset, by  $0.78\times$  at  $\varrho_\ell = 1$  and  $0.32\times$  at  $\varrho_\ell = 3$ . The ROM is sound for the near-circular  $\langle a \rangle$  population it was built for and **must not be used for a shape once  $\varrho_\ell$  exceeds about 1.5**; an anneal drives  $\varrho_\ell$  up, so Appendix D.6 requires  $\varrho_\ell$  to be recorded and the excursion flagged rather than assumed away.

**Step 9: the smeared discrete sink.** *Not implemented.* Equation (172): return a transferred population’s sink strength to the continuum as a nodal field built from where the loops actually are, at kernel width  $\max(h, L_s)$ . This is the missing cell of Table 8 — the regime  $\Theta_k \sim 1$ ,  $\Xi_k \lesssim 2$  that  $\langle c \rangle$  actually occupies — and it is  $O(N_{\text{seg}})$  against the  $O(N_{\text{seg}}^2)$  of Eq. (167). **Regression:** a zero-width kernel with no transferred population reproduces Eq. (38) exactly. **Validation goals:** (i) a transferred family’s contribution to  $k_m^2$  is recovered to the sampling error of the draw; (ii) the boundary shell is no longer compensated uniformly — this is the concrete defect Section 3.17 identifies, refusal being spatial while its compensation is a single scalar; (iii) the mobile field is insensitive to the kernel width above  $L_s$  and sensitive below it, which is the numerical form of the resolution argument.

## D.5 Phase V — observables and mechanics coupling

This phase did not exist in the first version of this plan, because the observables were carried inside the conservation subsection and the question of *which representation computes them* had not been separated out. It is now Section 5.

**Step 10: the continuum observables.** *Partly implemented.* The distortion tensor Eq. (198) and its rate are formed per point from the immobile state; the dispersed-barrier law Eq. (201) and the Taylor conversion Eq. (202) are implemented. **Validation goals:** (i) the trace of Eq. (198) equals the swelling computed independently from the conservation accumulators of Section 2.12 — these are the same atoms counted two ways and must agree; (ii) at zero deviatoric stress Eq. (200) vanishes identically, since the three variants are equally populated — a sharp test that the deviator is taken correctly; (iii)  $\Delta\sigma_y$  scales as  $M$  and not  $1/M$ . That is a regression test on a sign convention this document had inverted.

**Step 11: the discrete observables.** *Implemented, with one gap.*  $\beta^P$  and  $\dot{\beta}^P$  are written natively at every step, and the three hardening routes of Section 5.2 all run. **Validation goals:** (i)  $\text{tr } \dot{\beta}^P$  is zero for pure glide and non-zero for climb, which is the separation the trace is being used for; (ii) route D reproduces route A on a case where both are affordable — the seeding smoke test; (iii)  $\Upsilon_{h(k)}$ , by comparing  $\beta_D^P$  against Eq. (198) on one exported population.

**Goal (iii) has now been run, and it does not do what this plan claimed for it.** It was described as the cheapest unclaimed measurement in the document and as *determining*  $\Upsilon_{h(k)}$ . It is



cheap; it determines nothing. Both tensors rest on the same identity. The conversion gives a loop the area that stores its defects,  $A_\ell|\mathbf{b}|_\ell = m_\ell\Omega$ , and the continuum content of a family is by construction  $\sum_\ell A_\ell(\mathbf{b}\cdot\mathbf{n})/V$ , so for a climb loop —  $\mathbf{b}$  along  $\mathbf{n}$ , which every family here is —

$$\beta_D^P = \frac{1}{V} \sum_\ell A_\ell|\mathbf{b}|_\ell \mathbf{n} \otimes \mathbf{n} = \frac{\Omega}{V} \sum_\ell m_\ell \mathbf{n} \otimes \mathbf{n} = c_{sL}^k \mathbf{n} \otimes \mathbf{n},$$

and the ratio is identically one whatever the sizes are and however they are distributed. Measured: 1.0000000000000000 with *zero* spread across populations from 25 loops of 8000 defects to 4000 loops of 50, and 1.000000000000 on both habits of a real 200 nm export.

**What it is instead is a stringent end-to-end check on the handoff**, and worth keeping as one. Deliberately breaking the Burgers magnitude on one side moves the ratio by exactly the injected factor —  $\sqrt{2} \rightarrow 1.414213562373$ ,  $\frac{1}{2} \rightarrow 0.500000000000$  — which is precisely the class of error the  $\langle c \rangle$  Burgers correction removed, where cluster dynamics and dislocation dynamics disagreed by  $\sqrt{2}$  on one family and nothing complained. The ratio is one only when the two sides genuinely share  $|\mathbf{b}|$  and  $\Omega$ .

**A physical  $\Upsilon_{h(k)} \neq 1$  is not available from this comparison at all.** The formulation defines it as the ratio of the defect’s relaxation volume to  $\Omega$ , and the relaxation volume enters neither tensor: the material file carries `immobileSpeciesRelRelaxVol` (0.405 basal, 1.2 prismatic interstitial) and no observable reads it. Measuring  $\Upsilon$  means putting that number into one side and not the other, which is a modelling decision and not a measurement. **So the  $\Upsilon_{h(k)}$  that route 3 lists among the coefficients it calibrates is still an assumption**, and the route-2 budget buys  $\alpha_k$  alone.

**Step 12: the hybrid route.** *Partly implemented.* Route 3 of [Section 5.3](#): calibrate  $\alpha_k$  and  $\Upsilon_{h(k)}$  on small discrete runs below  $d_{\text{feas}}$ , then score *both* pieces of a split family with the same law and compose by [Table 17](#). The  $\alpha_k$  fit exists; the  $\Upsilon_{h(k)}$  measurement does not, and neither does the composition pass. **Validation goals:** (i) [Eq. \(204\)](#) verified numerically — an artificially split population reproduces the unsplit  $\Delta\tau$  to round-off, which is the check that quadrature and not addition is the rule; (ii) no observable uses a pre-transfer continuum field, tested by asserting that the distortion channels change by exactly  $f_{\text{kept}}$  when the transfer is disarmed; (iii)  $\rho_N$  is not advanced by [Eq. \(118\)](#) for a share that has gone discrete — the third silent failure mode of [Section 5.2](#); (iv)  $\alpha_k$  fitted at two doses agrees, or if it does not,  $\alpha_k(d_k)$  is carried instead. [Section 5.3](#) argues it will not agree and that the route-2 budget should be spent on a spread of small cases for that reason.

## D.6 Phase VI — the thermal anneal

[Section 4](#) is the part of the formulation with the widest gap between what is derived and what a march can run. Everything it computes — the binding energies of [Table 9](#), the lifetimes of [Table 11](#), the boundary-integral anneal of [Section 4.4](#), the remote concentration [Eq. \(191\)](#) and the distribution anneal of [Fig. 8](#) — runs in four standalone modules under `dislocluster_code.studies`, **none of which is imported anywhere outside that package**. They evaluate this section’s equations on a state; they do not advance one. [Section 4.6](#) names three things the march-level path needs and records that two of them belong here; steps 14 and 16 below are those two, and step 13 is a fourth that only became visible on reading the assembled system.

**This phase is placed before the refit deliberately.** A post-irradiation anneal is the most direct measurement of the binding and fault energies, and [Section 4.6](#) shows it is the only regime in which the floor current [Eq. \(127\)](#) is the *dominant* channel rather than a correction — so it is the calibration experiment for step 5, and step 17 should not be attempted before it has run.

**Step 13: emission into the mobile field.** *Not implemented, and the gap is structural.* [Equation \(27\)](#) is sourced by  $\mathbf{S}_{vL}$ , [Eq. \(113\)](#). In the binary it is absent: `solveMobileClusters` takes the single load vector `cascadeGlobalProduction` =  $\int G$ , and the three linear forms spliced into the Newton residual are each proportional to `mobileClusters`, so **the only term in the assembled mobile problem that does not vanish with  $\mathbf{c}_M$  is the cascade source**. The immobile population reaches the mobile field through `ImmobileSinks` alone, a matrix multiplying the unknown — that is  $-k_m^2 c_m$ , a pure removal. The emission that does exist, `ClusterDynamicsFEM.cpp:748`, subtracts content from the loop and returns it nowhere. **Consequence: switch the source off today and the pool drains to the Dirichlet values instead of settling at [Eq. \(191\)](#)**, so every loop shrinks at the



**maximum possible rate and the emitted vacancies are destroyed.** Implement Eq. (113) as a load vector assembled from the same  $Z_{sk,v}\rho_{sL}^k$  that `ImmobileSinks` already forms, so that the pair is  $\Pi_{sL}^k \bar{D}^v Z_{sk,v}(c_{sk}^{v,\text{eq}} - c^v)$  and vanishes at equilibrium *by construction* — assertion 3 at the head of this appendix, applied to a second pair of terms. **Regression:**  $c_{sk}^{v,\text{eq}} = 0$  reproduces step 12 bit-for-bit, the load vanishing while the sink matrix is untouched. **Validation goals:** (i) *detailed balance in the assembled system*, which is a stronger statement than step 3’s goal (i) on the nodal ODEs — on a fully periodic cell at  $G = 0$  with one family in a uniform state, the converged mobile field must equal that family’s own  $c_k^{v,\text{eq}}$  at every node; (ii) **Eq. (191) must come out of the solver**, and this is exact rather than approximate: periodicity kills  $\oint \mathbf{J}$ , so volume-averaging the steady equation gives  $\sum_k S_k c_k^{v,\text{eq}} = \langle c^v \rangle \sum_k S_k$ , and Eq. (191)’s denominator is  $k_v^2$  identically. A closed form the study module evaluates in microseconds is therefore a regression test on a 3-D finite-element solve, and it is the cheapest available proof that the zero-dimensional balance and the FEM are one construction; (iii) *irradiation is untouched* — under irradiation  $c^v \gg c^{v,\text{eq}}$ , so the new source is a small correction, and its size against the step 0 reference trajectory must be *measured* and not assumed negligible.

**Step 14: the anneal as a run mode.** *Not implemented.* From Section 4.6:  $G = 0$ , and a schedule specified in **time** rather than dose, since  $\Delta t = \Delta\gamma/G$  is undefined at  $G = 0$  — a log-spaced grid refined at the start, where the fastest family loses size quickest. `BOUNDARY['periodic_face_ids']` selects which annealing experiment is being simulated and is not a free choice: Dirichlet faces drain the emitted vacancies and the population anneals to extinction (a thin foil or a fine-grained specimen), a fully periodic cell conserves total vacancy content exactly and converges on Ostwald ripening instead (a well-annealed single-crystal interior). **Regression:** with  $G \neq 0$  and a time grid derived from the dose grid, an anneal-mode march reproduces the irradiation march bit-for-bit — **the anneal must be a value of the source term, not a second code path**, which is the only thing that keeps the two from drifting apart. **Validation goals:** (i) *the single-component limit* — one family, no network, Dirichlet faces, and the interior reduces to Eq. (181) exactly, which is what Eq. (193)’s use of  $Z_k^{\text{iso}}$  buys; (ii) *the ledger* Eq. (197), which on a periodic cell is a two-term identity that must close to solver tolerance and is the cheapest regression on the whole annealing path — it is also the check that catches vacancies lost when a loop is deleted at the floor, the step most likely to lose them; (iii) *the cadence runs backwards* — Section 4.6 shows the splitting coefficient *grows* through an anneal because Eq. (113) is exponential in loop size, so the `fem_every` independence study must be done at the *end* of the anneal, where under irradiation it is done at the start; (iv) *the fast solve is being asked a different question* — its drive is now the emission source alone and the field sits near  $c^{v,\text{eq}}$  rather than decades above it, so the positivity floor and the error mode that were settled for irradiation must be re-verified rather than inherited.

**Step 15: annealing a size distribution — the two representations.** *Requires steps 5, 6 and 14.* This is the measurement the whole phase is for. In the continuum a family loses loops through Eq. (127), an integral of the assumed log-normal below  $m_{\min}$  evaluated from two moments; in the discrete representation a loop reaching  $R_{\min}$  is deleted and its defects returned, one loop at a time, by counting. **They are the same physical process computed two entirely different ways, and Section 3.4 makes them agree exactly at  $t = 0$** , so every subsequent divergence is attributable. Table 24 specifies the cases; three points about their design carry more weight than the individual entries.

1. **A1 and A2 must both be run, and they differ only in what else is in the cell.** A1 can produce the ripening signature and A2 provably cannot (Eq. (195):  $\bar{c}_v = 1.46$  against a faulted floor of 2.53). Either case alone gives a wrong verdict on step 5 — A1 alone accepts an implementation that ripens unconditionally, A2 alone rejects a working one.
2. **B3 is the only case that isolates what a mean field cannot carry.** Neighbor re-absorption is a *variance* effect, largest at exactly the separations where the average is least representative, so it must be measured on a controlled pair rather than inferred from a population.
3. **Two defaults are traps, and both are now quantified.**  $L_{\text{cap}} = L_s$  under-reads the late shrinkage by  $L_s/R$ , which runs from 1 to 73 across the reference anneal, and is already 260–370× wrong in the isolated limit by Eq. (189); and the ROM’s shape is validated only to  $\varrho_\ell \lesssim 1.5$  (Table 14) while an anneal drives  $\varrho_\ell$  *up*. Both must be asserted on, not documented.

Table 24: Test cases for annealing a loop size distribution, in both representations. All share one volume, one mesh and one material file; B and C start from the state A leaves at the transfer dose, so the comparisons carry no fitted quantity. “Isolates” is the single thing the case is allowed to be evidence about.

|                                                                         | Cell and population                                                                                                                       | Isolates                                                                                                                             | Pass criterion                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Continuum representation — the two-moment closure of Section 2.9</i> |                                                                                                                                           |                                                                                                                                      |                                                                                                                                                                                                                                                               |
| A1                                                                      | periodic cube; the 0.1 dpa $\langle c \rangle$ family <b>alone</b> , network and $\langle a \rangle$ removed; $\sigma_{\text{In}} = 0.45$ | the closure, nothing else moving: $\bar{c}_v$ is the family’s own mean level, so $R^*$ falls inside the distribution by construction | $n_{sL}^k$ falls and $\bar{R}_{sL}^k$ rises — the signature two moments provably cannot give; content conserved to the floor current                                                                                                                          |
| A2                                                                      | Dirichlet faces; the <b>full</b> reference mixture, nine families plus $\rho_N$                                                           | that the signature is a property of the mixture, not of the model                                                                    | Eq. (194) has <i>no</i> root for $c_f$ ; $n_{sL}^k$ and $\bar{R}_{sL}^k$ both fall; solved $\bar{c}_v$ reproduces Table 15                                                                                                                                    |
| A3                                                                      | periodic cube; $\Sigma_k^{\text{net}}$ forced to zero                                                                                     | the floor current Eq. (127) with growth switched off                                                                                 | $n_{sL}^k$ falls at the computed $\Phi_{\text{min}}^k$ and $\bar{R}_{sL}^k$ rises by re-partitioning alone; $\Phi_{\text{min}}^k \rightarrow 0$ exponentially as $\Delta_k \rightarrow 1$                                                                     |
| <i>Discrete representation — Table 16 on a transferred population</i>   |                                                                                                                                           |                                                                                                                                      |                                                                                                                                                                                                                                                               |
| B1                                                                      | periodic cube; the A1 state transferred at 1 dpa (Section 3), sizes drawn by Eq. (142); Option 1                                          | <b>the closure, measured</b> — surviving loops by counting against $n_{sL}^k(t)$ by Eq. (127)                                        | the two densities agree, or the gap is reported as the closure error it is; $\varrho_\ell$ recorded and any excursion past 1.5 flagged                                                                                                                        |
| B2                                                                      | the same state; Option 2 above $\Theta_k \simeq 0.5$ , Option 1 below                                                                     | the switch of step 16, and the cost of not making it                                                                                 | Option 1-throughout differs from the switched run by more than the draw’s spread, or it does not — either sizes the switch. Reference $\langle c \rangle$ : $\Theta_k = 1.05 \rightarrow 0.5$ at $R = 27.4 \text{ nm}$ , $22.3 \text{ h}$ of $29.5 \text{ h}$ |
| B3                                                                      | two loops at separation $d$ , swept $2R \rightarrow R_{\text{cut}}$ ; Options 1 and 2                                                     | <b>neighbor re-absorption</b> , the one mechanism a monopole cannot carry                                                            | the two agree as $d \rightarrow R_{\text{cut}}$ and diverge as $d \rightarrow 2R$ ; that divergence is the variance $k_m^2$ carries only in the mean                                                                                                          |
| C                                                                       | both, same volume, same $t$ grid                                                                                                          | that the transfer is invertible <i>through</i> an anneal, not only at $t = 0$                                                        | re-binning the discrete population into $(n, c, q)$ recovers the continuum moments to the sampling error at $t = 0$ (step 6), the closure error after                                                                                                         |

**Step 16: the  $\Theta_k$ -triggered switch.** *Not implemented.* Table 16 selects between the two constructions on  $\Theta_k$  evaluated from the discrete population itself, and Eq. (196) is why the switch must be made *during* a run rather than once at the start:  $\Theta_k \rightarrow 0$  as the loops shrink while  $\Xi_k$  barely moves — on the reference  $\langle c \rangle$  family from 1.856 only to 1.501 — so the family crosses one cell boundary of Table 8 and not two. **Regression:** a switch threshold of  $\Theta_k = 0$  never fires and reproduces Option 2 throughout; a threshold of  $\infty$  reproduces Option 1 throughout. Both must be exact, which is what makes the switch auditable. **Validation goals:** (i) the two constructions agree across the switch to better than the discretization error, measured by switching at two different thresholds on one run — a discontinuity at the switch is the failure this goal exists to catch; (ii) **the switch pays for only the last quarter** — three quarters of the reference anneal is spent at  $\Theta_k > 0.5$  because  $\dot{R}$  is smallest when the loop is largest, so a cost argument for the switch must be made on the last quarter alone; (iii) the cutoff Eq. (170) survives the anneal,  $R_{\text{cut}}$  rising from 236 to 292 nm while the loop count falls to zero, **so Option 2’s cost falls monotonically** — and the justification for truncating at all survives with it, since  $\rho_N$  retains two thirds of  $k_m^2$  however completely the loops anneal. **Both statements fail together in a crystal whose network has also recovered**, which is the one case in which

Eq. (170) must be re-derived rather than re-used.

## D.7 Phase VII — calibration and performance

**Step 17: refit.** The 28-parameter set is stale on *five* independent counts, and they compound: the atomic volume correction, the choice of loop model, the  $\langle c \rangle$  Burgers vector, the replacement of the annealing lifetimes by a computed binding energy, and — once step 5 lands — the fact that the model now predicts a dispersion where it previously assumed one. **Goal:** a single refit against measured loop densities *and size distributions*, the latter now being something the model predicts rather than assumes, so the fit has more data to answer to than it did. **Test:** the refitted model reproduces the measured  $\langle a \rangle$  and  $\langle c \rangle$  densities across the dose and temperature range of Appendix A, and its predicted dispersions  $\Delta_k$  fall inside the measured spread rather than merely near the mean.

**Do not refit before steps 12 and 15.**  $\alpha_k$  and  $\Upsilon_{h(k)}$  enter every observable a fit would be scored against, and fitting microstructural parameters against a hardening curve while the constitutive coefficients are themselves free is how a parameter set absorbs an error in a coefficient. Step 15 is the other half of the same argument and it acts on the *data* side: an anneal measures the binding and fault energies more directly than any irradiation curve does, and Table 24’s A-cases are what turn the predicted dispersion from an assumption into something the fit must answer to. Refitting first would spend that leverage on absorbing a closure error.

**Step 18: performance.** The immobile state is 27 components per node after step 5. **Goal:** confirm that the cost argument of Section 6.5 survives the expansion. **Tests:** the slow sweep scales as  $27^3/18^3 = 3.4$  against the two-moment measurement and remains linear in node count; the fast solve is unchanged, since no mobile species was added; the climb solve is bounded by Eq. (170) rather than by the population size; and the end-to-end march time is still dominated by the fast solve. If the last fails, the operator split has stopped being the right decomposition and the fast-solve cadence Section 6.6 should be revisited before anything else is optimized.

## D.8 Summary

Table 25 collects the sequence. Two columns matter more than the others: “regression” is the property that makes a step safe to leave in place, and “status” is what has changed since this plan was first written — twelve of the nineteen steps are wholly or partly built.

**Phase VI is the one phase with nothing built in it**, and that is a statement about the software rather than about the physics: Section 4 is derived, verified against closed forms and reduced to four running modules, and none of it advances a march. The phase is also unusual in the sequence in that its first step, step 13, is a *correction* to a piece of the mobile solve that irradiation never exercised — a source term whose absence is invisible while  $c^v \gg c^{v,\text{eq}}$  and total once the source is switched off.

**Two steps to plan around, for opposite reasons.**

**Step 3** is the only one that cannot be reverted by a parameter, because it removes a channel rather than generalizing one, and everything after it inherits that. Doing it early — while the family set is still small and before the second moment multiplies the state — is deliberate, and its four validation goals are correspondingly sharper than elsewhere: detailed balance and the bulk limit are exact identities, not tolerances, and either one failing localizes the error to a sign.

**Step 11’s third goal** is the opposite case: it is not hard, not irreversible, and not expensive, and it is nonetheless the single measurement that the largest number of downstream numbers depend on. Until  $\Upsilon_{h(k)}$  is measured, every growth and creep number this model produces outside route 2 carries an assumed strain factor, and no amount of refitting elsewhere will reveal it.

Table 25: The implementation sequence. “Regression” is the parameter value at which the step reproduces its predecessor exactly. Status is as of this revision.

| Step | Change                                                                       | Regression                      | Status               |
|------|------------------------------------------------------------------------------|---------------------------------|----------------------|
| 0    | Lock and hash a reference trajectory                                         | —                               | done                 |
| 1    | Three prismatic vacancy families, <code>iSize</code> $8 \rightarrow 16$      | $n_{\text{fam}} = 4$            | <b>done</b>          |
| 2    | Character-resolved capture $X_{s,m}$                                         | $\chi = 1$                      | <b>done</b>          |
| 3    | Peripheral emission replaces $\tau_{avL}, \tau_{cvL}$                        | <code>emission_model</code> = 0 | <b>done</b>          |
| 4    | Basal chain $c_0 \rightarrow c_f \rightarrow c_p$ ; sigmoid removed          | <code>basal_chain</code> = 0    | <b>done</b>          |
| 5    | Second moment $q_{sL}^k$ and log-normal closure                              | <code>moments</code> = 0        | <b>done</b>          |
| 6    | Discrete conversion: drawn sizes, elliptical outlines                        | $e = 0, \Delta_k = 1$           | <b>done</b> / export |
| 7    | The trigger: $\phi^*$ gate, hold, feasibility veto                           | transfer disarmed               | <b>done</b>          |
| 8    | Climb: boundary integral, and the $\langle a \rangle$ ROM                    | no cutoff                       | <b>done</b> / module |
| 9    | Smeared discrete sink $k_{m,D}^2$                                            | no transferred population       | <b>done</b> / field  |
| 10   | Continuum observables $\beta^P, \Delta\tau$                                  | —                               | partial              |
| 11   | Discrete observables; $\Upsilon_{h(k)}$ shown <i>not</i> measurable this way | —                               | <b>done</b>          |
| 12   | Hybrid route: calibrate, then compose                                        | transfer disarmed               | partial              |
| 13   | Emission $\mathbf{S}_{vL}$ into the mobile field                             | $c_{sk}^{v,\text{eq}} = 0$      | —                    |
| 14   | Anneal run mode: $G = 0$ , schedule in time                                  | $G \neq 0$                      | —                    |
| 15   | Size-distribution anneal, both representations (Table 24)                    | —                               | —                    |
| 16   | The $\Theta_k$ -triggered switch between constructions                       | $\Theta_k = 0$ or $\infty$      | —                    |
| 17   | Refit the parameter set                                                      | —                               | —                    |
| 18   | Performance at 27 components per node                                        | —                               | —                    |

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